END OF WELL REPORT

AZADEGAN OIL FIELD

END OF WELL REPORT

AZN-10 (NEAR PAD #50), TYPE: IIF

Drilling Department

APRIL 2024

MSA/OEOC			
	Position	**Name**	**Date & Signature**
Prepared by	Drilling Engineering Department	Masoud Babakhani M.H.Haji Taleb	
Approved by	Project Head of Drilling Engineering	Mostafa Khedri	
	Project Head of Drilling Operation	Piran Farzaneh	
	Project Manager	Rashid Ansari	
PPL			
	Position	**Name**	**Date & Signature**
Check & Approved by	Drilling Operation Manager	Majid Nariman	
	Subsurface Engineering Manager	Mehdi Salimi Mehr	
	Project Manager	Ali Maghzi	
	Subsurface Project Manager	Alireza Hamoudi	

TABLE OF CONTENTS

1. General Information 10

1.1 Field Introduction 10

1.2 AZN-10 Well Location Map 12

2. Well History 13

2.1 Well Informaion 13

2.2 AZN-10 Well History 14

Workover Summary: 15

Dry Test Report: 20

Tie-Back Operation: 21

Tie-Back Dry Test Report: 24

Perforation Operation: 28

Completion Summary 30

2.3 AZN-10 Well Sketch 33

2.4 Bit Record Table 35

2.5 Hole Problem History 36

3. Time Analysis Report 37

3.1 Time vs. Depth Chart 37

3.2 Total Time Break-Down of Operation 38

WorkOver Section 38

Completion Section 39

3.3 NPT Break-Down 40

4. Log Reports 45

5. Bottom Hole Assembly Record 46

6. Casing Information 65

6.1 7" Tie-Back Tally 65

.7 ..... Cement Job & Plug Back Report 67

7.1 7" Tie-Back Cement Job Report 67

Required Material, Equipment & Personnel 69

Actual Operational Summary of Sequence 71

Field Records 72

7.2 Cement Plug Back Report 73

7.3 Operation Instruction 73

Pre job preparation 73

Job description 73

Job procedure 74

7.4 Execution of Operation 78

7.5 Operation Ticket 79

7.6 Job Log & Summary 80

7.7 Job Graph 81

7.8 Pumping Schedule 81

7.9 The Amount of Material Used in the Operation 82

8. Completion Report 83
8.1 Completion Tubing Information 83
8.2 Completion Tools Specification Checklist 83
8.3 Assembling Test Report 87
8.4 Completion Operation Checklist 107
8.5 Completion Sketch & Tally 111
8.6 Completion String Pressure Test 122
9. Wellhead Report 128
9.1 Final Well Status 128
9.2 Wellhead & X-mass tree Specification 129
9.3 Wellhead Schematic 132
9.4 X-mass Tree Specification (Section D) 133
9.5 X-mass Tree Pressure Test Report (Section D – befor install) 134
9.6 X-mass Tree Pressure Test Report (Section D – after install) 134
9.7 Wellhead Related Test Reports 136
10. Well Stimulation & Clean Up Report 140
10.1 Operation Summary 140
10.2 General Information 141
10.3 Well Basic Information 141
10.4 Total Operation Result 143
Executed Operation Flow Chart 143
Daily Operation Summary 147
Sequence of Operation 149
Operation Results 170
10.4.1 170
10.4.2 170
10.4.3 170
10.4.4 170
Summary of Pumping Data 288
Summary of Pumping Operation Results 289
Pumping Operation Results & Plots 291
Flowing / Static Survey Data & Interpretation 322
10.4.1 322
10.4.2 322
10.4.3 322
10.4.4 322
10.4.5 322
10.4.6 322
10.4.7 322
10.4.8 322
10.5 Failure & Problems 342
10.6 Lesson Learned 342

10.7 Appendix 343
Gauges Information 343
Well Testing Raw Data 343
Pumping Raw Data 343
N_2 Lifting Raw Data 343
CT Raw Data 343
Surface Equipment Layout 344
CT & SL BHA Diagram 345
10.8 Onsite QC Test Report of Stimulation Additives & Fluids 346
11. Drilling Fluid Information 347
11.1 Operation Remark 347
11.2 Operation Remark 348
11.3 Drilling Fluid type 349
11.4 Drilling Fluid type 350
11.5 Volume Analysis 353
11.6 Daily volume Breakdown 354
11.7 Loss Distribution 357
11.8 Solid Control Disscusion 360
11.9 Completion Fluid Information 360
12. List of Contractors 361
13. Appendix 362
13.1 DDR Summary 362
14. Correction & Revision 371

LIST OF TABLES

TABLE 2-1: AZN-10 WORKOVER WELL INFORMATION 13
TABLE 2-2: AZN-10 CASING INFORMATION 14
TABLE 2-3: AZN-10 WORKOVER INFORMATION 15
TABLE 2-4: AZN-10 DRY TEST INFORMATION 20
TABLE 2-5: AZN-10 TIE-BACK DRY TEST EVENT 25
TABLE 2-6: AZN-10 TIE-BACK DRY TEST INFORMATION 25
TABLE 2-7: AZN-10 TIE-BACK DRY TEST INFORMATION 27
TABLE 2-8: AZN-10 BIT RECORD TABLE 35
TABLE 3-1: AZN-10 WORKOVER TIME BREAKDOWN 38
TABLE 3-2: AZN-10 COMPLETION TIME BREAKDOWN 39
TABLE 3-3: AZN-10 NPT TIME BREAKDOWN 44
TABLE 4-1: AZN-10 LOGGING INFORMATION 45
TABLE 6-1: AZN-10 CASING INFORMATION 66
TABLE 7-1: AZN-10 TOTAL MATERIAL 69
TABLE 7-2: AZN-10 REQUIRED EQUIPMENT 70
TABLE 7-3: AZN-10 REQUIRED PERSONNEL 70
TABLE 7-4: AZN-10 FIELD RECORD TABLE 72
TABLE 7-5: AZN-10 PLUG BACK INFORMATION 73
TABLE 7-6: AZN-10 EQUIPMENT USED FOR THE JOB 78
TABLE 7-7: AZN-10 OPERATION CREW 78
TABLE 7-8: AZN-10 PUMPING SCHEDULE 81
TABLE 7-9: AZN-10 USED MATERIAL 82
TABLE 8-1: AZN-10 TUBING DATA 83
TABLE 8-2: AZN-10 EQUIPMENT CHECKLIST 85
TABLE 8-3: AZN-10 EQUIPMENT CHECKLIST 86
TABLE 8-4: AZN-10 OPERATION CHECKLIST 110
TABLE 9-1: AZN-10 FINAL WELL STATUS 128
TABLE 9-2: AZN-10 EQUIPMENT SPECIFICATION 131
TABLE 9-3: AZN-10 SECTION "D" SPECIFICATION 133
TABLE 9-4: AZN-10 SECTION "D" TEST REPORT BEFOR INSTALLATION 134
TABLE 9-5: AZN-10 SECTION "D" TEST REPORT AFTER INSTALLATION 135
TABLE 10-1: AZN-10 PRE AND POST ACID POTENTIAL OF THE WELL 140
TABLE 10-2: AZN-10 GENERAL INFORMATION 141
TABLE 10-3: AZN-10 WELL BASIC INFORMATION 142
TABLE 10-4: AZN-10 DAILY OPERATION SUMMARY 148
TABLE 10-5: AZN-10 SEQUENCE OF OPERATION 169
TABLE 10-6: AZN-10 PRE-ACID BULLHEAD OPERATION REPORT 292
TABLE 10-7: AZN-10 PUMPING PRE-FLUSH OPERATION REPORT 293
TABLE 10-8: AZN-10 PUMPING MAIN ACID OPERATION REPORT 306
TABLE 10-9: AZN-10 PUMPING OVER FLUSH OPERATION REPORT 310
TABLE 10-10: AZN-10 PUMPING POST-ACID BULLHEAD 1 OPERATION REPORT 311
TABLE 10-11: AZN-10 PUMPING POST-ACID BULLHEAD 2 OPERATION REPORT 312
TABLE 10-12: AZN-10 PUMPING PARAMETER 321
TABLE 10-13 PRE-STIMULATION FLOWING SURVEY REPORT (GAUGE NO: 90504) 323
TABLE 10-14 PRE-STIMULATION FLOWING SURVEY REPORT (GAUGE NO: 90506) 325
TABLE 10-15 POST STIMULATION FLOWING SURVEY REPORT (GAUGE NO: 90504) 333
TABLE 10-16 POST STIMULATION FLOWING SURVEY REPORT (GAUGE NO: 90506) 335
TABLE 10-17 PRE-ACID PRODUCTIVITY INDEX 342
TABLE 10-18 POST ACID PRODUCTIVITY INDEX 342
TABLE 10-19 COILED TUBING BHA DESIGN 345
TABLE 10-20 S/L BHA DESIGN 346

TABLE 11-1: AZN-10 DRILLING FLUID PROPERTIES 349
TABLE 11-2: AZN-10 DRILLING FLUID TYPE 350
TABLE 11-3: AZN-10 VOLUME ANALYSIS 353
TABLE 11-4: AZN-10 DAILY VOLUME BREAKDOWN 355
TABLE 11-5: AZN-10 LOSS DISTRIBUTION 358
TABLE 11-6: AZN-10 SOLID CONTROL 360
TABLE 11-7: AZN-10 COMPLETION FLUID INFORMATION 360
TABLE 12-1: AZN-10 CONTRACTORS INVOLVED 361
TABLE 13-1: AZN-10 DAILY DRILLING REPORT 370

LIST OF FIGURES

FIGURE 1-1: AZNS OILFIELD LOCATION MAP 11
FIGURE 1-2: AZN-10 WELL LOCATION MAP 12
FIGURE 2-1: AZN-10 DRY TEST DATA CHART MIDDLE B.T. 20
FIGURE 2-2: AZN-10 DRY TEST DATA CHART BOTTOM B.T 21
FIGURE 2-3: AZN-10 TIE-BACK DRY TEST DATA CHART MIDDLE B.T. 26
FIGURE 2-4: AZN-10 TIE-BACK DRY TEST DATA CHART BOTTOM B.T.. 26
FIGURE 2-5: AZN-10 TIE-BACK DRY TEST DATA TIME CHART 27
FIGURE 2-6: AZN-10 WELL SKETCH BEFORE WORKOVER 33
FIGURE 2-7: AZN-10 WELL SKETCH AFTER WORKOVER 34
FIGURE 3-1: AZN-10 WORKOVER WELL TIME VS. DEPTH 37
FIGURE 3-2: AZN-10 NPT BREAK-DOWN CHAR 40
FIGURE 6-1: AZN-10 TIE-BACK TALLY 65
FIGURE 7-1: AZN-10 WELL SCHEMATIC & CEMENT PROGRAM 67
FIGURE 7-2: AZN-10 CEMENT LAB REPORT 68
FIGURE 7-3: AZN-10 CEMENT JOB DESING & CALCULATION 76
FIGURE 7-4: AZN-10 CEMENT LAB REPORT 77
FIGURE 8-1: AZN-10 TECHNICAL SERVICE REPORT 88
FIGURE 8-2: AZN-10 TRSSSV TEST REPORT 89
FIGURE 8-3: AZN-10 TRSSSV CONNECTION GRAPGH 90
FIGURE 8-4: AZN-10 TRSSSV PRESSURE TEST GRAPGH 91
FIGURE 8-5: AZN-10 PERMANENT PACKER TEST REPORT 92
FIGURE 8-6: AZN-10 ANCHOR SEAL TEST REPORT 93
FIGURE 8-7: AZN-10 PERMANENT PACKER CONNECTION GRAPGH 94
FIGURE 8-8: AZN-10 SSD TEST REPORT 95
FIGURE 8-9: AZN-10 SSD CONNECTION & PRESSURE TEST GRAPGH 96
FIGURE 8-10: AZN-10 EXPANSION JOINT TEST REPORT 97
FIGURE 8-11: AZN-10 EXPANSION JOINT CONNECTION & PRESSURE TEST GRAPGH 98
FIGURE 8-12: AZN-10 LANDING NIPPLE TEST REPORT 99
FIGURE 8-13: AZN-10 LANDING NIPPLE CONNECTION & PRESSURE TEST GRAPGH 100
FIGURE 8-14: AZN-10 FIRST CIV TEST REPORT 101
FIGURE 8-15: AZN-10 FIRST CIM TEST REPORT 102
FIGURE 8-16: AZN-10 FIRST CIM CONNECTION & PRESSURE TEST GRAPGH 103
FIGURE 8-17: AZN-10 SECOND CIV TEST REPORT 104
FIGURE 8-18: AZN-10 SECOND CIM TEST REPORT 105
FIGURE 8-19: AZN-10 SECOND CIM CONNECTION & PRESSURE TEST GRAPGH 106
FIGURE 8-20: AZN-10 COMPLETION SKETCH 111
FIGURE 8-21: AZN-10 COMPLETION TALLY 121
FIGURE 8-22: AZN-10 TUBING PRESSURE TEST GRAPHS 125
FIGURE 8-23: AZN-10 TUBING PRESSURE & 3SV INFLOWTEST GRAPH 126
FIGURE 8-24: AZN-10 ANNULUS PRESSURE TEST GRAP 127
FIGURE 9-1: AZN-10 WELLHEAD SKETCH 132
FIGURE 9-2: AZN-10 X-MASS TREE PRESSURE TEST GRAPH BEFORE INSTALLATION 136
FIGURE 9-3: AZN-10 BODY & METAL SEAL OF X-MASS TREE PRESSURE TEST GRAPH 137
FIGURE 9-4: AZN-10 WELLHEAD MOM BEFOR INSTALLATION 138
FIGURE 9-5: AZN-10 WELLHEAD MOM AFTER INSTALLATION 139
FIGURE 10-1: AZN-10 EXECUTED OPERATION FLOW CHART 146
FIGURE 10-2: AZN-10 WELLHEAD SKETCH 283
FIGURE 10-3: AZN-10 WELLHEAD SKETCH 283
FIGURE 10-4: AZN-10 UPSTREAM PRESSURE GRAPH 284
FIGURE 10-5: AZN-10 UPSTREAM TEMPERATURE GRAPH 285
FIGURE 10-6: AZN-10 DOWNSTREAM PRESSURE GRAPH 286
FIGURE 10-7: AZN-10 DOWNSTREAM TEMPERATURE GRAPH 287
FIGURE 10-8: AZN-10 LIQUID PUMPING GRAPH (PR. & RATE) 313

FIGURE 10-9: AZN-10 LIQUID PUMPING GRAPH (VOLUME) 314
FIGURE 10-10: AZN-10 PRE-ACID BULLHEAD GRAPH (PR. & RATE) 315
FIGURE 10-11: AZN-10 PRE-ACID BULLHEAD GRAPH (VOLUME) 316
FIGURE 10-12: AZN-10 POST-ACID BULLHEAD 1 GRAPH (PR. & RATE) 317
FIGURE 10-13: AZN-10 POST-ACID BULLHEAD 1 GRAPH (VOLUME) 318
FIGURE 10-14: AZN-10 POST-ACID BULLHEAD 2 GRAPH (PR. & RATE) 319
FIGURE 10-15: AZN-10 POST-ACID BULLHEAD 2 GRAPH (VOLUME) 320
FIGURE 10-16: AZN-10 PRE-STIMULATION DOWN HOLE MEMORY GAUGE (P&T) PLOT (GAUGE NO: 90504) 326
FIGURE 10-17: AZN-10 PRE-STIMULATION PRESSURE VS TVD PLOT (GAUGE NO: 90504) 328
FIGURE 10-18: AZN-10 PRE-STIMULATION TEMPERATURE VS TVD PLOT (GAUGE NO: 90504) 329
FIGURE 10-19: AZN-10 PRE-STIMULATION PRESSURE VS TVD PLOT (GAUGE NO: 90506) 330
FIGURE 10-20: AZN-10 PRE-STIMULATION TEMPERATURE VS TVD PLOT (GAUGE NO: 90506) 331
FIGURE 10-21 POST STIMULATION FLOWING SURVEY DOWN HOLE MEMORY GAUGE (P&T) PLOT (GAUGE NO: 90504) 336
FIGURE 10-22 POST STIMULATION FLOWING SURVEY DOWN HOLE MEMORY GAUGE (P&T) PLOT (GAUGE NO: 90506) 337
FIGURE 10-23 POST STIMULATION FLOWING SURVEY PRESSURE VS TVD PLOT (GAUGE NO: 90504) 338
FIGURE 10-24 POST STIMULATION FLOWING SURVEY TEMPERATURE VS TVD PLOT (GAUGE NO: 90504) 339
FIGURE 10-25 POST STIMULATION FLOWING SURVEY PRESSURE VS TVD PLOT (GAUGE NO: 90506) 340
FIGURE 10-26 POST STIMULATION FLOWING SURVEY TEMPERATURE VS TVD PLOT (GAUGE NO: 90506) 341
FIGURE 11-1: AZN-10 MUD CHEMICAL USAGE 351
FIGURE 11-2: AZN-10 EFFECTIVE CONCENTRATION 352
FIGURE 11-3: AZN-10 VOLUME ANALYSIS 353
FIGURE 11-4: AZN-10 LOSS ANALYSIS 359

1. General Information

1.1 Field Introduction

Azadegan South Oil Field is situated in West Karoun, Khuzestan province on a large long-axis anticline with N-S trending with approximate dimensions about 20^{km} width and 60^{km} length near to Iran-Iraq border, It is located in The Northwest of The YADAVARAN Field and in The West of JUFEYR Field. The Average Monthly Temperature is 45$^{°c}$ in summer, and Highest Temperature is 52$^{°c}$. The Lowest Temperature is -4.4$^{°c}$ in winter. Maximum Rainfall is 78^{mm} in February, Winter Season.

The trap area of the south Azadegan is limited to 43.2x17.2^{km}. The Azadegan Oil field was discovered in 1999, when Well AZN-1 was drilled vertically down to the Cretaceous Fahliyan Formation with total depth of 4,643^{m}. The results of this well, revealed the reservoir characteristics of the field. Following to the result of the well no. AZN-1, another 5 exploration wells was drilled in different parts of this huge field down to Fahliyan Formation. In the later stage the well No.10 assumed as exploration well for evaluation of deeper formations. This well was drilled in 2008, down to the depth of 5340^{m} (Neyriz Fm.), mainly to evaluate the Jurassic formations.

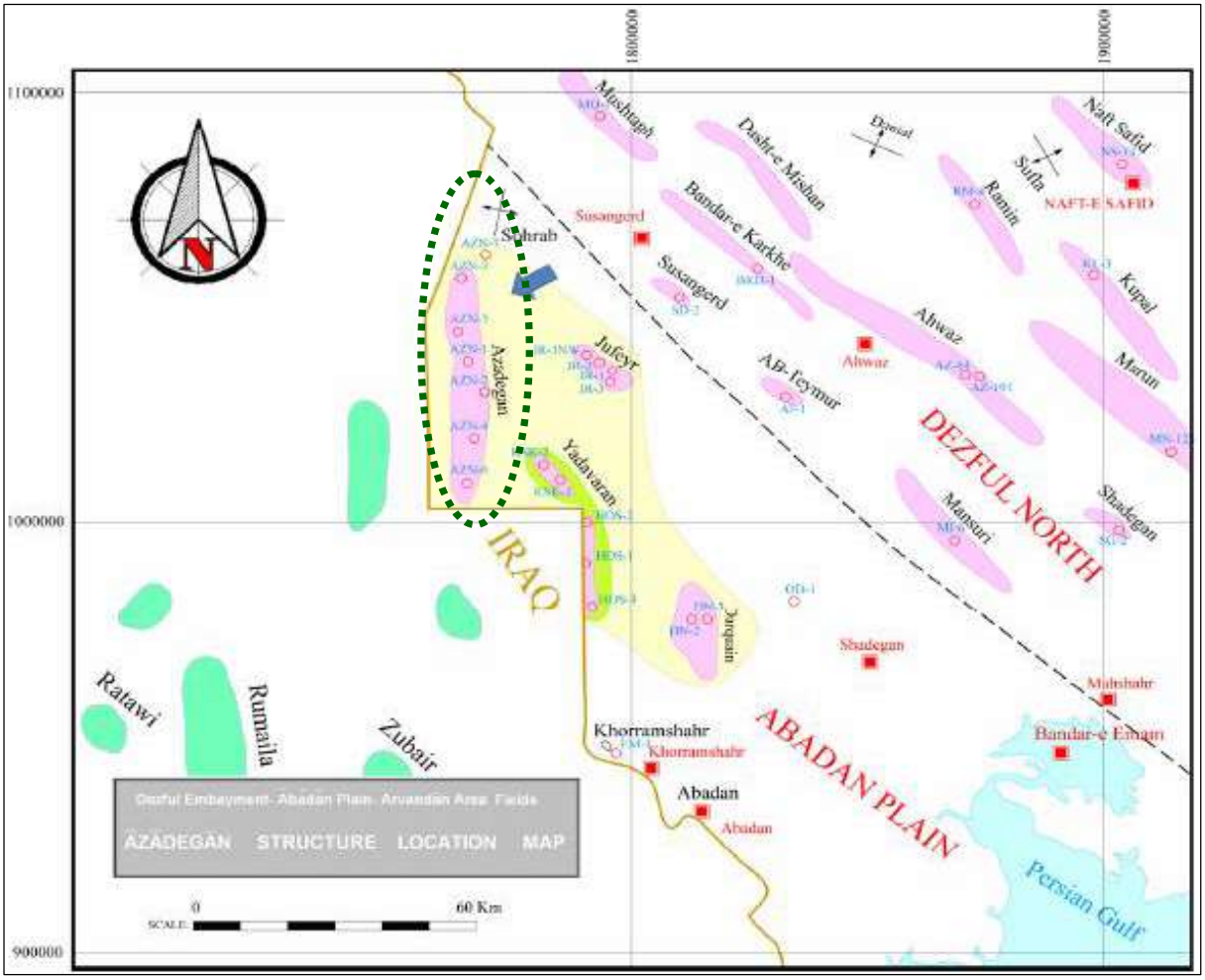


Figure 1-1: AZNS Oilfield Location Map

1.2 AZN-10 Well Location Map

This well is located near Pad-50 of South Azadegan Oil field in Block-II.

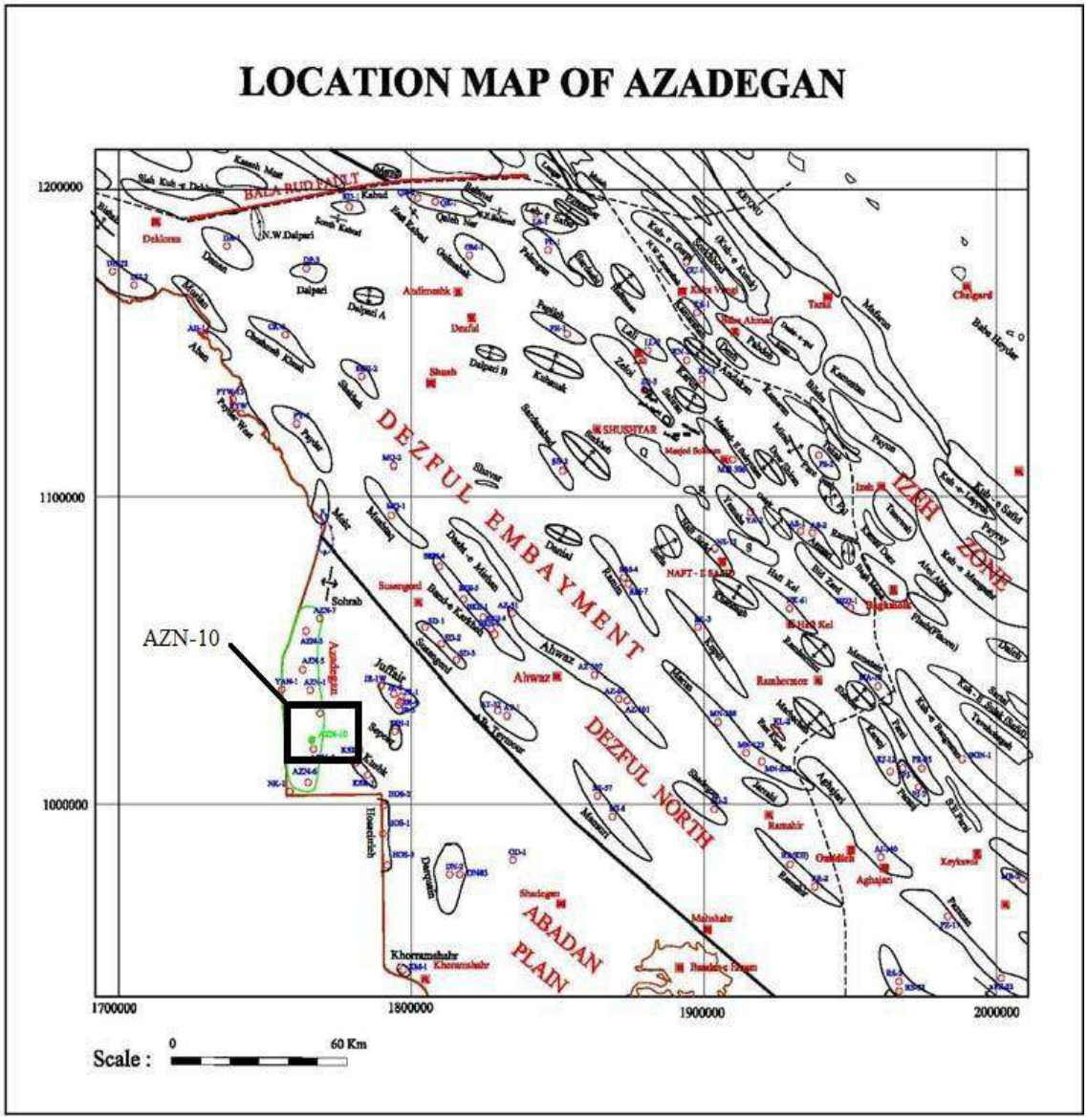


Figure 1-2: AZN-10 Well Location Map

2. Well History

2.1 Well Informaion

Rig name	201	**Owner**	PEDEC/PPL
Rig type	Land Rig 2000HP	**Contractor**	MSA/OEOC
Well Name	AZN-10	**Management Contractor**	MSA
Well Type	Vertical IIF	**Field**	South Azadegan
Completion Type	4-1/2" * 3-1/2", 10K	**RTE**	13.45
Planned TD	-	**GLE**	3.95
Actual TD	5340^{m}	-	
Surface Coordinate		N 3,451,636.1 – E 194,371.6	
Target Coordinate		N 3,451,650.6 – E 194,387.9	
Drilling & Suspension		**Completion**	
Spud Date	08/10/2023	**Spud Date**	16/12/2023
Released Date	29/12/2023	**Released Date**	29/12/2023
End of Drilling	-	**End of Drilling**	-
Rig Days Actual/Plan	83 / 37	**Rig Days Actual/Plan**	-

Size (in)	Shoe Depth (mMD)	Specification
20	151	K-55, 133^{ppf}
13-3/8	1502	N-80, 72^{ppf}
9-5/8	3854	T95, 53.5^{ppf}
7	3703.6 – 4922	T95, 35^{ppf}
7 tie-back	3527 – 3703.6	C110, 29^{ppf}
5	4781 – 5193.5	T95, 18^{ppf}

Table 2-1: AZN-10 Workover Well Information

2.2 AZN-10 Well History

The AZN-10 well was drilled f/25-June-2008 to 23-May-2010 to total depth 5340^{m} and abandoned with CMT plug. The 26" hole section was drilled to 152.5^{m} with Mw 65^{pcf} and cased by 20" casing to 151^{m} with K-55, 133# and equipped by 20-3/4" – 3000^{psi} wellhead & related BOP & accessories. The 17-1/2" hole section was drilled to 1502^{m} with Mw 74 to 90^{pcf}, because of tight hole, pressure drop observed several times & a fishing operation with overshot at 1264^{m}, this hole section was cased by 13-3/8" N-80, 72# 8RD & BTC casing to 1502^{m} and equipped by 13-5/8" – 5000^{psi} wellhead & related BOP & accessories. The 12" hole section was drilled to 3857m and Mw 85pcf, mud loss observed amount of 45 & 50BPH at 2690 & 273m, 68bph loss at 3002m, cased with 9-5/8" L-80, T95, 53.5#, SEC casing to 38545^{m} and equipped by 11" – 10,000^{psi} wellhead & related BOP & accessories. The 8-3/8" hole section was drilled to 4925.5^{m} With Mw 100 to 115^{pcf}, gas recorded in several intevals abou 2 to 14%, stuck pipe at 4430 & 4439^{m} and fishing operation included, pipe stuck at 4536^{m} and 90% gas was observed with return mud & 10 ^{bbl} increased in tank volume, control the well with 128^{pcf} Mw, drilling to 4925.5^{m} with 20-80% gas recorded while drilling and increasing Mw to 131 to 135^{pcf}, ran 7" liner T95, SSHC, 34#, BDS f/3707.5^{m} (L.Lap) to 4922^{m} (L.shoe) and dry test was OK. The 5-7/8" hole section was drilled to 5340^{m} and Mw decreased to 113^{pcf}, fish left in hole at 5190.32^{m}, performed F.B.D.S.T No. 1 to 4, ran whip Stock tool in several depth without any results from date 25-July-2009 to 3-Oct-2009, RIH with open ended pipes and put cement plug as balance at 5060^{m}, RIH with 5-7/8" bit K300 with Bent Sub (2°) & downhole motor 4-3/4" to 4990^{m} and cased with 5" liner T-95, VAM, 18# f/4781^{m} (L.Lap) to 5193.5^{m} (L.shoe) and LDC at 5169^{m}, performed dry test and F.B.D.S.T No. 5 to 8. Put CMT plugs at 4780^{m} , 3707^{m} & 1502^{m}. N/D BOPs & installed blind flange 11" – 10,000^{psi}, performed pressure tests, installed wellhead accessories, secured the well & released the rig at 23-March-2010.

Casing Spec.	20"	K-55, 133#	0 – 151^{m}
	13-3/8"	N-80, 72#	0 – 1502^{m}
	9-5/8"	T-95,53.5#	0 – 3854^{m}
	7"	T-95, 32#	3707.5 – 4922^{m}
	5"	T-95, 18#	4781 – 5193.5^{m}

Table 2-2: AZN-10 Casing Information

Summary of Workover Operation

The AZN-10 well was desuspend and completed by rig OEOC-201 from 0 8– October – 2023 to 29 – December – 2023 in order to produce from Fahliyan Reservoir as described in the following:

Workover Summary:

Date	08 – October – 2023
WorOver Depth	4250^{m} (Plug Back)
Formation	Fahliyan
Mud Weight	100 – 102^{pcf}

Table 2-3: AZN-10 WorkOver Information

The rig spud @00:00 & start work on BOP & well control equipment to rectify the leakage, tried to test annulus several times with low & high pressure w/no success. Continue perform BOP test against 11" plug tester while observe leakage at several points, work on to rectify all leakages successfully. Finally perform BOP pressure test as NIOC policy with 500 – 2500^{psi} for 5 to 10min, OK. All rams & C/M with related lines & valves test by 500 – 7500^{psi} for 5 to 10min, OK. L/D plug tester & P/U & M/U new BHA 1st DC with bit sub & install float valve inside the bit sub, M/U new 8-1/2" B.bit 517 open with appropriate TQ, 5 Stds 6-1/2" DC, 6-1/2" M.jar, 1Std DC & 1 Jt HWDP & RIH to 138^{m}. F/U the string, tried to Cir. but observed string plugged. Worked on unplug string several times with no succeed, POOH to surface & observed the plug about 0.5^{m} above top of float valve by debris & corrosion, unplugged string & L/D 6-1/2" Jar due to observe leakage f/mandrel while POOH the string. RIH back with same BHA to 150^{m}.

Cont. RIH to 520^{m} while F/U string, due to gel upped over Cir. 5min to shear the mud, Cont. RIH to 1120^{m}, observed took weight @1120^{m}, W&R/D to 1355^{m} the settlement. Tag the TOC @1381^{m} approved by put 10^{klbs}, reciprocated last Std to shear the mud, D/O hard CMT to 1405^{m} while checked D/O CMT with different parameters to take optimum rate, stop operation due to low ROP (less than 4^{mph}), commence POOH to change the bit to 910^{m} without F/U the well as settle down mud solid during trip operation. Cont. POOH to surface, bit dull grade 1-1-WT-A-E-I-NO-PR. P/U & M/U 8-1/2" used PDC bit Varel open, RIH with the same BHA to 1313^{m} while F/U string every 15 Stds. Cont. RIH with W/D to 1405^{m} & tag the TOC, resume D/O med to hard CMT f/1405 to 1497^{m}. Closed annulus BOP & Cont. D/O soft CMT to 1500^{m}, Cont. RIH with

W/D f/1542 to 1600[m] w/NG then RIH on elevator to 1790[m]. Took weight @1790[m], tried to W/D without any succeed, W&R f/1790 to 1830[m], Cont. RIH D/O BHA on elevator to 2660[m] w/NL-NG while F/U string each 10 Stds. Cont. RIH with W/D f/2660 to 2680[m] w/NG, Cont. RIH on elevator f/2680 to 3166[m], took weight @3166[m], tried to W/D w/no success, W&R f/3166 to 3168[m], Cont. RIH on elevator f/3168 to 3550[m], F/U the string & got return observed pressure up on SPP to 2000[psi], worked on string to get normal SPP. Cont. RIH Jt by Jt on elevator f/3550 to 3607[m], RIH with W&R to 3629[m] & tag TOC @3629[m].

Decided to POOH & R/B 4 Stds of 5" DPs, due to ELC. problem in SCR, TDS was out of service, worked on SCR to fix same while Cir. through & reciprocate string. POOH D/O BHA to 3520[m] & R/B 4 Stds 5" DPs, Cont. RIH f/3520 to 3629[m] while F/U string & perform Circ. through string, washed down last Std to tag TOC. Resume D/O soft to med CMT to 3633[m]. Stopped operation due to observed leakage on stand pipe ground line, off BTM string to 3620[m], changed mud pump ground manifold L/U, performed Circ. w/450[gpm] & 1800 – 2000[psi] & ran back to BTM. Cont. D/O med CMT f/3633 to 3640[m], RIH w/wash & ream f/3640 to 3693[m] & wash down f/3693 to 3706[m] w/NL-NG, tag 7" L.Lap @3706[m] & confirmed, CBU to clean & monitor the well NL-NG. POOH D/O BHA to surface while F/U the well w/NL-NG. P/U & M/U used 5-7/8" R.bit kingdram 217, w/bit sub & RIH D/O BHA to 2612[m] while F/U string every 10 Stds w/NL-NG. Cont. RIH to 4618[m] inside 7" liner, F/U string & RIH with wash down f/4618 to 4663[m], took weight & tag TOC @4663[m], checked CMT w/290 to 300[gpm] & 1800 – 2000[psi] & 15[klbs], no progress, displaced the well w/102[pcf] WBM @4663[m], tried to POOH due to failure of TDS torque wrench, unable to disconnect TDS & work on same with no success, disassemble TDS torque wrench & B/O connection from TDS double pin by rig tongs, POOH D/O BHA to surface w/NL-NG. R/B BHA & L/D 5-7/8" R.bit (1-1-WT-A-E-I-NO-BHA), M/U 5-7/8" C.C bit w/bit sub, & RIH to 4591[m] while F/U the string w/NL-NG. L/U CMT pump truck line to string & pressure test line to 3000[psi], OK, pumped 4[bbl] 110[pcf] spacer ahead, mixed & pumped 113[sxs] CMT "G" Delijan blend w/36.63[lb/sxs] silica flour, pumped 2[bbl] 110[pcf] spacer behind, displaced the well w/196[bbl] 102[pcf] mud as balance, NL-NG. B/O & L/D swage & line f/string, POOH 12 Stds f/4591 to 4250[m] while observed all Stds are wet, observed string is plug while try to flush the string, performed several attempts to unplug string by jerking, rotating & applying 2000[psi] pressure stepwise, no success. POOH the string f/4250 to 697[m], observed string plug by contaminated mud by slurry @697[m], POOH to surface while B/O & L/D DPs Jt by Jt, totally 73 Jts 3-1/2" DP, bit sub & C.C bit were plugged, 9 Jts of DPs successfully unplugged but 64 Jts 3-1/2" DP remain plug on rack, unplugged C.C bit & RIH w/bit sub to 4158[m] & break Cir. every 30 Stds, NL-NG. Took weight @4158[m], W/D to 4167[m] w/5 - 12[klbs] drag, 250 - 350[gpm] & 800 – 1400[psi], W&R f/4167 to 4188[m], D/O med to hard

CMT f/4188 to 4250^{m}, CBU w/NL-NG. POOH D/O BHA to surface while F/U hole w/NL-NG.

P/U & M/U 5-7/8" used R.bit kingdream 217 open, & RIH w/7" * 9-5/8" tandem scraper to 4123^{m} while scrap inside 9-5/8" CSG & 7" liner, F/U string every 15 Stds & break Circ. periodically w/NL-NG. In order to make required Stds to reach the TOC, POOH BHA f/4123 to 3993^{m} & R/B 3 Stds, continued RIH to 4122^{m}, washed down f/4140 to 4207^{m} w/80 – 100^{gpm} & 1900 – 2500^{psi} NL-NG. M/U new 5" DPs Std @4207^{m}, tried to establish Circ. observed string plugged, several attempts to unplug the string by applying pressure stepwise to 2800^{psi} without any succeed, decided to R/B last Std & POOH string to unplug, due to failure of TQ wrench of TDS decided to B/O the last Std by rig tongs, P/U string a single Jt & POOH to 4197^{m}, totally L/D 2 Jts, M/U double of Std on string, suddenly drawworks break failed, & slacked off TDS & block weight on double Jt which already on slips, bend double Jt & crack DP on slips, closed top & BTM rams on string, hanged up bell nipple, set 5" DP slips on hydrill by hammering, set safety clamps above slips, tried to P/U string gently, observed double Jts cut from crack point on slips but string remains on slips, L/D bended double Jts piece by piece with cutting, checked TDS all functions, changed clamps of TQ wrench, P/U & M/U taper tap 3"*5", engaged & tapped inside the string by applying 7$^{kft.ib}$ TQ, opened top & BTM rams, tried to P/U the string but taper tap jumped out by 200^{klbs} overpull on M/D without any movement of string, B/O & L/D taper tap & closed top & BTM rams again, P/U & M/U taper tap 4"*5", engaged & tap inside the string by applying 9$^{kft.ib}$ TQ, opened top & BTM rams, tried to P/U the string but taper tap jumped out by 14^{klbs} overpull on M/D without any movement of string, B/O & L/D taper tap & closed top & BTM rams again, P/U & M/U taper tap 3"*5", engaged & tap inside the string by applying 17$^{kft.ib}$ TQ, opened top & BTM rams, P/U string gently to 390^{klbs} on M/D, finally string retrieved f/slips, removed slips on RT & POOH the string by taper tap & applying normal weight on M/D f/4197 to 4188^{m}, removed slips on hydrill, installed bell nipple while set pump truck & L/U surface line. B/O last Jt of string & L/D taper tap, continued POOH tandem scraper on elevator to 3673^{m} while F/U hole w/NL-NG. Continued POOH to surface while trying to unplug the string several times by applying 2800^{psi} pressure stepwise without any succeed, observed string plugged by a bolt that belongs to TDS stabbing guide inside 4-3/4" bit sub, unplugged same. Observed 2 out of 6 blades of 7" scraper was loose, repaired & fixed same.

P/U & M/U 5-7/8" used R.bit kingdream 217 & RIH w/7" * 9-5/8" tandem scraper to 600^{m} while F/U the string & Circ. break Circ. every 15 Stds, Established Circ. mud in order to uniform Mw & checked mud condition, got min 55^{pcf} aerated mud @BTM up due to MPs malfunctions & dropped active pit Mw to 72^{pcf}, shut the one of MPs off & Cont. Circ. w/low GPM, worked on MP #1 & #2 respectively & overhaul same. After overhaul MPs observed MP's work effectively & return Mw gradually increased to 86^{pcf}

& active pit Mw to 90^{pcf} NL-NG. Decided to RIH to above 7" L.Lap & again performed mud condition, Cont. RIH f/1915 to 2691^{m}. Due to lack of enough DP to reach to TOC, M/U & RIH 5" DP to 3662^{m} in single while F/U string every 15 Stds. CBU in order to uniform Mw & check mud condition, Balance Mw in/out @93^{pcf}, Cont. RIH w/ 7" x 9-5/8" tandem scraper BHA f/3662 to 4150^{m} while scrapping inside 9-5/8" CSG & 7" LNR, F/U string every 15 Stds till getting return, RIH washed down w/7" * 9-5/8" tandem scraper BHA f/4150 to 4250^{m} while scrapping inside 9-5/8" CSG & 7" LNR. Performed CBU till homogenizing the mud properties & removing bubbles, meanwhile prepared 70^{bbl} $CaCl_2$ solution in an isolated mud pit to prevent mixing w/WBM. Cont. Mix $CaCl_2$ w/Drill water & prepare 70^{bbl} pumpable $CaCl_2$ solution 84^{pcf} in an isolated mud pit, Started to pump 70^{bbl} $CaCl_2$ solution to spot @4250^{m} w/low rate then 203^{bbl} WBM to displace the solution as balance by rig pump while observed the suction pit level during the operation as a reference, Started POOH 7" * 9-5/8" tandem scraper BHA f/4250 to 3625^{m} gently, meanwhile Circ. across wellhead w/NL-NG, Observed the well for 30min, NL-NG, Cont. POOH 7" * 9-5/8" tandem scraper BHA f/3625 to surface, meanwhile Circ. across wellhead w/ NL-NG, L/D BHA & bit with same running condition.

R/U logging equipment & tools while performed surface checks, all related tests done perfectly. M/U & RIH run #1 logging tools as 7" LNR cement evaluation by SBT-VDL-GR-CCL to 4247^{m} (3^{m} up to TOC), logged up f/4247 to 3700^{m}, took 100^{m} repeat log (interval: 4100 - 4000^{m}) as wellsite witness request, POOH to surface & L/D tools, logging data confirmed by wellsite witness, M/U & RIH run #2 logging tools as 7" LNR corrosion evaluation by CBIL-CDR-GR-CCL to 4247^{m} (3^{m} up to TOC), logged up f/4247 to 3700^{m}, took 70^{m} repeat log (interval: 3770 - 3700^{m}) as wellsite witness request, POOH to surface & L/D tools, logging data confirmed by wellsite witness, M/U & RIH run #3 logging tools as 9-5/8" CSG corrosion evaluation by CBIL-CDR-GR-CCL to 3706^{m},logged up f/3706 to surface, logging data not confirmed by wellsite witness due to noisy data, M/U & RIH run #4 back up logging tools as 9-5/8" CSG corrosion evaluation by CBIL-CDR-GR-CCL to 20^{m}, the tools got off, wait for contractor's logging crew while changed the tools w/main tools, RIH the logging tools & checked the results, OK. Cont. RIH run #4 logging tools as 9-5/8" CSG corrosion evaluation by CBIL-CDR-GR-CCL to 3706^{m} (7"L.Lap), logged up f/3706 to surface, took 2400^{m} repeat log (interval: 2400^{m} - surface) as wellsite witness request, logging data confirmed by wellsite witness, R/D the logging equipment.

P/U & M/U 9-5/8" dry test Assy. Checked the Assy. functionality & drill water weight (62.4^{pcf}) in the presence of dry test supervisor, OK, F/U the Assy. & RIH w/9-5/8" dry test BHA to 3691^{m} followed by 5" DP, checked the return level in degasser mud pit, changed handling equipment & elevator links, R/U & M/U chiksan joints, dry test chock manifold, bar drop Sub & safety valve, pressure test the line to 2500^{psi} & held for 15min,

OK. Set the packer in the presence of dry test supervisor, opened hydro-spring, observed string flow from dry test chock manifold for 3hrs. No bubble was observed, closed DCIP for 6hrs. Opened Circ. valve w/dropping the bar, reversed Circ. thru 9-5/8" dry test BHA till balance the mud in/out uniformed, stopped circulation, opened Rams, R/D chiksan joints, dry test chock manifold & safety valve, tried to close the hydro-spring & unset packer, successfully done, changed handling equipment & elevator links, POOH 9-5/8" dry test BHA f/3691 to surface meanwhile Circ. across wellhead w/NL-NG. L/D 9-5/8" dry test BHA, read & checked the memory gauge information, observed pressure increasing about 360^{psi} in the recorded data in close DCIP situation, set the plug tester, opened the wellhead side outlet, closed the hydrill & downstream valve, performed 3000^{psi} pressure test, OK, closed the top rams & gate valve, performed 7500^{psi}, OK, closed the bottom rams & gate valve, performed 7500^{psi} pressure, OK, unset the plug tester, Closed shear rams, performed 7500^{psi} pressure, OK, all steps held for 10min. P/U & M/U polish Assy. & RIH to 3705^{m} while F/U the string in every 15 Stds, Break Circ. & Obs. return on Sh.Shakers, performed BTMs up till clean Sh.Shakers while taking 240 - 250^{bbl} air-cut & $CaCl_2$-cut mud polish inside the 7" PBR gently while marking mill nose using LG w/10-12^{klbs} several times during polishing inside PBR, took weight (2 - 4^{klbs}) in passing 7" L.Lap, worked on 7" PBR internal space to smooth it gently down & passed the 7" PBR without rotating & pumping to final check in presence of contractor's supervisor, OK. POOH 7-5/16" polish Assy. f/3708.17 to surface on elevator, while Circ. across wellhead. Observed the LG mark on the polish mill nose in the presence of contractor's supervisor, OK. P/U & M/U tieback seal nipple PKR Assy. followed by 5" DPs, each Std must be RIH at least in 3min according to contractor's supervisor advice. Break Circ. in every 15 Stds & Obs. return in Sh.Shaker & RIH to 705^{m}. Circ. for 15min, set 8-1/4" tieback S.N PKR as per contractor's supervisor procedure, POOH 8-1/4" tieback S.N PKR Assy. setting tools f/3705 to surface on elevator, while Circ. across wellhead & using mud box. R/D setting tools.

P/U & M/U 9-5/8" dry test Assy. Checked the Assy. functionality & drill water weight (62.4^{pcf}) in the presence of dry test supervisor, OK, F/U the Assy. & RIH to 724^{m}, checked the return level in suction mud pit, operation stopped due to observe suddenly rabbit release from elevator & dropped into the string, decided POOH to surface & retrieve the rabbit, POOH 9-5/8" dry test BHA f/724 to 22^{m}, observed rabbit inside that, changed reversing sub & checked tools again in presence of dry test supervisor, OK. Cont. RIH w/ 9-5/8" dry test BHA f/22 to 3687^{m}, checked the return level in suction mud pit, R/U & M/U Chiksan joints, dry test chock manifold, bar drop Sub & Safety valve, pressure test the line to 2500^{psi} & held for 15min, OK. Set the packer in the presence of dry test supervisor, open hydro spring, Obs. string flow from dry test chock manifold for 3hrs. No bubble was observed, closed DCIP for 6hrs. Opened Circ. valve w/dropping the bar, reversed Circ. thru. 9-5/8" dry test BHA till balance the mud, in/out uniformed,

circulation stopped, opened rams, R/D Chiksan joints, dry test chock manifold & safety valve, tried to close the hydro spring & unset packer, successfully done. Started POOH 9-5/8" dry test BHA f/2497 to surface meanwhile Circ. across wellhead w/NL-NG. L/D 9-5/8" dry test BHA & checked the memory gauge info & analysis the fluid sample, set wear bushing & RIH w/securing to 516^{m}, performed Circ. while check in/out Mw, w/NL-NG.

Dry Test Report:

Row	Description	Time	Bottom Bt.	Middle Bt.
1	Initial Mud Static Pressure (psia)	8:33	--	8218
2	Open Well Pressure (psia)	1402/08/07 08:35	–	4228
		1402/08/07 11:35	–	4241
3	Close Well Pressure (psia)	1402/08/07 11:35	–	4242
		1402/08/07 17:35	–	4600
4	Final Hydrostatic Pressure (psia)	2:13	_	8446

Table 2-4: AZN-10 Dry Test Information

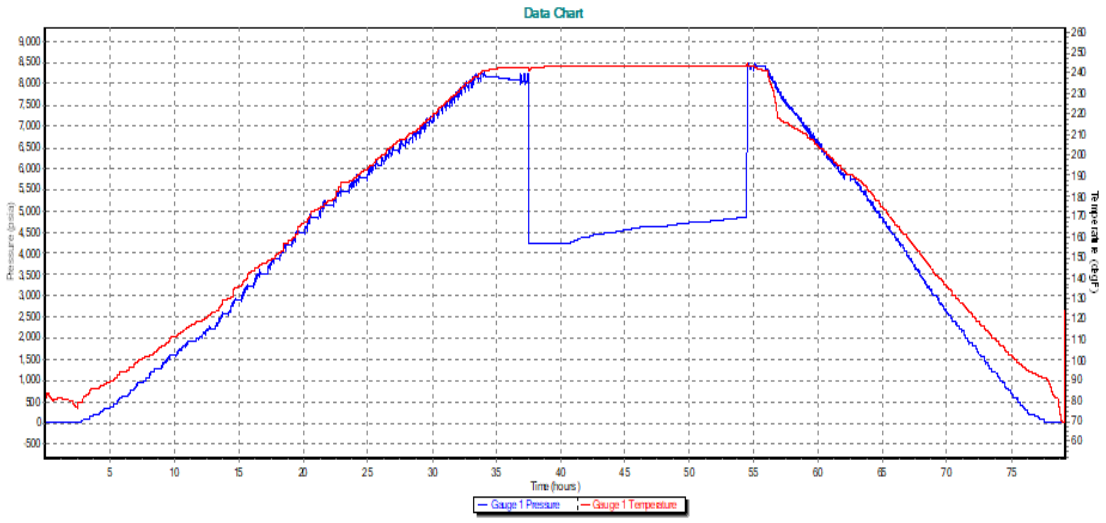


Figure 2-1: AZN-10 Dry Test Data Chart Middle B.T

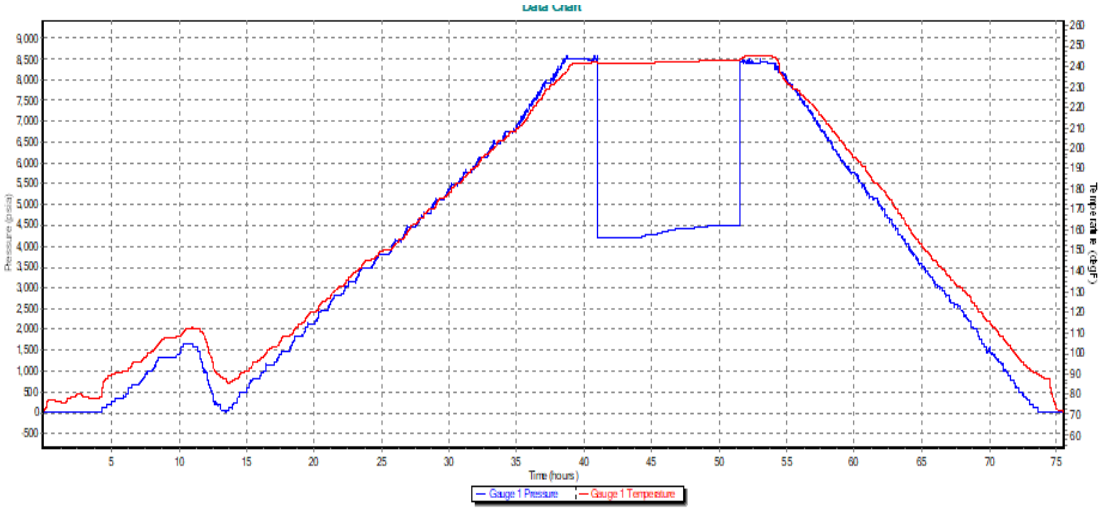


Figure 2-2: AZN-10 Dry Test Data Chart Bottom B.T

Tie-Back Operation:

Wait on instructions for the next operation by client from 06-Nov-23 to 10-Nov-23, Cont. Circ. & checked in/out Mw w/NL-NG, POOH securing BHA f/516^{m} to surface on elevator & L/D BHA & bit w/same condition while Circ. across wellhead. P/U & M/U polish Assy. 7-5/16" P.Mill new, 4*16/32 port & RIH to 111^{m}, F/U the BHA, Cont. RIH w/7-5/16" polish Assy. f/111 to 3697^{m} while F/U the string in every 15 Stds, break Circ. & Obs. return on Sh.Shakers, w/NL-NG. Performed CBU on top of 8-1/4" tie-back S.N PKR PBR till uniform the in/out mud w/NL-NG, Resumed wash down w/7-5/16" polish Assy. f/3697 to 3705.4^{m} gently & tag the 8-1/4" S.N PKR PBR LG, confirmed by applying WOB 5-7^{klbs}, then tried to make LG mark on 7-5/16" polish mill nose by applying some rotation w/controlled RPM. POOH f/LG depth to 3703.6^{m} (top of 8-1/4" tie-back S.N PKR PBR) gently, while polish inside PBR. POOH 7-5/16" polish Assy. f/3703.6 to 140^{m} on elevator, break Circ. across wellhead to F/U the well in every 10 Stds till Obs. return on Sh.Shakers & observed the well to flow check, static. Cont. POOH 7-5/16" polish Assy. to surface & L/D BHA & 7-5/16" polish mill, checked the LG mark on the P.Mill nose, OK. Retrieved wear bushing using plug tester, P/U & M/U 9-5/8" scraping Assy. 8-1/2" R.Bit open used, the functionality of all scraper blades got checked in the presence of the night OEOC tool pusher, OK. RIH to 3703^{m}, meanwhile scrape inside 9-5/8" CSG, F/U the BHA & break Circ. & Obs. return on Sh.Shakers, w/NL-NG. Performed CBU uniform the in/out mud & cleaned the probable debris, w/NL-NG. POOH 9-5/8" scrapping Assy. f/3703^{m} to surface, L/D BHA & bit with same condition. P/U & M/U 8-1/2" securing BHA B.Bit open used & RIH to 315^{m}. Wait on receive tie-back facilities from 10-Nov-23 to 19-Nov-23, slow Circ. w/NL-NG. RIH w/8-1/2" securing BHA f/315 to 3703^{m}, F/U the string in every 15 Stds w/NL-NG.

POOH 8-1/2" securing BHA f/3703 to 313^{m} while Circ. X-wellhead, w/NL-NG. POOH 8-1/2" securing BHA f/313^{m} to surface on elevator & L/D the bit w/same condition.

P/U & M/U 7" tie-back Assy. 7" S.N PKR + 7" LNR 29#, C-110, HSM3, while install centralizer, thread lock the 7" LDC, using joint-analyzed M/U system to apply appropriate TQ on connections. Cont. RIH w/7" tie-back Assy. to 3698^{m}, every 5" DP Std have to be run in 3min slip to slips, according to PPA supervisor advice while break Circ. in every 15 Stds, Circ. on top of L.Lap, P/U & M/U CMT head, Chiksans & L/U to pump truck, then load pump down plug, pumped 10^{bbl} chemical wash, 20^{bbl} 105^{pcf} spacer, 25^{bbl} 118^{pcf} CMT slurry, 3^{bbl} 118^{pcf} CMT slurry & displaced with 191^{bbl} 102^{pcf} mud, observed shearing the wiper plug studs w/2400^{psi} in the presence of PPA supervisor, then pumped 16^{bbl} mud, applying 10^{klbs} weight on string, released setting tools & set the PKR successfully. POOH 7" tie-back & checked the HKLD to confirm 7" tie-back Assy. released & lost the weight, OK. POOH 7" tie-back Assy. setting tool f/3241^{m} to surface, performed BOP test as NIOC policy.

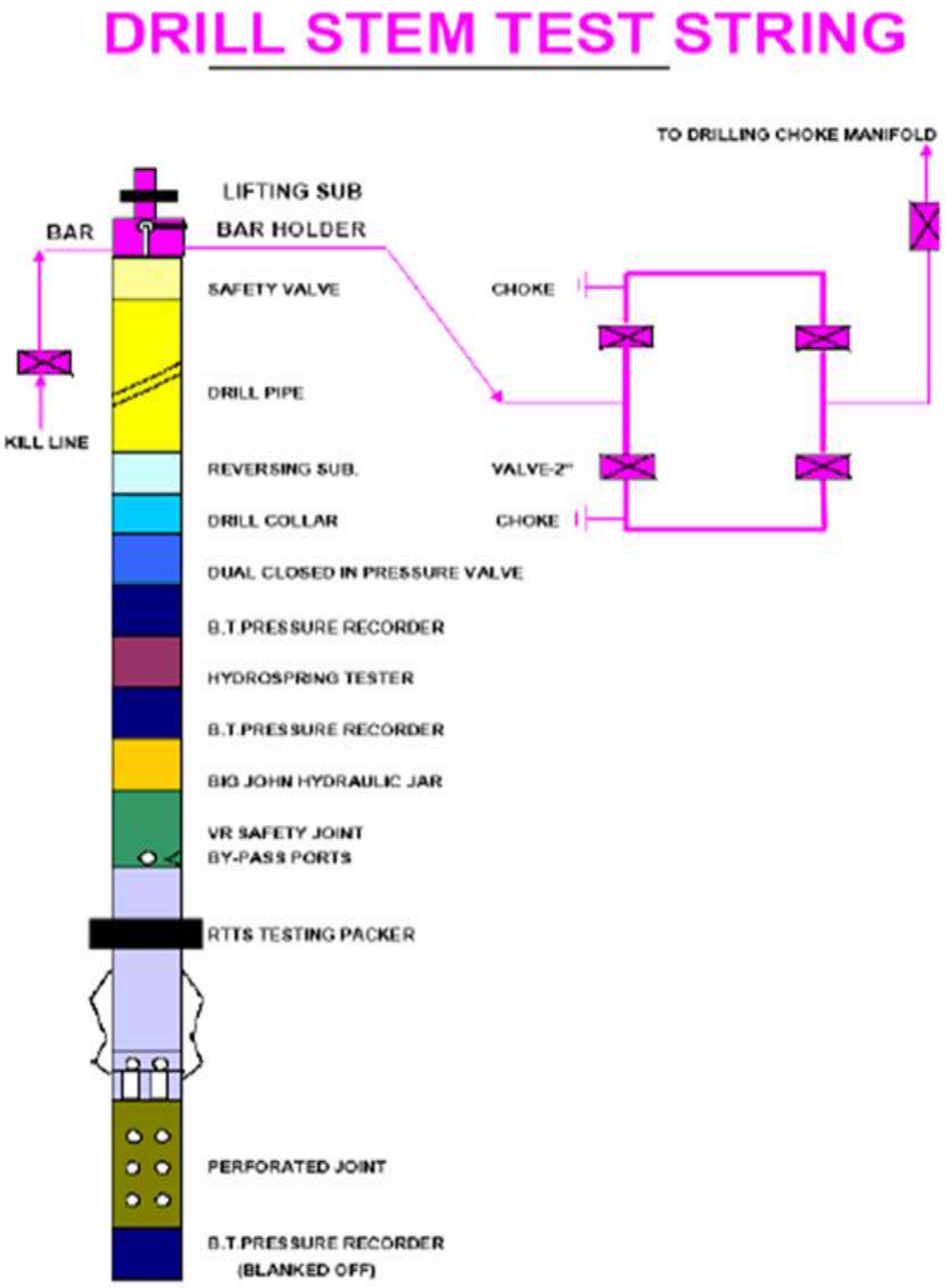
P/U & M/U new BHA w/8-1/2" used Varel PDC bit open & RIH to 3442^{m}, took weight @3442^{m}, D/O soft CMT f/3442 to 3467^{m}, resumed D/O medium to hard CMT f/3467 to 3527^{m}, new 7" L.Lap confirmed @3527^{m}. Commenced POOH f/3527 to surface & L/D BHA & bit with the same condition, P/U & M/U new BHA w/6" new R.bit 117 & RIH to depth of L.Lap @3527^{m} & tag TOC, commenced D/O hard CMT to 3530^{m} while D/O pack off bushing @3530^{m}, Cont. RIH w/rotation & pumping f/3530 to 3650^{m} while dressed the last interval several times before each connection & D/O hard CMT to 3702^{m}, pumping 20^{bbl} Hi-Vis Pill while D/O wiper plug & pump down plug (D/O LDC @3679^{m}). Performed CBU & pumped 30^{bbl} Hi-Vis Pill, commenced POOH BHA f/3702 to surface & L/D the BHA w/6" R.bit (1-1-W-A-E-I-NO-BHA).

P/U & M/U D/O BHA w/5-7/8" used R.bit 217 open & started RIH to 3702^{m} while F/U the string every 15 Stds w/NL-NG, took weight @3702^{m}, commenced D/O CMT 3702 to 3705^{m} while pass L.G & dressed the same length, Cont. RIH f/3705 to 4243^{m} w/no drag till took weight @4243^{m}, started POOH D/O BHA to surface while Circ. across wellhead & monitor the active mud pit level w/NL-NG, L/D the D/O BHA & 5-7/8" R.bit (1-1-WT-A-E-I-NO-BHA). P/U & M/U 9-5/8" * 7" tandem scraper & commenced RIH to 3700^{m} while scraping inside 9-5/8" CSG & 7" LNR several times & break Circ. every 15 Stds & CBU @3700^{m} w/NL-NG, started POOH 9-5/8" * 7" tandem scraper to 3523^{m}, pumped & spot 30^{bbl} Hi-Vis Pill above 7" L.Lap in order to set 9-5/8" dry test PKR element, Cont. POOH & L/D 9-5/8" * 7" tandem scraper & 5-7/8" R.bit (1-1-WT-A-E-I—NO-DST).

P/U & M/U 9-5/8" dry test Assy. & F/U same w/cushion water 62.4^{pcf} & commenced RIH to 3505^{m}, R/U dry test manifold & chiksan & L/U them, performed pressure test

the lines in two step (500 – 2500^{psi} for 15min), OK. Set PKR @3501^{m} by rotating string 7 turns forward & S/O 25^{klbs} string weight, observed hydro spring opening indication, checked annulus level, static. Performed dry test 7" tie-back for 3hrs. 1st open, observed no bubble, closed DCIP valve by rotating string 11 turns forward, OK & recorded 1st close period while checking annulus level, static. Opened DCIP valve by rotating string 14 turns forward, OK & recorded 2nd open period, observed bubble, closed DCIP valve by rotating string 14 turns forward, OK & recorded 2nd close period while dropping the bar & performed Rev. Circ. till in/out mud weight got uniformed. B/O & L/D dry test line & manifold, unset the PKR by applying 40^{klbs} over pull (330^{klbs} on M/D) step by step, OK. Commenced POOH dry test Assy. & L/D BHA & R/D same, memory gauge data is as follow: Initial open period for 3hrs. 3691 – 3633^{psi}, Initial close period for 6hrs. 3640 – 4120^{psi}. Secondary open period for 2hrs. 3686 – 3626^{psi}, Secondary close period for 2hrs. 3628 – 3760^{psi}. The sample properties of between DCIP & hydro spring is as follow: pH: 9.6, Mw: 106^{pcf}.

Tie-Back Dry Test Report:



TIME	EVENTS
28 – Nov – 23	
22:30	Connect Memory Gauges battery
29 – Nov – 23	
00:55	RIH DST tools
30 – Nov – 23	
10:01	Set packer & Hydro Spring opened (Open the well)
13:06	DCIP closed (Close the well)
19:21	Open DCIP
21:25	Close DCIP
23:25	Drop the Bar and start circulating
01 – Dec – 23	
00:07	Unset Packer
01:15	POOH DST String
22:39	String at Surface
23:12	Disconnect Memory Gauges
	End of job

Table 2-5: AZN-10 Tie-Back Dry Test Event

Calendar (yyyy/mm/dd hh:mm:ss)	Elapsed time (hours)	Pressure (psia)	Temperature (degF)	Comment
2023/11/30 10:01:02	35.51722	8321.966	229.564	Set Packer
2023/11/30 10:01:32	35.52556	3628.708	229.352	Open Hydrospring & Well
2023/11/30 13:06:42	38.61167	3631.778	229.051	Close DCIP, Well closed for the first time
2023/11/30 19:21:22	44.85611	3650.419	230.015	Open DCIP, Open the well for second time
2023/11/30 21:25:02	46.91722	3624.669	230.088	Close DCIP, Well closed for the second time
2023/12/01 00:06:42	49.61167	3770.070	230.377	Unset Packer
2023/12/01 01:20:02	50.83389	8167.647	232.740	Start POOH

Table 2-6: AZN-10 Tie-Back Dry Test Information

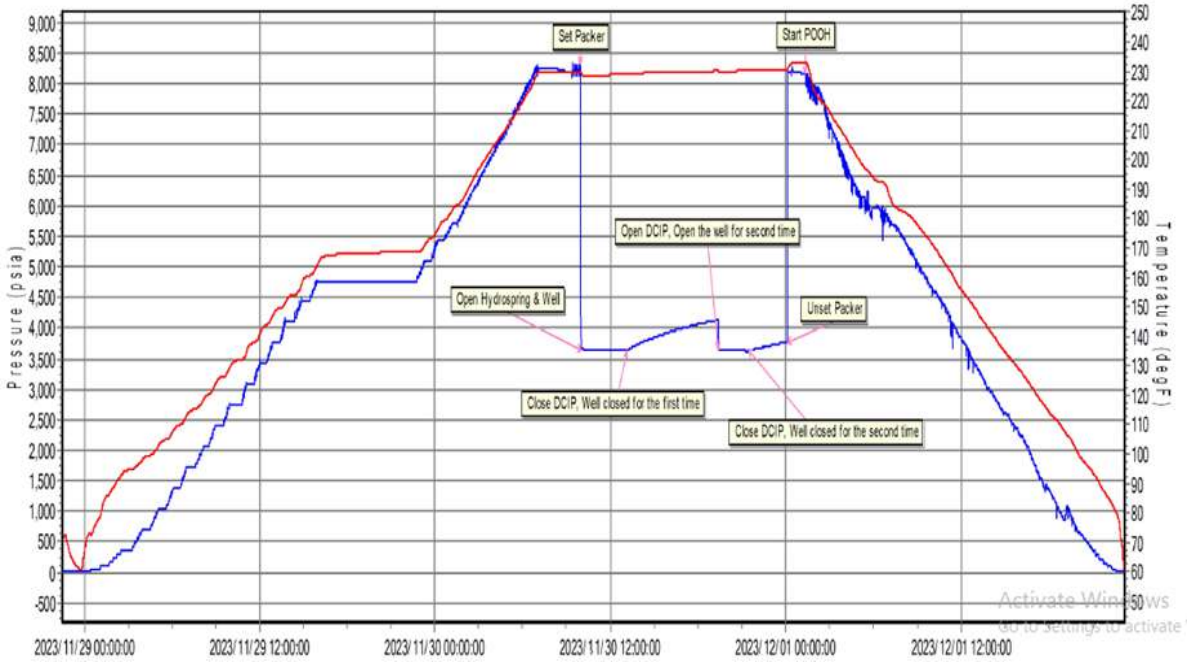


Figure 2-3: AZN-10 Tie-Back Dry Test Data Chart Middle B.T.

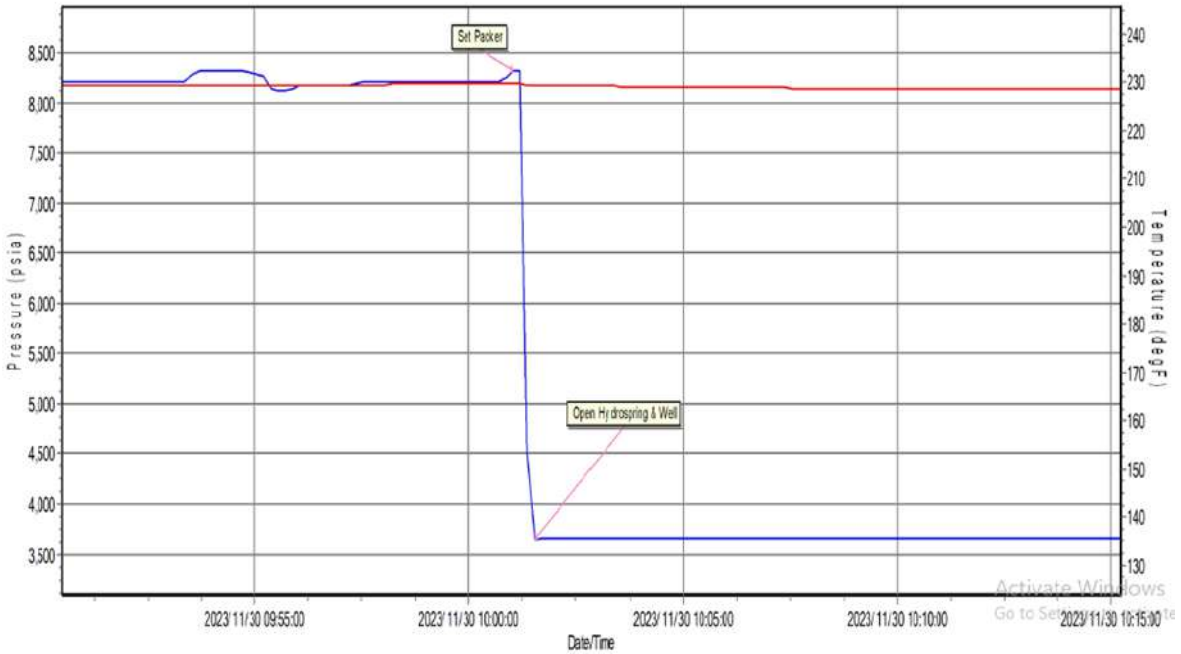


Figure 2-4: AZN-10 Tie-Back Dry Test Data Chart Bottom B.T..

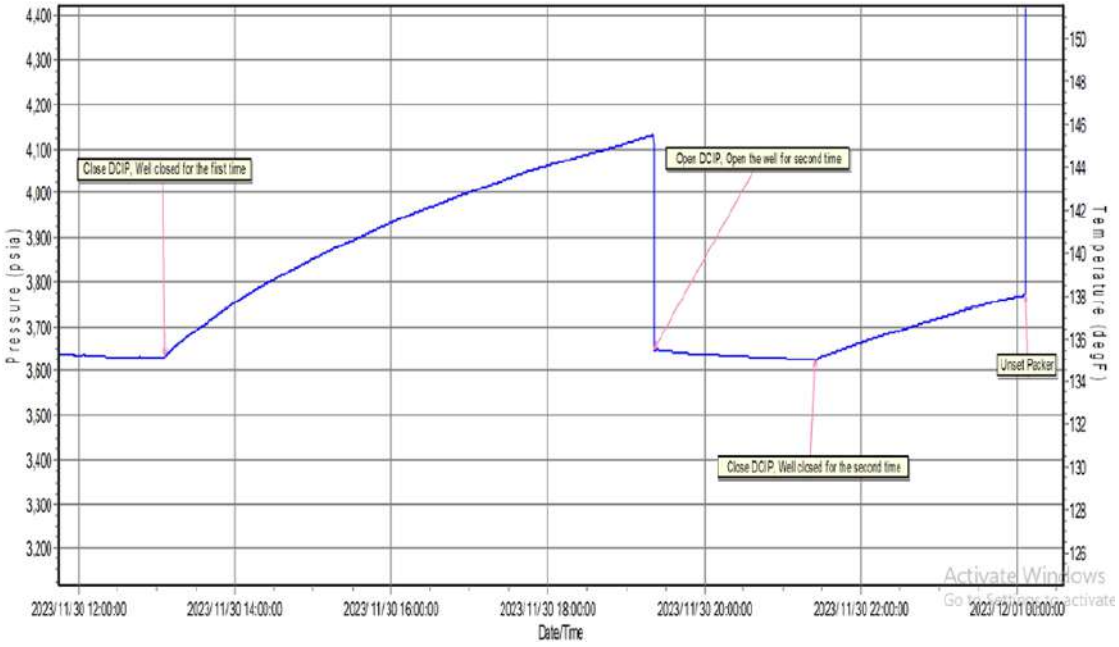


Figure 2-5: AZN-10 Tie-Back Dry Test Data Time Chart

Row	Description	Time	Bottom Bt.	Middle Bt.
1	Initial Mud Static Pressure (psi)	1402/09/09 10:00	8226	8174
2	Initial Open Well Pressure (psi)	1402/09/09 10:05	3691	3624
		1402/09/09 13:08	3633	3613
3	Initial Close Well Pressure (psi)	1402/09/09 13:09	3640	3616
		1402/09/09 19:20	4120	4104
4	Second Open Well Pressure (psi)	1402/09/09 19:21	3686	3624
		1402/09/09 21:27	3626	3604
5	Second Close Well Pressure (psi)	1402/09/09 21:28	3628	3606
		1402/09/09 23:51	3760	3737
6	Final Hydrostatic Pressure (psi)	1402/09/10 00:28	8188	8183

Table 2-7: AZN-10 Tie-Back Dry Test Information

Perforation Operation:

Performed BOP test as NIOC policy. Changed top rams 4-1/4" * 7" Var pipe ram. Install 11" plug tester, pressure test top rams to 7500^{psi}, OK. Retrieved plug tester, L/D mouse hole & install 7" shooting nipple. M/U W/L BOP on shooting nipple & pressure test same to 1000^{psi}, OK. R/U W/L tools & equipment, RIH CCL & 4-1/2" gun as dummy run to 3527^{m} (7" L.Lap) while Circ. CWH w/NL-NG, tried to pass the 7" L.Lap several times by reducing tension, No success. POOH W/L tool f/3527 to 1880^{m} while Circ. CWH, w/NL-NG. Started RIH CCL & 2 No. 4-1/2" gun as dummy run to 3527^{m} while Circ. CWH w/NL-NG, tried to pass the 7" L.Lap several times, No success. Started POOH W/L tools f/3527^{m} to surface & checked it visually, OK. L/D & R/D W/L equipment & shooting nipple, installed mouse hole, P/U & M/U magnet BHA w/5-3/16" magnet & RIH to 3700^{m} while F/U the string in every 15 Stds w/NL-NG, pass 7" L.Lap gently, performed break Circ. @ 3700^{m} for 30min. POOH 5-3/16" magnet BHA f/3700 to 3519^{m} on elevator, worked on junk sub by reciprocate the interval 3519 - 3535^{m} several times, tried to pass interval more, took weight @3528^{m}, applied rotation by 5^{RPM} & pass 3528^{m} several times, OK. Tried again without rotation, No Success. Performed CBU @3527- w/NL-NG. POOH 5-3/16" magnet BHA f/3527 to surface on elevator, while breaking Circ. till got return on Sh.Shakers & F/U the well in every 10 Stds w/NL-NG. B/O & L/D BHA & magnet, checked the magnet & junk sub & recovered 1^{lb} metal debris, P/U & M/U 11" plug tester & retrieved wear bushing, L/D mouse hole & installed 7" shooting nipple, M/U W/L BOP on shooting nipple & pressure test same to 1000^{psi}, OK. R/U W/L tools & equipment while Circ. CWH w/NL-NG, RIH CCL & 4-1/2" gun as dummy run to 4120^{m} while Circ. CWH, w/NL-NG then POOH to surface. P/U & M/U & RIH GR-CCL wireline to 4120^{m}, logged up f/4120 to 3500^{m} while Circ. CWH, w/NL-NG. POOH W/L tools f/3500^{m} to surface, R/D GR-CCL W/L tools while Circ. CWH w/NL-NG.

Connected detonator & RIH W/L perforation tools as 1st run to 4087^{m} while correlating depth by CCL-GR log, performed perforation f/4087 to 4082^{m}, POOH W/L tools to surface, Connected detonator & RIH W/L perforation tools as 2nd run to 4082^{m} while correlating depth by CCL-GR log, performed perforation f/4082 to 4077^{m}, POOH W/L tools to surface, Connected detonator & RIH W/L perforation tools as 3rd run to 4077^{m}, while correlating depth by CCL-GR log, performed perforation f/4077 to 4074^{m}, POOH W/L tools to surface, Connected detonator & RIH W/L perforation tools as 4th run to 4057^{m} while correlating depth by CCL-GR log, performed perforation f/4057 to 4051^{m}, POOH W/L tools to surface, Connected detonator & RIH W/L perforation tools as 5th run to 4051^{m} while correlating depth by CCL-GR log, performed perforation f/4051 to 4048^{m}, POOH W/L tools to surface, Connected detonator & RIH W/L perforation tools as 6th run to 4048^{m} while correlating depth by CCL-GR log, performed perforation

f/4048 to 4042m, POOH W/L tools to surface, Connected detonator & RIH W/L perforation tools as 7th run to 4040.6^{m} while correlating depth by CCL-GR log, performed perforation f/4040.6 to 4036.6^{m}, POOH W/L tools to surface, Connected detonator & RIH W/L perforation tools as 8th run to 4036.6^{m} while correlating depth by CCL-GR log, performed perforation f/4036.6 to 4034^{m}, POOH W/L tools to surface, Connected detonator & RIH W/L perforation tools as 9th run to 4030.5^{m} while correlate depth by CCL-GR log, performed perforation f/4030.5 to 4027^{m}, POOH W/L tools to surface w/normal W/L cable tension while Circ. CWH & monitor the well w/NL-NG. Checked the gun visually in the presence of Client witness, all bullets were shot successfully in each run. R/D W/L tools, P/U plug tester & set wear bushing, changed elevator & handling equipment.

P/U & M/U 5-7/8" M.M.J.mill & commenced RIH to 4077^{m} while F/U string in every 15 Stds till 3700^{m} w/NL-NG, took weight @4077^{m}, M/U TDS, ream down to 4101^{m} w/rotation, resumed RIH f/4101 to 4177^{m}, took weight @4177^{m}, M/U TDS, RIH to TD @4242^{m} w/rotation. POOH milling BHA f/TD to surface, retrieved wear bushing, L/D mouse hole & installed 7" shooting nipple, M/U W/L BOP & pressure test same to 1000^{psi} for 15min, OK. R/U wire line tools & equipment & RIH CCL to 4120^{m}, logged up f/4120 to 3700^{m}, confirmed perforation interval by client. POOH W/L tools to surface & R/D same while Circ. CWH w/NL-NG. Changed BOP top rams to 2-7/8" * 5" Var pipe ram, set plug tester & pressure test same in two steps (500 – 7500^{psi}) for 15min, OK. Retrieved plug tester & set wear bushing, P/U & M/U 8-1/2" B.bit & RIH to 496^{m}, POOH 8-1/2" B.bit & L/D Jt by Jt. M/U scraper BHA w/5-7/8" R.bit & RIH to BTM of perforated zone @4087^{m} in several runs. Pumped & Circ. out 40^{bbl} Hi-Vis Pill till clean the Sh.Shakers, pumped as balance 45^{bbl} $CaCl_2$ Brine 84^{pcf} to 3700^{m} w/NL-NG. POOH scraper BHA f/4087 to 3700^{m} on elevator while L/D 5" DPs Jt by Jt & preparing $CaBr_2$ completion fluid 104^{pcf}. R/U chiksan & L/U to string, pressure test the line, OK. Pumped 20^{bbl} Hi- Vis Pill, tried to displacing completion fluid, contractor pump truck got failed, changed pump truck by back up one, displaced the well from 3700^{m} by completion fluid 104^{pcf} 760^{bbl} in hole, 815^{bbl} was made, observed the well static. POOH scraper BHA f/3700 to surface on elevator, while L/D 5" DPs Jt by Jt & F/U well w/104^{pcf} completion fluid by T.Tank, w/NL-NG.

Completion Summary

Retrieved wear bushing, R/U casing running & completion equipment then test the same, all OK. Tried to M/U test connection, No success. Hydraulics system of casing running equipment not work properly, tried to fix it, OK. P/U & M/U 3-1/2" Mule Shoe + 3-1/2" BTM No-Go + 3-1/2" 1st Pup joint Jt of 3-1/2" TBG & RIH to 11.95m. P/U & M/U + 3-1/2" Pup joint 7" PKR + 3-1/2" Rach Latch + 3-1/2" Pup joint + 2nd Jt of 3-1/2" TBG & RIH f/11.95 to 29.38m w/NL-NG. P/U & M/U 3-1/2" Pup joint + 3-1/2" SSD + 3-1/2" Pup joint + 3rd Jt of 3-1/2" TBG & RIH f/29.38 to 43.41m w/NL-NG. R/U slick line & set 2-3/4" check valve in SSD (SSD in close position), R/D slick line. P/U & M/U 3-1/2" Pup joint + 3-1/2" Traveling Joint + 3-1/2" Pup joint + 4th Jt of 3-1/2" TBG & RIH f/43.41 to 62.23m w/NL-NG. P/U & M/U 3-1/2" Pup joint + 3-1/2" First CIV + 3-1/2" Pup joint + 5th Jt of 3-1/2" TBG & RIH f/62.23 to 76.23m w/NL-NG. P/U & M/U 3-1/2" Pup joint + Second 3-1/2" CIV + 3-1/2" Pup joint + 6th Jt of 3-1/2" TBG + 3-1/2" Pup joint & RIH f/62.23 to 80.61m w/ NL-NG. Prepared & installed CIL sheaves on monkey board & set CIL on CIVs, performed pressure test to 4000psi for 15min, OK. Resumed RIH 3-1/2" TBG f/80.61 to 223.32m w/NL-NG. Tried to M/U 3-1/2" TBG by Joint #A21, observed thread damage, No Success. L/D the same Joint. Cont. RIH 3-1/2" TBG f/223.32 to 251.85m w/NL-NG. Changed Elevator & handling equipment then resumed P/U & M/U XOS 4-1/2" * 3-1/2" & RIH 4-1/2" TBG f/252.87 to 262.4m w/NL-NG. R/U slick line & RIH w/2.8" gauge ring to top of SSD (drift string), OK. R/D slick line. Prepared for pressure test on completion string & R/U chiksans & pipes & L/U on completion string, performed pressure test the line & hold for 15min, then connected to completion string & performed 7000psi stepwise & hold for 30min, in presence of client witness, OK. Resumed RIH 4-1/2" TBG f/262.4 to 442.5m w/NL-NG. Cont. RIH 4-1/2" TBG f/442.5 to 1011m w/NL-NG. Prepared for pressure test on completion string & R/U chiksans & pipes & L/U on completion string, performed pressure test the line & hold for 15min, then connected to string & performed 7000psi stepwise & hold for 30min, in presence of client witness, OK. Cont. RIH 4-1/2" TBG f/1011 to 1769m w/NL-NG. Prepared for pressure test on string & R/U chiksans & pipes & L/U on string, performed pressure test the line & hold for 15min, then connected to string & performed 7000psi stepwise & hold for 30min, in presence of client witness, OK. Cont. RIH 4-1/2" TBG f/1769 to 2528m w/NL-NG. Prepared for pressure test on string, R/U chiksans & pipes & L/U on string. Cont. pressure test the line & hold for 15min, then connected to string & performed 7000psi stepwise & hold for 30min, in presence of client witness, OK. Cont. RIH 4-1/2" TBG f/2528 to 3604m w/NL-NG. P/U & M/U 4-1/2" Pup Joint + 4-1/2" Short Flow Coupling + TRSSV + 4-1/2" Long Flow Coupling + 4-1/2" Pup Joint, connected CL to DHSV & resumed RIH 4-1/2" TBG f/3610 to 3621m, prepared for pressure test on CL for 15min, 500 – 8000psi, OK. Then on completion string, R/U chiksans & pipes & L/U on string, performed pressure test the line & hold for 30min, then connected to

string & performed 7000[psi] stepwise & hold for 30min, in presence of client witness, OK. Then performed inflow test on TRSSV for 15min, on 1000[psi], OK. Performed equalize function of 3SV, OK. Cont. RIH 4-1/2" TBG f/3621 to 3659[m] w/NL-NG. P/U & M/U 4-1/2" Fluted Swage Adaptor + 11" * 4-1/2" TBG Hanger 10K WXN, pass CIL & CL through TBG Hanger, then P/U 4-1/2" Landing Joint & RIH to 3670[m]. R/U slick line & RIH with as run #1: Installed 3.437" Protection Sleeve of TRSSSV, run #2: Retrieved 2-3/4" WFB-2 Check valve from SSD, run #3: tried to Set 2-3/4" WRB-2 check valve in Landing Nipple, tried to pass SSD, no success, POOH to surface, observed Check valve left in the hole. RIH w/ No-Go Check valve to top of SSD, tried to pass, no success, POOH to surface, observed Check valve left in hole. RIH w/SB Pulling to retrieve Check valve, POOH to surface, observed Check valve not retrieved. Again, RIH w/SB Pulling to retrieve Check valve, POOH to surface, observed Check valve retrieved, RIH w/No-Go Check valve to top of SSD, tried to pass, no success, POOH to surface, observed Check valve left in hole. RIH w/SB Pulling to retrieve Check valve, POOH to surface, observed Check valve not retrieved. Operation stopped to change slick line crew, while installing Golpa flange & slick line BOP to secure the well, function test the BOP, OK. RIH to top of SSD, passed SSD, set Check Valve in No-Go & started POOH to surface. RIH w/2-75" Check valve to No-Go depth, installed Check valve as OEOC specialist approved. POOH to SSD depth, tried to pass SSD, no success. Requested for NIDC slick line specialist, passed SSD by supervision of NIDC specialist. POOH to surface, observed Check valve in SB pulling. Redressed Check valve & SB pulling, then RIH to No-Go, installed Check valve, POOH to surface. R/D slick line equipment, Golpa flange & BOP. R/U CMT line for set PKR. Set PKR as per completion engineer instruction with max 3500[psi] (Obs. Shearing indication @2500[psi]). Increased pressure to 7000[psi] & hold for 15min, OK. Bleed off pressure. Performed pull test with 15[klbs] overpull, OK. Land TBG HGR in spool & while Obs. shearing indication of expansion Jt shear pin. Performed final tests to 5000[psi] stepwise after 13min, Obs. suddenly return from annulus & dropped pressure to Zero. R/U slick line & ran SSD Check valve to 3035[m], Obs. string took weight, tried to pass this point, no success. POOH to surface, ran w/2.8" gauge cutter to 3029[m], Obs. string took weight, tried to pass this point several times, no success. POOH to surface, ran GS Pulling & retrieved Protection Sleeve f/TR3SV. POOH to surface, closed the TR3SV. Bleed off the pressure to Zero. R/U & installed 4-1/16" manual gate valve on Golpa flange, installed lubricator, test all barrier w/1000[psi] & set protection sleeve in TR3SV. Ran 2/8" impression block to 150[m] while increasing pressure in completion string to 850[psi], annulus pressure to 850[psi]. Cont. RIH w/2/8" impression block to 3014[m], POOH to surface & Obs. no mark on that. P/U & M/U 1-7/8" bailer & RIH to 3008[m], worked on catch sample, POOH to surface, Obs. no sample, pressure in completion string is 850[psi], annulus pressure is 850[psi]. Received 3 Solution Tank, prepared location & set the solution tank position, P/U & M/U chiksan & L/U to Golpa flange & CMT unit & connect to choke manifold, L/U choke line on

THS side outlet. Displaced the completion fluid in the hole w/WBM 104^{pcf} w/back pressure 500^{psi} w/NL-NG. Performed Circ. till uniform the in/out Mw 104^{pcf} while monitoring the well several times during the Circ. Pumped 70^{bbl} slug mud 106^{pcf} to balance the mud in completion string & annulus. L/U THS to choke manifold, R/D Golpa flange, P/U & M/U the safety valve on completion string, POOH the completion string for 1.5^{m} & applied 20^{klbs} overpull as pull test, OK. Prepared for run 2-1/2" blind box, R/D the safety flange, P/U & M/U the Golpa flange & slick line BOP. RIH w/2-1/2" blind box, POOH to surface. Circ. BTM's up till uniform in/out Mw, RIH w/SB R/T to retrieve check valve from No-Go LND Nipple, POOH to surface. No success. Again RIH w/SB R/T to retrieve check valve successfully. RIH w/2-3/4" plug & set to No-Go LND Nipple, POOH to surface. RIH w/equalizing prong to set at 2-3/4" plug, POOH to surface. Tried to perform string/annulus pressure, test w/700^{psi}, no success. RIH w/2-3/4" check valve to set in SSD, POOH to surface. Pressure test completion string w/5000^{psi}, RIH to retrieve check valve, POOH to surface. Prepared completion fluid while monitor the well & break Circ. & uniform the mud, meanwhile pressure test the annulus w/700^{psi}, OK. While preparing & improve 103.5^{pcf} $CaBr_2$ brine. Commenced displace the well w/180^{bbl} NaCl brine & after that pumped 103.5^{pcf} $CaBr_2$ brine. Cont. displace the annulus w/totally 620^{bbl} 103.5^{pcf} $CaBr_2$ brine (pumped 180^{bbl} 75^{pcf} brine as ahead). Opened S/L BOP & RIH w/shifting tools to close the SSD. Made sure SSD is close. POOH to surface. Decreased string pressure f/2500 to 500^{psi} & held same for 30min, OK (performed SSD negative pressure test). Bleed off pressure, opened the annulus preventer & Obs. annulus level, static. Wash & clean inside THS. Performed pressure test string against NO-GO LND Nipple Plug to 6800^{psi} for 25min, OK. Land TBG HGR in THS. Fasten tie-down screw. Performed final pressure test completion string w/6500^{psi} for 20min, OK. Performed pressure test annulus w/700^{psi} for 15min, OK. Retrieved 3.437" Protection Sleeve f/TR3SV. Disconnected CMT line f/string & connected to BOP stack. Pressure test tie-down screw w/annulus preventer to 2000^{psi} for 10min, OK. Back of landing Jt & L/D same. Installed 4" TWCV in TBG HGR. Pressure test 4" TWCV, HGR & tie-down against SBR to 7000^{psi} for 15min, OK. Disconnected BOP hydraulic hoses. N/D Bell Nipple. N/D 13-5/8" 10K BOP stack. N/U 11" * 4-1/16" 10K BMV MSP. Passed CIL & CL from BMV. L/U pump truck to BMV. Pressure test 3/8" CIL w/2000^{psi} for 15min, OK. Pressure test 1/4" CL w/8000^{psi} for 15min, OK. Performed cavity test, Hanger neck F-seal test & MM- Seal test w/8000^{psi} for 15min, OK. N/U 4-1/16" 10K MSP upper section of XMT on BMV. L/U test line to kill line. Body test XMT against 4" TWCV to 8000^{psi} for 15min, OK. Retrieved 4" TWCV from TBG HGR. Closed all valves of XMT except BMV & 2SV. Released the Rig 201 OEOC @1402-10-08, 23:59.

2.3 AZN-10 Well Sketch

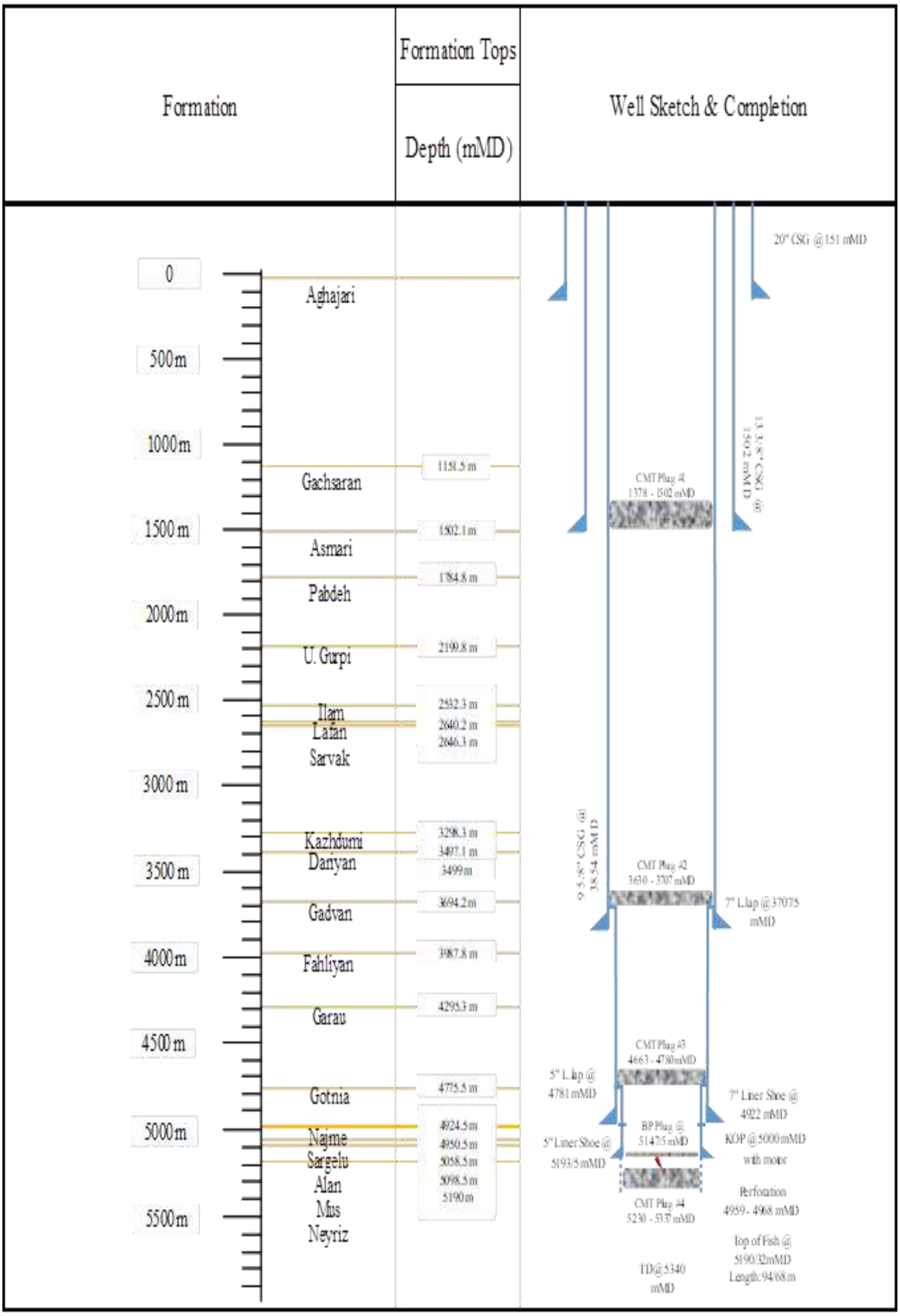


Figure 2-6: AZN-10 Well Sketch Before WorkOver

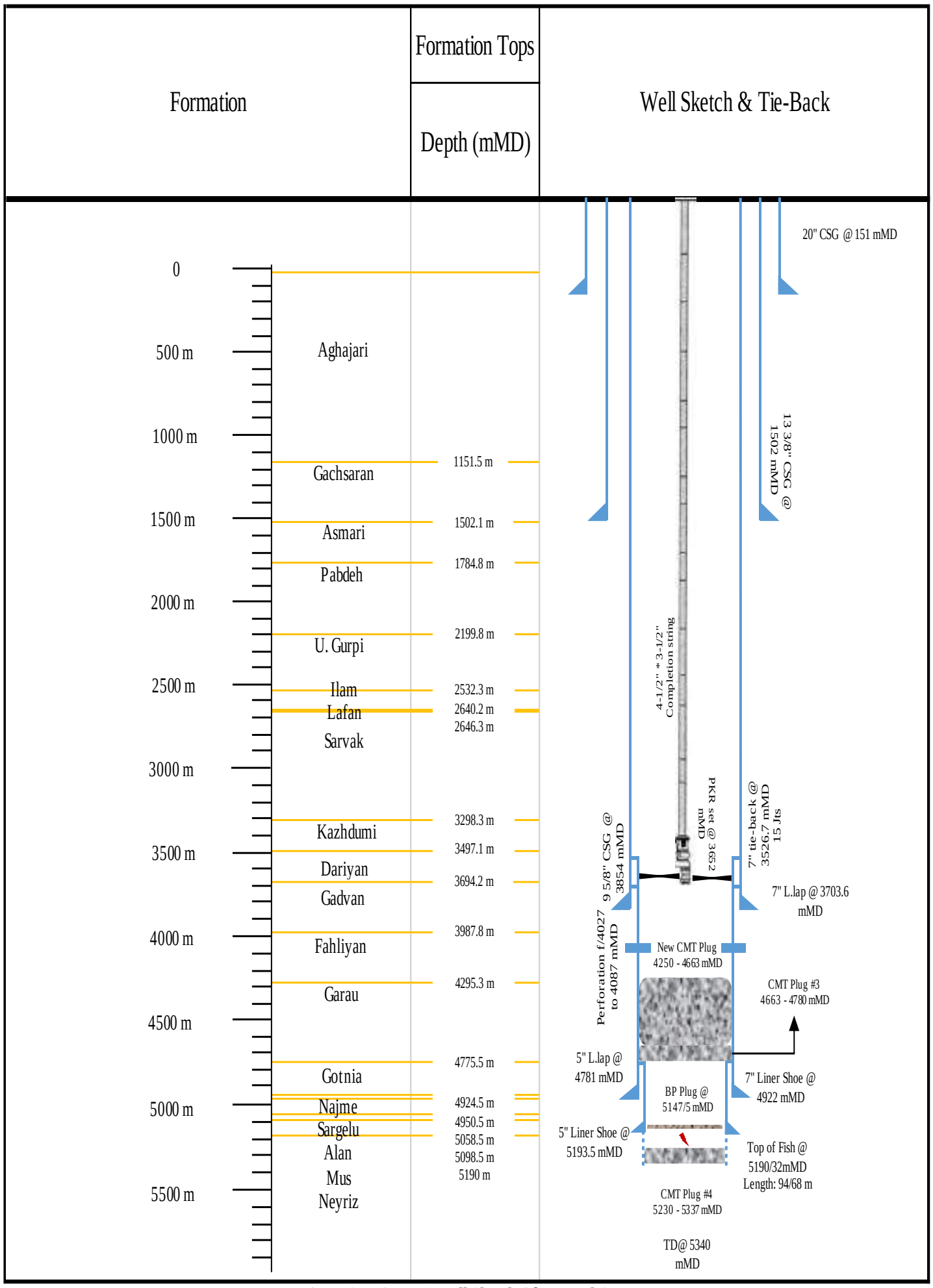


Figure 2-7: AZN-10 Well Sketch After WorkOver

2.4 Bit Record Table

Bit No	Size	Bit Manufacture	BHA No	Type	IADC Code	Serial No	Jets			CMT	Depth		Formation (Lithology)	Depth		Metres	Hrs	ROP	WOB (klb)	Pump Press	GPM		Mud Weight pcf		TFA	IR	OR	DC	Loc	B/S	G/16	OC	R Pulled
							No	x	Size		In	Out		In	Out																		
1	8-1/2	King Dream	1	B.Bit	M517	2003238-JH	Open			Yes	-	-		5340	5340								102		1								
1RR1	8-1/2	King Dream	1	B.Bit	M517	2003238-JH	Open			Yes	-	-		5340	5340	285	28.5		10,14		410		102		1	1	1	WT	A	E	I	NO	PR
2	8-1/2	Varel	2	PDC	616	600554	Open			Yes	-	-		5340	5340	297	113.5		10,15	260 - 650			102		1	1	1	WT	A	X	I	PN	BHA
3RR2	5 7/8	King Dream	3	R.Bit	217	2204638JH	Open			No	-	-		5340	5340	15.4	7.5						102		1	1	1	WT	A	E	I	NO	DST
4	5 7/8		4	C.C			Open			No	-	-		5340	5340								102		1								
4RR	5 7/8		5	C.C			Open			yes	-	-		5340	5340								102		1								
3RR3	5 7/8	King Dream	6	R.Bit	217	2204638JH	Open			No	-	-		5340	5340	94	11		3,12				102		1	1	1	WT	A	E	I	NO	DST
3RR4	5 7/8	King Dream	7	R.Bit	217	2204638JH	Open			No	-	-		5340	5340	94	11		3,12				102		1	1	1	WT	A	E	I	NO	DST
1RR2	8-1/2	King Dream	9	B.Bit	M517	2003238-JH	Open			No	-	-		5340	5340	285	28.5				300		102		1	1	1	WT	A	E	I	NO	PR
1RR3	8-1/2	King Dream	13	B.Bit	M517	2003238-JH	Open			No	-	-		5340	5340	285	28.5						102		1	1	1	WT	A	E	I	NO	BHA
1RR4	8-1/2	King Dream	15	B.Bit	M517	2003238-JH	Open			No	-	-		5340	5340	285	28.5						102		1	1	1	WT	A	E	I	NO	BHA
1RR5	8-1/2	King Dream	15	B.Bit	M517	2003238-JH	Open			No	-	-		5340	5340	285	28.5						102		1	1	1	WT	A	E	I	NO	BHA
RR	8-1/2	Varel	17	PDC (VTD616DGX)	616	600554	Open			No	-	-		5340	5340	355	105		1,7				102		1	1	1	HC	A	X	I	BT	BHA
5	6	King Dream	18	R.Bit	117	2105260	3	*	20	No	-	-		5340	5340	175	23		5,12				102		1	1	1	HC	A	E	I	NO	BHA
3RR6	5-7/8	King Dream	19	R.Bit	217	2204638JH	Open			No	-	-		5340	5340	102	13		1,10				102		1	1	1	WT	A	E	I	NO	DST
3RR7	5-7/8	King Dream	20	R.Bit	217	2204638JH	Open			No	-	-		5340	5340	102	13		1,10				102		1	1	1	WT	A	E	I	NO	DST
3RR7	5-7/8	King Dream	25	R.Bit	217	2204638JH	Open			No	-	-		5340	5340	102	13		1,10				102		1	1	1	WT	A	E	I	NO	DST

Table 2-8: AZN-10 Bit Record Table

2.5 Hole Problem History

Not Applicable.

3. Time Analysis Report

3.1 Time vs. Depth Chart

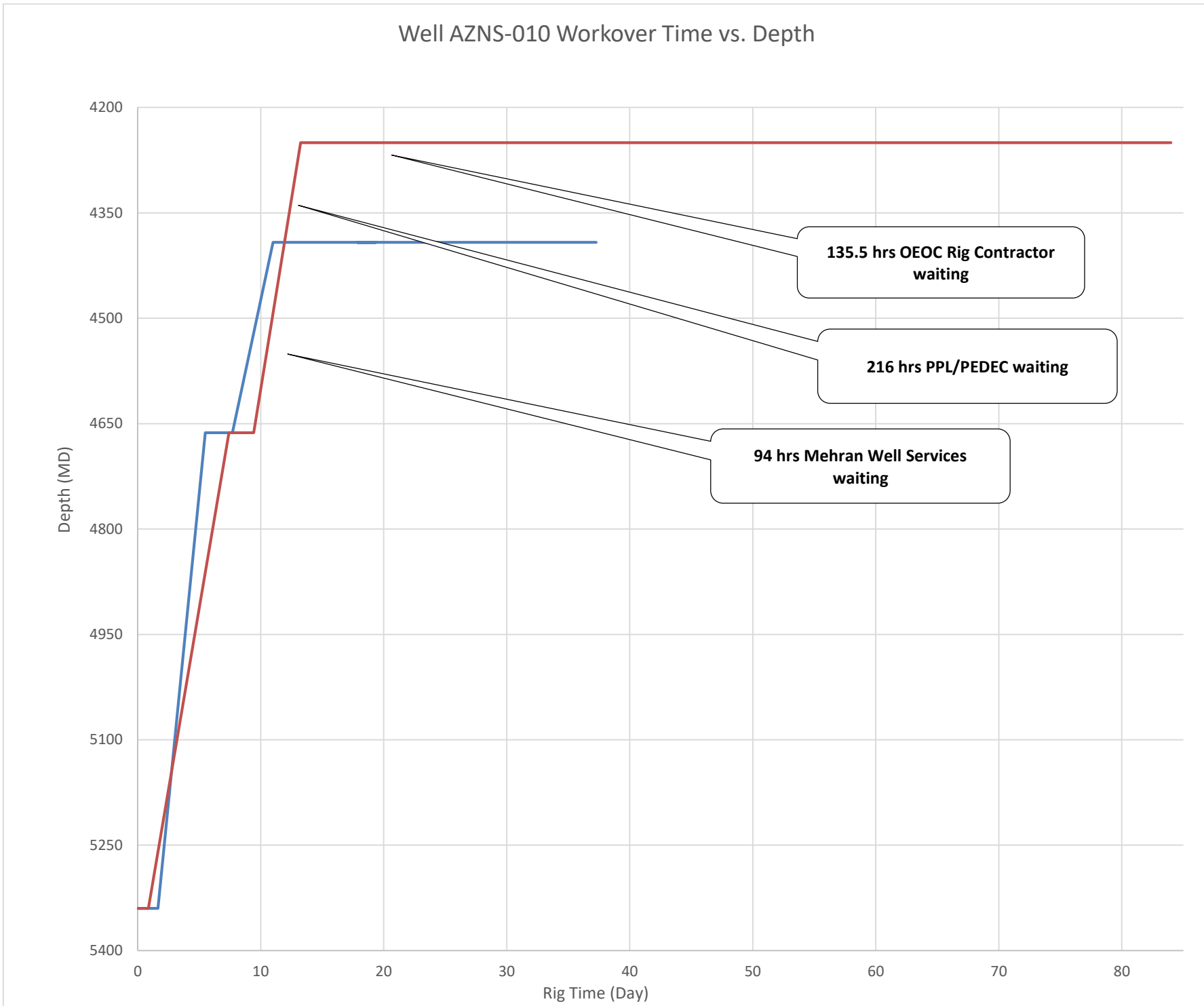


Figure 3-1: AZN-10 Workover Well Time vs. Depth

3.2 Total Time Break-Down of Operation

WorkOver Section

De-Complete, Work-over and Completion Operation TBD of Well: AZNS-10 Work-over	IADC Code	Depth (meters)	Time Duration (Hours)	Time Duration (Days)	Total Duration (Days)	Contingency (%) 10	Time Duration w/ contingency (Hours)	Time Duration w/ contingency (Days)	Total Duration w/ contingency (Days)	Actual Depth (meters)	Time Duration w/ contingency (Actual Hours)	Time Duration w/ contingency (Actual Days)	Total Duration w/ contingency (Actual Days)	Remark
Work-over														
Held P.JSM. Spud the Rig. Stand Back required 5" DPs, 5" HWDPs and 6-1/2" DC.	SPUD	5340	0.0	0.00	0.0	0.0	0.0	0.00	0.00	5340.00	0.00	0.00	0.00	
Check the wellhead for any damage or leakage. Record well head pressures; check the all annulus pressure. Check the return fluid & Bleed off any annulus pressure to Zero upon client's approval and recheck the well status. Note: If annulus pressure is not Zero after bleed off, follow contingency plan #1.	Wellhead	5340	4.0	0.17	0.2	0.2	4.4	0.18	0.2	5340.0	0.00	0.00	0.00	
Remove 11" blind flange & change wellhead section C with new one.	Wellhead	5340	18	0.75	0.9	0.8	19.8	0.83	1.0	5340.0	0.00	0.00	0.00	
N/U BOP and pressure test. Perform wellhead pressure integrity test.	BOP	5340	14	0.58	1.5	0.6	15.4	0.64	1.7	5340.0	21.00	0.88	0.88	
RIH with 8-1/2" BHA & drill out cement plug #1 F/1378m to 1502m.Continue drill out cement plug #2 F/3630m to 3707m, tag top of 7" liner, circulate till shakers clean & POOH to surface.	D/O	4663	84	3.50	5.0	3.9	92.4	3.85	5.5	4663.0	157.00	6.54	7.42	1.5 hrs. waiting for OEOC due to work on mud pump. 2 hrs. waiting for OEOC due to slip & cut. 19.5 hrs. waiting for OEOC
RIH with 5-7/8" Bit & tag cement plug #3 @ 4663m, Dress the CMT (drill out soft cement if any), CBU till shakers clean, POOH & L/D Bit.	D/O	4663	48	2.00	7.0	2.2	52.8	2.20	7.7	4663.0	48.50	2.02	9.44	2 hrs. OEOC waiting due to delay in RIH. 2 hrs. OEOC waiting due to work on TDS torque wrench.
RIH with 2-7/8" open end DPs, put CMT plug from 4663m to top of Garau (base of Fahliyan, 4392m), POOH to surface.	Plug Back	4392	72	3.00	10.0	3.3	79.2	3.30	11.0	4250.0	91.50	3.81	13.25	3.5 hrs. Spad Energy waiting due to delay to prepare CMT plug spacer & solution. 4 hrs. string plug in cement.
RIH with 7" & 9-5/8" tandem scrapper & scrap inside 9-5/8" (RTTS PKR depth, 3620 - 3680m) & 7" liner, POOH to surface.	Trip	4392	48	2.00	12.0	2.2	52.8	2.20	13.2	4250.0	119.00	4.96	18.21	64 hrs. OEOC waiting due to plug string. 2 hrs. OEOC waiting due to work on mud pump #2. 2.5 hrs. displace CaCl2
Perform BOP test.	BOP Test	4392	6	0.25	12.3	0.3	6.6	0.28	13.5	4250.0		0.00	18.21	
RIH with wireline logging tools (SBT or CAST/CBL/VDL) inside 7" liner, perform logging, POOH.	Log	4392	24	1.00	13.3	1.1	26.4	1.10	14.6	4250.0	8.00	0.33	18.54	
RIH with wireline corrosion logging tool	Log	4392	24	1.00	14.3	1.1	26.4	1.10	15.7	4250.0	52.00	2.17	20.71	21 hrs. Sealand waiting.
RIH RTTS PKR & perform 7" liner lap & cement plug dry test, POOH to surface. Note: If dry test is not ok, follow contingency plan #2.	Trip	4392	42	1.75	16.0	1.9	46.2	1.93	17.6	4250.0	79	3.29	24.00	33 hrs. OEOC dry test service waiting due to being behimd schedule.
R/U 7" tie-back liner running eqp. Run 7" tie-back liner f/3530m to L.lap, CMT 7" tie-back, P/U & M/U D/O BHA & RIH to D/O CMT, Perform 7" tie-back dry test.	CSG	4392	0	0.00	16.0	0.0	0.0	0.00	17.6	4250.0	766	31.92	55.92	New L.Lap Depth is 3703m. 8 hrs. OEOC waiting due to string droped. 216 hrs wait on receive tie-back facilities. 6 hrs waiting force majeure due to bad weather. 6 hrs OEOC waiting due to delay in trips. 6 hrs OEOC waiting due to work on TDS 10 hrs wait on client decision

Table 3-1: AZN-10 WorkOver Time Breakdown

Completion Section

RIH with 3-1/2" DP to 4392m (top of the new CMT plug), displace the 7" liner with completion fluid (CaBr2)	Completion	4392	38	1.58	17.6	1.7	41.8	1.74	19.3	4250.0	16	0.67	56.58	16 Wait on instructions for the next operation
Perform BOP test.	BOP Test	4392	6	0.25	16.3	0.3	6.6	0.28	17.9	4250.0	4	0.17	56.75	
Run perforation gun assembly & perforate 7" liner by TCP (F/4027m to 4057m, Fah-1a & 1b), POOH to surface	Completion	4392	60	2.50	18.8	2.8	66.0	2.75	20.6	4250.0	162.5	6.77	63.52	try to pass 7" L.Lap several times with no success. P/U magnet BHA (DDR-58-59) 3 hrs wait on client decision. 37.5 hrs. wait on day light.
Pick up 5 7/8" Mill & RIH W/ Clean out BHA, Clean inside 7" Liner to TD & displace the hole with CaBr2, POOH while L/D DP's.	Completion	4392	40	1.67	20.4	1.8	44.0	1.83	22.5	4250.0	47	1.96	65.48	3 hrs. PDK waiting due to pump truck failure.
RIH with GR-CCL log o confirm perforation intervals.	Completion	4392	12	0.50	20.9	0.6	13.2	0.55	23.0	4250.0	8.5	0.35	65.83	
RIH with DP's & DC's, POOH while L/D same.	Completion	4392	48	2.00	22.9	2.2	52.8	2.20	25.2	4250.0	89	3.71	69.54	14 hrs. RIH w/scraper BHA & 5-7/8" R.bit (2 hrs. OEOC waiting due to delay in operation & 2 hrs. OEOC Drilling Fluid Dep. waiting due to delay in preparing Comp. fluid)
Perform BOP test.	BOP Test	4392	6	0.25	23.2	0.3	6.6	0.28	25.5	4250.0	3	0.13	69.67	
Run 10K completion string as per program.	Completion	4392	96	4.00	27.2	4.4	105.6	4.40	29.9	4250.0	312	13.00	82.67	4 hrs. NDCO waiting dut to Hydraulics problem in casing running tools. 2 hrs. OEOC waiting due to o delay in operation and absent of crane operator. 19 Hrs. Spad Energywaiting due to failure in pump truck. 2 Hrs. OEOC waiting due to lack of crane.
Set BPV inside tubing hanger, N/D BOPs.	Completion	4392	14	0.58	27.8	0.6	15.4	0.64	30.5	4250.0	10	0.42	83.08	
Retrieve BPV, install TWCV in TBG Hanger,N/U X-mass Tree,Perform Required test as per program,Retrive TWCV	Completion	4392	8	0.33	28.1	0.4	8.8	0.37	30.9	4250.0	14	0.58	83.67	Rig released.
R/U Slick Line, Set protection sleeve in TRSSSV, R/D Slick Line. retrieve plug & prong from NO GO landing nipple, POOH.	Completion	4392	8	0.33	28.4	0.4	8.8	0.37	31.3	4250.0		0.00	83.67	
R/U CTU, Stimulation and Well Testing Eq. RIH w/ CT to Btm, displace the well with Diesel, POOH CT to surface.	Completion	4392	24	1.00	29.4	1.1	26.4	1.10	32.4	4250.0		0.00	83.67	
R/U Slick Line, Retrieve protection sleeve from TRSSSV, R/D Slick Line.	Completion	4392	4	0.17	29.6	0.2	4.4	0.18	32.5	4250.0		0.00	83.67	
Flow the well as pre-acid clean up till acceptable BSW & reading stable flow rate on 2 different choke size for 72 hrs.	Completion	4392	80	3.33	32.9	3.7	88.0	3.67	36.2	4250.0		0.00	83.67	
RIH with S/L, set protection Sleeve, POOH & R/D SL.	Completion	4392	Rig Less	0.00	32.9	0.0	0.0	0.00	36.2	4250.0		0.00	83.67	
RIH with C/T & perform acid wash as per program, POOH to surface.	Completion	4392	Rig Less	0.00	32.9	0.0	0.0	0.00	36.2	4250.0		0.00	83.67	
RIH with S/L, retrieve protection sleeve, POOH & R/D SL.	Completion	4392	Rig Less	0.00	32.9	0.0	0.0	0.00	36.2	4250.0		0.00	83.67	
Flow the well as post-acid clean-up till acceptable BSW & reading stable flow rate on 2 different choke size for min 96 hrs. NOTE: The ime is only for flowing duration (R/U, R/D & preparation time is excluded.)	Completion	4392	Rig Less	0.00	32.9	0.0	0.0	0.00	36.2	4250.0		0.00	83.67	
RIH with S/L, set protection sleeve.	Completion	4392	Rig Less	0.00	32.9	0.0	0.0	0.00	36.2	4250.0		0.00	83.67	
RIH with C/T to bottom & displace the well with diesel, POOH.	Completion	4392	Rig Less	0.00	32.9	0.0	0.0	0.00	36.2	4250.0		0.00	83.67	
RIH with S/L, install plug & prong into NO GO landing nipple, POOH.	Completion	4392	Rig Less	0.00	32.9	0.0	0.0	0.00	36.2	4250.0		0.00	83.67	
RIH with S/L, retrieve protection sleeve.	Completion	4392	Rig Less	0.00	32.9	0.0	0.0	0.00	36.2	4250.0		0.00	83.67	
Secure the well by set BPV in TBG Hanger, close all X-mas Tree valves except BMV, Release the Rig.	Completion	4392	Rig Less	0.00	32.9	0.0	0.0	0.00	36.2	4250.0		0.00	83.67	
Total Duration of Work-over:				*33.0*	*32.9*			*38.0*	*36.2*			*0.0*	*83.67*	
NOTE:														
Total Duration for Cut & Slip DL in all hole section.	Cut & Slip	4392	4	0.17	33.1	0.2	4.4	0.18	36.4	4250.0	8	0.33	84.00	
Total Duration for Rig Service (Consider 0.5hr per day) in all hole section.	Rig Service	4392	19	0.79	33.9	0.9	20.9	0.87	37.3	4250.0		0.00	84.00	
Total Duration of AZNS-010 Work-over Well:		4392			33.9				37.3					
Contingency Plans TBD of Well: AZNS-010 Work-over	IADC Code	Depth (meters)	Time Duration (Hours)	Time Duration (Days)	Total Duration (Days)	Contingency (%) 10	Time Duration w/ contingency (Hours)	Time Duration w/ contingency (Days)	Total Duration w/ contingency (Days)			0.0	84.00	

Table 3-2: AZN-10 Completion Time Breakdown

3.3 NPT Break-Down

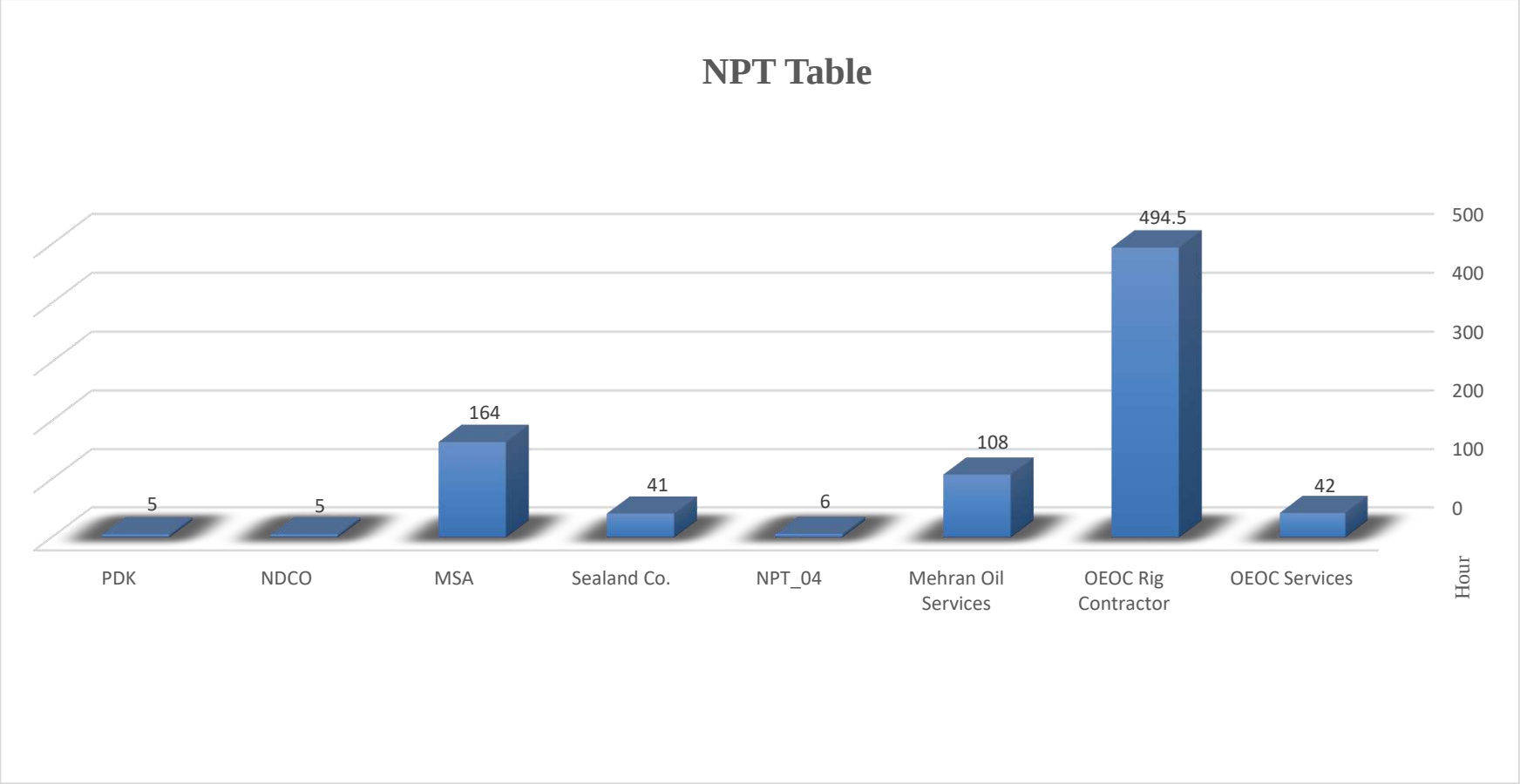


Figure 3-2: AZN-10 NPT Break-Down Char

Item	Company	Date	Problem event	Lost Time (hrs)
1	OEOC	1402.07.16	Plug string above top of float valve	4
2	OEOC	1402.07.19	Slip & cut drilling line.	2
3	OEOC		Work on mud pump #1, Delay in operation while tripping in due to low level of active pitfor wash DN & change active pit to dissander	8
4	OEOC	1402.07.23	Work on TDS	2
5	OEOC	1402.07.25	Failure balance CMT plug & plugged 21 Stds 3 – ½" DPs w/ hard cement	24
		1402.07.26		24
		1402.07.27		24
		1402.07.28		6
6	OEOC		Slip & cut drilling line.	2
7	OEOC	1402.07.29	Saver sub of TDS failure & change it	3
8	OEOC		Torque wrench of TDS failure & plugged string, Drawworks failure, Fishing	12
9	OEOC	1402.07.30	7" tandem scraper failure due to plug string by a bolt in 4 – ¾" bit sub, Repair 7" scraper & fix blade	24
		1402.08.01		24
10	OEOC	1402.08.02	Work on mud pump #2, Work on mud properties due to malfunction of mud pumps & aerated mud	12
11	Sealand Co.	1402.08.04	Not confirmed run #3 due to noisy data, Repeat run #3 & #4	24
		1402.08.05		17
12	OEOC	1402.08.09	Wait on decision	7
13	OEOC	1402.08.10	Delay in operation	2

14	OEOC	1402.08.13	Drop the rabbit into the string.	8
15	OEOC	1402.08.15	Rec. polish mill	2
		1402.08.16		13
16	OEOC		Leak f/ 7” liner lap & SN packer, Rec. liner & tie back	11
		1402.08.17		24
		1402.08.18		24
		1402.08.19		24
		1402.08.20		24
		1402.08.21		24
		1402.08.22		24
		1402.08.23		24
17	Mehran	1402.08.24	Wait on Rec. CMT & pump truck to run 7” tie back	24
		1402.08.25		24
		1402.08.26		24
		1402.08.27		24
		1402.08.28		12
18	NPT_04	1402.08.29	Wait on weather	6
19	OEOC	1402.09.02	Slip & cut drilling line.	2
20	OEOC	1402.09.07	Delay in operation	2
21	OEOC	1402.09.08	Work on TDS	7
22	OEOC	1402.09.11	Wait on decision	6

23	OEOC	1402.09.12	Try to pass logging tools 7” tie back w/ no success	24
		1402.09.13		21
24	OEOC	1402.09.14	Wait on OEOC (MSA CO.) to accept the risk & resposibility of the operation due to used non-standard BOP stack in perforation job	4.5
25	OEOC	1402.09.15	Wait on daylight	7.5
		1402.09.16		11.5
		1402.09.17		11
		1402.09.18		6
26	OEOC	1402.09.22	Delay in prepare completion fluid	6
27	PDK	1402.09.23	Pump truck failure	3
28	NDCO	1402.09.25	Delay in operation	1
			Hydraulic system failure	4
29	OEOC	1402.09.26	Delay in operation & absence of crane operator	2
30	OEOC Services	1402.09.29	Try to set check valve in No-Go w/ no success	12
		1402.09.29	Wait on arrive new slickline crew & pull out check valve from SSD	6
		1402.09.30	Wait on to set check valve in No-Go	24
31	MSA CO.	1402.10.01	Final string pressure test failure	24
		1402.10.02		24
		1402.10.03		24
		1402.10.04		24
		1402.10.05		3.5

32	PDK	1402.10.05	Pump truck failure	2
33	MSA CO.	1402.10.05	Final string pressure test failure	16.5
34	OEOC	1402.10.05	Lack of crane & prepare completion fluid	2
35	MSA	1402.10.06	Final string pressure test failure	24
		1402.10.07		24
Total NPT (hour)				865.5
Total NPT (day)				36

Table 3-3: AZN-10 NPT Time BreakDown

4. Log Reports

Run No.	Date	Depth Interval	Log Suite
1	25 – 10 – 23 1402 – 08 – 03	4247^{m} to 3700^{m}	VDL+SBT+nature GR+CCL
2	25 – 10 – 23 1402 – 08 – 03	4100^{m} to 4000^{m}	CBIL+CDR+GR+CCL
3	26 – 10 – 23 1402 – 08 – 03	3706^{m} to Surf.	CBIL – CDR – GR – CCL
4	26 – 10 – 23 1402 – 08 – 03	2400^{m} to surf.	CBIL – CDR – GR – CCL (repetation)
5	11 – 12 – 23 1402 – 09 – 20	4120^{m} to 3700^{m}	GR – CCL

Table 4-1: AZN-10 Logging Information

5. Bottom Hole Assembly Record

Tool Description	Length (m)	Cum. Length (m)
BHA No. #1		
8 – ½" R. bit	0.25	0.25
6 – 3/8" Bit sub	1.11	1.36
6 – ½" DC	134.86	136.22
5" HWDP	27.95	164.17
Total Length (m)		**164.17**
Drilling & Mud Properties		
WOB (lb)		10 – 14
RPM		30 – 70
Torque (klb)		1 – 2
SPP (psi)		-
Flow (gpm)		410.00
Mw (pcf)		94.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #2		
8 – ½" PDC	0.29	0.29
6 – 3/8" Bit sub	1.11	1.4
6 – ½" DC	134.86	136.26
5" HWDP	27.95	164.21

Total Length (m)		164.21
Drilling & Mud Properties		
WOB (lb)		5 – 20
RPM		40 – 60
Torque (klb)		1 – 4
SPP (psi)		1800 – 2000
Flow (gpm)		-
Mw (pcf)		99.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #3		
5 – 7/8” R. bit	0.20	0.20
4 – ¾” Bit sub	1.22	1.42
4 – ¾” DC	163.68	165.1
3 – ½” HWDP	55.22	220.32
3 – ½” DP	954.08	1174.4
Total Length (m)		**1174.4**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-

Mw (pcf)		102.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #4		
5 – 7/8" C.C bit	0.10	0.10
4 – ¾" Bit sub	1.225	1.325
3 – ½" DP	1445.81	1447.135
Total Length (m)		**1447.135**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		102
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #5		
5 – 7/8" C.C bit	0.10	0.10
4 – ¾" Bit sub	0.86	0.96
3 – ½" DP	722.90	723.86
Total Length (m)		**723.86**
Drilling & Mud Properties		
WOB (lb)		-

RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		102.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #6		
5 – 7/8” R. bit	0.20	0.20
7” Scraper	1.06	1.26
4 – ¾” Bit sub	0.85	2.11
4 – ¾” DC	163.68	165.79
3 – ½” DP	385.81	551.60
XOS (3-1/2" IF Pin x 4-1/2" IF Box)	0.81	552.41
5” DP	9.62	562.03
6 – ½” D.PIN (4 1/2" IF *4 1/2" REG)	0.69	562.72
9 – 5/8” Scraper	1.39	564.11
6 – ½”Bit sub (float in)	1.11	565.22
Total Length (m)		**565.22**
Drilling & Mud Properties		
WOB (lb)		3 – 12
RPM		-

Torque (klb)	1 – 3
SPP (psi)	-
Flow (gpm)	-
Mw (pcf)	100.00

Tool Description	Length (m)	Cum. Length (m)
BHA No. #7		
5 – 7/8” R. bit	0.20	0.20
7” Scraper	1.06	1.26
4 – ¾” Bit sub	0.85	2.11
4 – ¾” DC	163.84	165.95
3 – ½” DP	385.81	551.76
XOS (3-1/2" IF Pin x 4-1/2" IF Box)	0.81	552.57
5” DP	9.62	562.19
6 – ½” D.PIN (4 1/2" IF *4 1/2" REG)	0.69	562.88
9 – 5/8” Scraper	1.39	564.27
6 – ½” Bit sub (float in)	1.11	565.38
Total Length (m)		**565.38**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-

SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		100.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #8		
5" B. bit	1.26	1.26
5" Anchor Pipe	1.53	2.79
X.Over 3 1/2" EUE box -3 1/2"FH pin	0.40	3.19
8 – ¼” Middle Packer Element	1.33	4.52
8 – 1/5” Above Paker 9 5/8"	0.76	5.28
X.Over 3 – 1/2"IF box 4 – 1/2"IF pin	0.40	5.68
X.Over 3 – 1/2"FH box 3 – 1/2"IF pin	0.40	6.08
RH Release Safety Joint	0.73	6.81
Hydraulic Jar	1.00	7.81
5” Middle B.T(Second Memory Gauge)	1.25	9.06
HydroSpring	1.56	10.62
D.C.I.P	1.64	12.26
X.Over 4 – 1/2" IF box 3 – 1/2" IF pin	0.25	12.51
5” DP	9.62	22.13

Revers Sub	0.4	22.53
6 – ½" DC	80.58	103.11
Total Length (m)		**103.11**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		102.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #9		
8 – ½" R. bit	0.24	0.24
6 – 3/8" Bit sub	1.11	1.35
5" HWDP	27.95	29.30
Total Length (m)		**29.30**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		300

Mw (pcf)		102.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #10		
7 – 5/16” Polish mill	0.56	0.56
6 – ½” D. pin (4-1/2" IF * 4-1/2 REG)	0.69	1.25
6 – ½” Bit sub (float in)	1.11	2.36
6 – ½” DC	53.51	55.87
5” HWDP	52.82	108.69
Total Length (m)		**108.69**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		102.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #11		
8 – ¼” S.N Packer	3.59	3.59
P. Jnt	1.60	5.19
6 – ½” DC	134.86	140.05
5” HWDP	55.82	195.87

Total Length (m)		195.87
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		102.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #12		
5" B. bit	1.26	1.26
5" anchor pipe	1.53	2.79
Xover 3-1/2" EUE box * 3-1/2" FH pin	0.40	3.19
8 – ¼" Middle packer element	1.33	4.52
8 – 1/5" Above packer 9-5/8"	0.76	5.28
Xover 3-1/2" IF box * 4-1/2" IF pin	0.40	5.68
Xover 3-1/2" FH box * 3-1/2" IF pin	0.40	6.08
RH release safety Jt	0.73	6.81
HYDRAULIC JAR	1.00	7.81

5” MIDDLE B.T(second memory gauge)	1.25	9.06
HYDROSPRING	1.56	10.62
D.C.I.P	1.64	12.26
Xover 4-1/2" IF box * 3-1/2" IF pin	0.25	12.51
5” DP	9.62	22.13
Revers Sub	0.40	22.53
6 – ½” DC	80.58	103.11
Total Length (m)		**103.11**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		102.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #13		
8 – ½” R. bit	0.24	0.24
6 – 3/8” Bit sub	1.11	1.35
5” HWDP	27.95	29.30
Total Length (m)		**29.30**

Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		102.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #14		
7 – 5/16” Polish mill	0.49	0.49
6 – ½” D. pin (4-1/2" IF * 4-1/2 REG)	0.69	1.18
6 – ½” Bit sub (float in)	1.11	2.29
6 – ½” DC	53.51	55.80
5” HWDP	55.82	111.62
Total Length (m)		**111.62**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		102.00

Tool Description	Length (m)	Cum. Length (m)
BHA No. #15		
8 – ½" R. bit	0.24	0.24
9 – 5/8" Scraper	1.38	1.62
6 – 3/8" Bit sub (float in)	1.11	2.73
6 – ½" DC	107.62	110.35
5" HWDP	55.82	166.17
Total Length (m)		**166.17**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		102.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #17		
8 – ½" PDC bit	0.29	0.29
6 – 3/8" Bit sub	1.11	1.40
6 – ½" DC	107.62	109.02
5" HWDP	55.82	164.84
Total Length (m)		**164.84**

Drilling & Mud Properties		
WOB (lb)		1 – 7
RPM		50
Torque (klb)		1 – 3
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		102.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #18		
6” PDC bit	0.18	0.18
4 – ¾” Bit sub	1.23	1.41
4 – ¾” DC	109.13	110.54
3 – ½” HWDP	55.22	165.76
3 – ½” DP	86.98	252.74
XOS	0.79	253.53
Total Length (m)		**253.53**
Drilling & Mud Properties		
WOB (lb)		1 – 7
RPM		50.00
Torque (klb)		1 – 3
SPP (psi)		-
Flow (gpm)		-

Mw (pcf)		102.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #19		
5 – 7/8" R. bit	0.20	0.20
4 – ¾" Bit sub	1.23	1.43
4 – ¾" DC	109.13	110.56
3 – ½" HWDP	55.22	165.78
3 – ½" DP	722.90	888.68
XOS	0.79	889.47
Total Length (m)		**889.47**
Drilling & Mud Properties		
WOB (lb)		1 – 10
RPM		50
Torque (klb)		2 – 5
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		102.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #20		
5 – 7/8" R. bit	0.18	0.18
7" Scraper	1.01	1.19
4 – ¾" Bit sub	1.23	2.42

4 – ¾” DC	54.71	57.13
3 – ½” HWDP	55.22	112.35
3 – ½” DP	67.74	180.09
XOS	0.79	180.88
6 – ½” D. pin	0.69	181.57
9 – 5/8” Scraper	1.39	182.96
6 – ½” Bit sub	1.11	184.07
5” DP	9.62	193.69
Total Length (m)		**193.69**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		102.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #21		
5” B. bit	1.26	1.26
5” Anchor pipe	1.53	2.79
XOS	0.40	3.19
8 – ¼” Middle Packer Element	1.33	4.52

8 – 1/5” Above Packer 9-5/8”	0.76	5.28
XOS 3 1/2"IF BOX -4 1/2"IF PIN	0.40	5.68
XOS 3 1/2"FH BOX -3 1/2"IF PIN	0.40	6.08
RH.R.S.Jt	0.73	6.81
H. jar	1.00	7.81
5” M. bit	1.25	9.06
Hydro spring	1.56	10.62
D.C.I.P	1.64	12.26
X.Over 4-1/2” IF box * 3-1/2” IF pin	0.25	12.51
5” DP	9.62	22.13
Revers Sub	0.40	22.53
6 – ½” DC	107.62	130.15
Total Length (m)		**130.15**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		102.00

Tool Description	Length (m)	Cum. Length (m)
BHA No. #22		
5 – 3/16” Magnet	0.43	0.43
4 – ¾” D. pin	0.28	0.71
5” Junk sub	0.78	1.49
4 – ¾” Bit sub	1.23	2.72
4 – ¾” DC	27.48	30.20
3 – ½” HWDP	55.22	85.42
3 – ½” DP	722.90	808.32
XOS	0.79	809.11
Total Length (m)		**809.11**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		102.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #24		
5 – 7/8” Mill	0.93	0.93
4 – ¾” Bit sub (float in)	1.23	2.16

4 – ¾” DC	109.13	111.29
3 – ½” HWDP	55.22	166.51
3 – ½” DP	722.90	889.41
XOS	0.79	890.90
Total Length (m)		**890.20**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		102.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #25		
5 – 7/8” B. bit	0.20	0.20
4 – ¾” Bit sub (float in)	1.23	1.43
4 – ¾” DC	81.94	83.37
3 – ½” HWDP	55.22	138.59
3 – ½” DP	201.99	340.58
D. pin	0.28	340.86
5 – 3/8” Scraper	1.00	341.86
4 – ¾” Bit sub (float in)	0.86	342.72

3 – ½" DP	231.79	574.51
XOS	0.79	575.30
Total Length (m)		**575.30**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		104.00

6. Casing Information

6.1 7" Tie-Back Tally

MSA

Field:	South Azadegan	Well NO:	10	Rig No:	201	Drlg Co:	OEOC
Client:	PEDEC/ PPL	Job:	7" Liner Tie Back			Date:	11/14/2023

Grade	ID "	Drift "	Weight lb/ft	Burst psi	Collaspse psi	Yield klbs	Coupling OD in	TQ Optimum ft x lb	Nominal Dia. OD
C-110	6.184	6.125	29	11,220	8,530	929000	-	11480	7

Old 7" L.Lap	3703/6	Hook Weight :	66 k lb
New 7" L.Lap	3526/86	STRG CAP (bbl):	221
7" LDC :	3679.36	Mud Weight (pcf) :	102
Final 7"Over Lap:	1395.26	7" LNR Shoe (m):	4922
New 7"Over Lap	176.74		

Number on jt	Grade	Order No.	Jt Length (m)	Cum Length (m)	Dist to RT (m)	M/D weight (Klbs)	STRG CAP. (bbl)	Remarks	
								Centralizers	IN / OUT
					3703.6	66.000	0.000		
Seal Nipple	7" Sael Nipple, 29#, C-110, HSM3	-	0.300	0.300	3,703.300	66.023	0.037		
1	7" LNR, 29#, C-110, HSM3	1	11.844	12.144	3,691.456	66.924	1.480	Centralizers	
2	7" LNR, 29#, C-110, HSM3	2	11.841	23.985	3,679.615	67.825	2.923		
LDC	7" LDC, 29#, C-110, HSM3	-	0.258	24.243	3,679.357	67.854	2.955		
3	7" LNR, 29#, C-110, HSM3	3	11.445	35.688	3,667.912	68.725	4.350		
4	7" LNR, 29#, C-110, HSM3	4	11.085	46.773	3,656.827	69.569	5.701	Centralizers	
5	7" LNR, 29#, C-110, HSM3	5	11.207	57.980	3,645.620	70.422	7.067		
6	7" LNR, 29#, C-110, HSM3	6	11.478	69.458	3,634.142	71.296	8.466		
7	7" LNR, 29#, C-110, HSM3	7	11.534	80.992	3,622.608	72.174	9.872	Centralizers	
8	7" LNR, 29#, C-110, HSM3	8	11.049	92.041	3,611.559	73.015	11.219		
9	7" LNR, 29#, C-110, HSM3	9	11.400	103.441	3,600.159	73.883	12.608		
10	7" LNR, 29#, C-110, HSM3	10	11.842	115.283	3,588.317	74.784	14.052	Centralizers	
11	7" LNR, 29#, C-110, HSM3	11	11.653	126.936	3,576.664	75.671	15.472		
12	7" LNR, 29#, C-110, HSM3	12	11.842	138.778	3,564.822	76.573	16.915		
13	7" LNR, 29#, C-110, HSM3	13	11.637	150.415	3,553.185	77.458	18.334	Centralizers	
14	7" LNR, 29#, C-110, HSM3	14	11.828	162.243	3,541.357	78.358	19.775		
15	7" LNR, 29#, C-110, HSM3	15	11.816	174.059	3,529.541	79.258	21.216	Centralizers	
T.B +LG +POB	29#, C-110, HSM3	-	2.800	176.859	3526.741	79.468	21.557		
Neck	Neck of setting tools	-	2.500	179.359	3524.241	79.569	21.610		
5" DP	122Std + Double (49 "S" & 74std "G")	-	3,520.190	3,699.549	4.051	221.286	96.062		
5" Pup Jt	2no. 5" Pup Jt	-	6.200	3,705.749	-2.149	221.639	96.414		

Total joint on board	15
Total joint on rack	15
Total join RIH	15
Remain on Rack	0
Remain on Board	0

Liner weight in mud (Klbs)	13.47
Liner weight in Air (Klbs)	17.01
5" Grade G RIH (Std)	74.00
5" Grade S RIH (Std)	49.00

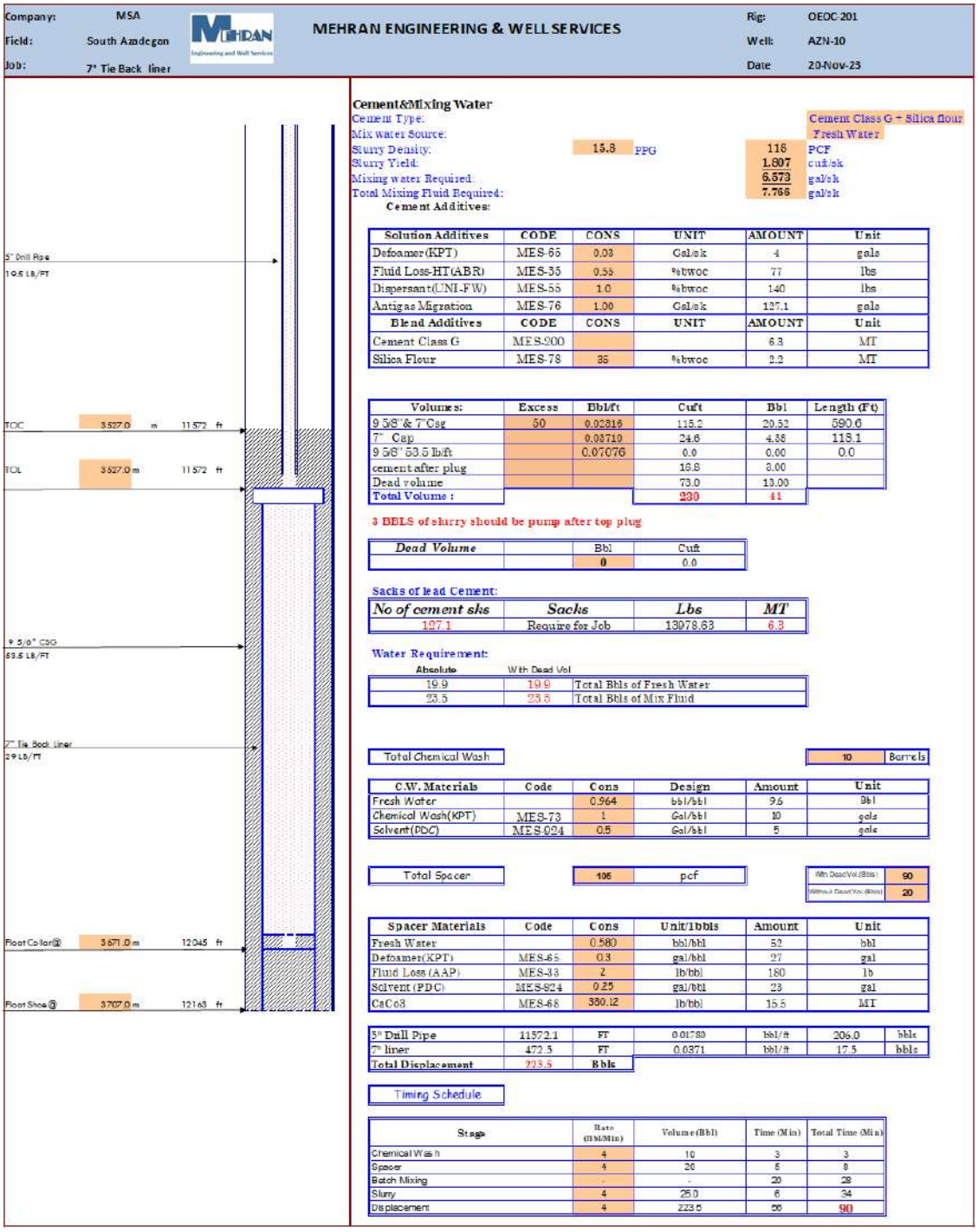
Figure 6-1: AZN-10 Tie-Back Tally

Casing Spec.	20"	K-55, 133#	0 – 151^{m}
	13-3/8"	N-80, 72#	0 – 1502^{m}
	9-5/8"	T-95,53.5#	0 – 3854^{m}
	7"	T-95, 32#	3707.5 – 4922^{m}
	7" Tie-Back	C-110, 29#	3527 – 3707.5^{m}
	5"	T-95, 18#	4781 – 5193.5^{m}

Table 6-1: AZN-10 Casing Information

7. Cement Job & Plug Back Report

7.1 7" Tie-Back Cement Job Report



Company: MSA
Field: South Azadegan
Job: 7" Tie Back liner

MEHRAN ENGINEERING & WELL SERVICES

Rig: OEOC 201
Well: AZN-10
Date: 20-Nov-23

5" Drill Pipe 19.5 LB/FT
TOC 3527.0 m 11572 ft
TOL 3527.0 m 11572 ft
9 5/8" CSG 53.5 LB/FT
7" Tie Back Liner 29 LB/FT
Float Collar@ 3671.0 m 12045 ft
Float Shoe@ 3707.0 m 12163 ft

Cement&Mixing Water

Cement Type:			Cement Class G + Silica flour
Mix water Source:			Fresh Water
Slurry Density:	15.8 PPG	118	PCF
Slurry Yield:		1.607	cuft/sk
Mixing water Required:		6.573	gal/sk
Total Mixing Fluid Required:		7.786	gal/sk

Cement Additives:

Solution Additives	CODE	CONS	UNIT	AMOUNT	Unit
Defoamer(KPT)	MES-65	0.03	Gal/sk	4	gals
Fluid Loss-HT(ABR)	MES-35	0.55	%bwoc	77	lbs
Dispersant(UNI-FW)	MES-55	1.0	%bwoc	140	lbs
Antigas Migration	MES-76	1.00	Gal/sk	127.1	gals
Blend Additives	**CODE**	**CONS**	**UNIT**	**AMOUNT**	**Unit**
Cement Class G	MES-200			6.3	MT
Silica Flour	MES-78	35	%bwoc	2.2	MT

Volumes:	Excess	Bbl/ft	Cuft	Bbl	Length (Ft)
9 5/8"& 7"Csg	50	0.02316	115.2	20.52	590.6
7" Cap		0.03710	24.6	4.38	118.1
9 5/8" 53.5 lb/ft		0.07076	0.0	0.00	0.0
cement after plug			16.8	3.00	
Dead volume			73.0	13.00	
Total Volume :			230	41	

3 BBLS of slurry should be pump after top plug

Dead Volume		Bbl	Cuft
		0	0.0

Sacks of lead Cement:

No of cement sks	Sacks	Lbs	MT
127.1	Require for Job	13978.63	6.3

Water Requirement:

Absolute	With Dead Vol	
19.9	19.9	Total Bbls of Fresh Water
23.5	23.5	Total Bbls of Mix Fluid

Total Chemical Wash — 10 Barrels

C.W. Materials	Code	Cons	Design	Amount	Unit
Fresh Water		0.964	bbl/bbl	9.6	Bbl
Chemical Wash(KPT)	MES-73	1	Gal/bbl	10	gals
Solvent(PDC)	MES-924	0.5	Gal/bbl	5	gals

Total Spacer — 105 pcf — With Dead Vol.(Bbls) 90 — Without Dead Vol.(Bbls) 20

Spacer Materials	Code	Cons	Unit/1bbls	Amount	Unit
Fresh Water		0.580	bbl/bbl	52	bbl
Defoamer(KPT)	MES-65	0.3	gal/bbl	27	gal
Fluid Loss (AAP)	MES-33	2	lb/bbl	180	lb
Solvent (PDC)	MES-924	0.25	gal/bbl	23	gal
CaCo3	MES-68	350.12	lb/bbl	15.5	MT

5" Drill Pipe	11572.1	FT	0.01780	bbl/ft	206.0 bbls
7" liner	472.5	FT	0.0371	bbl/ft	17.5 bbls
Total Displacement	223.5	Bbls			

Timing Schedule

Stage	Rate (Bbl/Min)	Volume (Bbl)	Time (Min)	Total Time (Min)
Chemical Wash	4	10	3	3
Spacer	4	20	5	8
Batch Mixing	-	-	20	28
Slurry	4	25.0	6	34
Displacement	4	223.5	56	90

Figure 7-1: AZN-10 Well Schematic & Cement Program

MEHRAN ENGINEERING & WELL SERVICES

MEHRAN CEMENTING LABORATORY

CEMENT Lab REPORT

Report No: Well: AZN-10 Date: 20-Nov-2023
Company: Field: South Azadegan
Well Depth: (TVD): 3707 m 12163 ft Job: 7" Tie Back liner
(TD): 3707 m 12163 ft Rig: OEOC-201

Cementing Additives	Mehran Code	Conc.	Units
Defoamer(KPT)	MES-65	0.03	Gal/sk
Fluid Loss-HT(ABR)	MES-35	0.55	%bwoc
Dispersant(UNI-FW)	MES-55	1.00	%bwoc
Antigas Migration	MES-76	1.00	Gal/sk

Blend Additives	Mehran Code	Conc.	Units
Cement Class G	MES-200		
Silica Flour	MES-78	35	%bwoc

Cement and Mixing Water
Cement Type: Cement Class G + Silica flour
Mix Water Source: Fresh Water
Slurry Density: 15.8 PPG 118 PCF
Slurry Yield: 1.807 cuft/sk
Mixing Water Required: 6.573 gal/sk
Total Mixing Fluid Required: 7.766 gal/sk

Temperature BHST: 118 °C BHCT: 89 °C
Temperature BHST: 244 °F BHCT: 193 °F Temperature Gradient: 1.35 °F / 100 ft
Heat up time: min Final Pressure : 9000 PSI

API Free Fluid: 0 % @ 2 Hrs.
API Fluid Loss: 30 cc 193 °F and 1000 Psi

Rheology Data										PV	YP
°F	300	200	100	60	30	6	3	10 sec Gel	10 min Gel	CP	lbs/100ft²
80	101	72.5	36	20.5	8.5	2.5	2	2	46	97.5	3.5
185	110	81	47	31	17.5	5	3	3	11	94.5	15.5

Pumping Time: 30(BC): 3:40 Hrs:Min 50(BC): 3:44 Hrs:Min 70(BC): 3:47 Hrs:Min

C.S (UCA Test): 3589 Psi in 24 Hrs @ 244 °F

PH: Prehydrated DB: Dry Blended %bwoc: % by weight of cement %bwow: % by weight of water

Note: Calc. based on 110 lbs sack of dry cement.
Remarks: 20 min Batch Mixing Time is considered
Choloride of Fresh Water: 1000 ppm

MEHRAN Lab Analyst: V.Dariushi MEHRAN Engineer: Pouya Najafi
CMT MANAGER: Soheil Saeedi

Notice: This report is for information only and the content is limited to the sample described. Mehran makes no warranties, express or implied whether of fitness for a particular purposemerchantability, or otherwise, as to the accuracy of the contents or results. Any user of this report agrees Mehran shall not be liable for any loss or damage, regardless of cause, includingany act or omission of Mehran, resulting from the use thereof.

Figure 7-2: AZN-10 Cement Lab Report

Required Material, Equipment & Personnel

Total Material				
Row	**Description**	**Contractual Reference**	**Unit**	**Volume**
1	Cement Class G	-	MT	6.3
2	Silica Flour	-	MT	2.2
3	Defoamer (KPT)	-	gal	31
4	Solvent (PDC)	-	gal	30
5	Fluid Loss H.T (ABR)	-	lb	110
6	Dispersant Uni (ABR)	-	lb	165
7	Fluid Loss (ABR)	-	lb	220
8	Antigas Migration	-	gal	127
9	$CaCO_3$	-	MT	12
10	Chemical Wash (KPT)	-	gal	10

Table 7-1: AZN-10 Total Material

Required Equipment					
Row	**Description**	**Contractual Reference**	**Quantity**	**Unit**	**Subtotal**
1	Cement Unit	-	2	each	2
2	Silo	-	1	each	1
3	Batch Mixer	-	1	each	1
4	Compressor	-	1	each	1
5	Pipe	-	1	set	1
Total Number of Equipment				each/set	6

Table 7-2: AZN-10 Required Equipment

Required Personnel					
Row	**Description**	**Contractual Reference**	**Quantity**	**Unit**	**Subtotal**
1	Cementing Supervisor	-	1	each	1
2	Cementing Operatopr	-	2	each	2
3	Bulk Plant Operator	-	1	each	1
4	Cementing Helper	-	4	each	4
5	Mechanic	-	1	each	1
Total Number of Personnel				each	9

Table 7-3: AZN-10 Required Personnel

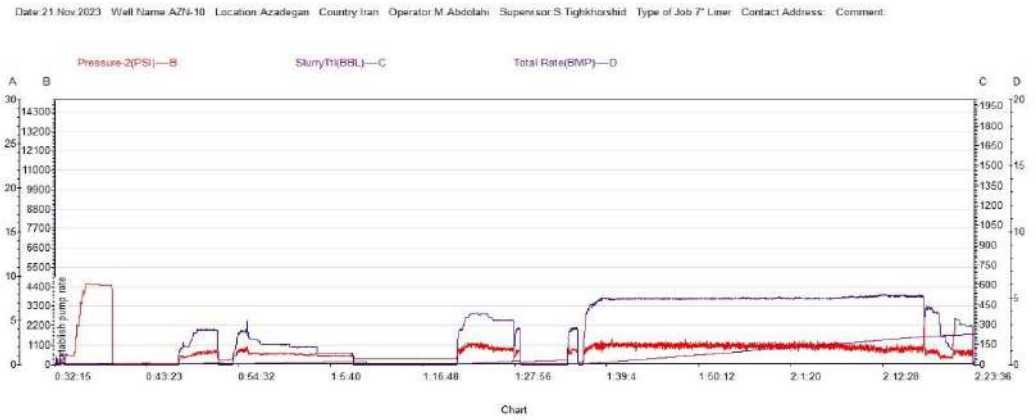
Actual Operational Summary of Sequence

1. Perform a Pre-Job safety meeting with all personnel involved on the operation as well as the client representative. Read and follow the risk assessment for this job.
2. Ensure communication between Mehran cementer and rig floor.
3. Pressure test cement line to 500^{psi} for 5 minutes then slowly increase the pressure to 4500^{psi} for 10 minutes & release the pressure to the cementing unit.
4. Pump 10^{bbl} chemical wash by cement unit with 4 BPM.
5. Pump 20^{bbl} 105^{pcf} spacer with 4 BPM.
6. Mix 41^{bbl} 118^{pcf} cement slurry in batch mixer.
7. Pump 25^{bbl} 118^{pcf} cement slurry with 4 BPM.
8. Stop pumping and Drop top plug, and displaced cement with cement unit (3^{bbl} cement slurry 118^{pcf}).
9. Continue displacement 207^{bbl} mud with Cement Unit (Totally 210^{bbl}).

Field Records

Time		Date	Pressure^{psi}	Rate^{bpm}	Operation Description
to	from				
20:00	23:00	20 – Nov	-	-	Mixing cement solution
18:00	22:00	20 – Nov	-	-	Mixing spacer solution
00:00	00:15	21 – Nov	-	-	Prejob safety meeting
00:30	00:45	21 – Nov	4500	-	Flush the surface line & pressure test the line to 4500^{psi}, Ok
01:07	01:10	21 – Nov	-	4	Start pump 10^{bbl} chemical wash by CMT unit
01:10	01:15	21 – Nov	-	4	Start 20^{bbl} spacer 105^{pcf} by rig pump
00:45	01:15	21 – Nov	-	-	Mix 41^{bbl} CMT slurry 118^{pcf} in bathc mixer
01:15	01:22	21 – Nov	-	4	Pump 25^{bbl} CMT slurry 118^{pcf}
01:22	01:25	21 – Nov	1000	3	Stop pumping & drop top plug & displace CMT with CMT unit (3^{bbl} CMT slurry 118^{pcf})
01:25	02:30	21 – Nov	1000 – 2500	2 – 5	Continue displacement 207^{bbl} mud with CMT unit
	02:30	21 – Nov			End of the job

Table 7-4: AZN-10 Field Record Table



7.2 Cement Plug Back Report

Slurry Data		
Name	**Volume (bbl)**	**Density (pcf)**
Ahead spacer	4	110
Slurry	35.5	118
Behind spacer	2	110

Table 7-5: AZN-10 Plug Back Information

7.3 Operation Instruction

Pre job preparation

1. Bulk plant R/U & preparations (prepare the checklist).
2. Cementing pump truck R/U & preparations (prepare the checklist).
3. Preparing mixing fluid (solution) with good cooperation by the rig crew.
4. Cementing high pressure line R/U to the rig floor.
5. Setting up HSE measurments for the operation.

Job description

1. L/U cement line do DPs.
2. Pressure test cement line with 3000^{psi} for 10min. Ok.
3. Pumped 5^{bbl} ahead spacer 110^{pcf} by pump truck with 1^{bpm}.
4. Mixed & pumped 113^{sxs} cement G Delijan (S.V: 35.5^{bbl}, S.W: 118^{pcf}) as slurry with 5^{bpm}, max 1500^{psi}.
5. Pumped 1^{bbl} behind spacer 110^{pcf} by pump truck with 1^{bpm}.

6. Displace with 196^{bbl} 102^{pcf} mud with 3 – 5^{bpm}, max 1500^{psi}.
7. POOH.

Job procedure

1. Checking equipments & ensuring of proper condition.
2. Taking work permission.
3. Holding pre-job safety meeting & tool box under supervision of HSE officer.
4. R/U cement unit, silo & compressor & prepare for cementing operation.
5. L/U cement unit to the rig floor & secure the lines.
6. Clean up pump truck displacement tank, preparing chemicals for mixing fluid.
7. Fill tanks with 19^{bbl} fresh water for mixing fluids.
8. Start mixing cement additives & preparing mixing fluids according to below:

Main cement solution mixing procedure.

- Add retarder – R02 & allow mixing for 1min.
- Add retarder – R12 & allow mixing for 15min.
- Add dispersant – D01 & allow mixing for 15min.
- Add anti foam – A02 & allow mixing for 5min.

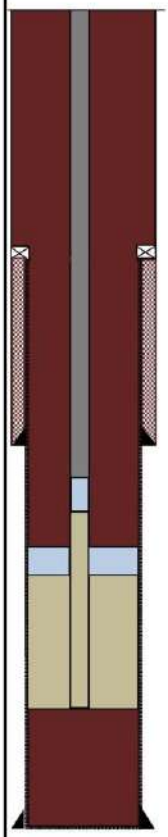
9. Test the lines with 500psi for 5min. & 3000psi for 10min.
10. Run DPs to 4591^{m}, then circulate to condition the hole.
11. Start to pumping 3.97^{bbl} 110^{pcf} ahead spacer by rate 2^{bpm}.
12. Mix & pump 35.51^{bbl} cement slurry 118^{pcf} by rate 4^{bpm}.
13. Pump 1.97^{bbl} 110^{pcf} spacer behind by rate 2^{bpm}.
14. Displace with mud at the optimum rate. Slow down the pumping rate to 2^{bpm} for the last 10^{bbl}.
15. Bleed off pressure & monitor U-Tube return.
16. Pull out the DPs above the top of the cement, flush the line & hold 50^{psi} pressure on top of the cement unit the cement sets. (optional)
17. R/D cement lines.
18. Wait on cement for 24 hrs.

CEMENTING JOB DESIGN AND CALCULATIONS

Rev.02

Client	MSA	Rig Name	201	Job Type	CMT Plug @ 4591 m
Field	Azadegan	Well Name	AZNS-010	Design Date	16-Oct-23

JOB VOLUMES



Surface
0 m

3 1/2 " DP

7 " liner Lap
3,707.5 m

9 5/8 " shoe
3,854.0 m

T.O.C
4,291.0 m

B.O.C
4,591.0 m

7 " liner shoe
4,922.0 m

liner	
OD (in)	ID (in)
7	6.094

Drill Pipe	
OD (in)	ID (in)
3.5	2.6875

Spacer	
Ahead (bbl)	Behind (bbl)
3.97	1.95

TOTAL SLURRY VOLUME	**35.51** bbl
TOTAL DISPLACEMENT VOLUME	**92.02** bbl

Note:Displacement Volume should be calculated on Rig

SLURRY DESIGN

CEMENT SLURRY

Material	Conc.	UOM	Qty for Job*	
Class "G" Cement	110	lb/sk	112.91	sx
Silica Flour	36.63	lb/sk	4,136.1	lb
Retarder (SE-R02)	0.80	lb/sk	90.3	lb
Retarder (SE-R12)	0.60	lb/sk	67.7	lb
Dispersant (SE-D01)	1.10	lb/sk	124.2	lb
Anti-foam (SE-A20)	0.03	gal/sk	3.4	lb

Slurry Parameters		
Slurry Volume	199.37	cu ft
	35.51	bbl
Slurry Density	118	pcf
Slurry Yield	1.7657	cu ft/sk
Mix Water Req	7.0670	gal/sk
Mixing Fluid Req	7.3420	gal/sk
Dead Volume	0	bbl

Mix Water Required:	**19.0 bbl**
Mixing Fluid Required:	**19.7 bbl**

Mix Water Type	Fresh Water

CEMENTING JOB DESIGN AND CALCULATIONS

Rev.02

Client	MSA	Rig Name	201	Job Type	CMT Plug @ 4591 m
Field	Azadegan	Well Name	AZNS-010	Design Date	16-Oct-23

Temperature Profile

Temperature Gradient	1.42	°F/100ft
BHST	297.25	°F
BHCT	238	°F

MUD REMOVAL

Spacer

Material	Conc.	Qty for Job
Fresh Water	35 gal/bbl	907.14 gal
Chemical Spacer (SE-P03)	10.0 lb/bbl	259.18 lb
NACL	125 lb/bbl	3239.79 lb
Anti-Foam (SE-A20)	0.1 gal/bbl	2.59 gal
Surfactant (SE-S40)	2.0 gal/bbl	51.84 gal
Barite	270 lb/bbl	6997.94 lb

Dead Volume: 20 bbls	
Spacer Volume	25.92 bbl
Spacer Density	110.0 pcf

* No excess's for losses, dead volumes, etc

PUMPING SCHEDULE

DESCRIPTION	Volume (bbls)	Rate (bpm)	Pump	Job Time (min)
Spacer ahead	3.97	2	Rig	2.0
Pump Cement Slurry	35.51	4	Unit	8.9
Spacer behind	1.95	2	Rig	1.0
Displacement	82.02	5	Unit	16.4
Displacement	10.00	2	Unit	5.0
Pull Out Drill Pipe				20.0

Total Job Time	53
Cementing Time	51

Thickening Time (30 BC): 177 min

Figure 7-3: AZN-10 Cement Job Desing & Calculation

Cementing Lab Report

LA-FO-00

Rev.02

Report No:
Date: 16-Oct-23 1402/07/24
Well Depth: (TVD): 4591.0 m 15063.071 ft
(MD): 4591.0 m 15063.071 ft

Field: Azadegan
Client: MSA
Well: AZNS-010
Rig: 201

Job Type : CMT Plug @ 4591
OH Size: inch
Last Csg/Liner: 5 inch
Mud Weight: 105 pcf

Cementing Additives	SPAD Code	Conc.	Units
Cement	Class "G Delijan	110.00	lb/sk
Silica Flour	FS95	36.63	lb/sk
Retarder	SE-R02	0.80	lb/sk
Retarder	SE-R12	0.60	lb/sk
Dispersant	SE-D01	1.10	lb/sk
Antifoam	SE-A20	0.03	gal/sk

Cement and Mixing Water

Cement Type: Class G Delijan

Slurry Density:	15.775 PPG	118 PCF
Slurry Yield:	0.314 bbl/sk	1.7657 cuft/sk
Mixing Water Required:	0.1683 bbl/sk	7.0670 gal/sk
Total Mixing Fluid Required:	0.1748 bbl/sk	7.3420 gal/sk

Temperature BHCT: 114 °C BHST: 147.4 °C
Temperature BHCT: 238 °F BHST: 297.25 °F
Temperature Gradient: 1.42 °F / 100 ft

API Free Fluid: 0 % @ 2 hrs 0 @250 CC
Fluid Loss: N.A cc/30 min @ 238 °F and 1000 Psi
API Fluid Loss: N.A cc/30 min @ 238 °F and 1000 Psi

Rheology Data

°F	°c	300	200	100	60	30	6	3	10 sec Gel	10 min Gel	PV CP	YP lbs/100ft²
80	26.6	107	73	41			13	10	16	32	99.00	8.00
238	114	27	18	12			4	3	6	14	22.50	4.50

Pumping Time: 30(BC): 2:57 Hrs:Min 50(BC): 3:28 Hrs:Min 70(BC): 3:39 Hrs:Min

Compressive Strength: -- Psi in 24 Hrs @ 297.25 °F 3000 psi
Crash test C.S: 4323.5 Psi in 24 Hrs @ 297.25 °F 3000 psi

Water Annalysis: Chlorides: 400 mg/liter

PH: Prehydrated DB: Dry Blended % bwoc: % by weight of cement % bwow: % by weight of water

Note: Calc. based on 110 lbs sack of dry cement.
Remarks:

Pilot Test Data

Function	Code	Concentration		Lab QTY (gr)
Cement	G-Delijan	110	lb/sk	598.74
Silica Flour	FS95	36.63	lb/sk	199.38
Retarder	SE-R02	0.8	lb/sk	4.35
Retarder	SE-R12	0.6	lb/sk	3.27
Dispersant	SE-D01	1.1	lb/sk	5.99
Anti-Foam Agent	SE-A20	0.03	gal/sk	1.36
Water		7.07	gal/sk	321.02

Lab Volume (ml)	600
Slurry Density (pcf)	118
Yield (cuft/sk)	1.7657
Mix Water (gal/sk)	7.0670
Mix Fluid (gal/sk)	7.3420

Figure 7-4: AZN-10 Cement Lab Report

7.4 Execution of Operation

Item	Description	Qty
1	Cement pump truck & related connections (PD – 05, PD – 06)	2 ea.
2	Air compressor (PD – C – 04)	1 ea.
3	Silo (PD – S – 07)	1 ea.

Table 7-6: AZN-10 Equipment Used for the Job

Item	Description	Qty
1	Cementing supervisor	1
2	Cementing operator	1
3	Cementing helper	5
4	Cementing helper (bulk)	1

Table 7-7: AZN-10 Operation Crew

7.5 Operation Ticket

FIELD SERVICES TICKET

Client:	MSA	Contract No:	Invoice No:
Service Company:	SPAD ENERGY		
Field : AZADGAN	Well No: AZNS-10	Rig No: OEOC 201	Date: 19- Oct- 2023
Division:		Period:	16-Oct-2023 till 19-Oct-2023
Service Requested:	PERSONEL-EQUIPMENT CMT Plug Operation		

ITEM	Description	Amount		STB		OPT		Total
		Qty	Unit	Qty	Unit	Qty	Unit	
1	Cementing Supervisor	1	per	3	Day	1	Day	4
3	Cementing Operator	2	per	3	Day	1	Day	4
4	Helper	6	per	3	Day	1	Day	4
5	PumpTruck PD - 05	1	UNIT	3	Day	1	Day	4
6	PumpTruck PD - 06	1	UNIT	2	Day	1	Day	3
7	Silo PD-S-07	1	UNIT	4	Day	1	Day	5
9	Air Compressor PD-C-04	1	UNIT	4	Day	1	Day	5
11	MOB/ DMOB PumpTruck PD - 05	1	UNIT	1				
12	MOB/ DMOB PumpTruck PD - 05	1	UNIT	1				
13	MOB/ DMOB Silo SE-S-07	1	UNIT	1				
14	MOB/ DMOB Air Compressor PD-C-04	1	UNIT	1				

SPAD Energy supervisor:

Client Representative:

Date:

	DATE			TIME	
On location	16	Oct	2023		
Operation Start	16	Oct	2023		
End of operation	17	Oct	2023		
Left Location	19	Oct	2023		
On Base					

Remark:

* شرایط کاملا خوبی در منطقه حکمران
* میزان ۳۰۰ لیتر گازوئیل جهت مصرف تجهیزات تحویل شده است
* پمپ کراک PD-06 دوروز بدلیل داشتن حضور داشته است
* ۲۵ ساعت در طی عملیات بدلیل انتظار در نظر گرفته شده است

7.6 Job Log & Summary

JOB SUMMARY

Client:	MSA	Rig No:	201	JOB PERIOD DATE	
Service Company:	SPAD ENERGY	Well No:	AZNS-10	From:	16-Oct-2013
Field:	S. AZADEGAN	Hole Section:	8 1/2"	To:	17-Oct-2013
Service Requested:	CEMENTING Plug				

JOB LOG

Date	Time	Volume (bbl)	Density (pcf)	Rate (bpm)	Pressure (psi)	Description
16-Oct-23	10:00					Start Mix Spacer
17-Oct-23	4:45					Start mix solution 19.5 bbl solution in CMT unit
17-Oct-23	5:15					End of mix the solution
17-Oct-23	5:17					Safty and operation meeting
17-Oct-23	6:00					Test the line and both unit W500/3000 psi
17-Oct-23	6:05	5				Pump 5 bbl spacer by CMT unit
17-Oct-23	6:11		118		1000	Start mix CMT slurry 118 pcf by CMT unit
17-Oct-23	6:22	35.5	118	5	1500	End of pumping after 35.5 bbl CMT slurry 118pcf
17-Oct-23	6:25	1				Pump 1 bbl spacer by CMT unit
17-Oct-23	6:48					Start Displace by CMT unit
17-Oct-23	7:45	196	102	5	1500	End of Displace the CMT slurry W/ 196 bbl mud
17-Oct-23						End of th CMT Plug job

Material Spacer

Consume			Mixing Information		
Materials	Unit	QTY for Job		Spacer Volume	25.92bbl
Fresh water	Gal	907		Spacer Density	110 pcf
Chemical Spacer SE-P03	Lb / 55Lb sks	259.18 / 5 sks			
Nacl (Provided by MSA)	Lb	3239.79			
Antifoam(SE-A20)	Gal / 5gal can	2.59 gal / 1 can			
Surfactant	Gal / 55gal Drum	51.84gal /1 Drum	Total Dead Volume		
Barite (Provided by MSA)	Lb	6997.94	20 bbl		

Material CMT Slurry

Consume			Mixing Information		
Materials	Unit	QTY for Job	Total Mixing Water in CMT unit	Cement Slurry Volume	35.51 bbl
Class "G" Cement Delijan	Lb /[illegible]Lb Sack	19800 / 180sks	19 bbl	Cement Slurry Density	118 pcf
Silica Flour	Lb	6,600.00	Total Mixing Fluid in CMT unit	Slurry Yield	1.7657 Cu ft/sks
Retarder SE-R-02	Lb / 55Lb sks	90.3 lb / 2 sks	19.7 bbl	Mixing Water Required	7.0670 gal / sks
Retarder SE-R-12	Lb / 55Lb sks	67.7lb / 2 sks		Mixing Fluid Required	7.3420 gal / sks
Dispersant SE-D01	Lb / 55Lb sks	124.2 lb / 3 sks			
Antifoam(SE-A20)	Gal / 5gal can	3.4 gal / 1 sks			
			Total Dead Volume		
			0		

Client Representative: MSA EPDS CONTRACTOR Rig No: Date: ۱۴۰۲/۸/۱

SPAD ENERGY supervisor:

* مورد با تایید خانم دکتر تهرانیان قرار دارد

7.7 Job Graph



7.8 Pumping Schedule

Pumping Schedule				
Name	**Flow Rate (bbl/min)**	**Volume (bbl)**	**Stage Time (min)**	**Comments**
Ahead spacer	1	5	5	-
Cement slurry	5	35.5	7	-
Behind spacer	1	1	1	-
Displacement	3 – 5	196	55	-

Table 7-8: AZN-10 Pumping Schedule

7.9 The Amount of Material Used in the Operation

Slurry type: Spacer				
Weight: 110 pcf		Volume: 26 bbl		D.V: 20 bbl
Row	**Material Name**	**Consumble**	**Unit**	**Description**
1	Fresh water	907.14	gal	Pumped by pump truck
2	Chemical spacer (SE – P03)	259.18	lb	
3	NaCl	3239.79	lb	
4	Anti foam (SE – A20)	2.59	gal	
5	Surfactant (SE – S40)	51.84	gal	
6	Barite	6997.94	lb	
Cement slurry				
Weight: 118 pcf	Slurry Vol: 35.5 bbl	Mix Water Vol: 19 bbl	Mix Fluid Vol: 19.7 bbl	Slurry Yield: 1.7657 cu.ft/sks
Row	**Material Name**	**Consumble**	**Unit**	**Description**
1	Class "G" cement Delijan	112.91	sks	
2	Silica flour (FS95)	4136.1	lb	
3	Retarder (SE – R02)	90.3	lb	
4	Retarder (SE – R12)	67.7	lb	
5	Dispersant (SE – D01)	124.2	lb	
6	Anti foam (SE – A20)	3.4	gal	

Table 7-9: AZN-10 Used Material

8. Completion Report

8.1 Completion Tubing Information

Tubing Size(in)	Weight (ppf)	Grade	Ultimate Strengths	Connection	Collapse Pressure (psi)	Burst Pressure (psi)	Interval (m)
4 – 1/2	18.9	L – 80	233 KLB	HSM3	13830	13380	0 – 3703
3 – 1/2	10.2	L – 80	440 KLB	HSM3	12120	11560	3703 – 3955

Table 8-1: AZN-10 Tubing Data

8.2 Completion Tools Specification Checklist

Item	Equipment Description	
1	**CONTROL LINE "WEIQIXIN":**	
	THINKNESS:	0.065"
	WORK PRESSURE:	15000 PSI
	O.D.:	0.25"
	TYPE:	SEAMLESS
	MATERIAL:	INC 825
	LENGTH:	120m/Roll
2	**UPPER FLOW COUPLING " WEIQIXIN" :**	
	OD:	**5.5"**
	ID:	3.640"
	THREADS:	4-1/2" 18.9 PPF, HSM3
	WORKING PRESSURE:	10000 PSI
	LENGTH:	6 FT
	MATERIAL:	13CR-110KSI
	LOWER FLOW COUPLING " WEIQIXIN" :	
	OD:	5.550"
	ID:	3.640"
	THREADS:	4-1/2" 18.9 PPF, HSM3
	WORKING PRESSURE:	10000 PSI
	LENGTH:	4 FT
	MATERIAL:	13CR-110KSI
3	**TUBING RETRIEVABL SAFETY VALVE " WEIQIXIN" :**	
	SETTING DEPTH:	61.32^{m}
	WORKING PRESSURE:	10000 PSI
	TEMPERATURE:	325 DEG F

	MAX O.D.:	7.5"
	MIN. I.D.:	3.437"
	PROFILE:	AF
	SELF-EQUALIZING FEATURE	HYDRAULIC CONTROL
	MATERIAL:	13CR-110KSI
	THREADS:	4-1/2" 18.9 PPF HSM3
	PROTECTION SLEEVE	
	PROFILE	AF
	MIN ID	2.9"
4	**TRAVELING JOINT " WEIQIXIN" :**	
	STROKE LENGTH	10 FT
	SHEAR PIN SET:	15000LB
	ELASTOMER:	AFLAS
	WORKING PRESSURE:	10000 psi
	TEMPERATURE:	300 DEG F
	TOOL OD:	5.315"
	TOOL ID:	2.922"
	THREADS:	3-1/2" 10.20 PPF HSM3 B*P
	MATERIAL:	13CR-110KSI
	TRANSFER TORQUE FEATURE	YES
5	**SLIDING SIDE DOOR " WEIQIXIN" :**	
	PROFILE:	X
	MAX.OD:	4.551"
	MIN ID:	2.750"
	PRESSURE RATING:	10000 PSI
	TEMPERATURE:	300 DEG F
	THREADS:	3-1/2" 10.20 PPF HSM3 B*P
	MATERIAL:	13CR-110KSI
	OPEN METHOD	DOWNWARD OPEN
6	**COMPLETION PACKER " WEIQIXIN" :**	
	TYPE:	PERMANENT
	CASING :	7" 32 PPF
	OD:	5.689"
	ID:	2.922"
	WORKING PRESSURE:	10000 PSI
	TEMPERATURE:	300 DEG F
	THREADS:	3-1/2" 10.20 PPF HSM3 B*P
	MATERIAL:	13CR-110KSI
	ELEMENT MATERIAL:	AFLAS
	4.0" ANCHOR SEAL	
	MAX OD (PACKING):	4"
	ID:	2.922"
	WORKING PRESSURE:	10000PSI
	TEMPERATURE:	300 DEG F
	THREADS:	3-1/2" 10.20 PPF HSM3 B
	TYPE:	RATCH LATCH

7	**LANDING NIPPLE " WEIQIXIN" :**	
	PROFILE TYPE:	XN
	WORKING PRESSURE:	10000 PSI
	MAX O.D.:	4.5"
	SEAL BORE:	2.75”
	MIN ID:	2.635"
	THREADS:	3-1/2" 10.20 PPF HSM3 B*P
	TEMPERATURE:	300 DEG F
	MATERIAL:	13CR-110KSI
8	**RE-ENTRY GUIDE SHOE " WEIQIXIN" :**	
	OD:	4.5"
	MIN ID:	2.922"
	TOP CONNECTION	3-1/2" 10.20 PPF HSM3 BOX
	MATERIAL:	13CR-110KSI
9	**3-1/2" CHEMICAL INJECTION" WEIQIXIN" :**	
	TYPE	TUBING RETRIVABLE
	MAX OD	5.00"
	MIN ID	2.922"
	MATERIAL:	13CR-110KSI
	TEMPERATURE	300 DEG F
	CONNECTION	3-1/2" 10.20 PPF HSM3 B*P
	CIV CONNECTION	NPT 3/8" BOX FOR CIV
	1.00"CIV VALVE	
	TYPE	DUAL CHECK VALVE WITH RUPTURE DISC
	WORKING PRESSURE	15000PSI
	MARERIAL	13CR-110KSI
	TEMPERATURE:	300 DEG F
	TOP CONNECTION	3/8" AUTOCLAVE
	BOTTOM CONNECTION	3/8" NPT
	RUPTURE DISC PRESSURE	6000PSI

Table 8-2: AZN-10 Equipment Checklist

Item	Name	Description
1	**Tubing Hanger**	11" BOWL x 4-1/2" 18.9 PPF BOX TPCQ-EX CONNECTION BOTTOM & 4-1/2" 4-2G ACME BOX TOP WITH 4-1/2" EXTENDED NECK, WITH ONE CONTINUOUS 1/4" NPT C.LINE EXIT FOR SSSV, ONE 1/4" LINE EXIT FOR DHPG, INSTALLATION COMPLETE WITH RELATED FITTINGS AND PLUGS, WITH TWO CHANNEL FOR 3/8" CHEMICAL, C/W ALIGNMENT PREPATION FOR ESP. BPV: 4" TYPE H THREAD PREPARATION, THREE FS SEAL TO PACK OFF THE UPPER NECK AND TWO S SEAL TO PACK OFF THE BODY OF HANGER, MATERIAL: 410SS. M.C: FF-0.5 T.C: U PSL:3G PR:2 COMPANY: MSP
2	**Fluted Adaptor**	4 1/2" 18.9 PPF, L-80 HSM3 PIN * TPCQ EX PIN, OD: 5.5'' ID: 3.64'' SLOTS: 6 * 1 INCH WIDTH

Table 8-3: AZN-10 Equipment Checklist

8.3 Assembling Test Report

Technical Service Report

Client	MSA	Contractor	WEIQIXIN
Service Subject	Completion Tools Installation Service for Well AZN-10		
Service Duration	2023/12/15-2023/12/30	Service Location	South Azadegan, IRAN

Service Personnel

No.	Name	Position	No.	Name	Position
1	Wu Xiaomin	Supervisor	3	Ma Hongjian	Engineer
2	Yan Aimin	Supervisor			

Job Description

Date	Service Details
Dec.15,2023	Arrive at the rig site, well AZN-10, check the Completion Tools.
Dec.16,2023	RIH Guide Shoe & Landing Nipple & Packer & SSD, RIH slick line drift the string to above Landing Nipple,set Check Valve in SSD.Continue RIH Expansion Joint & two CIM and CIV,connect the chemical injection lines with the CIV,pressure test the chemical injection lines 2000Psi for 15mins,testing Ok,bleed down the pressure to1000Psi and then RIH.Make up Crossover.
Dec.17,2023	After RIH one 4-1/2" tubing, RIH slick line drift the string to above SSD,Ok! Pressure test RIH string 7000Psi for 15mins,testing Ok!Continue RIH completion string.After RIH 80 joints 4-1/2" tubing,pressure test RIH string 7000Psi,testing Ok! Continue RIH completion string.
Dec.18,2023	After RIH 160/240 joints 4-1/2" tubing,pressure test RIH string 7000Psi,testing Ok! Continue RIH completion string.
Dec.19,2023	RIH TRSSV,connect the control line with TRSSV,pressure test 8000Psi for 15mins,testing Ok!Bleed off the control line pressure to 4000Psi and continue RIH completion string.After RIH one 4-1/2" tubing above TRSSV,pressure test completion string 7000Psi, testing Ok! Perform inflow test TRSSV flapper,testing Ok!Continue RIH completion string.P/U & M/U Fluted Swage Adapter and Tubing Hanger.
Dec.20,2023	Pass two 3/8" chemical injection lines and one 1/4" control line through Tubing Hanger.Land Tubing Hanger in T. H. S.RIH slick line set protection sleeve in TRSSV.RIH slick line pull out check valve from SSD.
Dec.21,2023	RIH slick line set check valve in Landing Nipple(it is difficult every time pass through SSD).POOH Tubing Hanger 1.5m aboveT. H. S.
Dec.22,2023	Set the Packer 500Psi×5mins, 2000Psi×5mins,3000Psi×10mins,3500psi×10mins, Ok!Continue test completion string 7000Psi,testing Ok! Confirm the Packer is set,pull the string to 10klbs overpull,Ok! Slack off completion string weight 15klbs,shear 3 shear pins of Expansion Joint. Land Tubing Hanger in T. H. S.Pressure test completion string 5000Psi,heard a sound and then completion fluid flowed out from annulus while keep the pressure.RIH slick line to set 2.75"WPX plug in SSD,but can't pass at depths 3035m. RIH slick line with 2.8" gauge cutter,but can't pass at the same depths.

Supervisor
South Azadegan Project

Technical Service Report

Client	MSA	Contractor	WEIQIXIN
Service Subject	Completion Tools Installation Service for Well AZN-10		
Service Duration	2023/12/15-2023/12/30	Service Location	South Azadegan, IRAN

Service Personnel

No.	Name	Position	No.	Name	Position
1	Wu Xiaomin	Supervisor	3	Ma Hongjian	Engineer
2	Yan Aimin	Supervisor			

Job Description

Date	Service Details
Dec.23,2023	RIH slickine with LIB,do not have the trail on the LIB.RIH slickine with bailer,pull out and check find have nothing inside the bailer.
Dec.24,2023	Tubing pressure&annulus pressure is 800Psi.Reverse circulation use mud with weight 106PCF,then the pressures to 0Psi.
Dec.25,2023	Confirm the Packer is set,pull the string to 26klbs overpull(string weight 257klbs),Ok!Test the Packer with 1200Psi×15mins in annulus, Ok!RIH slick line pull out check valve from Landing Nipple.
Dec.26,2023	RIH slick line set 2.75"WPXN plug and prong in Landing Nipple.RIH slick line set check valve in SSD,pressure test string 5000Psi,testing Ok!RIH slick line pull out check valve from SSD.
Dec.27,2023	Pressure test plug and prong 900Psi,testing Ok! Reverse displace the mud with 30bbl Hi-Vis & 180bbl Nacl brine & 600bbl 103PCF $CaBr_2$ completion fluid.
Dec.28,2023	RIH slick line close SSD with B shifting tools.Pressure test completion string 6500Psi,testing Ok! RIH slick line pull out protection sleeve from TRSSV.Land Tubing Hanger in T. H. S. Pressure test completion string 6500Psi,testing Ok! Pressure test the Packer with 700Psi×15mins in annulus, testing Ok! Release the pressure in control line close TRSSV. L/D land joint ,install TWCV in Tubing Hanger, close the top BOP rams, pressure test tubing hanger,testing Ok!
Dec.29,2023	L/D BOP. N/U X-tree adapter,feeding through adapter with 1/4" control line and chemical injection lines. Pressure test TRSSV 8000Psi×15mins,testing OK! Pressure test two CIV 2000Psi×15mins,testing OK!Block up two chemical injection lines.
Dec.30,2023	L/D control line from rig floor.take all of service equipment in truck and back to warehouse in ahwaz.

Client approval

Prepared By	Approved By
2023.12.30	
WQX Supervisor	Supervisor

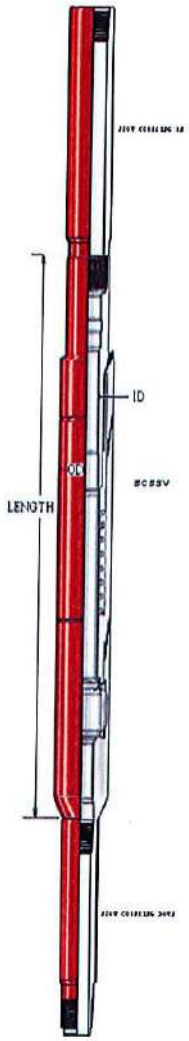


Figure 8-1: AZN-10 Technical Service Report

WEIQIXIN PETROLEUM MACHINERY(TIANJIN)CO.LTD

惟其信 WEIQIXIN

SCSSV Test Report



Type:	210W 343-AF04	
Serial Number:	2306 23038-02	
Connection Size:	4-1/2" 18.90 PPF HSM3 BOX × PIN	
Working Pressure	10,000Psi	
OD:	7.500"	
ID:	3.437"	
Length:	1.780 m	
Material:	S13CR-110 KSI	
Hydraulic Control Pressure	Full Open	Full Closed
	1800 Psi	900 Psi
8,000 Psi Hydraulic Control Pressure No Drop in 15 Minutes	ok	
8,000 Psi Pressure No Drop in 15 Minutes	ok	
8,000 Psi flapper Pressure No Drop in 15 Minutes	ok	
Top Connection	Length:	1.625 m
Flow Coupling UP:	Serial Number:	2306 Y230414-02
	Length:	1.725 m
Flow Coupling DOWN:	Serial Number:	2306 Y230415-01
	Length:	1.115 m
Bottom Connection:	Length:	1.621m
Well Name and Location:	AZNS-10	

MSA Project Manager:
Name/Title:
Date: E. Parvaresh

WQX:
Name/Title:
Date: 2023.12.11

Figure 8-2: AZN-10 TRSSSV Test Report

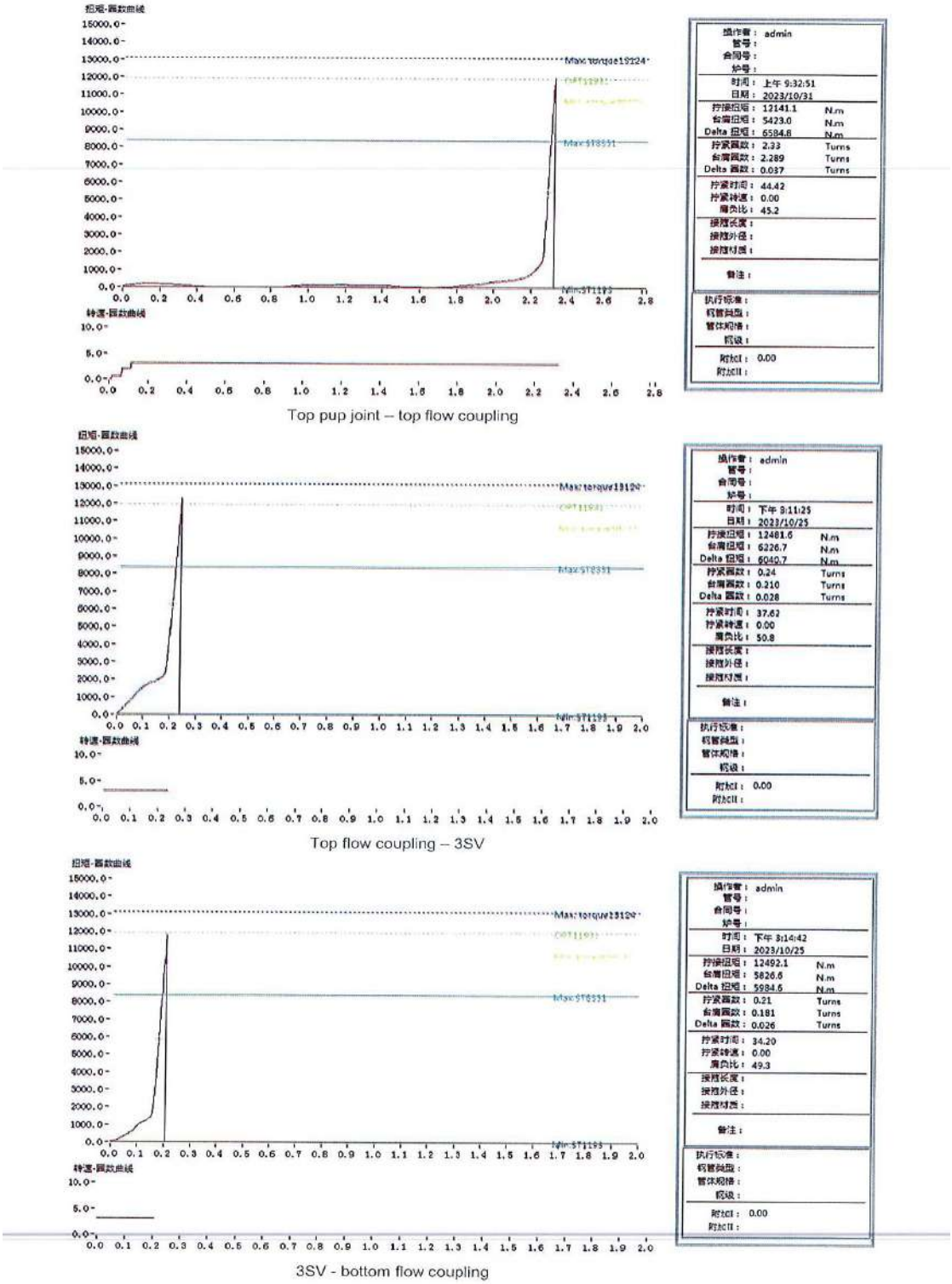


Figure 8-3: AZN-10 TRSSSV Connection Grapgh

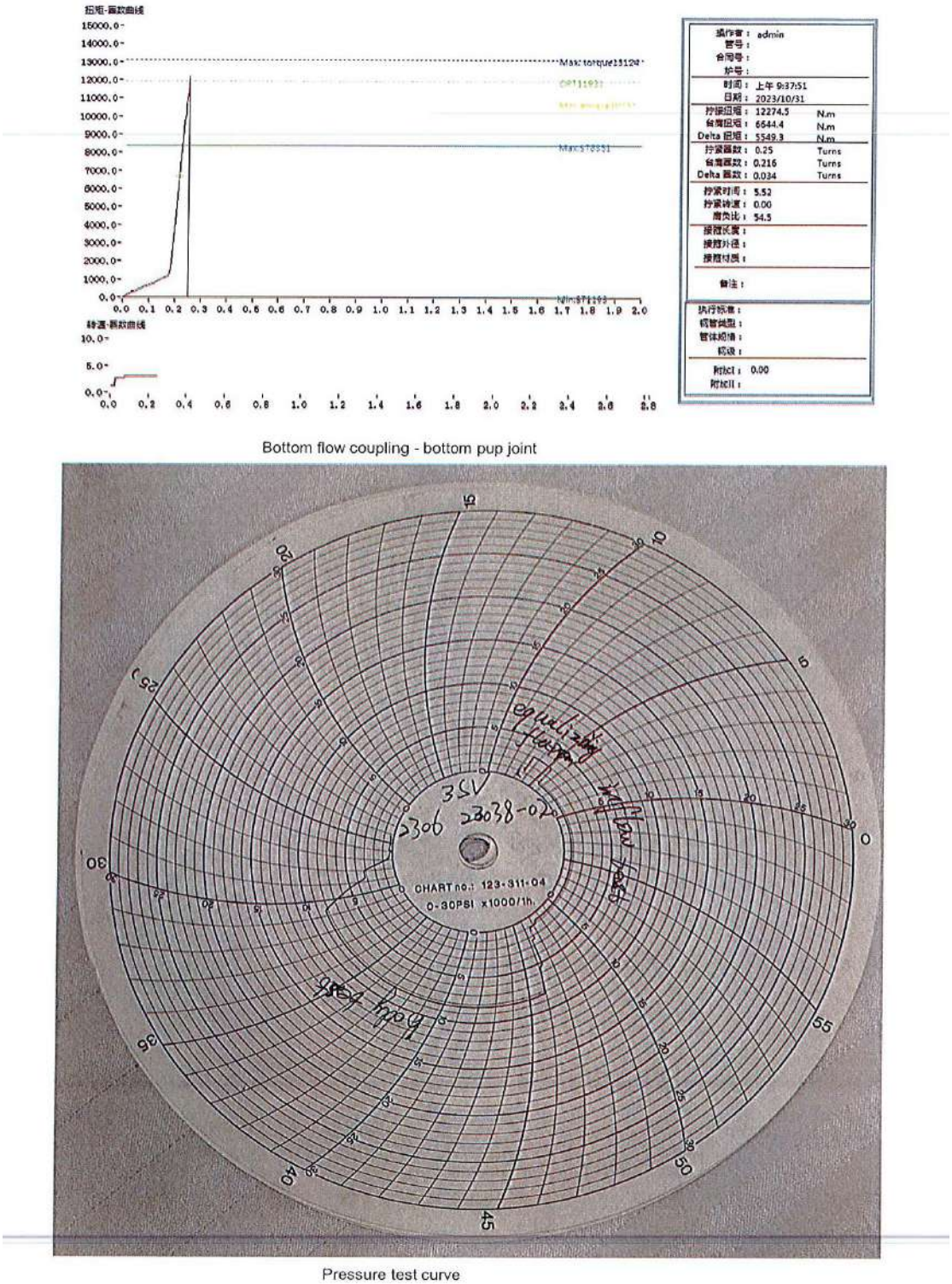
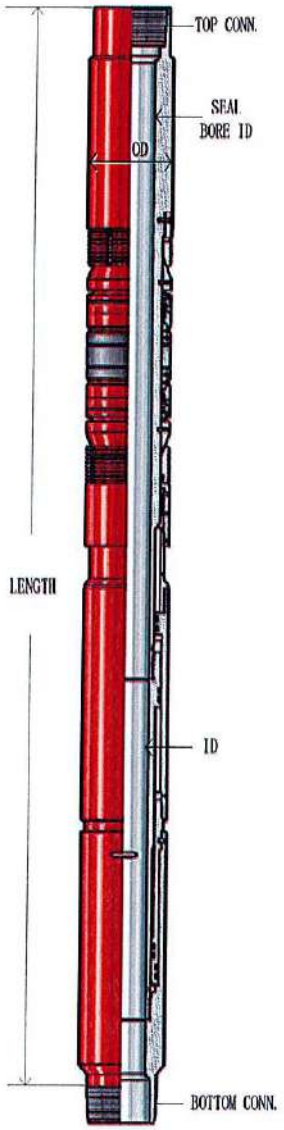


Figure 8-4: AZN-10 TRSSSV Pressure Test Grapgh

WEIQIXIN PETROLEUM MACHINERY(TIANJIN)CO.LTD

WSDP Hydraulic Set Permanent Packer Test Report



Type:	336W150-000	
Serial Number:	2306 23062-02	
Connection Size:	LEFT HAND X 3-1/2" 10.20 PPF HSM3 PIN	
Working Pressure	10,000 Psi	
OD:	5.910"	
ID:	2.922"	
Length:	4.440 m	
Material:	S13CR-110KSI	
Temperature:	300 DEG F	
Element :	Aflas	
Setting Pressure:	3,500 Psi	
Seal Bore Size:	Upper Seal Bore ID:4.00" & Lower Seal Bore ID:2.922"	
Top Connection:	Length:	1.824m
Bottom Connection:	Length:	1.822 m
Well Name and Location:	AZNS-10	

MSA Project Manager:
Name/Title:
Date: E.Parvaresh

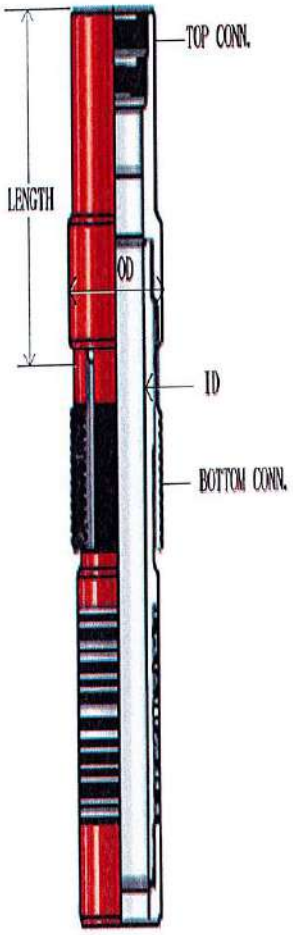
WQX:
Name/Title:
Date: 2023.12.11

Figure 8-5: AZN-10 Permanent Packer Test Report

WEIQIXIN PETROLEUM MACHINERY(TIANJIN) CO.LTD

惟其信 WEIQIXIN

Model WK Anchor Tubing Seal Report

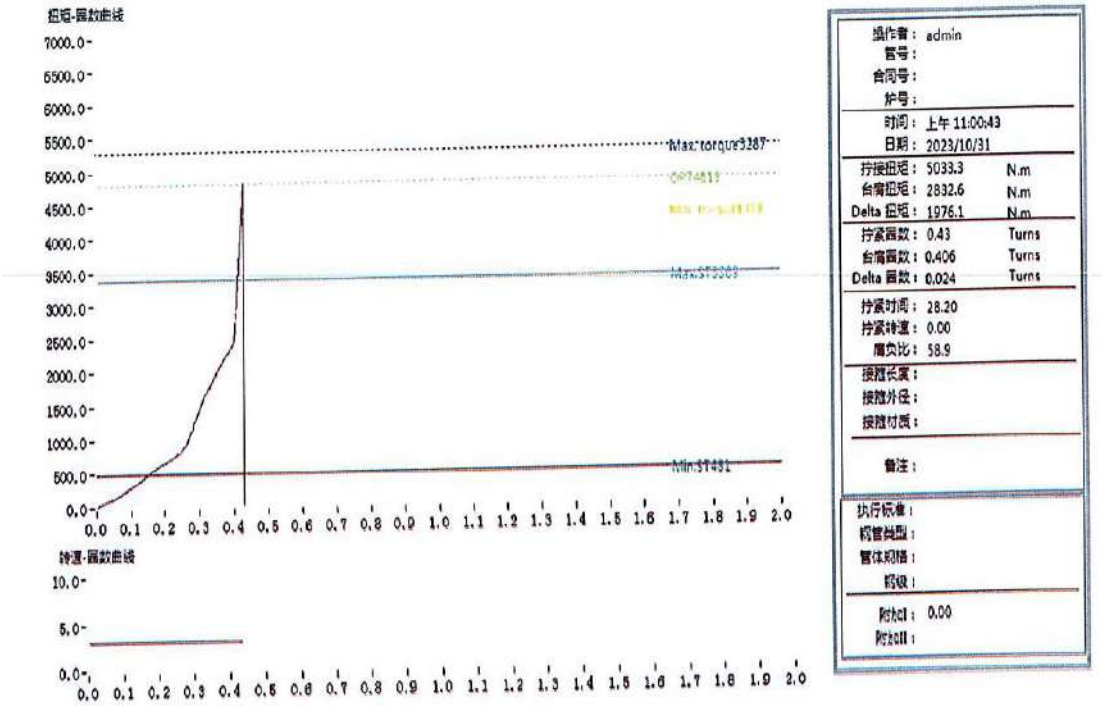


Type:	360W150-000
Serial Number:	2306 23063-03
Connection Size:	3-1/2" 10.20 PPF HSM3 BOX X MULE SHOE GUIDE
OD:	5.689"
ID:	2.922"
Seal OD:	4.000"
Working pressure:	10,000Psi
Length:	230mm
Material:	S13CR-110 KSI
Well Name and Location	AZNS-10

MSA Project Manager:
Name/Title:
Date:

WQX:
Name/Title:
Date: 2023.12.11

Figure 8-6: AZN-10 Anchor Seal Test Report



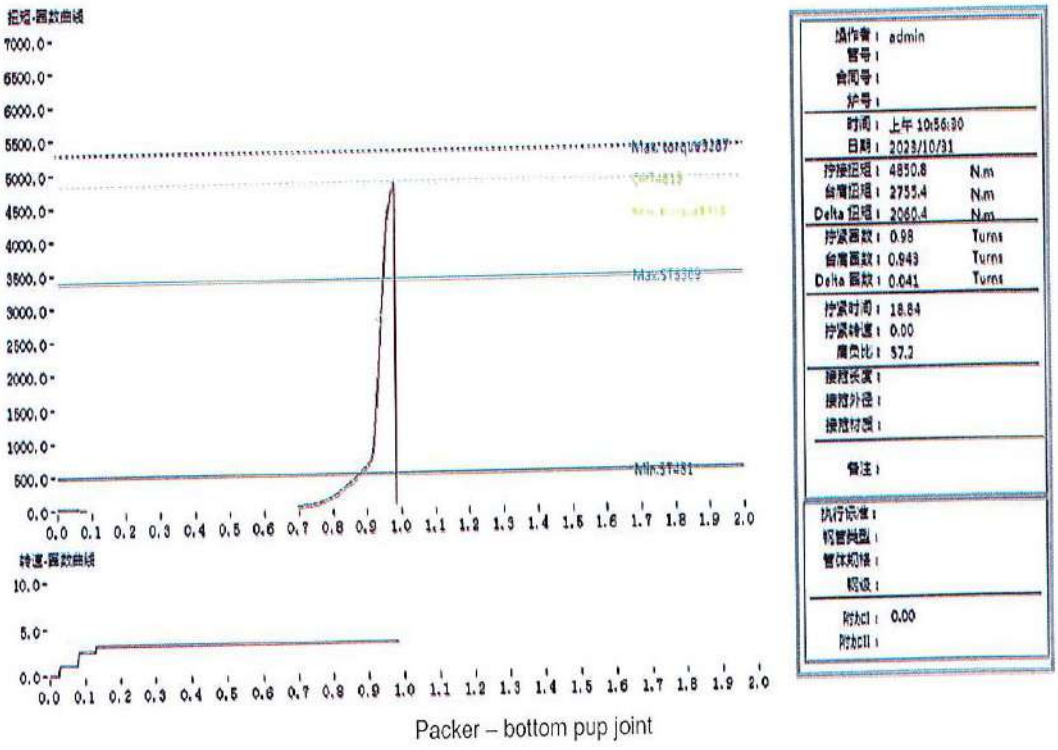
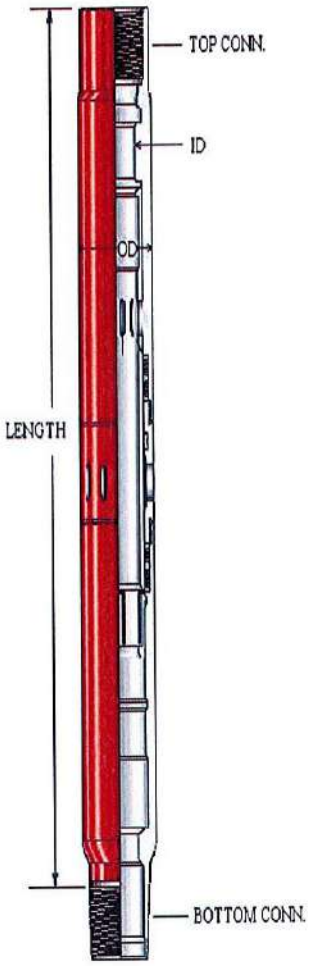


Figure 8-7: AZN-10 Permanent Packer Connection Grapgh

WEIQIXIN PETROLEUM MACHINERY(TIANJIN) CO.LTD

惟其信 WEIQIXIN

Sliding Side Door Test Report

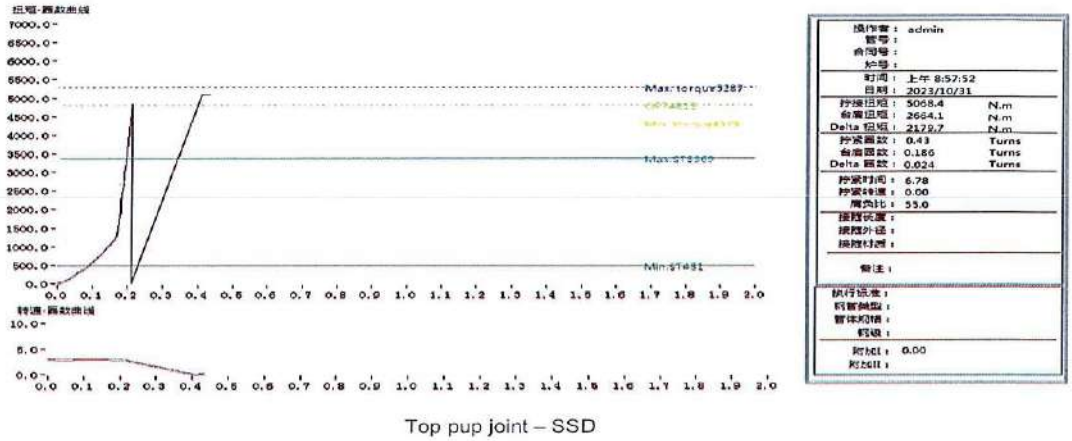


Type:	750W275-X39	
Serial Number:	2306 23039-02	
Connection Size:	3-1/2" 10.20 PPF HSM3 BOX× PIN	
Working Pressure	10,000 Psi	
OD:	4.551"	
ID:	2.750"	
Seal Bore ID :	2.750"	
Length:	1.195m	
Temperature	300 DEG F	
Material:	S13CR-110KSI	
Profile:	X	
Shifting Direction:	DOWNWARD OPEN	
Top Connection:	Length:	1.622 m
Bottom Connection:	Length:	1.617 m
8,000 Psi Pressure No Drop in 15 Minutes	Ok	
8,000 Psi Blanking Plug Pressure No Drop in 15 Minutes	Ok	
Well Name and Location:	AZNS-10	

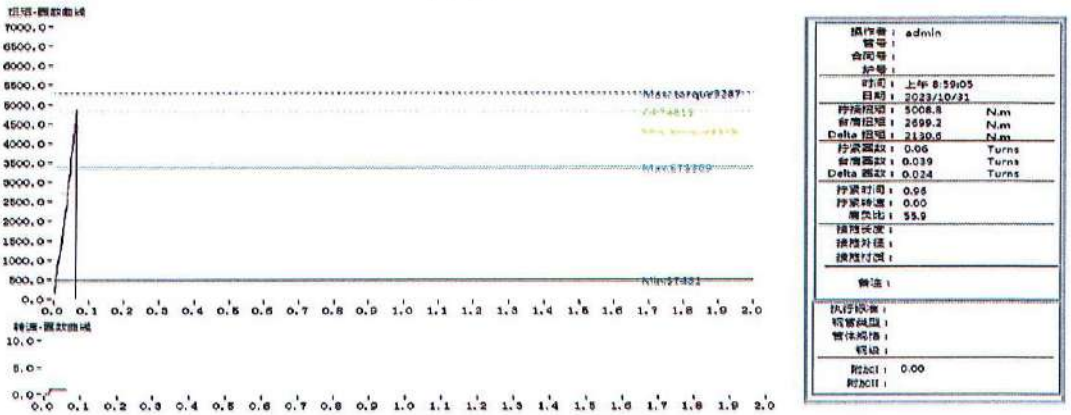
MSA Project Manager:
Name/Title:
Date:

WQX:
Name/Title:
Date: 2023.12.11

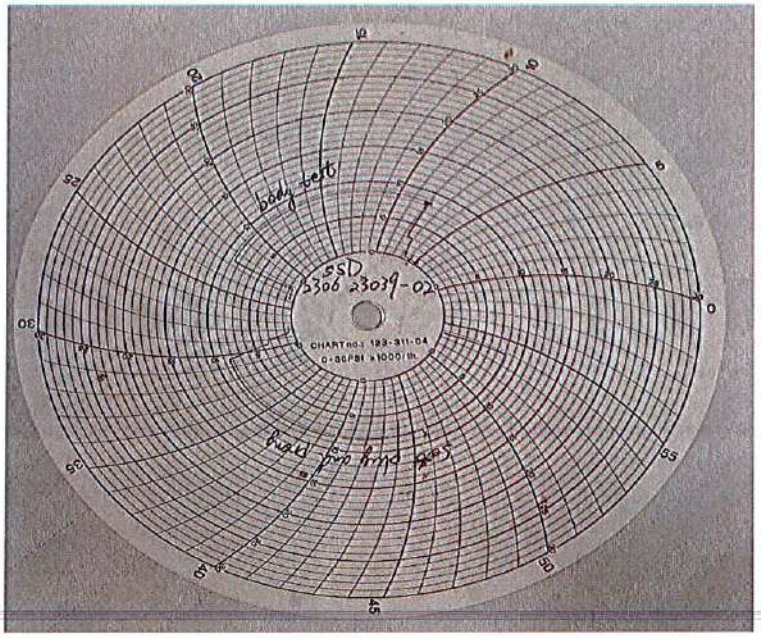
Figure 8-8: AZN-10 SSD Test Report



Top pup joint – SSD



SSD – bottom pup joint



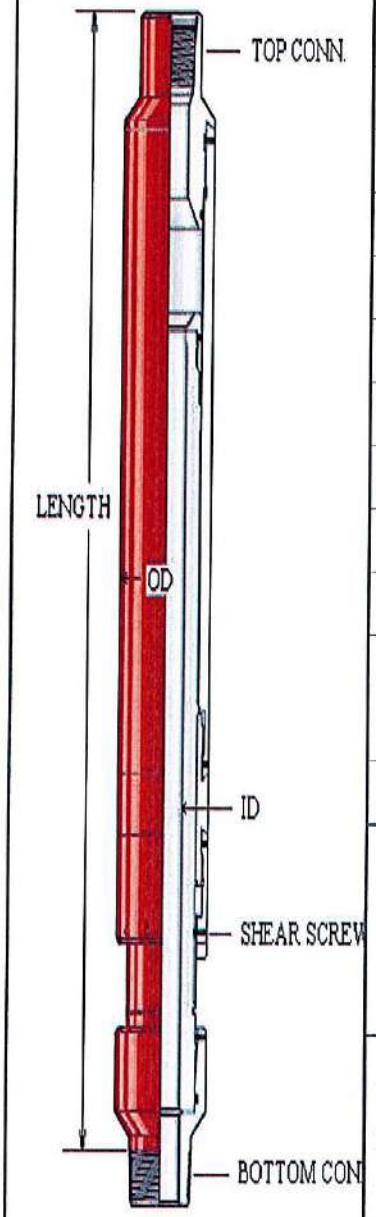
Pressure test curve

Figure 8-9: AZN-10 SSD Connection & Pressure Test Grapgh

WEIQIXIN PETROLEUM MACHINERY(TIANJIN) CO.LTD

惟其信 WEIQIXIN

Expansion Joint Test Report

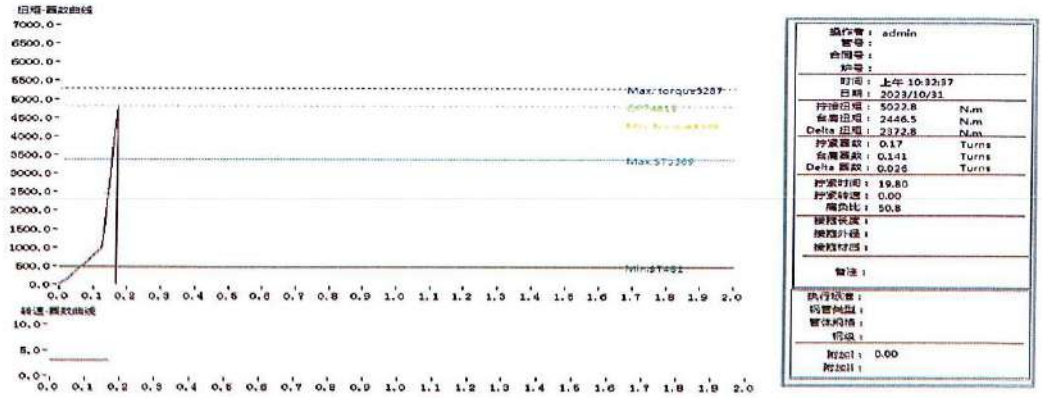


Type:	370W350-030	
Serial Number:	2306 23040-01	
Connection Size:	3-1/2" 10.20 PPF HSM3 BOX× PIN	
Working Pressure	10,000 Psi	
OD:	5.315"	
ID:	2.922"	
Length:	7.320 m	
Material	S13CR-110KSI	
Share Screw:	3	
Top Connection:	Length:	1.820 m
Bottom Connection:	Length:	1.822 m
8,000 Psi Pressure No Drop in 15 Minutes	ok	
Well Name and Location:	AZNS-10	

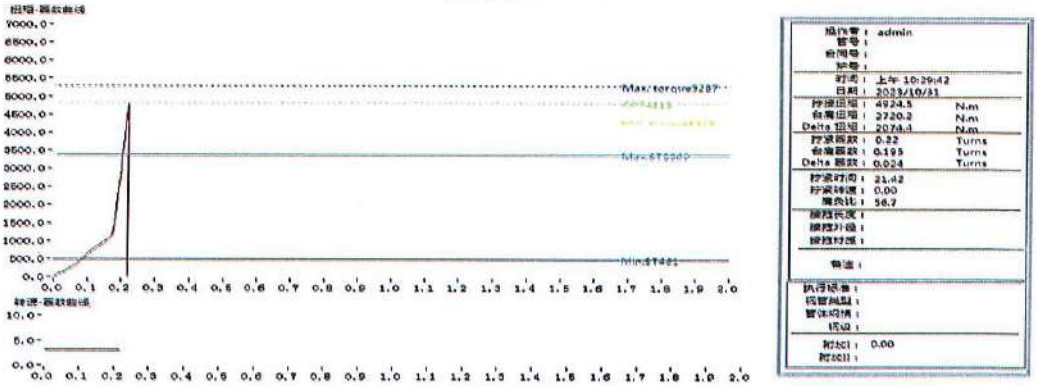
MSA Project Manager:
Name/Title:
Date:

WQX:
Name/Title:
Date: 2023.12.11

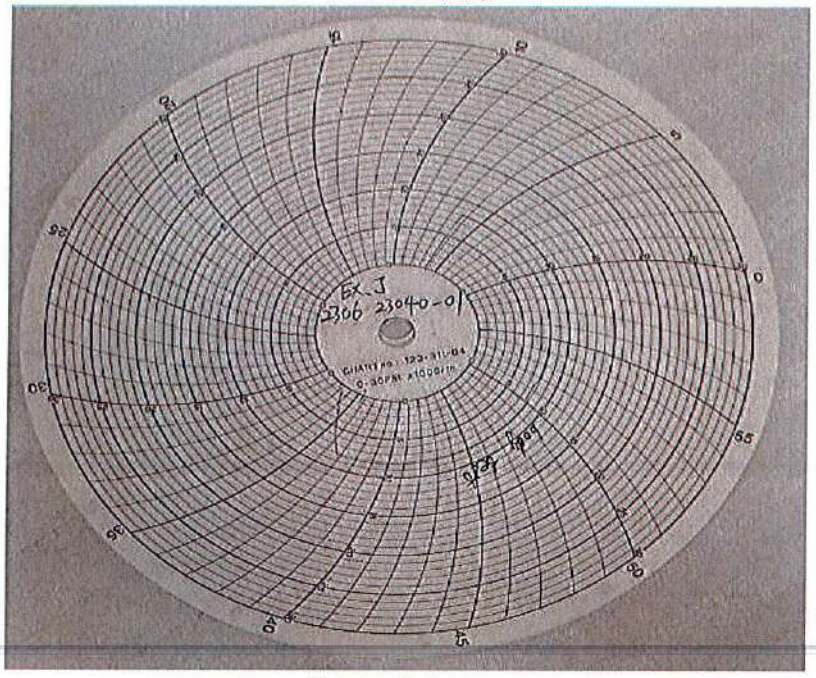
Figure 8-10: AZN-10 Expansion Joint Test Report



Top pup joint – EX.J



EX.J – bottom pup joint



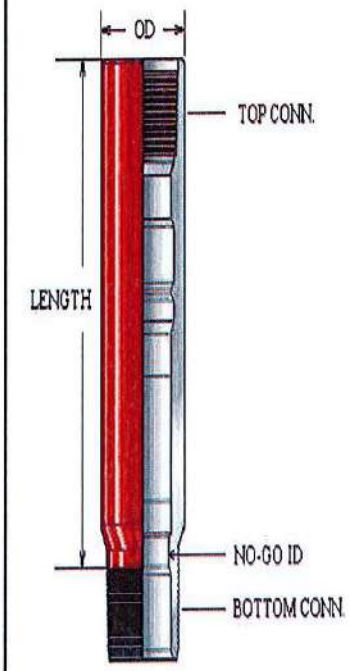
Pressure test curve

Figure 8-11: AZN-10 Expansion Joint Connection & Pressure Test Grapgh

WEIQIXIN PETROLEUM MACHINERY(TIANJIN) CO.LTD

惟其信 WEIQIXIN

Landing Nipple Test Report

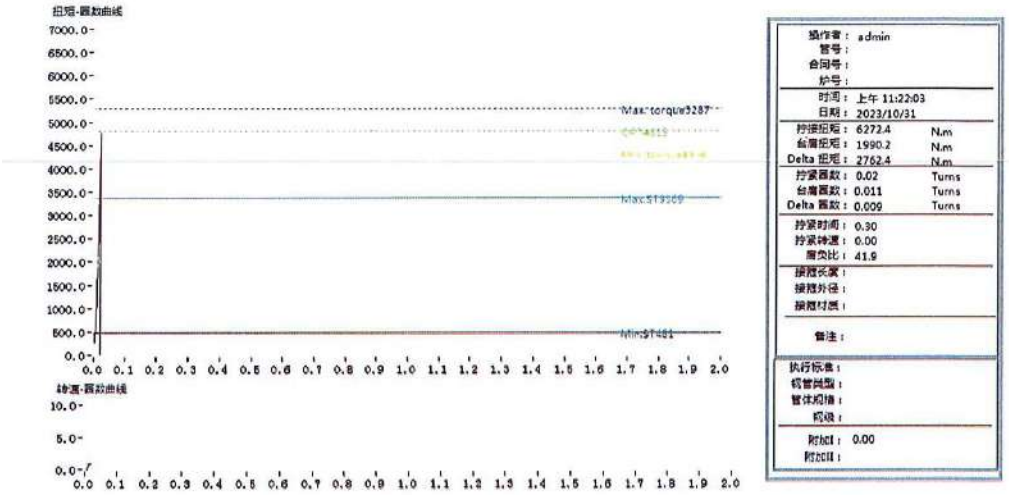


Type:	706W275-N24	
Serial Number:	2306 Y230416-02	
Connection Size:	3-1/2"10.22 PPF HSM3 BOX × PIN	
OD:	4.50"	
ID:	2.75"	
NO-GO ID:	2.635"	
Length:	0.370m	
Material:	S13CR-110 KSI	
Top Connection:	Length:	1.825m
8,000 Psi Check Valve Pressure No Drop in 15 Minutes	Ok	
Well Name and Location	AZNS-10	

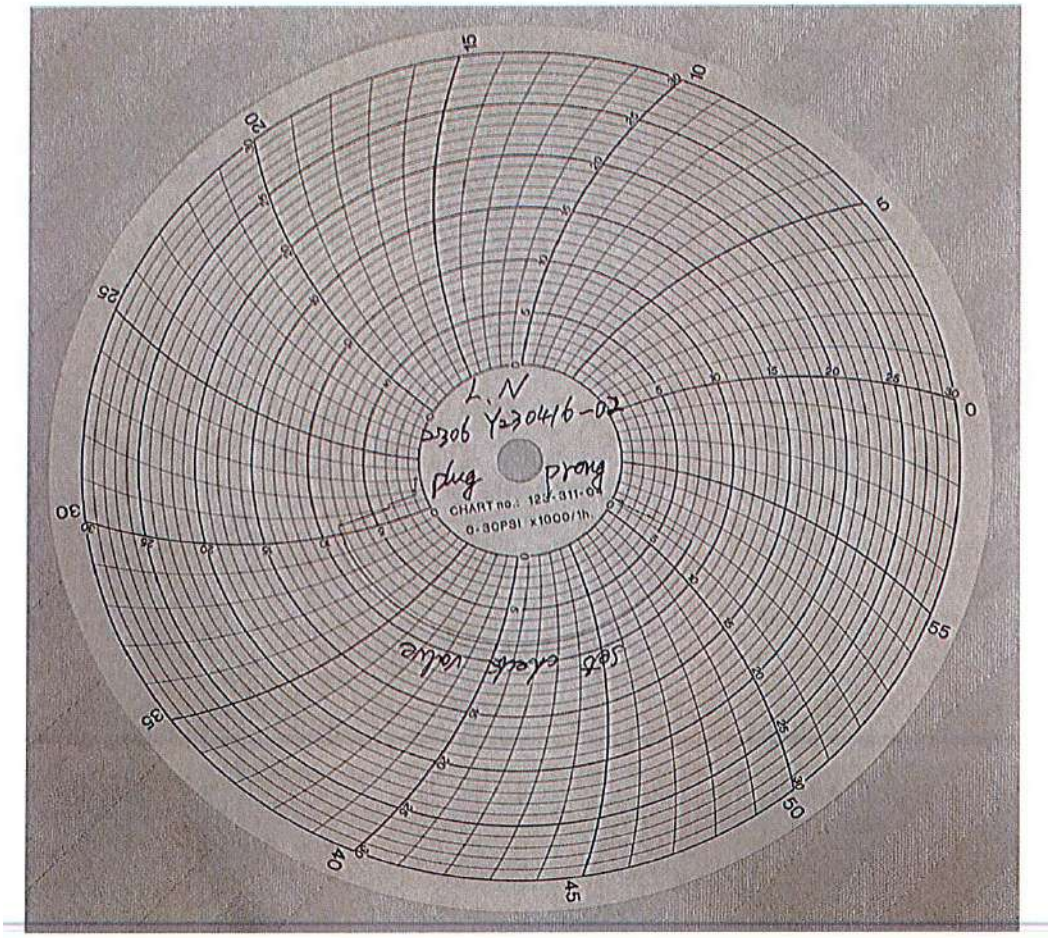
MSA Project Manager:
Name/Title:
Date:

WQX:
Name/Title:
Date: 2023. 12. 11

Figure 8-12: AZN-10 Landing Nipple Test Report



Top pup point – L.N



Pressure test curve

Figure 8-13: AZN-10 Landing Nipple Connection & Pressure Test Grapgh

WEIQIXIN PETROLEUM MACHINERY(TIANJIN) CO.LTD

惟其信 WEIQIXIN

Chemical Injection Valve Test Report



Type:	802W100-601
Serial Number:	2308 23068-07
Connection Size:	3/8" JAM NUT BOX x 3/8" NPT PIN
OD:	1"
ID:	0.299"
Rupture DISK:	6000Psi
Spring open pressure:	500Psi
Length:	310mm
Material:	S13CR-110 KSI
8,000 Psi Pressure No Drop in 15 Minutes	Ok
Well Name and Location	AZNS-10

MSA Project Manager:
Name/Title:
Date: E. Purvaresh

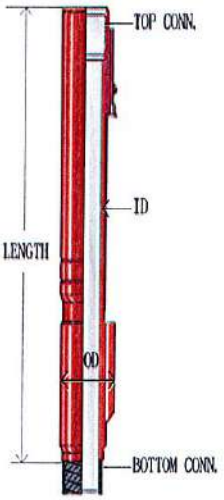
WQX:
Name/Title:
Date: 2023.12.11

Figure 8-14: AZN-10 First CIV Test Report

WEIQIXIN PETROLEUM MACHINERY(TIANJIN) CO.LTD

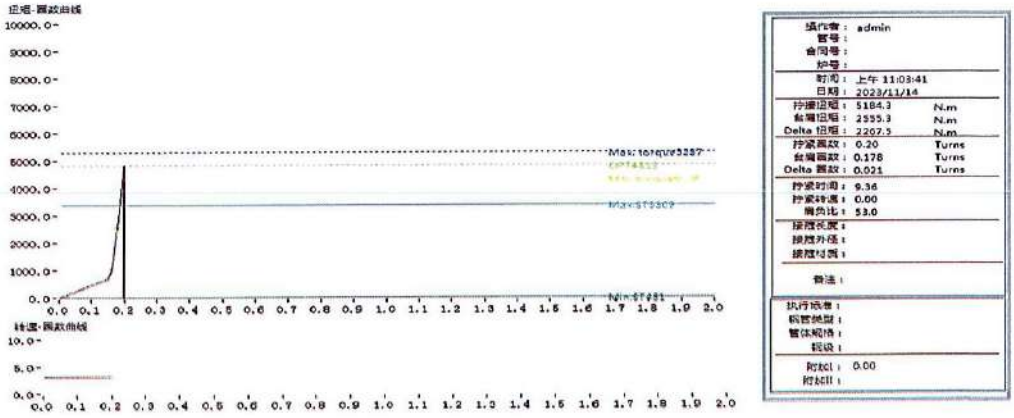
惟其信 WEIQIXIN

Chemical Injection Mandrel Test Report

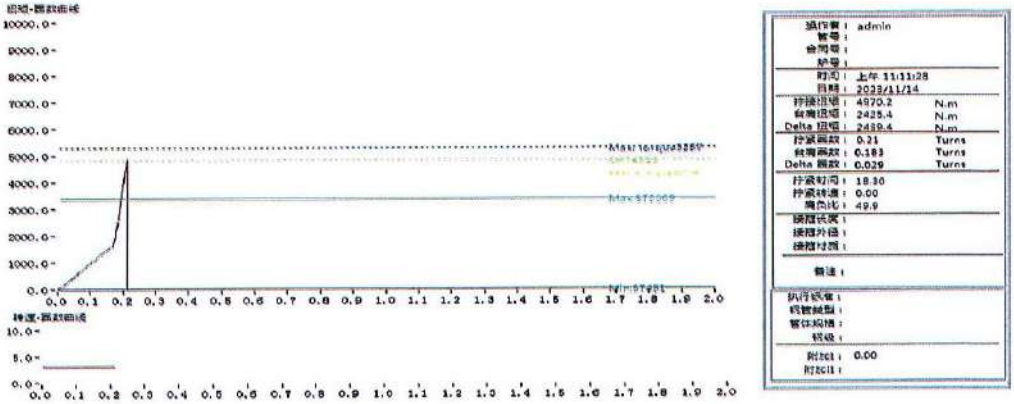


Type:	800W350-028	
Serial Number:	2308 Y230708-01	
Connection Size:	3-1/2" 10.2PPF HSM3 BOX × PIN	
OD:	5.000"	
ID:	2.922"	
Working pressure:	10,000Psi	
Temperature:	300 DEG F	
Length:	0.729m	
Top Connection:	Length:	1.829 m
Bottom Connection:	Length:	1.825 m
Material:	S13CR-110KSI	
8,000 Psi Pressure No Drop in 15 Minutes	Ok	
Well Name and Location	AZNS-10	
MSA Project Manager: Name/Title: Date:	E-Parvaresh	
WQX: Name/Title: Date:	2023.12.11	

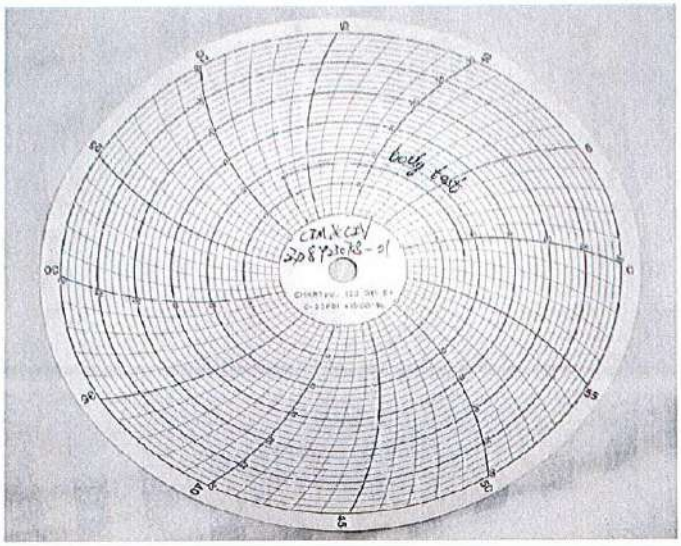
Figure 8-15: AZN-10 First CIM Test Report



Top pup joint - CIM



CIM – bottom pup joint



Pressure test curve

Figure 8-16: AZN-10 First CIM Connection & Pressure Test Grapgh

WEIQIXIN PETROLEUM MACHINERY(TIANJIN) CO.LTD

惟其信 WEIQIXIN

Chemical Injection Valve Test Report



Type:	802W100-601
Serial Number:	2308 23068-08
Connection Size:	3/8" JAM NUT BOX x 3/8" NPT PIN
OD:	1"
ID:	0.299"
Rupture DISK:	6000Psi
Spring open pressure:	500Psi
Length:	310mm
Material:	S13CR-110 KSI
8,000 Psi Pressure No Drop in 15 Minutes	Ok
Well Name and Location	AZNS-10

MSA Project Manager:
Name/Title:
Date: E. Purgaresh

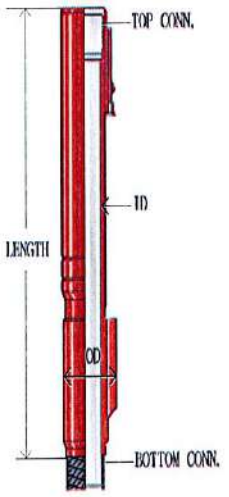
WQX:
Name/Title:
Date: 2023.12.11

Figure 8-17: AZN-10 Second CIV Test Report

WEIQIXIN PETROLEUM MACHINERY(TIANJIN) CO.LTD

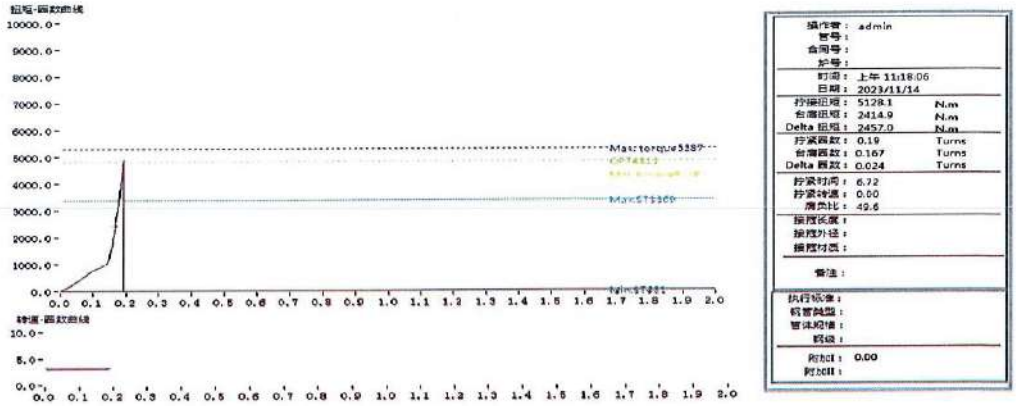
惟其信 WEIQIXIN

Chemical Injection Mandrel Test Report

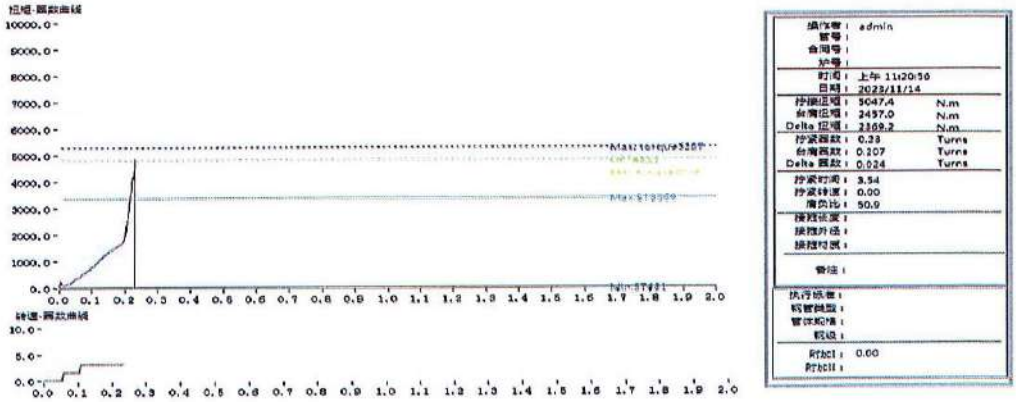


Type:	800W350-028	
Serial Number:	2308 Y230708-02	
Connection Size:	3-1/2" 10.2PPF HSM3 BOX × PIN	
OD:	5.000"	
ID:	2.922"	
Working pressure:	10,000Psi	
Temperature:	300 DEG F	
Length:	0.727m	
Top Connection:	Length:	1.825 m
Bottom Connection:	Length:	1.827 m
Material:	S13CR-110KSI	
8,000 Psi Pressure No Drop in 15 Minutes	Ok	
Well Name and Location	AZNS-10	
MSA Project Manager: Name/Title: Date:	E. Rezvanpour 2023.12.11	
WQX: Name/Title: Date:		

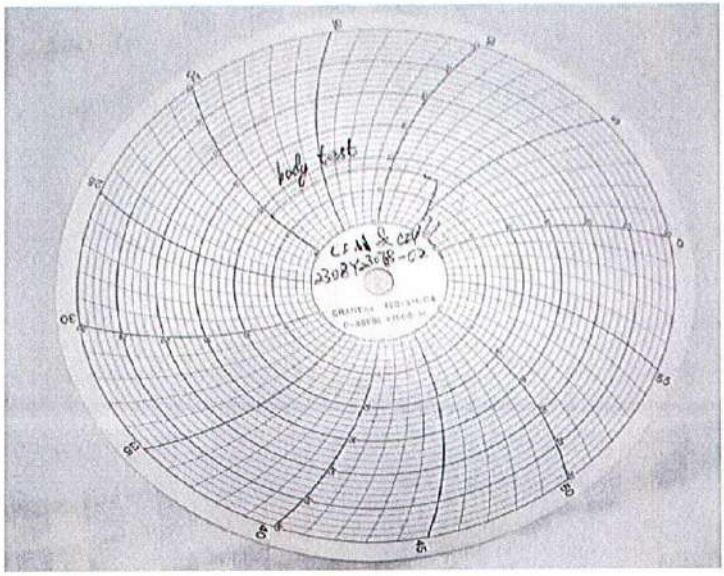
Figure 8-18: AZN-10 Second CIM Test Report



Top pup joint - CIM



CIM – bottom pup joint



Pressure test curve

Figure 8-19: AZN-10 Second CIM Connection & Pressure Test Grapgh

8.4 Completion Operation Checklist

Item	Operation Name	Remark	Check Mark✓
1	Scraper Trip	Packer depth should be scraped	√
2	Pre-job preparation	Pre-connection All Down Hole Equipment With Upper & Lower Pup Joints On Confirmed Thread-makeup at Workshop, And Pressure Test Prior To Being Transported To Well Site.	√
		Retrieve Wear Bushing, M/U Jet Sub And Flush Bop Stack & Well Head With Fresh Water.	√
		All Crossovers, Pup Joints For Space Out Measured Put In Right Place.	√
		Lift Control Line And Protectors On Rig Floor.	√
		Hold A Pre-Job Safety Meeting For Running Upper Completion String.	√
3	Assembling and RIH	M/U 3-1/2" Re-entry Guide Shoe And 3-1/2" BTM No Go Landing Nipple	√
		Check The Conditions Landing Nipple To Make Sure There Is No Damage, Observe There Is No Any Trash Trapped Inside.	√
		Assemble Landing Nipple.	√
		RIH 1 Joint Tubing As Per Tally	√
		P/U And M/U Hydraulic-Set PERMANENT Production Packer.	√
		Much More Care Should Be Taken When Packer Is Passing Well Head And Liner Hanger While RIH,	√
		Assemble Ratch Latch If Packer	-
		Slack Off Packer Assembly, Double Check The Quantity Of Shear Pins,	√

		Remove The Tape On The Element, Check If Any Damages On It.	
		RIH 1 Joint Tubing As Per Tally	√
		M/U Pup Joint And SSD (close Position) To Tubing String With check valve Inside.	√
		Pressure Test Tubing String 7000psi/15min Every 50 JTs.	√
		RIH Completion String and M/U traveling joint as per running tally.	√
		M/U 2 NO. CIV as per running tally& Press Test to 2000 Psi.	-
		Cont. Running Tubing as per Tally while Install Cross protector for the 2 NO. 3/8"control line.	-
		M/U TRSSSV	√
		Install control line to TRSSSV and pressure test control line 8000psi x 15mins. And then keep 5000psi pressure while running to ensure TRSSSV open. Install TRSSSV protector for the 1/4" control line. Continue to run 4-1/2" tubing as per running tally.	√
4	M/U and Space out Tubing Hanger	Pick up & M/U tubing hanger with double pin crossover. M/U landing joint. Check string weight upward and downward.	√
		Roughly cut control line about 2 meters above tubing hanger, wrap 2-3 turns and then feed through Tubing Hanger.	√
		Check string weight upward and downward and Space out Tubing Hanger 1.5 m (Half stroke) above tubing head spool	√
Item	**Operation Name**	**Remark**	**Check Mark✓**
5	Set production	R/Up Slick Line. Remove Check valve inside SSD	√
		RIH and Set check vlve in Landing Nipple. POOH to surface.	√

	packer	Pressurize inside tubing to 500psi to ensure traveling joint is full open. Increase Pressure gradually to SET PACKER. Increase pressure to 7000 Psi to Test String.	√
7	Land Tubing hanger	Move down completion string and apply required weight to shear traveling joint load pins and Land Tubing hanger in T.H.S. Slack off completely & Record the weight.	√
		Drain the BOP stack through the gate valves of tubing head before landing hanger in THS. Leave the valves open.	√
		Tie Down Screws on THS should be fully open before Operation	√
8	N/D BOP & N/U X-mas tree	Tighten tie down screws.	√
		Test inside string w/ 5000 psi, 15 min.	√
		R/Up Slick Line. RIH with slick line and Ret. Check valve from Landing Nipple. POOH to surface.	√
		Test Annulus w/ 2000 Psi 15min.	√
		Close BOP ram and test above tubing hanger (annulus between BOP and landing joint) to 1000psi.	√
		B/O & L/D landing joint.	√
		Set 4” 2Way in TBG Hanger & Test it, Back off and L/D Running Tool.	√
		N/D BOP.	√
		Feed control line, through head bonnet.	√
		Check hanger for any damage and pressure test control line to 5000psi	√
		Pressure test chemical injection line & rapture the disc after installing	-

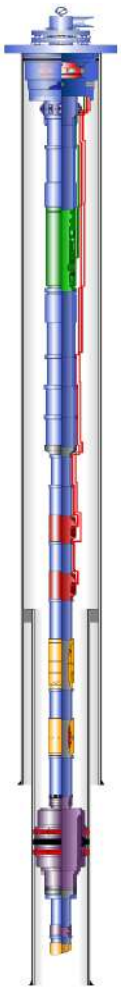
		Bonnet.	
		N/U x-mass tree as program.(Bottom Master)	√
		Perform required tests for X-mass tree. (Bottom Master)	√
		Connect from X-mass tree production valve to the pump unit & well test choke manifold with T-type connection & test all lines to 7500 Psi.	√
		Remove 4" two way, install 4" one way	√
		Close TRSSSV	√
		Release the rig. Clean the well site.	√

Table 8-4: AZN-10 Operation Checklist

8.5 Completion Sketch & Tally

AZN-10 Completion Sketch

Contractor		Liner & Tubing Data						Client	
Field	AZADEGAN	Description	7"Liner	7" tie-back	3 1/2" Tubing	4 1/2" Tubing	Tubing	7" Liner Lap:	3707.5
Well Number:	AZN-10	OD[in]	7.00	7/00	3.50	4.50		7" Liner Shoe:	4922
Rig :	OEOC 201	Nominal weight [lb/ft]	35.00	29.00	10.20	18.90		7" Tie-Back L.Lap:	3527
RTE-GLE	9	Grade	T-95	C-110	L-80	L-80		5' Liner Lap:	4781
RTE-THF	9.15	Thread	BDS	HSM3	HSM3	HSM3		5" Liner Shoe	5193.5
Rig Spudded Date :	7/10/2023	Inside diameter [in]	6.004	6.184	2.92	3.640		Total Depth:	5340
End of Operation		API Drift Diameter	5.879	6.059	2.80	3.515		X-mass Tree:	MSP-T TYPE
								Size & Rating:	4 1/16"-10K

Schematic		Description	Max OD	Min ID	Length(m)	BTM(m)	TOP(m)
	1	Elevation (floor - TBG Head Flange)				9.15	
	2	TBG HANGER 11" * 4 1/2" ACME Box up * 4 1/2" TPCQ-EX Box Down , M/C: FF-0.5 T/C: U PSL: 3G PR: 2, 4" Type "H" Two Way	11	3.68	0.150	9.30	9.15
	3	4-1/2" FLUTED SWAGE ADAPTER D/P, 18.9# TPCQ-EX PIN ON TOP X HSM3 PIN ON DOWN* SUPER 13CR-110KSI	5.5	3.64	1.018	10.32	9.30
	4	5 JOINTs TUBING 4 1/2",HSM3 ,18.9#, L-80, Type-1	5.118	3.64	47.648	57.97	10.32
	5	PUP JOINT 4 1/2",HSM3 ,18.9#, L-80, Type-1	5.118	3.64	1.625	59.59	57.97
	6	4-1/2" FLOW COUPLING,18.9#,SUPER 13CR-110KSI,HSM3 , 6 FT	5.5	3.64	1.725	61.32	59.59
	7	4-1/2" X 3.437" TRSSV-10 TUBING RETRIEVABLE SURFACE CONTROLLED SAFETY VALVE WITH SELF-EQUALIZING FEATURE, WITH AF PROFILE	7.5	3.437	1.780	63.10	61.32
	8	4-1/2" FLOW COUPLING, 18.9#,SUPER 13CR-110KSI,HSM3 , 4' FT	5.5	3.64	1.115	64.21	63.10
	9	PUP JOINT 4 1/2",HSM3 ,18.9#, L-80, Type-1	5.118	3.64	1.623	65.83	64.21
	10	353 JOINTs TUBING 4-1/2"HSM3 18.9#, L-80	5.118	3.64	3351.513	3417.35	65.83
	11	X-over 4-1/2"HSM3, 18.9# BOX * 3-1/2"HSM3 , 10.2# PIN,	5.118	2.922	1.022	3418.37	3417.35
	12	18 JOINTs TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	171.240	3589.61	3418.37
	13	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.824	3591.43	3589.61
	14	3 1/2" CIV TUBING RETRIEVABLE CHEMICAL INJECTION MANDRELHSM3 10.2#,SUPER 13CR-110KSI W/CHECK VALVE, DUAL TYPE & RUPTURE DISC	5	2.922	0.730	3592.16	3591.43
	15	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.822	3593.99	3592.16
	16	1 JOINT TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	9.517	3603.50	3593.99
	17	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.825	3605.33	3603.50
	18	3 1/2" CIV TUBING RETRIEVABLE CHEMICAL INJECTION MANDRELHSM3 , 10.2#,SUPER 13CR-110KSI W/CHECK VALVE, DUAL TYPE & RUPTURE DISC	5	2.922	0.730	3606.06	3605.33
	19	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.832	3607.89	3606.06
	20	1 JOINT TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	9.522	3617.41	3607.89
	21	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.820	3619.23	3617.41
	22	3-1/2" EXP. JOINT,HSM3 , SUPER 13CR-110KSI, 10'STK (Half Closed) Splind Rotational Lock System Any Position, Shear Pin Set 20,000 LBS,	5.315	2.922	5.755	3624.99	3619.23
	23	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.822	3626.81	3624.99
	24	1 JOINT TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	9.507	3636.32	3626.81
	25	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.625	3637.94	3636.32
	26	3-1/2" NON-ELASTOMERIC SLIDING SIDE DOOR, SUPER 13CR-110KSI, HSM3 With Type X Selective Profile (PROFILE: 2.750" X, DOWNWARD OPEN)	4.551	2.75	1.195	3639.14	3637.94
	27	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.700	3640.84	3639.14
	28	1 JOINT TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	9.217	3650.05	3640.84
	29	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.725	3651.78	3650.05
	30	3-1/2" ANCHOR SEAL ASSEMBLY,HSM3, 10.2#,SUPER 13CR-110KSI, 4" PACKING OD	4.00	2.922	0.230	3652.01	3651.78
	31	7" HYDRAULIC SET PERMANENT PRODUCTION PACKER, MODEL WSDP TO BE RUN IN 7.00" 29-32 PPF CASING, SUPER 13CR-110KSI, UPPER SEAL BORE: 4"	5.91	2.922	4.445	3656.45	3652.01
	33	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.822	3658.27	3656.45
	34	1 JOINT TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	9.510	3667.78	3658.27
	35	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.825	3669.61	3667.78
	36	3-1/2" BOTTOM NO-GO LANDING,HSM3 , 10.2#, SUPER 13CR-110KSI W/ Seal Bore 2.75" TYPE XN	4.5	2.635	0.370	3669.98	3669.61
	37	3-1/2" WIRLINE RE-ENTRY GUIDE, 10.2 #,HSM3 Con,SUPER 13CR-110KSI	4.25	2.922	0.240	3670.22	3669.98
		END OF COMPLETION STRING				3670.22	

Tubing Retrievable Saftey Valve										63.10	61.32
Traveling Joint										3624.99	3619.23
CIV 1										3592.16	3591.43
CIV 2										3606.06	3605.33
SSD										3639.14	3637.94
7" HYDRAULIC SET PERMANENT PRODUCTION PACKER										3656.45	3652.01
NO-GO Landing Nipple										3669.98	3669.61
Entry Guide										3670.22	3669.98

Figure 8-20: AZN-10 Completion Sketch

MSA

Field :	Azadegan	Well NO:	AZNS10	Rig NO:	OEOC201	Drig CO:	OEOC
PEDEC, PPL		Completion String					

Grade	ID "	Drift "	Weight lb/ft	Burst psi	Collaspse psi	Yield klbs	Coupling "	TQ ft x lb	Nominal Dia. OD
L-80	2.922	2.797	10.2	11,560.0	12120	233	3.962	3200-3550-3900	3-1/2"
L-80	3.640	3.515	18.9	13,380.0	13820	440	5.137	7920-8800-9680	4-1/2"

BTM NO GO Depth:	3669.61	Formation Name :	Fahliyan	Hook Weight :	65 k lb
BTM of String	3670.22	Formation Top :	3987	VOLUME (TOTAL)	150.63
Packer Depth :	3652.01			Mud Weight (pcf) :	104

Number on jt	Grade	Run In No.	Jt Length (m)	Cum Length (m)	Dist to RT (m)	M/D weight (Klbs)	VOL(BBL)	Remarks	
								joint no	IN / OUT
					3670.22	65.00			
3-1/2" WIRLINE RE-ENTRY GUIDE, 10.2 #,HSM3 Con,SUPER 13CR-110KSI, WEIQIXIN Co	3-1/2" Mule Shoe		0.240	0.24	3669.98	65.01	0.01		
13-1/2" BOTTOM NO-GO LANDING,HSM3 , 10.2#, SUPER 13CR-110KSI W/ Seal Bore 2.75" TYPE XN, WEIQIXIN Co	3-1/2" BTM NO GO		0.370	0.61	3669.61	65.02	0.02		
3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	Pup Joint 3-1/2", 10.2#,HSM3		1.825	2.44	3667.785	65.06	0.07		
A1	TUBING 3-1/2", 10.2#, L-80,HSM3	1	9.510	11.95	3658.275	65.31	0.33	1	
3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	Pup Joint 3-1/2", 10.2#,HSM3		1.822	13.77	3656.453	65.36	0.37		
7" HYDRAULIC SET PERMANENT PRODUCTION PACKER, MODEL WSDP TO BE RUN IN 7.00" 29-32 PPF CASING, SUPER 13CR-110KSI, UPPER SEAL BORE: 4", WEIQIXIN Co	PKR		4.445	18.21	3652.008	65.48	0.50		
3-1/2" ANCHOR SEAL ASSEMBLY,HSM3, 10.2#,SUPER 13CR-110KSI, 4" PACKING OD, WEIQIXIN Co	Ratch Lach		0.230	18.44	3651.778	65.49	0.50		
3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	Pup Joint 3-1/2", 10.2#,HSM3		1.725	20.17	3650.053	65.53	0.55		
A2	TUBING 3-1/2", 10.2#, L-80,HSM3	2	9.217	29.38	3640.836	65.77	0.80	2	
3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	Pup Joint 3-1/2", 10.2#,HSM3		1.700	31.08	3639.136	65.82	0.85		
3-1/2" NON-ELASTOMERIC SLIDING SIDE DOOR, SUPER 13CR-110KSI, HSM3 With Type X Selective Profile (PROFILE: 2.750" X, DOWNWARD OPEN), WEIQIXIN Co	SSD		1.195	32.28	3637.941	65.85	0.88		
3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	Pup Joint 3-1/2", 10.2#,HSM3		1.625	33.90	3636.316	65.89	0.92		
A3	TUBING 3-1/2", 10.2#, L-80,HSM3	3	9.507	43.41	3626.809	66.14	1.18	3	
3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	Pup Joint 3-1/2", 10.2#,HSM3		1.822	45.23	3624.987	66.19	1.23		
3-1/2" EXP. JOINT,HSM3 , SUPER 13CR-110KSI, 10'STK (Half Open) Splind Rotational Lock System Any Position, Shear Pin Set 20,000 LBS, WEIQIXIN Co	Traveling Jt		5.755	50.99	3619.232	66.34	1.39		
3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	Pup Joint 3-1/2", 10.2#,HSM3		1.820	52.81	3617.412	66.39	1.44		
A4	TUBING 3-1/2", 10.2#, L-80,HSM3	4	9.522	62.33	3607.89	66.64	1.70	4	
3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	Pup Joint 3-1/2", 10.2#,HSM3		1.832	64.16	3606.058	66.69	1.75		
3 1/2" CIV TUBING RETRIEVABLE CHEMICAL INJECTION MANDRELHSM3 , 10.2#,SUPER 13CR-110KSI W/CHECK VALVE, DUAL TYPE & RUPTURE DISC, WEIQIXIN Co	CIV		0.730	64.89	3605.328	66.71	1.77		
3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	Pup Joint 3-1/2", 10.2#,HSM3		1.825	66.72	3603.503	66.76	1.82		
A5	TUBING 3-1/2", 10.2#, L-80,HSM3	5	9.517	76.23	3593.986	67.01	2.07	5	
3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	Pup Joint 3-1/2", 10.2#,HSM3		1.822	78.06	3592.164	67.06	2.12		
3 1/2" CIV TUBING RETRIEVABLE CHEMICAL INJECTION MANDRELHSM3 , 10.2#,SUPER 13CR-110KSI W/CHECK VALVE, DUAL TYPE & RUPTURE DISC, WEIQIXIN Co	CIV		0.730	78.79	3591.434	67.08	2.14		
3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	Pup Joint 3-1/2", 10.2#,HSM3		1.824	80.61	3589.61	67.12	2.19		
A6	TUBING 3-1/2", 10.2#, L-80,HSM3	6	9.520	90.13	3580.09	67.37	2.45	6	
A7	TUBING 3-1/2", 10.2#, L-80,HSM3	7	9.519	99.65	3570.571	67.63	2.71	7	
A8	TUBING 3-1/2", 10.2#, L-80,HSM3	8	9.526	109.18	3561.045	67.88	2.97	8	
A9	TUBING 3-1/2", 10.2#, L-80,HSM3	9	9.512	118.69	3551.533	68.13	3.23	9	
A10	TUBING 3-1/2", 10.2#, L-80,HSM3	10	9.509	128.20	3542.024	68.38	3.49	10	

A11	TUBING 3-1/2", 10.2#, L-80,HSM3	11	9.509	137.71	3532.515	68.63	3.75	11	
A12	TUBING 3-1/2", 10.2#, L-80,HSM3	12	9.522	147.23	3522.993	68.88	4.01	12	
A13	TUBING 3-1/2", 10.2#, L-80,HSM3	13	9.508	156.74	3513.485	69.13	4.27	13	
A14	TUBING 3-1/2", 10.2#, L-80,HSM3	14	9.512	166.25	3503.973	69.38	4.52	14	
A15	TUBING 3-1/2", 10.2#, L-80,HSM3	15	9.504	175.75	3494.469	69.63	4.78	15	
A16	TUBING 3-1/2", 10.2#, L-80,HSM3	16	9.517	185.27	3484.952	69.88	5.04	16	
A17	TUBING 3-1/2", 10.2#, L-80,HSM3	17	9.522	194.79	3475.43	70.13	5.30	17	
A18	TUBING 3-1/2", 10.2#, L-80,HSM3	18	9.512	204.30	3465.918	70.38	5.56	18	
A19	TUBING 3-1/2", 10.2#, L-80,HSM3	19	9.507	213.81	3456.411	70.63	5.82	19	
A20	TUBING 3-1/2", 10.2#, L-80,HSM3	20	9.507	223.32	3446.904	70.88	6.08	20	
A22	TUBING 3-1/2", 10.2#, L-80,HSM3	21	9.512	232.83	3437.392	71.13	6.34	21	
1	TUBING 3-1/2", 10.2#, L-80,HSM3	22	9.512	242.34	3427.88	71.39	6.59	22	
3	TUBING 3-1/2", 10.2#, L-80,HSM5	23	9.510	251.85	3418.37	71.64	6.85	23	
X-over 4-1/2"HSM3, 18.9# BOX * 3-1/2"HSM3 , 10.2# PIN,	XOS 4-1/2", HSM3 B X 3-1/2" HSM3, P	0	1.022	252.87	3417.348	71.44	6.62		
11	TUBING 4-1/2", 18.9#, L-80,HSM3	1	9.528	262.40	3407.82	71.85	7.00	1	Drift & Pr. Test
12	TUBING 4-1/2", 18.9#, L-80,HSM3	2	9.523	271.92	3398.297	72.32	7.40	2	
13	TUBING 4-1/2", 18.9#, L-80,HSM3	3	9.526	281.45	3388.771	72.78	7.80	3	
14	TUBING 4-1/2", 18.9#, L-80,HSM3	4	9.526	290.98	3379.245	73.25	8.20	4	
15	TUBING 4-1/2", 18.9#, L-80,HSM3	5	9.521	300.50	3369.724	73.71	8.61	5	
16	TUBING 4-1/2", 18.9#, L-80,HSM3	6	9.525	310.02	3360.199	74.18	9.01	6	
17	TUBING 4-1/2", 18.9#, L-80,HSM3	7	9.356	319.38	3350.843	74.63	9.40	7	
18	TUBING 4-1/2", 18.9#, L-80,HSM3	8	9.563	328.94	3341.28	75.10	9.81	8	
19	TUBING 4-1/2", 18.9#, L-80,HSM3	9	9.523	338.46	3331.757	75.56	10.21	9	
20	TUBING 4-1/2", 18.9#, L-80,HSM3	10	9.261	347.72	3322.496	76.02	10.60	10	
21	TUBING 4-1/2", 18.9#, L-80,HSM3	11	9.435	357.16	3313.061	76.48	11.00	11	
22	TUBING 4-1/2", 18.9#, L-80,HSM3	12	9.425	366.58	3303.636	76.94	11.40	12	
23	TUBING 4-1/2", 18.9#, L-80,HSM3	13	9.442	376.03	3294.194	77.40	11.80	13	
24	TUBING 4-1/2", 18.9#, L-80,HSM3	14	9.527	385.55	3284.667	77.86	12.20	14	
25	TUBING 4-1/2", 18.9#, L-80,HSM3	15	9.378	394.93	3275.289	78.32	12.59	15	
26	TUBING 4-1/2", 18.9#, L-80,HSM3	16	9.518	404.45	3265.771	78.79	13.00	16	
27	TUBING 4-1/2", 18.9#, L-80,HSM3	17	9.522	413.97	3256.249	79.25	13.40	17	
28	TUBING 4-1/2", 18.9#, L-80,HSM3	18	9.523	423.49	3246.726	79.72	13.80	18	
29	TUBING 4-1/2", 18.9#, L-80,HSM3	19	9.520	433.01	3237.206	80.18	14.20	19	
30	TUBING 4-1/2", 18.9#, L-80,HSM3	20	9.524	442.54	3227.682	80.65	14.60	20	
31	TUBING 4-1/2", 18.9#, L-80,HSM3	21	9.520	452.06	3218.162	81.11	15.01	21	
32	TUBING 4-1/2", 18.9#, L-80,HSM3	22	9.441	461.50	3208.721	81.57	15.41	22	

33	TUBING 4-1/2", 18.9#, L-80,HSM3	23	9.521	471.02	3199.2	82.04	15.81	23	
34	TUBING 4-1/2", 18.9#, L-80,HSM3	24	9.522	480.54	3189.678	82.50	16.21	24	
35	TUBING 4-1/2", 18.9#, L-80,HSM3	25	9.525	490.07	3180.153	82.97	16.61	25	
36	TUBING 4-1/2", 18.9#, L-80,HSM3	26	9.523	499.59	3170.63	83.43	17.01	26	
37	TUBING 4-1/2", 18.9#, L-80,HSM3	27	9.527	509.12	3161.103	83.90	17.42	27	
38	TUBING 4-1/2", 18.9#, L-80,HSM3	28	9.412	518.53	3151.691	84.36	17.81	28	
39	TUBING 4-1/2", 18.9#, L-80,HSM3	29	9.544	528.07	3142.147	84.82	18.22	29	
40	TUBING 4-1/2", 18.9#, L-80,HSM3	30	9.342	537.42	3132.805	85.28	18.61	30	
41	TUBING 4-1/2", 18.9#, L-80,HSM3	31	9.336	546.75	3123.469	85.73	19.01	31	
42	TUBING 4-1/2", 18.9#, L-80,HSM3	32	9.533	556.28	3113.936	86.20	19.41	32	
43	TUBING 4-1/2", 18.9#, L-80,HSM3	33	9.342	565.63	3104.594	86.65	19.80	33	
44	TUBING 4-1/2", 18.9#, L-80,HSM3	34	9.534	575.16	3095.06	87.12	20.21	34	
45	TUBING 4-1/2", 18.9#, L-80,HSM3	35	9.532	584.69	3085.528	87.59	20.61	35	
46	TUBING 4-1/2", 18.9#, L-80,HSM3	36	9.525	594.22	3076.003	88.05	21.01	36	
47	TUBING 4-1/2", 18.9#, L-80,HSM3	37	9.339	603.56	3066.664	88.51	21.40	37	
48	TUBING 4-1/2", 18.9#, L-80,HSM3	38	9.521	613.08	3057.143	88.97	21.81	38	
49	TUBING 4-1/2", 18.9#, L-80,HSM3	39	9.435	622.51	3047.708	89.43	22.20	39	
50	TUBING 4-1/2", 18.9#, L-80,HSM3	40	9.535	632.05	3038.173	89.90	22.61	40	
51	TUBING 4-1/2", 18.9#, L-80,HSM3	41	9.357	641.40	3028.816	90.35	23.00	41	
52	TUBING 4-1/2", 18.9#, L-80,HSM3	42	9.251	650.66	3019.565	90.81	23.39	42	
53	TUBING 4-1/2", 18.9#, L-80,HSM3	43	9.528	660.18	3010.037	91.27	23.80	43	
54	TUBING 4-1/2", 18.9#, L-80,HSM3	44	9.522	669.71	3000.515	91.74	24.20	44	
55	TUBING 4-1/2", 18.9#, L-80,HSM3	45	9.526	679.23	2990.989	92.20	24.60	45	
56	TUBING 4-1/2", 18.9#, L-80,HSM3	46	9.528	688.76	2981.461	92.67	25.00	46	
57	TUBING 4-1/2", 18.9#, L-80,HSM3	47	9.536	698.30	2971.925	93.13	25.41	47	
58	TUBING 4-1/2", 18.9#, L-80,HSM3	48	9.523	707.82	2962.402	93.60	25.81	48	
59	TUBING 4-1/2", 18.9#, L-80,HSM3	49	9.518	717.34	2952.884	94.06	26.21	49	
60	TUBING 4-1/2", 18.9#, L-80,HSM3	50	9.522	726.86	2943.362	94.53	26.61	50	Pr. Test
61	TUBING 4-1/2", 18.9#, L-80,HSM3	51	9.532	736.39	2933.83	94.99	27.01	51	
62	TUBING 4-1/2", 18.9#, L-80,HSM3	52	9.522	745.91	2924.308	95.46	27.42	52	
63	TUBING 4-1/2", 18.9#, L-80,HSM3	53	9.398	755.31	2914.91	95.92	27.81	53	
64	TUBING 4-1/2", 18.9#, L-80,HSM3	54	9.411	764.72	2905.499	96.38	28.21	54	
65	TUBING 4-1/2", 18.9#, L-80,HSM3	55	9.460	774.18	2896.039	96.84	28.61	55	
66	TUBING 4-1/2", 18.9#, L-80,HSM3	56	9.529	783.71	2886.51	97.30	29.01	56	
67	TUBING 4-1/2", 18.9#, L-80,HSM3	57	9.532	793.24	2876.978	97.77	29.41	57	
68	TUBING 4-1/2", 18.9#, L-80,HSM3	58	9.527	802.77	2867.451	98.23	29.82	58	
69	TUBING 4-1/2", 18.9#, L-80,HSM3	59	9.414	812.18	2858.037	98.69	30.21	59	
70	TUBING 4-1/2", 18.9#, L-80,HSM3	60	9.530	821.71	2848.507	99.16	30.62	60	
71	TUBING 4-1/2", 18.9#, L-80,HSM3	61	9.527	831.24	2838.98	99.62	31.02	61	
72	TUBING 4-1/2", 18.9#, L-80,HSM3	62	9.527	840.77	2829.453	100.09	31.42	62	
73	TUBING 4-1/2", 18.9#, L-80,HSM3	63	9.433	850.20	2820.02	100.55	31.82	63	
74	TUBING 4-1/2", 18.9#, L-80,HSM3	64	9.528	859.73	2810.492	101.01	32.22	64	
75	TUBING 4-1/2", 18.9#, L-80,HSM3	65	9.526	869.25	2800.966	101.48	32.62	65	
76	TUBING 4-1/2", 18.9#, L-80,HSM3	66	9.526	878.78	2791.44	101.94	33.03	66	
77	TUBING 4-1/2", 18.9#, L-80,HSM3	67	9.251	888.03	2782.189	102.40	33.42	67	
78	TUBING 4-1/2", 18.9#, L-80,HSM3	68	9.526	897.56	2772.663	102.86	33.82	68	
79	TUBING 4-1/2", 18.9#, L-80,HSM3	69	9.533	907.09	2763.13	103.33	34.22	69	
80	TUBING 4-1/2", 18.9#, L-80,HSM3	70	9.530	916.62	2753.6	103.79	34.62	70	
81	TUBING 4-1/2", 18.9#, L-80,HSM3	71	9.574	926.19	2744.026	104.26	35.03	71	
82	TUBING 4-1/2", 18.9#, L-80,HSM3	72	9.528	935.72	2734.498	104.72	35.43	72	

83	TUBING 4-1/2", 18.9#, L-80,HSM3	73	9.536	945.26	2724.962	105.19	35.83	73	
84	TUBING 4-1/2", 18.9#, L-80,HSM3	74	9.441	954.70	2715.521	105.65	36.23	74	
85	TUBING 4-1/2", 18.9#, L-80,HSM3	75	9.528	964.23	2705.993	106.12	36.64	75	
86	TUBING 4-1/2", 18.9#, L-80,HSM3	76	9.523	973.75	2696.47	106.58	37.04	76	
87	TUBING 4-1/2", 18.9#, L-80,HSM3	77	9.528	983.28	2686.942	107.05	37.44	77	
88	TUBING 4-1/2", 18.9#, L-80,HSM3	78	9.456	992.73	2677.486	107.51	37.84	78	
89	TUBING 4-1/2", 18.9#, L-80,HSM3	79	9.528	1002.26	2667.958	107.97	38.24	79	
90	TUBING 4-1/2", 18.9#, L-80,HSM3	80	9.519	1011.78	2658.439	108.44	38.64	80	
91	TUBING 4-1/2", 18.9#, L-80,HSM3	81	9.546	1021.33	2648.893	108.90	39.05	81	
92	TUBING 4-1/2", 18.9#, L-80,HSM3	82	9.531	1030.86	2639.362	109.37	39.45	82	
93	TUBING 4-1/2", 18.9#, L-80,HSM3	83	9.528	1040.39	2629.834	109.83	39.85	83	
94	TUBING 4-1/2", 18.9#, L-80,HSM3	84	9.398	1049.78	2620.436	110.29	40.25	84	
95	TUBING 4-1/2", 18.9#, L-80,HSM3	85	9.575	1059.36	2610.861	110.76	40.65	85	
96	TUBING 4-1/2", 18.9#, L-80,HSM3	86	9.448	1068.81	2601.413	111.22	41.05	86	
97	TUBING 4-1/2", 18.9#, L-80,HSM3	87	9.519	1078.33	2591.894	111.69	41.45	87	
98	TUBING 4-1/2", 18.9#, L-80,HSM3	88	9.527	1087.85	2582.367	112.15	41.86	88	
99	TUBING 4-1/2", 18.9#, L-80,HSM3	89	9.419	1097.27	2572.948	112.61	42.25	89	
100	TUBING 4-1/2", 18.9#, L-80,HSM3	90	9.527	1106.80	2563.421	113.08	42.66	90	
101	TUBING 4-1/2", 18.9#, L-80,HSM3	91	9.530	1116.33	2553.891	113.54	43.06	91	
102	TUBING 4-1/2", 18.9#, L-80,HSM3	92	9.485	1125.81	2544.406	114.00	43.46	92	
103	TUBING 4-1/2", 18.9#, L-80,HSM3	93	9.527	1135.34	2534.879	114.47	43.86	93	
104	TUBING 4-1/2", 18.9#, L-80,HSM3	94	9.195	1144.54	2525.684	114.92	44.25	94	
105	TUBING 4-1/2", 18.9#, L-80,HSM3	95	9.188	1153.72	2516.496	115.37	44.64	95	
106	TUBING 4-1/2", 18.9#, L-80,HSM3	96	9.040	1162.76	2507.456	115.81	45.02	96	
107	TUBING 4-1/2", 18.9#, L-80,HSM3	97	9.527	1172.29	2497.929	116.27	45.42	97	
108	TUBING 4-1/2", 18.9#, L-80,HSM3	98	9.527	1181.82	2488.402	116.74	45.82	98	
109	TUBING 4-1/2", 18.9#, L-80,HSM3	99	9.524	1191.34	2478.878	117.20	46.23	99	
110	TUBING 4-1/2", 18.9#, L-80,HSM3	100	9.530	1200.87	2469.348	117.67	46.63	100	
111	TUBING 4-1/2", 18.9#, L-80,HSM3	101	9.530	1210.40	2459.818	118.13	47.03	101	
112	TUBING 4-1/2", 18.9#, L-80,HSM3	102	9.184	1219.59	2450.634	118.58	47.42	102	
113	TUBING 4-1/2", 18.9#, L-80,HSM3	103	9.523	1229.11	2441.111	119.05	47.82	103	
114	TUBING 4-1/2", 18.9#, L-80,HSM3	104	9.531	1238.64	2431.58	119.51	48.22	104	
115	TUBING 4-1/2", 18.9#, L-80,HSM3	105	9.248	1247.89	2422.332	119.96	48.61	105	
116	TUBING 4-1/2", 18.9#, L-80,HSM3	106	9.410	1257.30	2412.922	120.42	49.01	106	
117	TUBING 4-1/2", 18.9#, L-80,HSM3	107	9.529	1266.83	2403.393	120.89	49.41	107	
118	TUBING 4-1/2", 18.9#, L-80,HSM3	108	9.415	1276.24	2393.978	121.35	49.81	108	
119	TUBING 4-1/2", 18.9#, L-80,HSM3	109	9.528	1285.77	2384.45	121.81	50.21	109	
120	TUBING 4-1/2", 18.9#, L-80,HSM3	110	9.526	1295.30	2374.924	122.28	50.62	110	
121	TUBING 4-1/2", 18.9#, L-80,HSM3	111	9.533	1304.83	2365.391	122.74	51.02	111	
122	TUBING 4-1/2", 18.9#, L-80,HSM3	112	9.526	1314.36	2355.865	123.21	51.42	112	
123	TUBING 4-1/2", 18.9#, L-80,HSM3	113	9.528	1323.88	2346.337	123.67	51.82	113	
124	TUBING 4-1/2", 18.9#, L-80,HSM3	114	9.525	1333.41	2336.812	124.14	52.23	114	
125	TUBING 4-1/2", 18.9#, L-80,HSM3	115	9.529	1342.94	2327.283	124.61	52.63	115	
126	TUBING 4-1/2", 18.9#, L-80,HSM3	116	9.526	1352.46	2317.757	125.07	53.03	116	
127	TUBING 4-1/2", 18.9#, L-80,HSM3	117	9.549	1362.01	2308.208	125.54	53.43	117	
128	TUBING 4-1/2", 18.9#, L-80,HSM3	118	9.549	1371.56	2298.659	126.00	53.84	118	

129	TUBING 4-1/2", 18.9#, L-80,HSM3	119	9.547	1381.11	2289.112	126.47	54.24	119	
130	TUBING 4-1/2", 18.9#, L-80,HSM3	120	9.520	1390.63	2279.592	126.93	54.64	120	
131	TUBING 4-1/2", 18.9#, L-80,HSM3	121	9.529	1400.16	2270.063	127.40	55.04	121	
132	TUBING 4-1/2", 18.9#, L-80,HSM3	122	9.121	1409.28	2260.942	127.84	55.43	122	
133	TUBING 4-1/2", 18.9#, L-80,HSM3	123	9.528	1418.81	2251.414	128.31	55.83	123	
134	TUBING 4-1/2", 18.9#, L-80,HSM3	124	9.039	1427.85	2242.375	128.75	56.21	124	
135	TUBING 4-1/2", 18.9#, L-80,HSM3	125	9.528	1437.37	2232.847	129.22	56.62	125	
136	TUBING 4-1/2", 18.9#, L-80,HSM3	126	9.519	1446.89	2223.328	129.68	57.02	126	
137	TUBING 4-1/2", 18.9#, L-80,HSM3	127	9.334	1456.23	2213.994	130.14	57.41	127	
138	TUBING 4-1/2", 18.9#, L-80,HSM3	128	9.523	1465.75	2204.471	130.60	57.81	128	
139	TUBING 4-1/2", 18.9#, L-80,HSM3	129	9.527	1475.28	2194.944	131.07	58.22	129	
140	TUBING 4-1/2", 18.9#, L-80,HSM3	130	9.532	1484.81	2185.412	131.53	58.62	130	Pr. Test
141	TUBING 4-1/2", 18.9#, L-80,HSM3	131	9.193	1494.00	2176.219	131.98	59.01	131	
142	TUBING 4-1/2", 18.9#, L-80,HSM3	132	9.523	1503.52	2166.696	132.45	59.41	132	
143	TUBING 4-1/2", 18.9#, L-80,HSM3	133	9.373	1512.90	2157.323	132.90	59.81	133	
144	TUBING 4-1/2", 18.9#, L-80,HSM3	134	9.525	1522.42	2147.798	133.37	60.21	134	
145	TUBING 4-1/2", 18.9#, L-80,HSM3	135	9.529	1531.95	2138.269	133.83	60.61	135	
146	TUBING 4-1/2", 18.9#, L-80,HSM3	136	9.523	1541.47	2128.746	134.30	61.01	136	
147	TUBING 4-1/2", 18.9#, L-80,HSM3	137	9.523	1551.00	2119.223	134.76	61.41	137	
148	TUBING 4-1/2", 18.9#, L-80,HSM3	138	9.523	1560.52	2109.7	135.23	61.82	138	
149	TUBING 4-1/2", 18.9#, L-80,HSM3	139	9.529	1570.05	2100.171	135.69	62.22	139	
150	TUBING 4-1/2", 18.9#, L-80,HSM3	140	9.534	1579.58	2090.637	136.16	62.62	140	
151	TUBING 4-1/2", 18.9#, L-80,HSM3	141	9.526	1589.11	2081.111	136.62	63.02	141	
152	TUBING 4-1/2", 18.9#, L-80,HSM3	142	9.518	1598.63	2071.593	137.09	63.43	142	
153	TUBING 4-1/2", 18.9#, L-80,HSM3	143	9.524	1608.15	2062.069	137.55	63.83	143	
154	TUBING 4-1/2", 18.9#, L-80,HSM3	144	9.527	1617.68	2052.542	138.02	64.23	144	
155	TUBING 4-1/2", 18.9#, L-80,HSM3	145	9.527	1627.21	2043.015	138.48	64.63	145	
156	TUBING 4-1/2", 18.9#, L-80,HSM3	146	9.529	1636.73	2033.486	138.95	65.04	146	
157	TUBING 4-1/2", 18.9#, L-80,HSM3	147	9.531	1646.27	2023.955	139.41	65.44	147	
158	TUBING 4-1/2", 18.9#, L-80,HSM3	148	9.529	1655.79	2014.426	139.88	65.84	148	
159	TUBING 4-1/2", 18.9#, L-80,HSM3	149	9.525	1665.32	2004.901	140.34	66.24	149	
160	TUBING 4-1/2", 18.9#, L-80,HSM3	150	9.522	1674.84	1995.379	140.81	66.64	150	
161	TUBING 4-1/2", 18.9#, L-80,HSM3	151	9.524	1684.37	1985.855	141.27	67.05	151	
162	TUBING 4-1/2", 18.9#, L-80,HSM3	152	9.528	1693.89	1976.327	141.74	67.45	152	
163	TUBING 4-1/2", 18.9#, L-80,HSM3	153	9.333	1703.23	1966.994	142.20	67.84	153	
164	TUBING 4-1/2", 18.9#, L-80,HSM3	154	9.526	1712.75	1957.468	142.66	68.25	154	
165	TUBING 4-1/2", 18.9#, L-80,HSM3	155	9.525	1722.28	1947.943	143.13	68.65	155	
166	TUBING 4-1/2", 18.9#, L-80,HSM3	156	9.523	1731.80	1938.42	143.59	69.05	156	
167	TUBING 4-1/2", 18.9#, L-80,HSM3	157	9.529	1741.33	1928.891	144.06	69.45	157	
168	TUBING 4-1/2", 18.9#, L-80,HSM3	158	9.526	1750.86	1919.365	144.52	69.85	158	
169	TUBING 4-1/2", 18.9#, L-80,HSM3	159	9.548	1760.40	1909.817	144.99	70.26	159	
170	TUBING 4-1/2", 18.9#, L-80,HSM3	160	9.525	1769.93	1900.292	145.45	70.66	160	
171	TUBING 4-1/2", 18.9#, L-80,HSM3	161	9.529	1779.46	1890.763	145.92	71.06	161	
172	TUBING 4-1/2", 18.9#, L-80,HSM3	162	9.528	1788.99	1881.235	146.38	71.46	162	
173	TUBING 4-1/2", 18.9#, L-80,HSM3	163	9.524	1798.51	1871.711	146.85	71.87	163	
174	TUBING 4-1/2", 18.9#, L-80,HSM3	164	9.424	1807.93	1862.287	147.31	72.26	164	

175	TUBING 4-1/2", 18.9#, L-80,HSM3	165	9.529	1817.46	1852.758	147.77	72.67	165	
176	TUBING 4-1/2", 18.9#, L-80,HSM3	166	9.525	1826.99	1843.233	148.24	73.07	166	
177	TUBING 4-1/2", 18.9#, L-80,HSM3	167	9.525	1836.51	1833.708	148.70	73.47	167	
178	TUBING 4-1/2", 18.9#, L-80,HSM3	168	9.548	1846.06	1824.16	149.17	73.88	168	
179	TUBING 4-1/2", 18.9#, L-80,HSM3	169	9.525	1855.59	1814.635	149.63	74.28	169	
180	TUBING 4-1/2", 18.9#, L-80,HSM3	170	9.548	1865.13	1805.087	150.10	74.68	170	
181	TUBING 4-1/2", 18.9#, L-80,HSM3	171	9.523	1874.66	1795.564	150.56	75.08	171	
182	TUBING 4-1/2", 18.9#, L-80,HSM3	172	9.208	1883.86	1786.356	151.01	75.47	172	
183	TUBING 4-1/2", 18.9#, L-80,HSM3	173	9.525	1893.39	1776.831	151.48	75.87	173	
184	TUBING 4-1/2", 18.9#, L-80,HSM3	174	9.525	1902.91	1767.306	151.94	76.28	174	
185	TUBING 4-1/2", 18.9#, L-80,HSM3	175	9.535	1912.45	1757.771	152.41	76.68	175	
186	TUBING 4-1/2", 18.9#, L-80,HSM3	176	9.420	1921.87	1748.351	152.87	77.08	176	
187	TUBING 4-1/2", 18.9#, L-80,HSM3	177	9.414	1931.28	1738.937	153.33	77.47	177	
188	TUBING 4-1/2", 18.9#, L-80,HSM3	178	9.522	1940.81	1729.415	153.79	77.88	178	
189	TUBING 4-1/2", 18.9#, L-80,HSM3	179	9.526	1950.33	1719.889	154.26	78.28	179	
190	TUBING 4-1/2", 18.9#, L-80,HSM3	180	9.528	1959.86	1710.361	154.72	78.68	180	
191	TUBING 4-1/2", 18.9#, L-80,HSM3	181	9.523	1969.38	1700.838	155.19	79.08	181	
192	TUBING 4-1/2", 18.9#, L-80,HSM3	182	9.546	1978.93	1691.292	155.66	79.49	182	
193	TUBING 4-1/2", 18.9#, L-80,HSM3	183	9.525	1988.45	1681.767	156.12	79.89	183	
194	TUBING 4-1/2", 18.9#, L-80,HSM3	184	9.524	1997.98	1672.243	156.59	80.29	184	
195	TUBING 4-1/2", 18.9#, L-80,HSM3	185	9.586	2007.56	1662.657	157.05	80.70	185	
196	TUBING 4-1/2", 18.9#, L-80,HSM3	186	9.398	2016.96	1653.259	157.51	81.09	186	
197	TUBING 4-1/2", 18.9#, L-80,HSM3	187	9.523	2026.48	1643.736	157.98	81.49	187	
198	TUBING 4-1/2", 18.9#, L-80,HSM3	188	9.520	2036.00	1634.216	158.44	81.90	188	
199	TUBING 4-1/2", 18.9#, L-80,HSM3	189	9.527	2045.53	1624.689	158.91	82.30	189	
200	TUBING 4-1/2", 18.9#, L-80,HSM3	190	9.155	2054.69	1615.534	159.35	82.69	190	
201	TUBING 4-1/2", 18.9#, L-80,HSM3	191	9.421	2064.11	1606.113	159.81	83.08	191	
202	TUBING 4-1/2", 18.9#, L-80,HSM3	192	9.525	2073.63	1596.588	160.28	83.49	192	
203	TUBING 4-1/2", 18.9#, L-80,HSM3	193	9.381	2083.01	1587.207	160.74	83.88	193	
204	TUBING 4-1/2", 18.9#, L-80,HSM3	194	9.532	2092.55	1577.675	161.20	84.28	194	
205	TUBING 4-1/2", 18.9#, L-80,HSM3	195	9.527	2102.07	1568.148	161.67	84.69	195	
206	TUBING 4-1/2", 18.9#, L-80,HSM3	196	9.182	2111.25	1558.966	162.12	85.07	196	
207	TUBING 4-1/2", 18.9#, L-80,HSM3	197	9.528	2120.78	1549.438	162.58	85.48	197	
208	TUBING 4-1/2", 18.9#, L-80,HSM3	198	9.523	2130.31	1539.915	163.05	85.88	198	
209	TUBING 4-1/2", 18.9#, L-80,HSM3	199	9.525	2139.83	1530.39	163.51	86.28	199	
210	TUBING 4-1/2", 18.9#, L-80,HSM3	200	9.546	2149.38	1520.844	163.98	86.68	200	
211	TUBING 4-1/2", 18.9#, L-80,HSM3	201	9.524	2158.90	1511.32	164.44	87.09	201	
212	TUBING 4-1/2", 18.9#, L-80,HSM3	202	9.533	2168.43	1501.787	164.91	87.49	202	
213	TUBING 4-1/2", 18.9#, L-80,HSM3	203	9.526	2177.96	1492.261	165.37	87.89	203	
214	TUBING 4-1/2", 18.9#, L-80,HSM3	204	9.531	2187.49	1482.73	165.84	88.29	204	
215	TUBING 4-1/2", 18.9#, L-80,HSM3	205	9.531	2197.02	1473.199	166.30	88.70	205	
216	TUBING 4-1/2", 18.9#, L-80,HSM3	206	9.433	2206.45	1463.766	166.76	89.09	206	
217	TUBING 4-1/2", 18.9#, L-80,HSM3	207	9.529	2215.98	1454.237	167.23	89.50	207	
218	TUBING 4-1/2", 18.9#, L-80,HSM3	208	9.524	2225.51	1444.713	167.69	89.90	208	
219	TUBING 4-1/2", 18.9#, L-80,HSM3	209	9.428	2234.94	1435.285	168.15	90.30	209	

220	TUBING 4-1/2", 18.9#, L-80,HSM3	210	9.528	2244.46	1425.757	168.62	90.70	210	Pr. Test
221	TUBING 4-1/2", 18.9#, L-80,HSM3	211	9.526	2253.99	1416.231	169.08	91.10	211	
222	TUBING 4-1/2", 18.9#, L-80,HSM3	212	9.523	2263.51	1406.708	169.55	91.50	212	
223	TUBING 4-1/2", 18.9#, L-80,HSM3	213	9.523	2273.04	1397.185	170.01	91.91	213	
224	TUBING 4-1/2", 18.9#, L-80,HSM3	214	9.413	2282.45	1387.772	170.47	92.30	214	
225	TUBING 4-1/2", 18.9#, L-80,HSM3	215	9.526	2291.97	1378.246	170.94	92.71	215	
226	TUBING 4-1/2", 18.9#, L-80,HSM3	216	9.553	2301.53	1368.693	171.41	93.11	216	
227	TUBING 4-1/2", 18.9#, L-80,HSM3	217	9.526	2311.05	1359.167	171.87	93.51	217	
228	TUBING 4-1/2", 18.9#, L-80,HSM3	218	9.523	2320.58	1349.644	172.34	93.91	218	
229	TUBING 4-1/2", 18.9#, L-80,HSM3	219	9.528	2330.10	1340.116	172.80	94.32	219	
230	TUBING 4-1/2", 18.9#, L-80,HSM3	220	9.518	2339.62	1330.598	173.27	94.72	220	
231	TUBING 4-1/2", 18.9#, L-80,HSM3	221	9.518	2349.14	1321.08	173.73	95.12	221	
232	TUBING 4-1/2", 18.9#, L-80,HSM3	222	9.518	2358.66	1311.562	174.20	95.52	222	
233	TUBING 4-1/2", 18.9#, L-80,HSM3	223	9.528	2368.19	1302.034	174.66	95.92	223	
234	TUBING 4-1/2", 18.9#, L-80,HSM3	224	9.528	2377.71	1292.506	175.13	96.33	224	
235	TUBING 4-1/2", 18.9#, L-80,HSM3	225	9.518	2387.23	1282.988	175.59	96.73	225	
236	TUBING 4-1/2", 18.9#, L-80,HSM3	226	9.523	2396.76	1273.465	176.06	97.13	226	
237	TUBING 4-1/2", 18.9#, L-80,HSM3	227	9.523	2406.28	1263.942	176.52	97.53	227	
238	TUBING 4-1/2", 18.9#, L-80,HSM3	228	9.523	2415.80	1254.419	176.98	97.94	228	
239	TUBING 4-1/2", 18.9#, L-80,HSM3	229	9.523	2425.32	1244.896	177.45	98.34	229	
240	TUBING 4-1/2", 18.9#, L-80,HSM3	230	9.528	2434.85	1235.368	177.92	98.74	230	
241	TUBING 4-1/2", 18.9#, L-80,HSM3	231	9.523	2444.38	1225.845	178.38	99.14	231	
242	TUBING 4-1/2", 18.9#, L-80,HSM3	232	9.523	2453.90	1216.322	178.84	99.54	232	
243	TUBING 4-1/2", 18.9#, L-80,HSM3	233	9.528	2463.43	1206.794	179.31	99.95	233	
244	TUBING 4-1/2", 18.9#, L-80,HSM3	234	9.528	2472.95	1197.266	179.78	100.35	234	
245	TUBING 4-1/2", 18.9#, L-80,HSM3	235	9.528	2482.48	1187.738	180.24	100.75	235	
246	TUBING 4-1/2", 18.9#, L-80,HSM3	236	9.528	2492.01	1178.21	180.71	101.15	236	
247	TUBING 4-1/2", 18.9#, L-80,HSM3	237	9.523	2501.53	1168.687	181.17	101.56	237	
248	TUBING 4-1/2", 18.9#, L-80,HSM3	238	9.373	2510.91	1159.314	181.63	101.95	238	
249	TUBING 4-1/2", 18.9#, L-80,HSM3	239	9.428	2520.33	1149.886	182.09	102.35	239	
250	TUBING 4-1/2", 18.9#, L-80,HSM3	240	9.528	2529.86	1140.358	182.55	102.75	240	
251	TUBING 4-1/2", 18.9#, L-80,HSM3	241	9.528	2539.39	1130.83	183.02	103.15	241	
252	TUBING 4-1/2", 18.9#, L-80,HSM3	242	9.533	2548.92	1121.297	183.48	103.56	242	
253	TUBING 4-1/2", 18.9#, L-80,HSM3	243	9.553	2558.48	1111.744	183.95	103.96	243	
254	TUBING 4-1/2", 18.9#, L-80,HSM3	244	9.558	2568.03	1102.186	184.42	104.36	244	
255	TUBING 4-1/2", 18.9#, L-80,HSM3	245	9.555	2577.59	1092.631	184.88	104.77	245	
256	TUBING 4-1/2", 18.9#, L-80,HSM3	246	9.553	2587.14	1083.078	185.35	105.17	246	
257	TUBING 4-1/2", 18.9#, L-80,HSM3	247	9.553	2596.70	1073.525	185.82	105.57	247	
258	TUBING 4-1/2", 18.9#, L-80,HSM3	248	9.543	2606.24	1063.982	186.28	105.98	248	
259	TUBING 4-1/2", 18.9#, L-80,HSM3	249	9.543	2615.78	1054.439	186.75	106.38	249	
260	TUBING 4-1/2", 18.9#, L-80,HSM3	250	9.353	2625.13	1045.086	187.21	106.78	250	
261	TUBING 4-1/2", 18.9#, L-80,HSM3	251	9.408	2634.54	1035.678	187.66	107.17	251	
262	TUBING 4-1/2", 18.9#, L-80,HSM3	252	9.271	2643.81	1026.407	188.12	107.56	252	
263	TUBING 4-1/2", 18.9#, L-80,HSM3	253	9.418	2653.23	1016.989	188.58	107.96	253	
264	TUBING 4-1/2", 18.9#, L-80,HSM3	254	9.443	2662.67	1007.546	189.04	108.36	254	
265	TUBING 4-1/2", 18.9#, L-80,HSM3	255	9.445	2672.12	998.101	189.50	108.76	255	

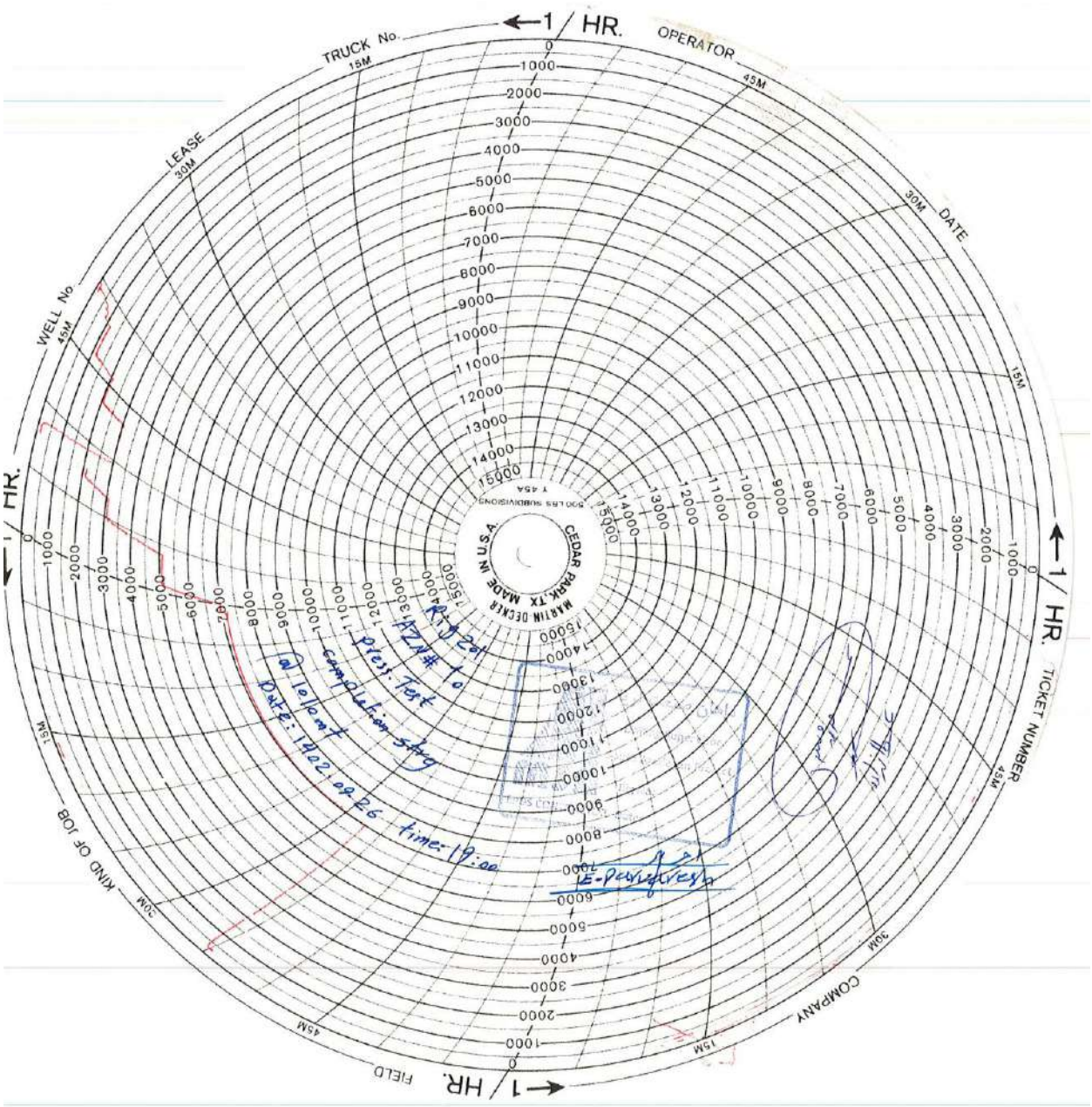
266	TUBING 4-1/2", 18.9#, L-80,HSM3	256	9.378	2681.50	988.723	189.96	109.16	256	
267	TUBING 4-1/2", 18.9#, L-80,HSM3	257	9.193	2690.69	979.53	190.41	109.54	257	
268	TUBING 4-1/2", 18.9#, L-80,HSM3	258	9.518	2700.21	970.012	190.87	109.95	258	
269	TUBING 4-1/2", 18.9#, L-80,HSM3	259	9.518	2709.73	960.494	191.34	110.35	259	
270	TUBING 4-1/2", 18.9#, L-80,HSM3	260	9.525	2719.25	950.969	191.80	110.75	260	
271	TUBING 4-1/2", 18.9#, L-80,HSM3	261	9.528	2728.78	941.441	192.27	111.15	261	
272	TUBING 4-1/2", 18.9#, L-80,HSM3	262	9.520	2738.30	931.921	192.73	111.55	262	
273	TUBING 4-1/2", 18.9#, L-80,HSM3	263	9.525	2747.82	922.396	193.20	111.96	263	
274	TUBING 4-1/2", 18.9#, L-80,HSM3	264	9.531	2757.36	912.865	193.66	112.36	264	
275	TUBING 4-1/2", 18.9#, L-80,HSM3	265	9.526	2766.88	903.339	194.13	112.76	265	
276	TUBING 4-1/2", 18.9#, L-80,HSM3	266	9.525	2776.41	893.814	194.59	113.16	266	
277	TUBING 4-1/2", 18.9#, L-80,HSM3	267	9.183	2785.59	884.631	195.04	113.55	267	
278	TUBING 4-1/2", 18.9#, L-80,HSM3	268	9.530	2795.12	875.101	195.50	113.95	268	
279	TUBING 4-1/2", 18.9#, L-80,HSM3	269	9.528	2804.65	865.573	195.97	114.36	269	
280	TUBING 4-1/2", 18.9#, L-80,HSM3	270	9.522	2814.17	856.051	196.43	114.76	270	
281	TUBING 4-1/2", 18.9#, L-80,HSM3	271	9.565	2823.73	846.486	196.90	115.16	271	
282	TUBING 4-1/2", 18.9#, L-80,HSM3	272	9.410	2833.14	837.076	197.36	115.56	272	
283	TUBING 4-1/2", 18.9#, L-80,HSM3	273	9.538	2842.68	827.538	197.83	115.96	273	
284	TUBING 4-1/2", 18.9#, L-80,HSM3	274	9.418	2852.10	818.12	198.29	116.36	274	
285	TUBING 4-1/2", 18.9#, L-80,HSM3	275	9.408	2861.51	808.712	198.75	116.76	275	
286	TUBING 4-1/2", 18.9#, L-80,HSM3	276	9.568	2871.08	799.144	199.21	117.16	276	
287	TUBING 4-1/2", 18.9#, L-80,HSM3	277	9.565	2880.64	789.579	199.68	117.57	277	
288	TUBING 4-1/2", 18.9#, L-80,HSM3	278	9.543	2890.18	780.036	200.15	117.97	278	
289	TUBING 4-1/2", 18.9#, L-80,HSM3	279	9.538	2899.72	770.498	200.61	118.37	279	
290	TUBING 4-1/2", 18.9#, L-80,HSM3	280	9.538	2909.26	760.96	201.08	118.77	280	
291	TUBING 4-1/2", 18.9#, L-80,HSM3	281	9.525	2918.79	751.435	201.54	119.18	281	
292	TUBING 4-1/2", 18.9#, L-80,HSM3	282	9.528	2928.31	741.907	202.01	119.58	282	
293	TUBING 4-1/2", 18.9#, L-80,HSM3	283	9.528	2937.84	732.379	202.47	119.98	283	
294	TUBING 4-1/2", 18.9#, L-80,HSM3	284	9.541	2947.38	722.838	202.94	120.38	284	
295	TUBING 4-1/2", 18.9#, L-80,HSM3	285	9.523	2956.91	713.315	203.40	120.79	285	
296	TUBING 4-1/2", 18.9#, L-80,HSM3	286	9.398	2966.30	703.917	203.86	121.18	286	
297	TUBING 4-1/2", 18.9#, L-80,HSM3	287	9.528	2975.83	694.389	204.33	121.59	287	
298	TUBING 4-1/2", 18.9#, L-80,HSM3	288	9.528	2985.36	684.861	204.79	121.99	288	
299	TUBING 4-1/2", 18.9#, L-80,HSM3	289	9.528	2994.89	675.333	205.26	122.39	289	
300	TUBING 4-1/2", 18.9#, L-80,HSM3	290	9.529	3004.42	665.804	205.72	122.79	290	Pr. Test
301	TUBING 4-1/2", 18.9#, L-80,HSM3	291	9.408	3013.82	656.396	206.18	123.19	291	
302	TUBING 4-1/2", 18.9#, L-80,HSM3	292	9.523	3023.35	646.873	206.65	123.59	292	
303	TUBING 4-1/2", 18.9#, L-80,HSM3	293	9.413	3032.76	637.46	207.11	123.99	293	
304	TUBING 4-1/2", 18.9#, L-80,HSM3	294	9.540	3042.30	627.92	207.57	124.39	294	
305	TUBING 4-1/2", 18.9#, L-80,HSM3	295	9.553	3051.85	618.367	208.04	124.80	295	
306	TUBING 4-1/2", 18.9#, L-80,HSM3	296	9.533	3061.39	608.834	208.50	125.20	296	
307	TUBING 4-1/2", 18.9#, L-80,HSM3	297	9.533	3070.92	599.301	208.97	125.60	297	
308	TUBING 4-1/2", 18.9#, L-80,HSM3	298	9.531	3080.45	589.77	209.43	126.00	298	
309	TUBING 4-1/2", 18.9#, L-80,HSM3	299	9.533	3089.98	580.237	209.90	126.41	299	

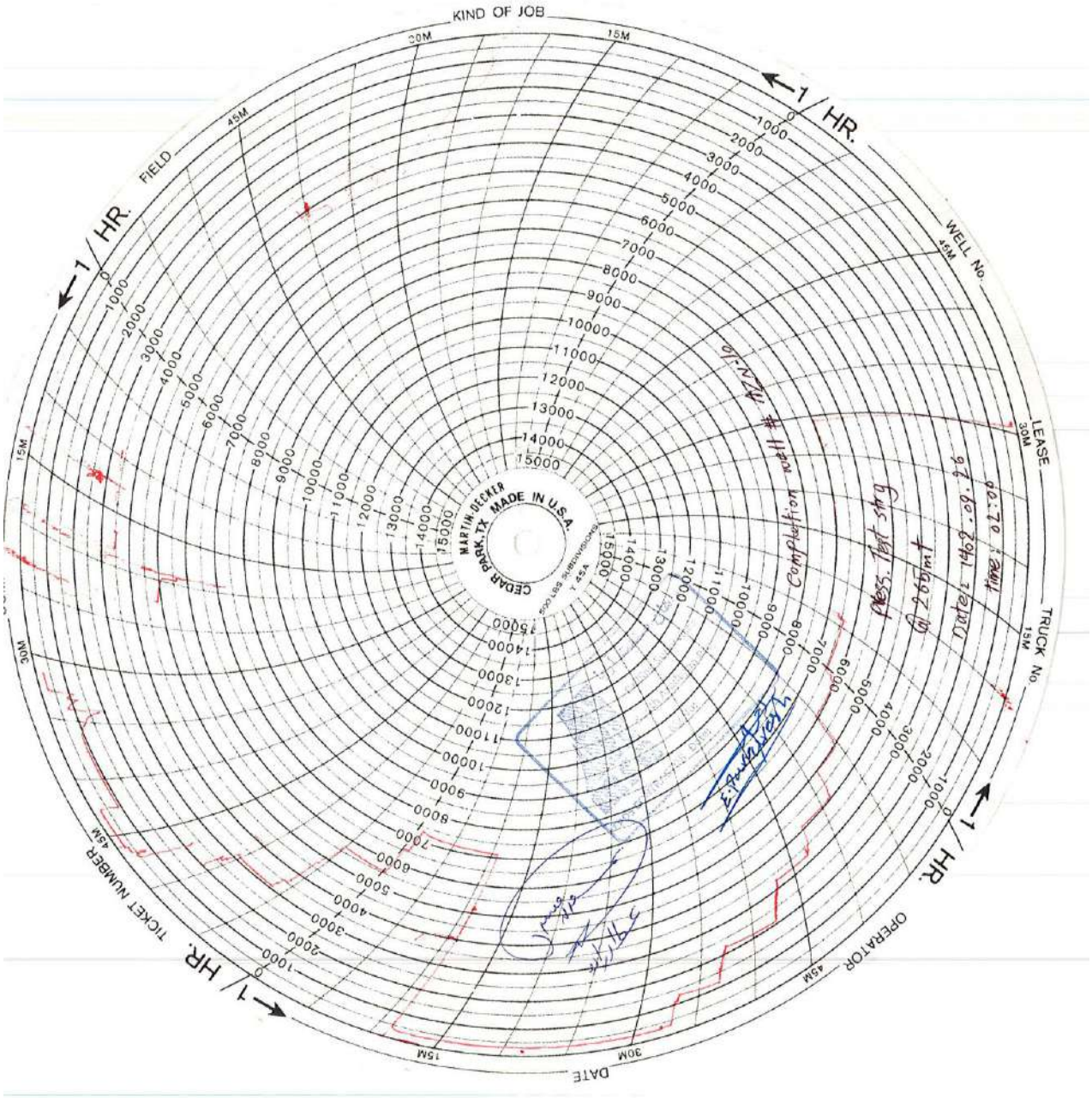
310	TUBING 4-1/2", 18.9#, L-80,HSM3	300	9.538	3099.52	570.699	210.37	126.81	300	
311	TUBING 4-1/2", 18.9#, L-80,HSM3	301	9.518	3109.04	561.181	210.83	127.21	301	
312	TUBING 4-1/2", 18.9#, L-80,HSM3	302	9.514	3118.55	551.667	211.30	127.61	302	
313	TUBING 4-1/2", 18.9#, L-80,HSM3	303	9.515	3128.07	542.152	211.76	128.01	303	
314	TUBING 4-1/2", 18.9#, L-80,HSM3	304	9.523	3137.59	532.629	212.22	128.42	304	
315	TUBING 4-1/2", 18.9#, L-80,HSM3	305	9.518	3147.11	523.111	212.69	128.82	305	
316	TUBING 4-1/2", 18.9#, L-80,HSM3	306	9.523	3156.63	513.588	213.15	129.22	306	
317	TUBING 4-1/2", 18.9#, L-80,HSM3	307	9.531	3166.16	504.057	213.62	129.62	307	
318	TUBING 4-1/2", 18.9#, L-80,HSM3	308	9.528	3175.69	494.529	214.08	130.03	308	
319	TUBING 4-1/2", 18.9#, L-80,HSM3	309	9.528	3185.22	485.001	214.55	130.43	309	
320	TUBING 4-1/2", 18.9#, L-80,HSM3	310	9.531	3194.75	475.47	215.02	130.83	310	
321	TUBING 4-1/2", 18.9#, L-80,HSM3	311	9.525	3204.28	465.945	215.48	131.23	311	
322	TUBING 4-1/2", 18.9#, L-80,HSM3	312	9.528	3213.80	456.417	215.95	131.63	312	
323	TUBING 4-1/2", 18.9#, L-80,HSM3	313	9.523	3223.33	446.894	216.41	132.04	313	
324	TUBING 4-1/2", 18.9#, L-80,HSM3	314	9.530	3232.86	437.364	216.88	132.44	314	
325	TUBING 4-1/2", 18.9#, L-80,HSM3	315	9.528	3242.38	427.836	217.34	132.84	315	
326	TUBING 4-1/2", 18.9#, L-80,HSM3	316	9.533	3251.92	418.303	217.81	133.24	316	
327	TUBING 4-1/2", 18.9#, L-80,HSM3	317	9.553	3261.47	408.75	218.27	133.65	317	
328	TUBING 4-1/2", 18.9#, L-80,HSM3	318	9.525	3271.00	399.225	218.74	134.05	318	
329	TUBING 4-1/2", 18.9#, L-80,HSM3	319	9.528	3280.52	389.697	219.20	134.45	319	
330	TUBING 4-1/2", 18.9#, L-80,HSM3	320	9.403	3289.93	380.294	219.66	134.85	320	
331	TUBING 4-1/2", 18.9#, L-80,HSM3	321	9.523	3299.45	370.771	220.13	135.25	321	
332	TUBING 4-1/2", 18.9#, L-80,HSM3	322	9.520	3308.97	361.251	220.59	135.65	322	
333	TUBING 4-1/2", 18.9#, L-80,HSM3	323	9.518	3318.49	351.733	221.06	136.06	323	
334	TUBING 4-1/2", 18.9#, L-80,HSM3	324	9.523	3328.01	342.21	221.52	136.46	324	
335	TUBING 4-1/2", 18.9#, L-80,HSM3	325	9.526	3337.54	332.684	221.99	136.86	325	
336	TUBING 4-1/2", 18.9#, L-80,HSM3	326	9.518	3347.05	323.166	222.45	137.26	326	
337	TUBING 4-1/2", 18.9#, L-80,HSM3	327	9.525	3356.58	313.641	222.92	137.66	327	
338	TUBING 4-1/2", 18.9#, L-80,HSM3	328	9.525	3366.10	304.116	223.38	138.07	328	
339	TUBING 4-1/2", 18.9#, L-80,HSM3	329	9.523	3375.63	294.593	223.85	138.47	329	
340	TUBING 4-1/2", 18.9#, L-80,HSM3	330	9.523	3385.15	285.07	224.31	138.87	330	
341	TUBING 4-1/2", 18.9#, L-80,HSM3	331	9.526	3394.68	275.544	224.78	139.27	331	
342	TUBING 4-1/2", 18.9#, L-80,HSM3	332	9.523	3404.20	266.021	225.24	139.68	332	
343	TUBING 4-1/2", 18.9#, L-80,HSM3	333	9.528	3413.73	256.493	225.71	140.08	333	
344	TUBING 4-1/2", 18.9#, L-80,HSM3	334	9.523	3423.25	246.97	226.17	140.48	334	
345	TUBING 4-1/2", 18.9#, L-80,HSM3	335	9.528	3432.78	237.442	226.64	140.88	335	

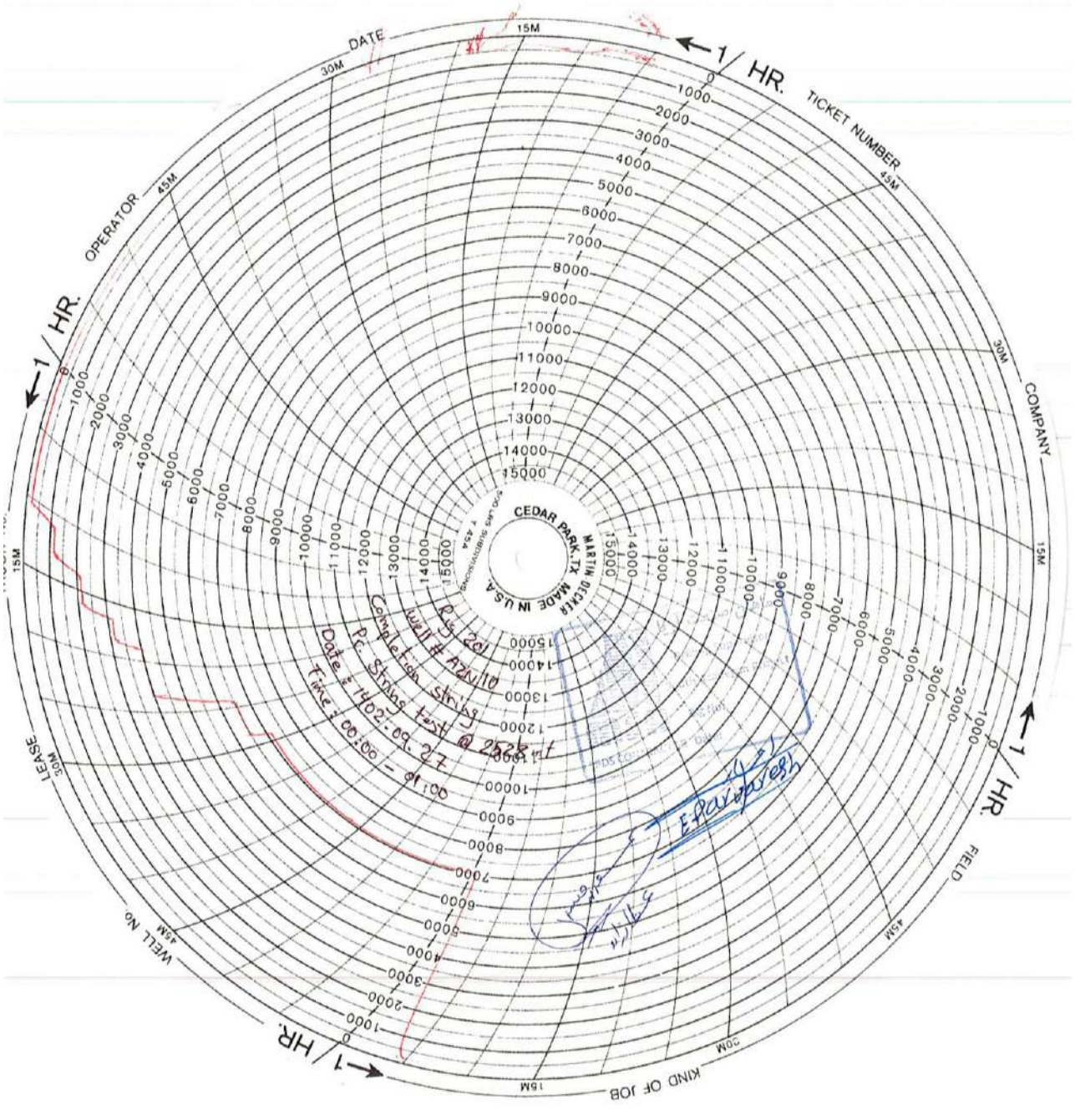
346	TUBING 4-1/2", 18.9#, L-80,HSM3	336	9.531	3442.31	227.911	227.10	141.28	336	
347	TUBING 4-1/2", 18.9#, L-80,HSM3	337	9.531	3451.84	218.38	227.57	141.69	337	
348	TUBING 4-1/2", 18.9#, L-80,HSM3	338	9.528	3461.37	208.852	228.03	142.09	338	
349	TUBING 4-1/2", 18.9#, L-80,HSM3	339	9.528	3470.90	199.324	228.50	142.49	339	
350	TUBING 4-1/2", 18.9#, L-80,HSM3	340	9.526	3480.42	189.798	228.96	142.89	340	
351	TUBING 4-1/2", 18.9#, L-80,HSM3	341	9.526	3489.95	180.272	229.43	143.30	341	
352	TUBING 4-1/2", 18.9#, L-80,HSM3	342	9.525	3499.47	170.747	229.89	143.70	342	
353	TUBING 4-1/2", 18.9#, L-80,HSM3	343	9.522	3509.00	161.225	230.36	144.10	343	
354	TUBING 4-1/2", 18.9#, L-80,HSM3	344	9.523	3518.52	151.702	230.82	144.50	344	
355	TUBING 4-1/2", 18.9#, L-80,HSM3	345	9.526	3528.04	142.176	231.29	144.91	345	
356	TUBING 4-1/2", 18.9#, L-80,HSM3	346	9.523	3537.57	132.653	231.75	145.31	346	
357	TUBING 4-1/2", 18.9#, L-80,HSM3	347	9.543	3547.11	123.11	232.22	145.71	347	
358	TUBING 4-1/2", 18.9#, L-80,HSM3	348	9.543	3556.65	113.567	232.68	146.11	348	
359	TUBING 4-1/2", 18.9#, L-80,HSM3	349	9.543	3566.20	104.024	233.15	146.52	349	
360	TUBING 4-1/2", 18.9#, L-80,HSM3	350	9.548	3575.74	94.476	233.62	146.92	350	
361	TUBING 4-1/2", 18.9#, L-80,HSM3	351	9.545	3585.29	84.931	234.08	147.32	351	
362	TUBING 4-1/2", 18.9#, L-80,HSM3	352	9.548	3594.84	75.383	234.55	147.73	352	
363	TUBING 4-1/2", 18.9#, L-80,HSM3	353	9.548	3604.39	65.835	235.01	148.13	353	
PUP JOINT 4 1/2",HSM3 ,18.9#, L-80, Type-1	PUP JOINT 4 1/2",HSM3		1.623	3606.01	64.212	235.09	148.20		
4-1/2" FLOW COUPLING, 18.9#,SUPER 13CR-110KSI,HSM3 , 4' FT, WEIQIXIN Co	Short Flow Coupling		1.115	3607.12	63.097	235.15	148.24		
TRSSV-10 Tubing Retrievable Saftey Valve, 4 1/2", 18.9# W/ 3.437" Seal bore,HSM3 , SUPER 13CR-110KSI, 10K WP With AF Profile, Flapper Type, WEIQIXIN Co	TRSSV		1.780	3608.90	61.317	235.24	148.32		
4-1/2" FLOW COUPLING,18.9#,SUPER 13CR-110KSI,HSM3 , 6 FT, WEIQIXIN Co	Long Flow Coupling		1.725	3610.63	59.592	235.32	148.39		
PUP JOINT 4 1/2",HSM3 ,18.9#, L-80, Type-1	PUP JOINT 4 1/2",HSM3		1.625	3612.25	57.967	235.40	148.46		
364	TUBING 4-1/2", 18.9#, L-80,HSM3	354	9.528	3621.78	48.439	235.86	148.86	354	
365	TUBING 4-1/2", 18.9#, L-80,HSM3	355	9.533	3631.31	38.906	236.33	149.27	355	Pr. Test
366	TUBING 4-1/2", 18.9#, L-80,HSM3	356	9.533	3640.85	29.373	236.79	149.67	356	
367	TUBING 4-1/2", 18.9#, L-80,HSM3	357	9.525	3650.37	19.848	237.26	150.07	357	
368	TUBING 4-1/2", 18.9#, L-80,HSM4	358	9.529	3659.90	10.319	237.72	150.47	358	
4-1/2" FLUTED SWAGE ADAPTER D/P, 18.9# TPCQ-EX PIN ON TOP X HSM3 PIN ON DOWN* SUPER 13CR-110KSI, WEIQIXIN Co	Fluted Swage		1.018	3660.92	9.301	237.31	150.11		
TBG HANGER MSP, 10K, 11" * 4 1/2" ACME Box up * 4 1/2" TPCQ-EX Box Down , M/C: FF-0.5 T/C: UPSL: 3G PR: 2, 4" Type "H" Two Way	TBG HGR		0.150	3661.07	9.151	237.32	150.12		
Tubing Hanger Running Tool	Tubing Hanger Running Tool		0.550	3661.62	8.601	237.34	150.14		
L/Joint	L/Joint		11.603	3673.22	-3.002	237.91	150.63		Final Press Test

Figure 8-21: AZN-10 Completion Tally

8.6 Completion String Pressure Test




KIND OF JOB
1 / HR.
FIELD
WELL No.
LEASE
TRUCK No.
OPERATOR
DATE
TICKET NUMBER
MARTIN-DECKER
CEDAR PARK, TX MADE IN U.S.A.


DATE
1/HR.
TICKET NUMBER
COMPANY
FIELD
KIND OF JOB
WELL No.
LEASE
OPERATOR
CEDAR PARK, TX
MADE IN U.S.A.
Well # AZN-10
Completion String
P.C. String test @ 2526 psi
Date: 14021209
Time: 00:00 - 01:00

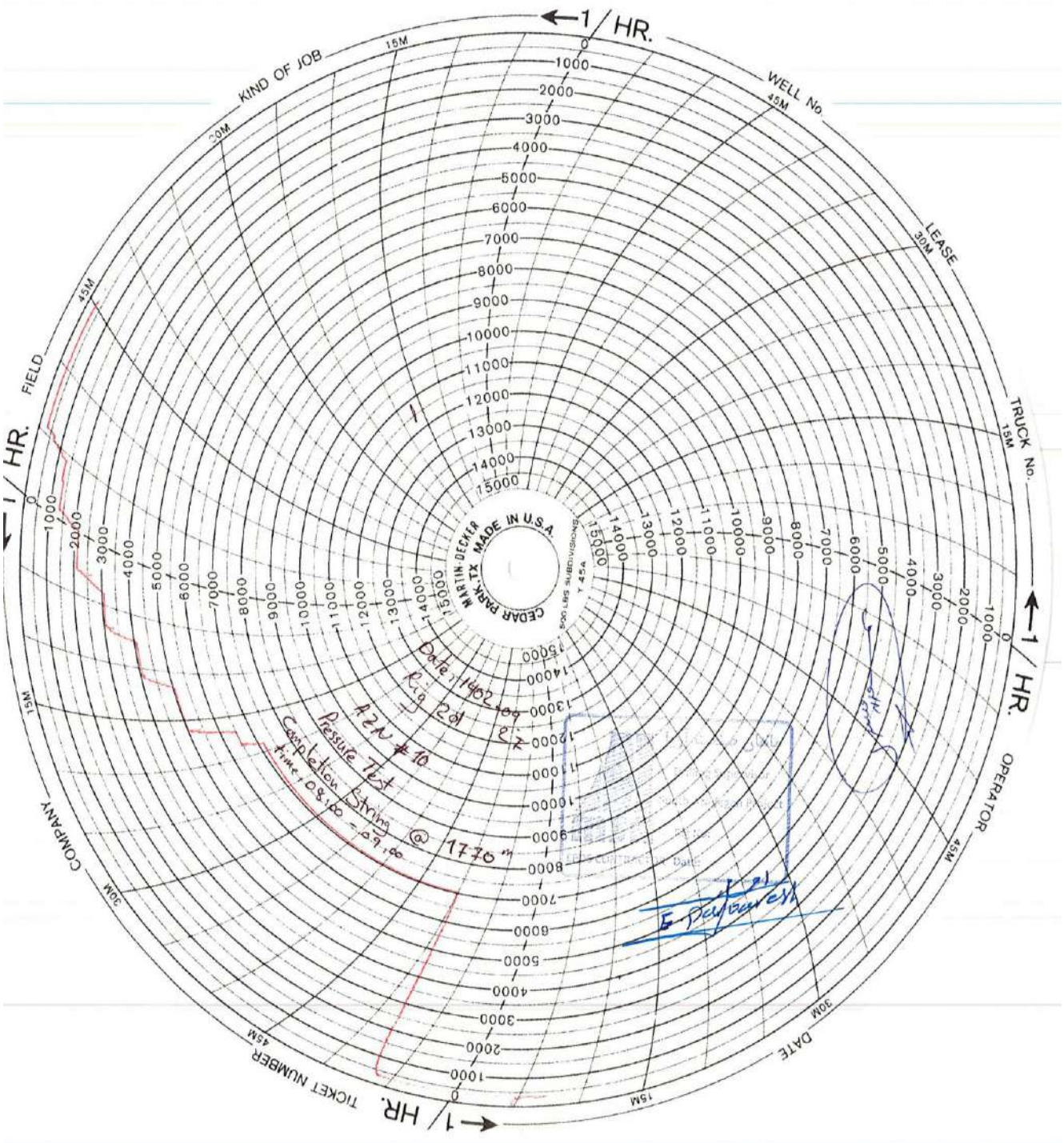


Figure 8-22: AZN-10 Tubing Pressure Test Graphs

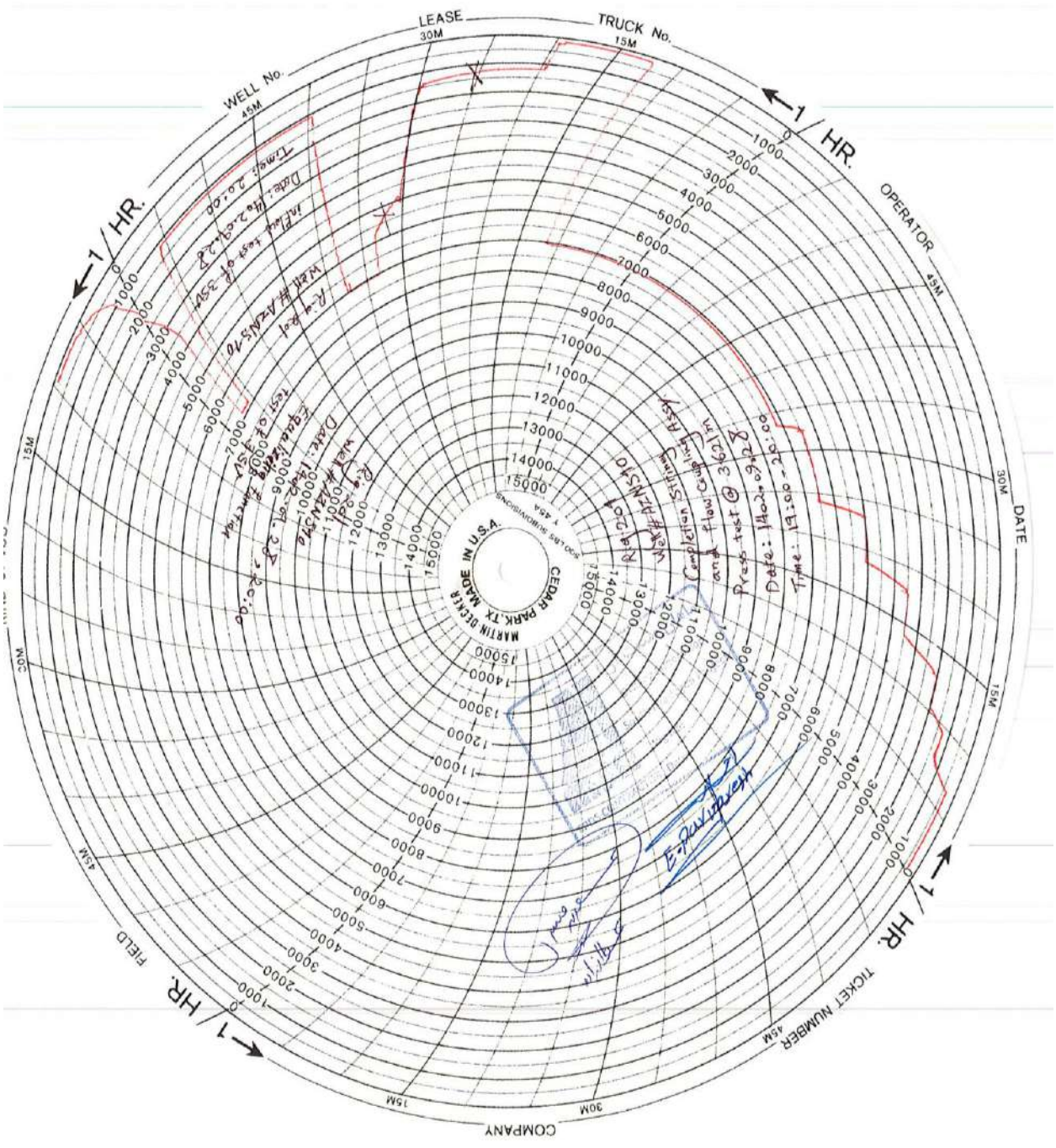


Figure 8-23: AZN-10 Tubing Pressure & 3SV InflowTest Graph

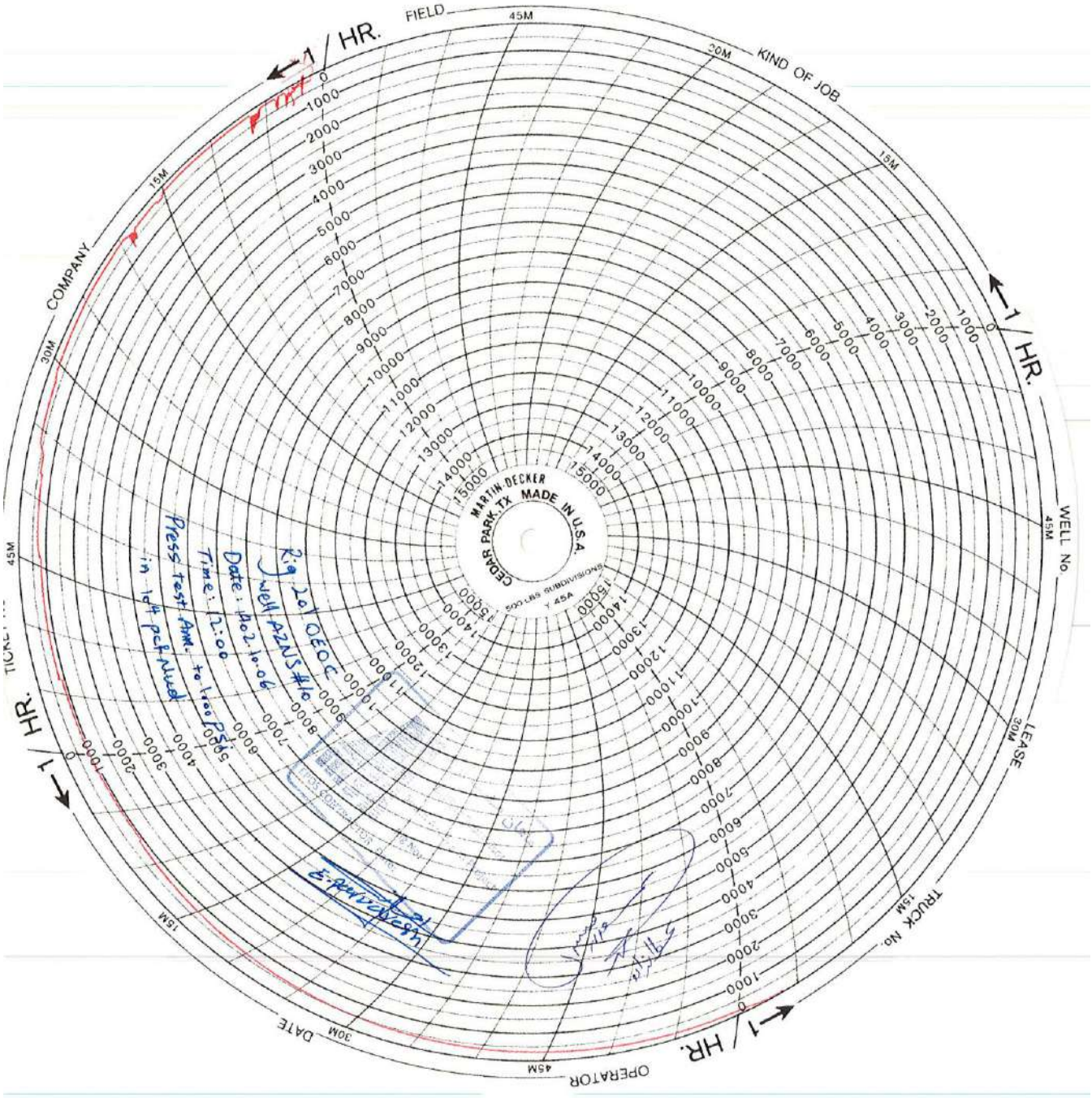


Figure 8-24: AZN-10 Annulus Pressure Test Grap

9. Wellhead Report

9.1 Final Well Status

Well Head

Item	Specification (size, pressure rating, M/C)	Test Pressure	Test Duration
C.H.H	20-3/4'' 3K, DD-0.5 PSL1, PR1	Already Installed	
C.H.S	20-3/4'' 3K * 13-5/8'' 5K , DD-0.5, PSL1, PR1	Already Installed	
T.H.S. (MSP Co.)	13-5/8'' 5K * 11''-10K, FF-0.5, PSL3, PR2	4500 psi	15 MIN
X-MASS TREE(BTM – MSP Co.)	4-1/16''-10K, FF-0.5, PSL3G, PR2	8000 psi	15 MIN

Wellbore Completion Fluids

Well bore	Fluid Type	Weight (pcf)
Cased Hole	$CaBr_2$	103-104

Well Head Pressure Recor

Annulus	Pressure (psi)	Date / Time
Wellhead	0	1402.10.08
Tubing- 9 5/8"	0	1402.10.08
9 5/8" – 13 3/8"	0	1402.10.08
13 3/8" – 20"	0	1402.10.08

Barriers

Barrier Type	Depth	Pressure Test	Date Tested
Close TRSSSV	61.32 m	500 psi (inflow test after finish flow the well)	1402.10.01

Temporary Installed Equipment:

After Completing the well, Plug & Prong have been installed in No-Go landing nipple

Table 9-1: AZN-10 Final Well Status

9.2 Wellhead & X-mass tree Specification

Well Type	Specification	Check and Remark

MSP Co. Well Head 10,000 Psi for Fahliyan Wells

TUBING HEAD ADAPTOR

BOTTOM STUDDED FLANGE 11", 10000 PSI, BX158 TOP STUDDED FLANGE 4-1/16" 10000 PSI BX155 FULL BORE PREPARED WITH TWO S-SEAL AND ONE METAL TO METAL SEAL TO PACK TOP NECK OF TUBING HANGER C/W RELATED TEST AND INJECTION PORTS PREPARTION. C/W TWO CONTINUOUS FEED THROUGH TYPE 1/4" NPT CONTROL LINE EXITS PREPARTION (ONE FOR 1/4" TRSSSV CL AND ONE FOR 1/4" DOWNHOLE PRESSURE GAUGE) AND TWO 3/8" NPT CONTINUOUS FEED THROUGH TYPE LINE PORTS FOR CHEMCIAL INJECTION LINE EXIT INCLUDING RELATED SWAGE LOCK FITTINGS C/WITH ONE 4-1/16", 10K WP FULL BORE MANUAL GATE VALVE FORGED BODY AND BONNET, ALLOY STAINLESS STEEL 410SS GATE & SEAT AISI 410 COATED WITH TUNGSTEN CARBIDE USING HVOF METHOD STEM: INCONEL 718 WITH BACK SEAT CAPABILITY, NON RISING STEM METAL TO METAL SEAL BETWEEN SEAT AND GATE C/W HAND WHEEL FORGED BODY ALLOY STEEL AISI 410 MAXIMUM HEIGHT: 800MM

M/C:FF-0.5 T/C:U PSL:3G PR:2

CROSS T-TYPE SOLID BLOCK

CROSS T-TYPE SOLID BLOCK, BOTTOM FLANGED 4-1/16" API 10000 PSI WP TOP FLANGED STUDDED 4-1/16" API 10000 PSI WP FOR TREE CAP WITH TWO STUDDED SIDE OUTLETS 4-1/16" 10000PSI WP ELEVATED AT 900 DEG AND TWO 4-1/16", 10000 PSI MANUAL GATE VALVE AT THE TOP AND BOTTOM AND ONE 4-1/16", 10000 PSI WP SAFETY VALVE EQUIPPED WITH HYDRAULIC WIRE CUT ACTUATOR AS UPPER MASTER VALVE MATERIAL OF BODY,BONNET: AISI 410

SEAT, GATE: AISI 410SS HARD FACED WITH TUNGSTEN CARBIDE UASING HVOP METHOD MATERIAL OF STEM: INCONEL 718

	M/C:FF-0.5 T/C:U PSL:3G PR:2G. MANUAL GATE VALVE API 4-1/16" 10000 PSI WP. FLANGED ENDS AND FULL BORE METAL TO METAL SEAL BETWEEN BODY/SEAT/GATE WITH BACK SEAT CAPABILITY, NON RISING STEM C/W HANDWHEELS MATERIAL OF BODY,BONNET,SEAT,GATE: AISI 410SS MATERIAL OF STEM: INCONEL 718 GATE&SEAT HARD FACED WITH TUNGSTEN CARBIDE BY HOVF METHOD M/C:FF-0.5 T/C:U PSL:3G PR:2 X-MAS TREE CAP 4-1/16" API 10000 PSI WP, FLANGED BOTTOM (4-1/16", BORE) 8-3/8"-4ACME-2G THREAD TOP, BLANKING PLUG TAPPED WITH 1/2" NPT THREAD, FORGED BODY, MATERIAL: STAINLESS STEEL 410 M/C:FF-0.5 T/C:U PSL:3G PR:2	

Table 9-2: AZN-10 Equipment Specification

9.3 Wellhead Schematic

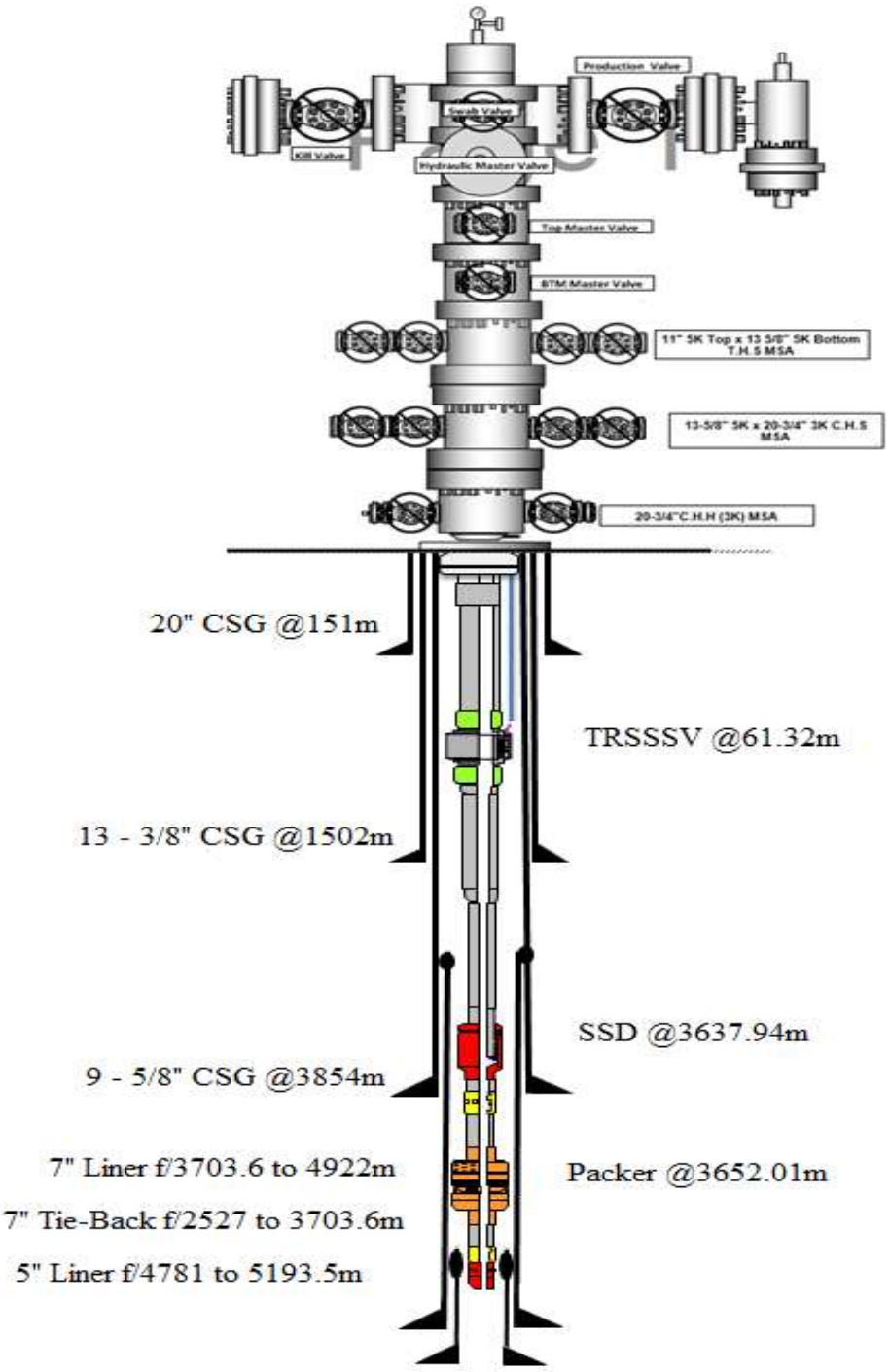


Figure 9-1: AZN-10 Wellhead Sketch

9.4 X-mass Tree Specification (Section D)

Item	Equipment	M/C	T/C	PSL	Qty
1	BTM Master Valve 11” * 4 1/16” 10K	FF-0.5	U	3G	1
2	TOP Master Valve 4 1/16” 10K	FF-0.5	U	3G	1
3	Surface Safety Valve 4 1/16” 10K	FF-0.5	U	3G	1
4	Deeping Valve 4 1/16” 10K	FF-0.5	U	3G	1
5	Wing gate valve-1 (Kill Valve) 4 1/16" 10K	FF-0.5	U	3G	1
6	Wing gate valve-2 (production valve) 10K	FF-0.5	U	3G	1
7	41/16" Choke valve 4 1/16" 10K	FF-0.5	U	3G	1
8	Christ Mas Tree	FF-0.5	U	3G	1

Table 9-3: AZN-10 Section "D" Specification

9.5 X-mass Tree Pressure Test Report (Section D – befor install)

Item	Test	Check Mark and Result
1	Pressure test for the kill Valve W/ 1000 & 8000 Psi / 15 min	OK
2	Pressure test for the SSV W/ 1000 & 8000 Psi / 15 min	OK
3	Pressure test for the top master valve W/ 1000 & 8000 Psi / 15 min	OK
4	Pressure test for Swab valve W/ 1000 & 8000 Psi / 15 min	OK
5	Pressure test for the Production valve W/ 1000 & 8000 Psi / 15 min	OK

Table 9-4: AZN-10 Section "D" Test Report Befor Installation

9.6 X-mass Tree Pressure Test Report (Section D – after install)

Item	Test	Check mark and Result
1	Pressure test X-MAS TREE solid block and bottom section W/1000 &8000 psi for 15min	OK
2	Pressure test for the SSV Exit block W/ 8000 Psi / 15 min	OK
3	Pressure test neck of tubing hanger metal seal W/1000 &8000 psi for 15min	OK

4	Pressure test for Swab valve W/ 4500 Psi / 15 min	OK
5	Pressure test for Production valve W/4500 Psi / 15 min	OK
6	Pressure test cavity of FS.seal and Ring Gasket W 8000 Psi / 15 min	OK
7	Pressure test CIV line to 2000psi	OK

Table 9-5: AZN-10 Section "D" Test Report After Installation

9.7 Wellhead Related Test Reports

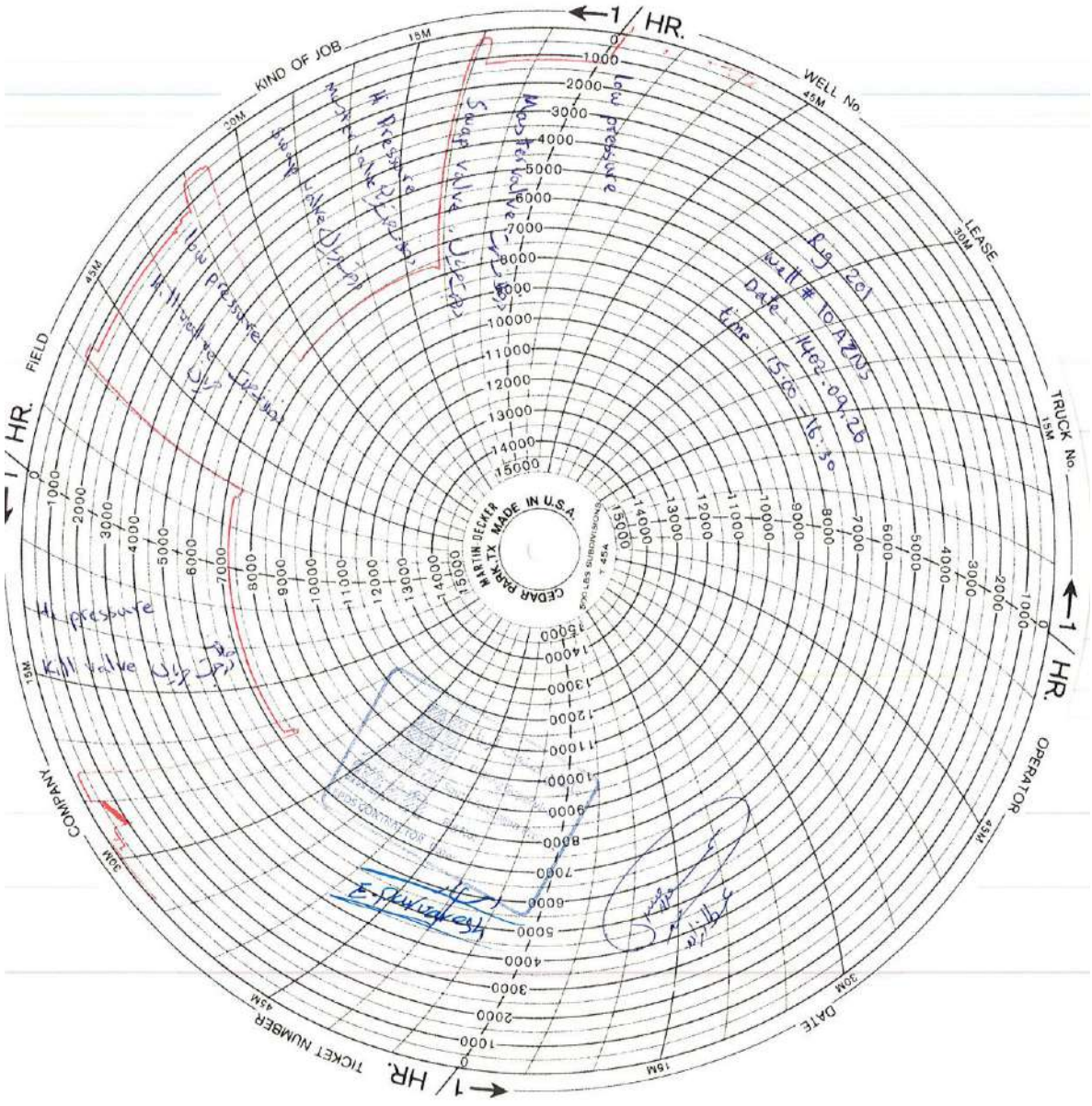


Figure 9-2: AZN-10 X-Mass Tree Pressure Test Graph before Installation

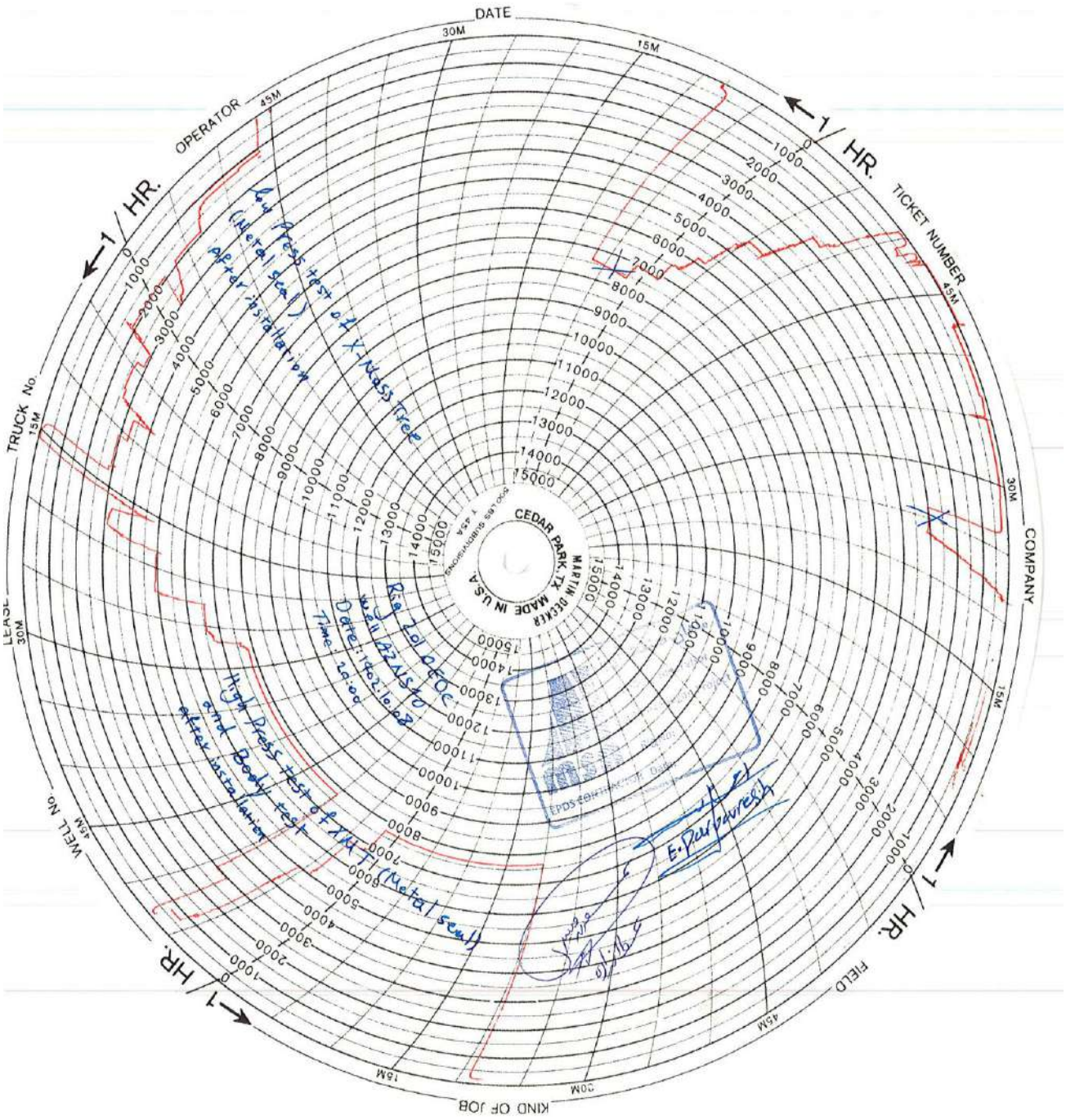


Figure 9-3: AZN-10 Body & Metal Seal of X-Mass Tree Pressure Test Graph

تاریخ: ۱۴۰۲/۱۰/۰۸

پیوست: دارد

بسمه تعالی

موضوع: صورتجلسه تست سطحی تاج چاه ۱۰ ازادگان جنوبی، قبل از نصب در دکل ۲۰۱ عملیات اکتشاف نفت

بدینوسیله در مورخه ۱۴۰۲/۰۹/۲۶ تست های سطحی تاج چاه X-MASS TREE SOLID BLOCK T TYPE 4-1/16" 10K, MSP DRILEX مرتبط به چاه شماره ۰۴۰ آزادگان جنوبی با حضور نماینده محترم کارفرما (پتروپارس) در محل دستگاه ۲۰۱ عملیات اکتشاف نفت، قبل از نصب بر روی تاج چاه به شرح ذیل انجام پذیرفت:

- BODY PRESSURE TEST OF X-MASS TREE SOLID BLOCK W/ 1000PIS, 5MIN & 8000 PSI, 15MIN "OK" SN: 008929L N180001
- INFLOW PRESSURE TEST OF 4-1/16" PRODUCTION (WING) VALVE W/ 1000PIS, 5MIN & 8000 PSI, 15MIN "OK" SN: 002929B N180048
- INFLOW PRESSURE TEST OF 4-1/16" DIPPING (SWAB) VALVE W/ 1000PIS, 5MIN & 8000 PSI, 15MIN "OK". SN: 008929K N180001
- INFLOW & OUT FLOW PRESSURE TEST OF 4-1/16" KILL VALVE W/ 1000PIS, 5MIN & 8000 PSI, 15MIN "OK" SN: 002929B S210020
- INFLOW PRESSURE TEST OF 4-1/16" SURFACE SAFETY VALVE W/ 1000PIS, 5MIN & 8000 PSI, 15MIN "OK" SN: 0269167C N180001
- INFLOW & OUT FLOW PRESSURE TEST OF 4-1/16" TOP MASTER VALVE W/ 1000PIS, 5MIN & 8000 PSI, 15MIN "OK"
- PRESSURE TEST OF 2SV ACTUATOR W/ 2500 PSI, 15 MIN "OK"

– گراف مربوط به تست فشار پیوست صورتجلسه گردید.

– لازم به ذکر است طبق قرارداد، روی تاج چاه کاهنده ثابت نصب گردید.

نماینده محترم تکمیل چاه کارفرما

شرکت نفت کیش

ناظر عملیات حفاری

شرکت ماهان صنعت آریا

ماهان صنعت آریا
Drilling Supervisor
South Azadegan Project
Rig No:
EPDS CONTRACTOR Date:

E. Parvaresh

Figure 9-4: AZN-10 Wellhead MOM Befor Installation

تاریخ: ۱۴۰۲/۱۰/۰۸

پیوست: دارد

بسمه تعالی

موضوع: صورتجلسه تست های پس از نصب تاج و ولهد چاه ۱۰ ازادگان جنوبی بعد از نصب در دکل ۲۰۱ عملیات اکتشاف نفت

بدینوسیله در مورخه ۱۴۰۲/۱۰/۰۸ تست های تاج چاه X-MASS TREE SOLID BLOCK T TYPE 4-1/16" 10K, MSP DRILEX و ولهد چاه شماره ۱۰ ازادگان جنوبی با حضور نماینده محترم تکمیل چاه کارفرما و نماینده عملیات حفاری ماهان صنعت آریا در محل دستگاه حفاری ۲۰۱ عملیات اکتشاف نفت بعد از نصب بر روی چاه به شرح ذیل انجام پذیرفت:

- BODY PRESSURE TEST OF X-MASS TREE SOLID BLOCK W/ 10K 11" x 4-1/16" 10K TUBING HEAD ADAPTOR (BTM MASTER VALVE) 10K MSP/DRILEX W/ 8000 PSI FOR 15MIN "OK" SN: 008929L N180001
- PRESSURE TEST TBG HGR METAL SEAL TO 800PSI FOR 5MIN, 8000PSI FOR 15MIN "OK"
- PRESS TEST F-SEAL & RING GASKET BETWEEN T.H.S & TUBING HEAD ADAPTOR (BTM MASTER VALVE) TO 8000 PSI FOR 15 MIN "OK"
- PRESSURE TEST NECK OF TBG HGR F-SEAL TO 8000PSI FOR 15MIN "OK"
- PRESS TEST 1/4" CONTROL LINE TO 8000 PSI FOR 15MIN "OK"
- PRESS TEST 3/8"CONTROL INJECTION LINES TO 2000 PSI "OK"

- فشار تمامی دالیزها بررسی گردید که فشار آنها صفر بود.
- بعد از اتمام نصب تاج چاه درون آویزه لوله مغزی TWO WAY CHECK VALVE جهت انگیزش چاه خارج گردید.
- خروجی کنترل لاین شیر ایمنی درون چاهی در جهت شیر اصلی پایینی می باشد و همچنین CIV شماره یک در جهت ویست پیت و CIV شماره دو در خلاف جهت ویست پیت می باشد و همچنین بر روی دو CIV و کنترل لاین پلاگ کپ نصب گردید.
- لازم به ذکر است روی تاج چاه کاهنده ثابت نصب گردید و سایزهای مختلف CHOKE BEAN کاهنده ثابت تاج چاه طبق قرداد می بایست تحویل کارفرما گردد.
- گراف مربوط به تست فشار پیوست صورتجلسه گردید.

ناظر عملیات حفاری

شرکت ماهان صنعت آریا

Drilling Supervisor

South Azadegan Project

Rig No:

EPDS CONTRACTOR Date:

نماینده محترم تکمیل چاه کارفرما

شرکت نفت گیش

Figure 9-5: AZN-10 Wellhead MOM After Installation

10. Well Stimulation & Clean Up Report

10.1 Operation Summary

Objective of operation was to increase injectivity by injecting 3.36 gal/ft of 15 % HCl via Coiled Tubing and conditioning the formation for stimulation operation, and reduce formation damage and wash out the remained debris and residuals of completion and drilling operations. Based on mentioned objective, operation was designed by stimulating well by using 15% HCl, Flowing pressure survey was designed to investigate well condition.

In this operation, the well was cleaned up by flowing with 8/64" to 30/64" choke size while rate measurement in same choke size. To perform acid wash, RIH CT to reach the bottom of producing interval. After displacing the wellbore by diesel, Pre-flush, main acid (15% HCL), and over-flush were injected by CT, and finally, the stimulation fluid was squeezed. After acidizing, the well was cleaned up and a flowing pressure/temperature survey was carried out.

Surface testing data were analyzed to investigate the effect of stimulation operation. According to prestimulation clean up results show that wellhead pressure and oil rate were 3303 PSI and 1446 BPD at 16/64" choke size. On the other hand, data gathered from post stimulation clean up shows that well head pressure and oil rate have increased to 3631 PSI and 3368 BPD at 16/64" choke size. The above data illustrates that stimulation operation has led to a 328-PSI increase in wellhead pressure and a 1922-BPD increase in oil rate. On top of that, the calculated pre- and post-stimulation PI of the well were about 0.66 and 3.261 STB/(D.PSI) at the datum depth. Generally, the stimulation operation was carried out successfully.

Pre/Post Acid	Pre – Acid	Post – Acid
Choke Size (1/64" Inch)	16	16
Wellhead Pressure (PSIG)	3303	3631
Flow Rate (STB/D)	1446	3368
PI (STB/D/PSI)	0.66	3.261

Table 10-1: AZN-10 Pre and Post Acid Potential of the Well

10.2 General Information

General information	
SERVICE:	Well testing, acidizing, Coiled Tubing, S/L, oil transfer
SERVICE COMPANY:	OEOC
OPERATION DATE:	03/February/2024 – 22/February/2024
STATE BEFORE THE OPERATION:	New Drill
WELL GEOMETRY:	Deviated
WELL TYPE:	Oil Producer
FLUID TYPE:	OIL & GAS
COMPLETION METHOD:	7” Perforated
CASING SIZE:	20”, 13 3/8”, 9 5/8”, 7”
TUBING SIZE:	4 ½” 18.9 ppf L-80 (0–3413.6 mMD); 3 ½” 10.2 ppf L-80 (3413.6 – 3665.3 mMD)
PRODUCING INTERVAL:	4087 – 4027 mMD

Table 10-2: AZN-10 General Information

10.3 Well Basic Information

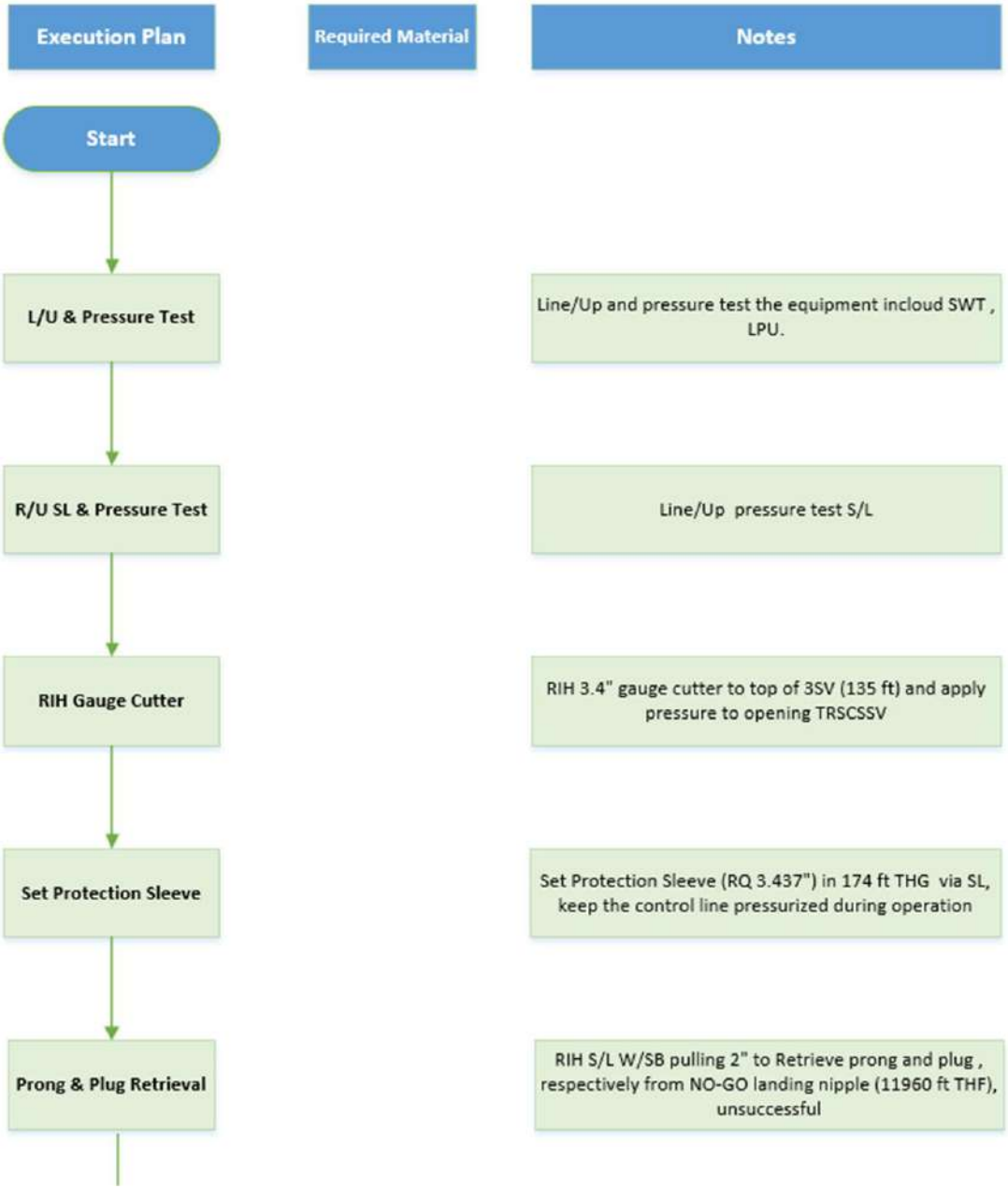
WELL BASIC INFORMATION	
Well Name	AZN - 10
Well Type	Highly Dev. Sarvak Prod. Well
Well Pad	Near Pad No.50
Well Objective	To remove possible damage and enhance well’ PI

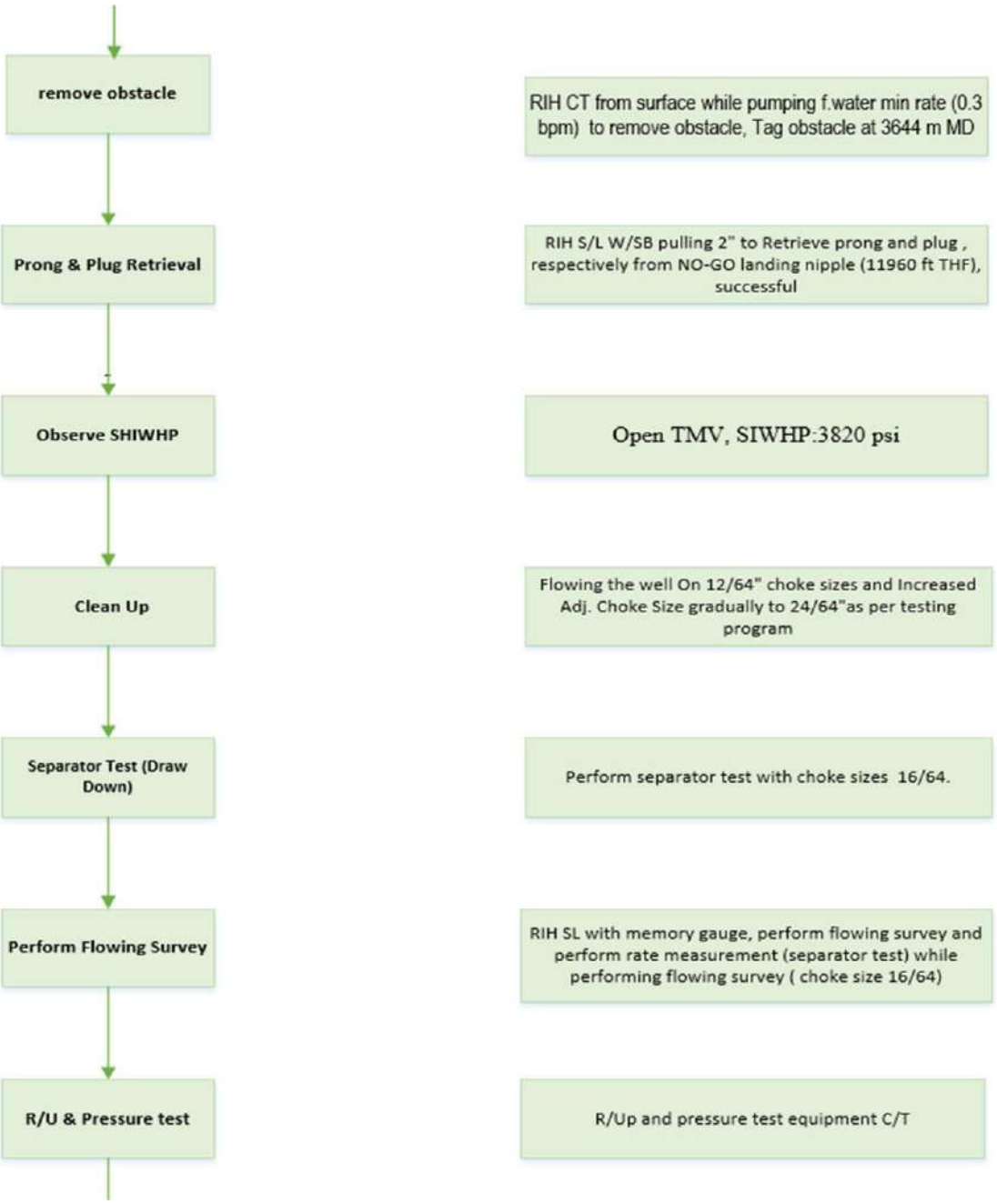
Reference Depth (Rotary Table Elevation from mean sea level)	13.45 MAMSL
Total Depth	5340 m MD
Target Formation	Fahliyan (Fah-1a, Fah-1b & Fah-1c)
Oil Density (API)	37.67 as per last post acid flow
Minimum ID	2.562" @ 3963.02 mMD, 3 ½" Bottom No Go Landing Nipple
Total Depth	4200 MD (-4186.55 m TVDss)
Producing Formation	Fahliyan (Fah-1b, Fah-1c, Fah-2b, and Fah-3a)
Formation Lithology	Limestone 90%, Dolomite 8.5%, Clay 1.5%
Clay Type	Illite, Kaolinite, Montmorillonite
Formation/Layer Permeability (md)	Fah (1a) = 8; Fah (1a) = 6 ; Fah (1b) = 20; Fah (1c) = 25
Formation/Layer Porosity (%)	Fah (1a) = 6.9; Fah (1b) = 3.8 ; Fah (1b) = 13.4; Fah (1c) = 13.2
H2S Content (%)	0.35 mole %
Asphaltene content (%)	Less than 0.5 wt %
Surface Temperature	About 150 ˚F
Reservoir Temperature	270 ˚F @ Datum Depth (Oil Datum Depth - 4,000 mss)
Reservoir Pressure	9276^{psi} @Datum Depth: 4000 mss (Based on Dynamic Model)
GOR (SCF/STB)	900 as per last post acid flow
Formation Fracture Gradient (PSI/ft)	0.962 psi/ft

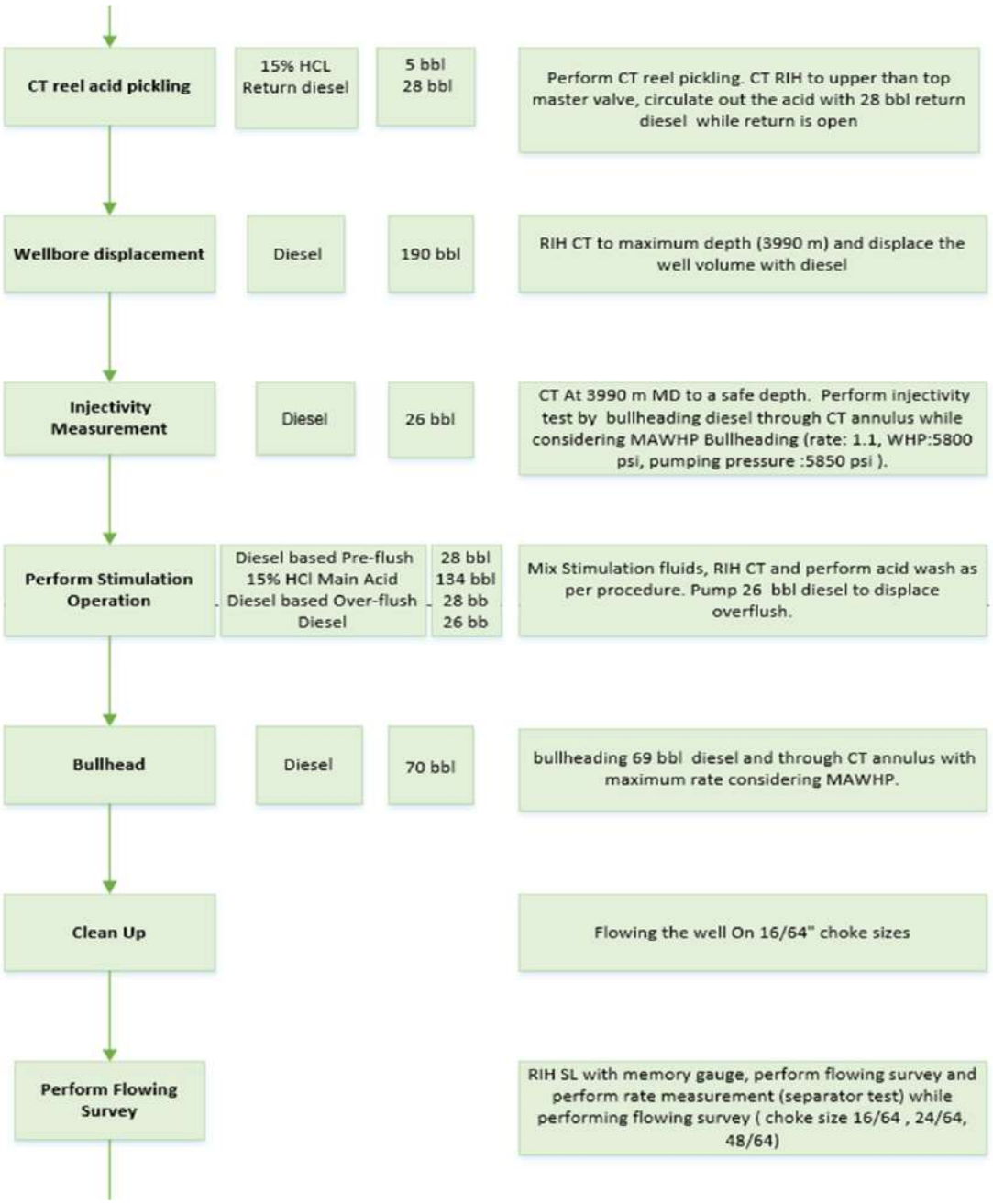
Table 10-3: AZN-10 Well Basic Information

10.4 Total Operation Result

Executed Operation Flow Chart




remove obstacle
RIH CT from surface while pumping f.water min rate (0.3 bpm) to remove obstacle, Tag obstacle at 3644 m MD
Prong & Plug Retrieval
RIH S/L W/SB pulling 2" to Retrieve prong and plug , respectively from NO-GO landing nipple (11960 ft THF), successful
Observe SHIWHP
Open TMV, SIWHP:3820 psi
Clean Up
Flowing the well On 12/64" choke sizes and Increased Adj. Choke Size gradually to 24/64"as per testing program
Separator Test (Draw Down)
Perform separator test with choke sizes 16/64.
Perform Flowing Survey
RIH SL with memory gauge, perform flowing survey and perform rate measurement (separator test) while performing flowing survey (choke size 16/64)
R/U & Pressure test
R/Up and pressure test equipment C/T


CT reel acid pickling
15% HCL
Return diesel
5 bbl
28 bbl
Perform CT reel pickling. CT RIH to upper than top master valve, circulate out the acid with 28 bbl return diesel while return is open
Wellbore displacement
Diesel
190 bbl
RIH CT to maximum depth (3990 m) and displace the well volume with diesel
Injectivity Measurement
Diesel
26 bbl
CT At 3990 m MD to a safe depth. Perform injectivity test by bullheading diesel through CT annulus while considering MAWHP Bullheading (rate: 1.1, WHP:5800 psi, pumping pressure :5850 psi).
Perform Stimulation Operation
Diesel based Pre-flush
15% HCl Main Acid
Diesel based Over-flush
Diesel
28 bbl
134 bbl
28 bb
26 bb
Mix Stimulation fluids, RIH CT and perform acid wash as per procedure. Pump 26 bbl diesel to displace overflush.
Bullhead
Diesel
70 bbl
bullheading 69 bbl diesel and through CT annulus with maximum rate considering MAWHP.
Clean Up
Flowing the well On 16/64" choke sizes
Perform Flowing Survey
RIH SL with memory gauge, perform flowing survey and perform rate measurement (separator test) while performing flowing survey (choke size 16/64 , 24/64, 48/64)

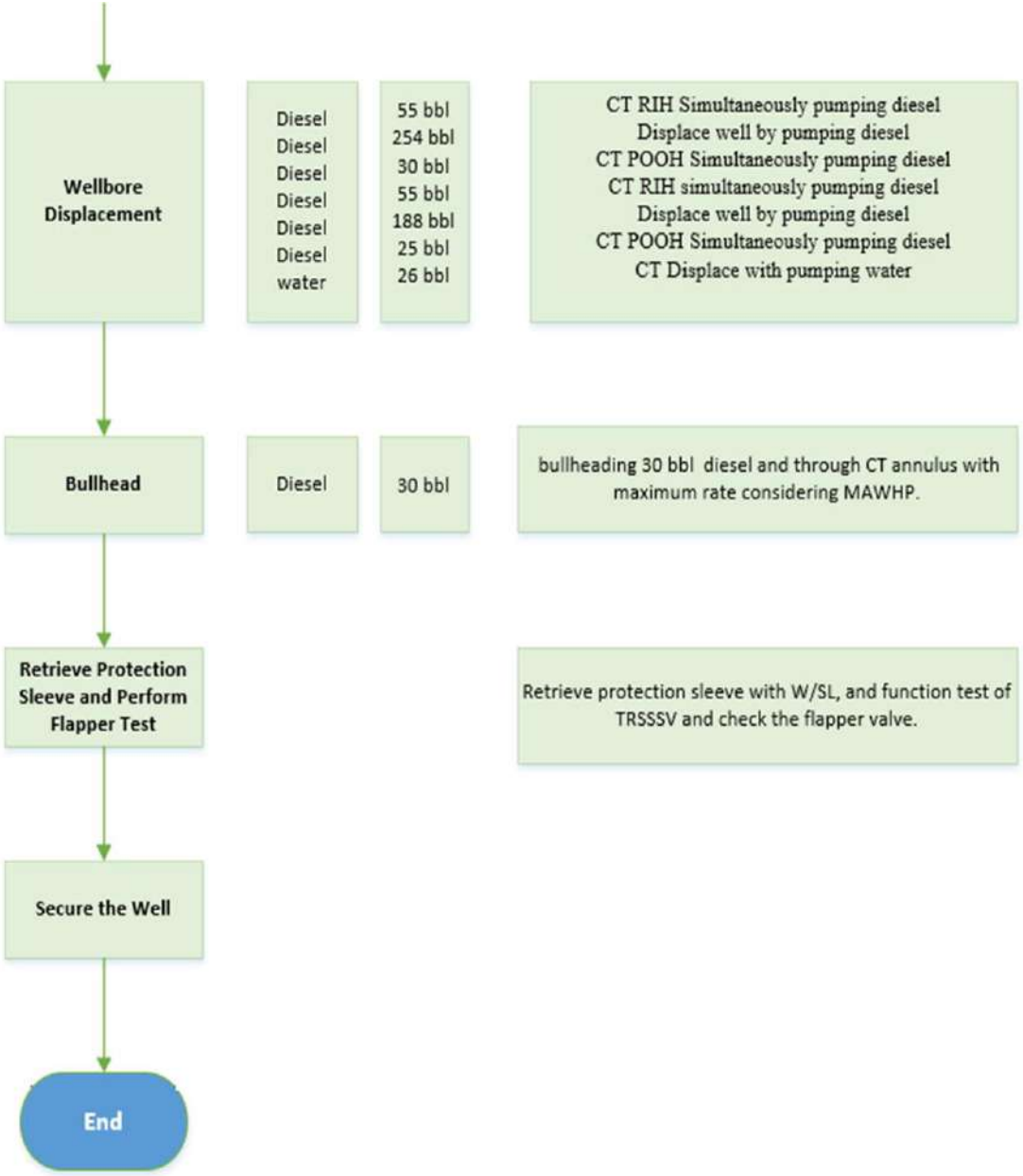


Figure 10-1: AZN-10 Executed Operation Flow Chart

Daily Operation Summary

Date	Operation Summary
03 – Fe – 2024	Rig up and PT of SWT equipment, rig up and PT of S/L, Perform P/T of TMV.
04 – Fe – 2024	Open TMV & RIH of S/L with gauge cutter (3.4 in) to the 3SV depth (135 ft.), Opening TRSCSSV flapper with 6000 psi, RIH S/L for set protection sleeve (AF, OD: 3.437", ID :2.9"), RIH S/L W/SB pulling 2" to retrieve prong (unsuccessful), CT BOP/UP.
05 – Fe – 2024	RIH CT from surface while pumping f.water min rate (0.3 bpm) to remove obstacle, R/U Sl and retrieve prong, RIH S/L W/GS pulling 2.9" to retrieve plug (unsuccessful).
06 – Fe – 2024	Open TMV (SIWHP:3820 psi), Open C/M by 12/64 adj, flow to separator and storage tank.
07 – Fe – 2024	Close TMV for R/U the SL, perform pressure test of S/L BOP, perform inflow test of TMV, Open TMV, Open C/M by 20/64 adj and flow to separator, divert to fix choke 24/64, RIH S/L W/gage cutter 2.2" to TD check.
08 – Fe – 2024	S/L at 4070m (maximum accessible depth), Close the well due to F/U oil storage tanks, Open the TMV, SIWHP:5068 psi, Open the well from C/M (16/64).
09 – Fe – 2024	Perform rate measurement (CS: 16/64 fix), RIH S/L to perform FGS, Perform rate measurement (CS: 16/64 fix), Close TMV and reg up and PT of CT,
10 – Fe – 2024	Open C/M by 16/64 fix Choke and flowing directly to burner boom, close the well from C/M for oil transportation operation.
11 – Fe – 2024	Perform acid pickling, displacement CT reel with Return diesel.
12 – Fe – 2024	Open the C/M with 8/64 adj and flow to separator and increase gradually to 28/64 adj, CT RIH while the LPU was pumping the diesel fluid, Perform injectivity test by 26 bbl diesel, pumping pre flush (28bbl), Closed the well from C/M, CT at perforation and pumping main acid (134 bbl), CT at 4027 and pumping overflush (28 bbl), Bullheading diesel (70 bbl), Perform post acid injectivity test, Open the C/M with 8/64 adj and increase CS to 24/64.
13 – Fe – 2024	Close TMV and rig up and PT of SL, Open C/M by 16/64 adj choke and flowing through separator, Perform rate measurement (CS: 16/64 fix), RIH SL with memory gauge
14 – Fe – 2024	Perform rate measurement (CS: 16/64), SL at surface, Perform rate measurement (CS: 16/64)
15 – Fe – 2024	Close TMV and Rig up and PT of CT, Open C/M by 16/64 adj choke and flowing through separator and increase CS up to 48/64.

16 – Fe – 2024	Perform rate measurement (CS: 48/64), Close the well from CM for the oil transportation operation.
17 – Fe – 2024	Open C/M by 16/64" adj choke and flowing through separator, increase adj c/s up to 44/64, Close the well from CM
18 – Fe – 2024	Open C/M by 12/64" adj choke and flowing through separator, increase adj c/s up to 44/64, CT RIH started while pumping Diesel with min. rate.
19 – Fe – 2024	Displace the well by diesel, Closed the well from C.M.
20 – Fe – 2024	Open the well from C.M on 16/64" adj C/S and divert to separator, RIH CT and pumping diesel
21 – Fe – 2024	CT at 1000 m & pumping liquid switch to water for displace CT, CT at surface and start bullheading (30 bbl diesel), Rig up and PT of CT, RIH S/L for retrieve protection sleeve, Open the well from TMV to CM (Observed SIWHP: 4314 PSI), 3SV closed, inflow test performed, rig up lubricator and 1-way BPV installed, Secure the well.

Table 10-4: AZN-10 Daily Operation Summary

Sequence of Operation

DATE (dd/mm/yyyy)	TIME (hh:mm)	Sequence
03-February -2024	08:25	Perform Pressure Test (Online) Separator gas line with 450 psi by water is (OK).
03-February -2024	10:20	The surface test of the slick line wire was done and was confirmed.
03-February -2024	10:42	Performed pressure test for Upstream Chock valve's & line's: (6500 psi) for 15min is (ok).
03-February -2024	11:30	Repairs are being done on the downstream chock valve @ (11:30) to (13:00).
03-February -2024	13:05	Performed pressure test for TMV. (6500 psi), is (Not ok).
03-February -2024	13:10	Performed Function test for slick line BOP is (ok).
03-February -2024	13:25	Performed pressure test for well head production valve. (6500 psi) is (Ok).
03-February -2024	13:45	Performed pressure test for well head kill valve. (6500 psi), is (Ok).
03-February -2024	13:45	Performed pressure test for well head kill valve. (6500 psi), (Not ok).
03-February -2024	14:40	Perform pressure test for Burner boom line tubing's (3 Line's) is (OK).
03-February -2024	15:45	Perform Pressure Test for Separator input valve up to 500 psi is (OK).
03-February -2024	16:05	Perform pressure test of Separator by 500 psi is (OK).
03-February -2024	16:45	Perform P/T of downstream (from C. M to inlet valve's surge tanks) by 500 psi, 15 min, ok.
03-February -2024	18:00	Perform P/T of TMV, *pressure drop: 66 psi/min, not ok.
03-February -2024	18:53	Perform P/T of TMV, *pressure drop: 51 psi/min, not ok.
03-February -2024	19:10	Perform P/T of TMV, 6600 psi, ok* by analogue gauge from X-cap.
03-February -2024	21:55	Perform P/T of TMV, 6500 psi (by digital gauge), 15 min, OK.
03-February -2024	23:10	Perform Re-pressure test of UP stream C/M, (6000 psi), 10 min, OK.

04-February -2024	06:00	Start R/U Slick Line Lubricator and B.O.P Equipment.
04-February -2024	06:30	Disconnected X-mass three Cap by Slick line group.
04-February -2024	06:35	Rigged Up Dual B.O.P.
04-February -2024	06:40	Rigged Up S/L Lubricator.
04-February -2024	07:00	Safety meeting held.
04-February -2024	07:30	Perform Slick line Pull Test 1000 lb. is (OK).
04-February -2024	08:00	Perform Pressure Test Lubricator by Pump Truck with Water Is 6500 psi is (OK).
04-February -2024	09:10	Perform Function test of burner boom, OK.
04-February -2024	09:20	Open TMV & start RIH of S/L by gauge cutter 3.4 in.
04-February -2024	09:23	S/L @ (135 ft) and start pullout.
04-February -2024	09:27	S/L @ surface and closed swab valve.
04-February -2024	10:05	Open 3SV (with 6000 psi).
04-February -2024	10:10	RIH S/L for set protection sleeve (AF, OD: 3.437", ID :2.9").
04-February -2024	10:20	Slick line at (174 ft) and POOH To the Surface (Set Protection Sleeve Successful).
04-February -2024	10:30	Slick line at Surface.
04-February -2024	12:24	Start RIH S/L W/SB pulling 2" to retrieve prong (company: WQX). BHA Run: Rope socket 1-7/8” +12 ft stem 1-7/8” +3 ft H.jar 1 7/8"+4ft M.J 1-7/8” + 1 ft SB pulling 2".
04-February -2024	12:30	Try to pass 11960 ft (from THG) W/400 lb overpull, unsuccessful.
04-February -2024	12:45	Start to POOH S/L.
04-February -2024	13:40	S/L at Surface.

04-February -2024	16:25	the CT BOP/UP.
04-February -2024	16:55	Perform function test of CT BOP, OK.
04-February -2024	19:22	Perform P/T of CT blind rams, 6500 psi, 10 min, OK.
04-February -2024	19:25	Perform pull test of CT connector sub, 10 klbs, OK.
04-February -2024	19:50	The CT injector head up.
04-February -2024	19:55	Start pumping water for displacement the CT reel from 19:55 to 20:20 (v: 26 bbl water).
04-February -2024	20:25	perform pressure test the CT stripper from 20:25 to 20:40 (4800 psi) (not ok).
04-February -2024	20:35	Observe leakage of CT injector head quick union.
04-February -2024	21:10	Perform pressure test the CT stripper and body test the BOP from 21:10 to 21:50 (6500 psi) (ok) (2 step).
04-February -2024	22:05	Perform pressure test the CT Check Valve from 22:05 to 22:10 (1000 psi) (ok).
04-February -2024	22:10	Perform pressure test the CT pipe rams from 22:10 to 23:00 (6500 psi) (not ok) (2 step).
04-February -2024	23:00	Start repairing the BOP pipe rams.
05-February -2024	00:39	Perform p/T of CT pipe ram 6375 psi "OK".
05-February -2024	00:48	Open the top master valve.
05-February -2024	00:50	Start RIH CT from surface while pumping f.water min rate (0.3 bpm) to remove obstacle.
05-February -2024	04:23	Tag obstacle at 3644 m MD (drop CT weight 2 klbs) and increase rate of pumping to 1 bpm.
05-February -2024	04:50	Tag plug prong at 3668 m MD.
05-February -2024	05:20	Finish pumping 50 bbl f. water R/1bpm Start POOH CT from 3668 m MD.
05-February -2024	08:36	CT at surface and stop Pumping (total vol pumped 235bbl).

05-February -2024	08:43	Closed TMV.
05-February -2024	11:00	The SL R/U (11:00-12:15).
05-February -2024	12:00	Performed the SL pull test up to 1000lbs "OK".
05-February -2024	12:32	Performed the SL lubricator and body test up to 6500psi "OK".
05-February -2024	12:52	The SL RIH. BHA: rope socket 1-7/8", 12ft stem 1-7/8", hydraulic jar 1-7/8", 4ft mechanical jar, SP pulling2".
05-February -2024	14:00	The SL catch the prong at 1212ft, then the SWHP increased from 76 Psi to 3200 Psi
05-February -2024	14:03	The SL POOH.
05-February -2024	14:28	The SL received to surface and closed the TMV and released the pressure from C/M to the oil storage tank.
05-February -2024	14:40	flush the up-chock line and the BOP with pumping 2bbl water. SL lubricator down at 14:40 (retrieved the prong successfully).
05-February -2024	15:21	Start RIH S/L W/GS pulling 2.9" to retrieve plug. S/L BHA: Run: rope socket 1 7/8"+12ft stem 1 7/8"+3ft H.jar1 7/8"+4ft M.jar 1 7/8" +1 ft GS pulling 2.7".
05-February -2024	16:18	The SL tag the plug at 12014 ft and catch it, and try to retrieving the plug (try 10 times).
05-February -2024	17:12	The SL start to POOH.
05-February -2024	18:00	The SL received to surface, and closed the TMV.
05-February -2024	18:02	released the pressure from C/M to the oil storage tank.
05-February -2024	18:05	flush the up-chock line and the BOP with pumping 2bbl water.
05-February -2024	18:10	SL lubricator down at but the plug didn’t retrieve.

05-February -2024	19:10	Perform pressure test of lubricator 6500 psi during 10 min (observed and record the pressure from deta header).
05-February -2024	19:20	opened the well from TMV.
05-February -2024	19:20	Start RIH S/L W/GS pulling 2.9" to retrieve the plug. S/L BHA: Run#3: ropesocket 1 7/8"+10ft stem 1 7/8"+3ft H.jar1 7/8"+6ft M.jar 1 7/8" +1 ft GS pulling 2.7"
05-February -2024	20:20	The SL tag the plug at 12014ft and catch it and try to retrieve the plug (try 45 times).
05-February -2024	22:30	The SL start to POOH.
05-February -2024	23:45	The SL at surface.
05-February -2024	23:47	closed the TMV and released the pressure through the C/M to the oil storage tanks.
05-February -2024	23:50	Flush the lines and lubricator and BOP with pumping 2 bbl water. the lubricator down (retried the plug successfully).
06-February -2024	08:01	Open TMV (SIWHP:3820 psi).
06-February -2024	08:06	Open C/M by 12/64 adj choke through storage tank.
06-February -2024	09:26	Close the well from C/M due to leakage & plug release line of side valve "section C".
06-February -2024	11:16	Open C/M by 12/64 adj choke through storage tanks.
06-February -2024	19:31	Check vent port, no outlet.
07-February -2024	14:07	Close C/M.
07-February -2024	14:10	Close TMV for R/U the SL
07-February -2024	15:15	Perform pull test of S/L rope socket, 1000 lbs, OK.
07-February -2024	17:37	Perform pressure test of S/L BOP, Not OK.

07-February -2024	18:44	Perform pressure test of S/L BOP, 5500 psi, 10 min, OK.
07-February -2024	19:00	Perform inflow test of TMV at 19:00 to 19:50
07-February -2024	21:15	S/L Rig up the lubricator.
07-February -2024	21:39	Perform P/T of S/L lubricator 6000 psi,10min, ok.
07-February -2024	21:43	Open TMV.
07-February -2024	21:59	Open C/M by 20/64 adj, flow to separator.
07-February -2024	22:35	Divert to fix choke 24/64.
07-February -2024	22:38	Start RIH S/L W/gage cutter 2.2" to TD check. S/L BHA: Run: rope socket 1 7/8"+10 ft stem 1 7/8"+4ft M.jar 1 7/8"+3ft H.jar1 7/8"+1 ft gauge cutter 2.2".
08-February -2024	00:25	S/L at 4070m (maximum accessible depth), start to POOH.
08-February -2024	01:55	S/L at surface.
08-February -2024	02:08	Close TMV.
08-February -2024	02:10	Close C/M.
08-February -2024	02:10	Close the well due to F/U oil storage tanks.
08-February -2024	14:08	Power on 3*memory Up: sample rate: 35,16 kpsi, S/N: 90504, Down: sample rate:3s,16 kpsi, S/N: 90506.
08-February -2024	15:00	Perform pull test of S/L, 900 lbs, OK.
08-February -2024	15:20	Perform pressure test of S/L lubricator against C/M & TMV, not OK but no external leakage.
08-February -2024	15:40	Failure of WT C/M.
08-February -2024	19:50	Perform P/T of C/M U/S, 6000 psi, 10 min, OK.

08-February -2024	19:54	Open the TMV, SIWHP:5068 psi.
08-February -2024	20:00	Open the well from C/M.
09-February -2024	04:31	Rate Measurement, Time: (04:31-04:51), Choke Size: 16/64" Fix UCP: 3333 PSI, DCP: 217 PSI, Oil Rate: 1424 STBD, Gas Rate: 1.374944 MMSCFD, GOR: 965.5 SCF/STB, Oil Specific Gravity: 0.838m, API @ 60 F: 37.3, Weight: 52 PCF, BS&W: 0 % Further Info: Sep.P: 213 PSI, Sep.T: 76 F, Orifice Diff. P: 45.5 in H2O, Orifice Plate Size: 1.5 in, Sep. Gas Line ID: 5.761 in, Beta ratio: 0.26, Sensor range: 0-100 in H2O, UCT: 97 F, DCT: 90 F, ANN P: 0 psi, Gas Specific Gravity: 0.799, H2S: 5000 ppm, CO2: 3 %, Surge tank press: 5 psi, Shrinkage factor: -.
09-February -2024	06:27	Start RIH S/L to perform FGS.
09-February -2024	08:34	S/L reach to 13114 ft.
09-February -2024	09:16	Rate Measurement, Time: (09:16-09:36), Choke Size: 16/64" Fix UCP: 3313 PSI, DCP: 213.65 PSI, Oil Rate: 1446 STBD, Gas Rate: 1.405249 MMSCFD, GOR: 971 SCF/STB, Oil Specific Gravity: 0.831 @ 60 F, API @ 60 F: 37.1, Weight: 52 PCF, BS&W: 0 % Further Info: Sep.P: 213 PSI, Sep.T: 76 F, Orifice Diff. P: 45.29 in H2O, Orifice Plate Size: 1.50 in, Sep. Gas Line ID: 5.761 in, Beta ratio: 0.26, Sensor range: 0-100 in H2O, UCT.: 99.9 F, DCT: 96.2 F, ANN P: 0 psi, Gas Specific Gravity: 0.779, H2S: 5000 ppm, CO2: 3 %, Surge tank press: 5 psi, Shrinkage factor: -
09-February -2024	10:22	Rate Measurement, Time: (10:22-10:42), Choke Size: 16/64" Fix UCP: 3310.36 PSI, DCP: 214.30 PSI, Oil Rate: 1435 STBD, Gas Rate: 1.404364 MMSCFD, GOR: 978 SCF/STB, Oil Specific Gravity: 0.831 @ 60 F, API @ 60 F: 37.1, Weight: 52 PCF, BS&W: 0 % Further Info: Sep.P: 216 PSI, Sep.T: 76 F, Orifice Diff. P: 45.66 in H2O, Orifice Plate Size: 1.50 in, Sep. Gas Line ID: 5.761 in, Beta ratio: 0.26, Sensor range: 0-100 in H2O, UCT: 100.19 F, DCT: 97.5 F, ANN P: 0 psi, Gas Specific Gravity: 0.779, H2S: 5000 ppm, CO2: 3 %, Surge

		tank press: 5 psi, Shrinkage factor: -
09-February -2024	12:19	Rate Measurement, Time: (12:19-12:39), Choke Size: 16/64" Fix UCP: 3303.36 PSI, DCP: 201 PSI, Oil Rate: 1446 STBD, Gas Rate: 1.417025 MMSCFD, GOR: 979.9 SCF/STB, Oil Specific Gravity: 0.831 @ 60 F, API @ 60 F: 37.1, Weight: 52 PCF, BS&W: 0 % Further Info: Sep.P: 199 PSI, Sep.T: 86 F, Orifice Diff. P: 51.05 in H2O, Orifice Plate Size: 1.50 in, Sep. Gas Line ID: 5.761 in, Beta ratio: 0.26Sensor range: 0-100 in H2O, UCT.: 101.33 F, DCT: 99 F, ANN P: 0 psi, Gas Specific Gravity: 0.779, H2S: 5000 ppm, CO2: 3 %, Surge tank press: 5 psi, Shrinkage factor: -
09-February -2024	13:11	S/L at surface.
09-February -2024	13:14	Close TMV.
09-February -2024	16:35	Rig Up CT BOP.
09-February -2024	16:45	Perform function test of CT BOP, OK.
09-February -2024	18:12	Perform P/T of CT BOP Blind rams 6000psi ,15mins 18:12 _18:27, OK.
09-February -2024	18:35	Observe leakage from master valve.
09-February -2024	19:37	Perform pressure test of TMV, ok.
09-February -2024	20:45	Perform P/T of CT body and stripper, 6500, 10 min, OK.
09-February -2024	21:45	Perform P/T of CT pipe ram, 6100, 10 min, OK.
09-February -2024	22:10	Perform P/T of CT double check valve, 1000 psi, 10 min, OK.
10-February -2024	10:22	Open C/M by 16/64 fix Choke and flowing directly to burner boom.
10-February -2024	10:30	Switch flow to 12/64 adj Choke and increase gradually to 16/64 adj Choke.
10-February -2024	11:27	close the well from C/M for oil transportation operation.
11-February -2024	16:46	Open C/M and flowing through separator to burner boom.

11-February -2024	17:00	start mixing 5 bbl the acid pickling at 17:00 to 18:10 (HCL15%)
11-February -2024	17:32	Close C/M.
11-February -2024	17:43	closed TMV.
11-February -2024	17:45	OPEN C/M.
11-February -2024	17:45	pumping the pickling acid (HCL 15%) from 17:45 to 17:55.
11-February -2024	17:55	displacement CT reel with Return diesel from 17:55 to 18:30 (V:28bbl).
11-February -2024	21:50	Open the TMV.
12-February -2024	00:36	Open the C/M with 8/64 adj and flow to separator.
12-February -2024	00:56	Increase choke size to 20/64 adj.
12-February -2024	01:01	Increase choke size to 22/64.
12-February -2024	01:17	Increase choke size gradually to 28/64 adj.
12-February -2024	01:30	CT RIH while the LPU was pumping the diesel fluid (r: 0.3- p: 2200psi) at 01:30 (with BHA: check valve double flapper + wash nuzzle)
12-February -2024	01:42	Increase choke size to 30/64 adj.
12-February -2024	05:30	Observe Drag at 3670 m to 3800m depth.
12-February -2024	06:30	Observe Drag at 3902m depth.
12-February -2024	06:52	Observe Drag at 4061m depth.
12-February -2024	07:00	CT reached the maximum depth (4084m = TD).
12-February -2024	07:00	CT POOH.
12-February -2024	07:10	Decrease choke size to 24/64 adj.

12-February -2024	07:20	Start mixing pre/over flush.
12-February -2024	07:40	CT reached 3990m depth. (total pumping: 190 bbl)
12-February -2024	07:49	Perform injectivity test by 26 bbl diesel, while the well was closed (rate: 1.1, WHP:5800 psi, pumping pressure :5850 psi).
12-February -2024	08:15	Start mixing Acid.
12-February -2024	11:20	Finished mixing main acid in Tank#1.
12-February -2024	12:20	Finished mixing main acid in Tank#2.
12-February -2024	13:55	Opened the well from C/M.
12-February -2024	14:50	pumping pre flush at 14:50 to 15:20(R:0.8. P: 3000.V: 28 bbl)
12-February -2024	15:22	Closed the well from C/M.
12-February -2024	15:24	Preflush pumping has finished (total pumping: 28 bbl).
12-February -2024	15:26	Start main acid pumping.
12-February -2024	16:10	CT arrived to bottom of perforation at 4087 m Md while 26 bbl acid pumped.
12-February -2024	16:32	Continue pumping acid and then POOH CT to top of perforation 4027 m Md.
12-February -2024	18:06	Finished 134 bbl main acid, switched to overflush while CT stopped at 4027 m MD.
12-February -2024	18:53	Finished pumping 28 bbl overflush and switched to diesel. Start to RIH CT to spot overflush in perforation interval.
12-February -2024	19:35	Finished pumping 26 bbl diesel and start to POOH to safe depth.
12-February -2024	19:45	CT stopped at 3980 m.
12-February -2024	19:48	Perform post acid injectivity test, max rate 7.0 at 6500 psi.
12-February -2024	20:06	Finish bullhead (69 bbl diesel).

12-February -2024	20:07	The CT start POOH with pumping minimum rate of diesel.
12-February -2024	20:21	Open the C/M with 8/64 adj.
12-February -2024	20:46	Increase choke size gradually to 24/64 adj and divert flow to separator and surg and the oil storage tanks.
13-February -2024	00:17	CT at surface and total pumping 62 bbl.
13-February -2024	00:20	Dipping valve closed.
13-February -2024	0:30	Release annulus pressure from 480 psi to 188 psi, no oil contaminates, weight: 97 pcf, vol:80 lit.
13-February -2024	00:50	Release annulus pressure from 509 psi to 201psi, no oil contaminates, weight: 96 pcf, vol:70 lit.
13-February -2024	01:50	Release annulus pressure from 480 psi to 185psi, no oil contaminates, weight: 94 pcf, vol:80 lit.
13-February -2024	08:35	Close CM.
13-February -2024	08:45	Close TMV and release pressure.
13-February -2024	09:08	Rig down CT injector.
13-February -2024	09:50	Rig down CT BOP.
13-February -2024	12:10	SL pull test performed up to 1000 lb.
13-February -2024	12:20	SL BOP function test performed.
13-February -2024	12:26	Power on 2 memory gauges.
13-February -2024	13:15	SL lubricator pressure tested against CM and TMV up to 6400 psi, unsuccessfully.
13-February -2024	13:35	SL lubricator pressure tested against CM and TMV up to 6400 psi, OK.
13-February -2024	14:05	Open C/M by 16/64 adj choke and flowing through separator.
13-February -2024	14:08	Switch flow to 16/64 fix.

13-February -2024	15:05	Annulus section B pressure released from 580 psi to 100 psi (Sample Weight= 73 pcf, Duration= 8 min, Tot. Vol.= 8 lit).
13-February -2024	21:30	Release annulus pressure section B from 480 psi to 100 psi (Sample weight: 74 pcf, vol: 4lit, Duration=2min.
13-February -2024	22:10	Rate Measurement, Time: (22:10-22:25), Choke Size: 16/64" Fix UCP: 4351 PSI, DCP: 215 PSI, Oil Rate: 1808 STBD, Gas Rate: 1.707 MMSCFD, GOR: 944 SCF/STB, Oil Specific Gravity: 0.836 @ 60 F, API @ 60 F: 37.5, Weight: 52 PCF, BS&W: 0, *Further Info: *Sep.P: 204PSI, Sep.T: 103 F, Orifice Diff. P: 76 in H2O, Orifice Plate Size: 1.50 in, Sep. Gas Line ID: 5.761 in, Beta ratio: 0.26, Sensor range: 0-100 in H2O, UCT.: 114 F, DCT: 110 F, ANN P: 0 psi, Gas Specific Gravity: 0.792, H2S: 5000 ppm, CO2: 3, Surge tank press: 5 psi, Shrinkage factor: -
13-February -2024	22:30	Rate Measurement, Time: (22:30-22:45), Choke Size: 16/64" Fix. UCP: 4350 PSI, DCP: 215 PSI, Oil Rate: 1808 STBD, Gas Rate: 1.707 MMSCFD, GOR: 944 SCF/STB, Oil Specific Gravity: 0.836 @ 60 F, API @ 60 F: 37.6, Weight: 52 PCF, BS&W: 0 *Further Info: *Sep.P: 204PSI, Sep.T: 103 F, Orifice Diff. P: 76 in H2O, Orifice Plate Size: 1.50 in, Sep. Gas Line ID: 5.761 in, Beta ratio: 0.26, Sensor range: 0-100 in H2O, UCT: 114 F, DCT: 110 F, ANN P: 0 psi, Gas Specific Gravity: 0.793, H2S: 5000 ppm, CO2: 3, Surge tank press: 5 psi, Shrinkage factor: -
13-February -2024	23:20	SL RIH commenced BHA: Rope Socket 1-7/8" + 12 ft Stem 1-7/8" + 3 ft H. Jar 1-7/8" + Shock Absorber 1-1/2" + 5 ft 2*Memory Gauges 1-1/4".
14-February -2024	01:17	S/L at 3997 m.
14-February -2024	02:15	Rate Measurement, Time: (02:15-02:30), Choke Size: 16/64" Fix UCP: 4340 PSI, DCP: 214 PSI, Oil Rate: 1779 STBD, Gas Rate: 1.707 MMSCFD, GOR: 959 SCF/STB, Oil Specific Gravity: 0.838 @ 60 F, API @ 60 F: 37.3, Weight: 52 PCF, BS&W: 0 *Further Info: *Sep.P: 204PSI, Sep.T: 102 F, Orifice Diff. P: 76 in H2O, Orifice Plate Size: 1.50 in, Sep. Gas Line ID: 5.761 in, Beta ratio: 0.26, Sensor range: 0-100 in H2O, UCT: 114 F,

		DCT: 110 F, ANN P: 0 psi, Gas Specific Gravity: 0.793, H2S: 5000 ppm, CO2: 3, Surge tank press: 5 psi, Shrinkage factor: -
14-February -2024	02:39	Start POOH S/L.
14-February -2024	03:10	Rate Measurement, Time: (03:10-03:25), Choke Size: 16/64” Fix UCP: 4337 PSI, DCP: 214 PSI, Oil Rate: 1808 STBD, Gas Rate: 1.702 MMSCFD, GOR: 941 SCF/STB, Oil Specific Gravity: 0.839 @ 60 F, API @ 60 F: 37.2Weight: 52 PCF, BS&W: 0 *Further Info: *Sep.P: 205 PSI, Sep.T: 102 F, Orifice Diff. P: 76 in H2O, Orifice Plate Size: 1.50 in, Sep. Gas Line ID: 5.761 in, Beta ratio: 0.26, Sensor range: 0-100 in H2O, UCT: 114 F, DCT: 110 F, ANN P: 0 psi, Gas Specific Gravity: 0.792, H2S: 5000 ppm, CO2: 3, Surge tank press: 5 psi, Shrinkage factor: -
14-February -2024	06:52	SL at surface.
14-February -2024	06:57	Close TMV.
14-February -2024	07:20	Rig down S/L BOP.
14-February -2024	07:35	Open TMV.
14-February -2024	18:08	Rate Measurement, Time: (18:08-18:18) Choke Size: 24/64” Fix UCP: 3631.43 PSI, DCP: 275.16 PSI, Oil Rate: 3368 STBD, Gas Rate: 3253698 MMSCFD, GOR: 966 SCF/STB, Oil Specific Gravity: 0.833 @ 60 F, API @ 60 F: 37.5, Weight: 52 PCF, BS&W: 0 *Further Info: *Sep.P: 258.16 PSI, Sep.T: 122.8 F, Orifice Diff. P: 72.65 in H2O, Orifice Plate Size: 2.0 in, Sep. Gas Line ID: 5.761 in, Beta ratio: 0.34, Sensor range: 0-1000 in H2O, UCT: 132 F, DCT: 126 F, ANN P: 0 psi, Gas Specific Gravity: 0.802, H2S: 4000 ppm, CO2: 3, Surge tank press: 15 psi, Shrinkage factor: -
14-February -2024	18:25	Rate Measurement, Time: (18:25-18:35) Choke Size: 24/64” Fix UCP: 3629.5 PSI, DCP: 276.2 PSI, Oil Rate: 3346 STBD, Gas Rate: 3248766 MMSCFD, GOR: 970 SCF/STB, Oil Specific Gravity: 0.832 @ 60 F, API @ 60 F: 37.7, Weight: 52 PCF, BS&W: 0 *Further Info: *Sep.P: 258 PSI, Sep.T: 122.7 F, Orifice Diff. P: 72.55 in H2O, Orifice Plate

		Size: 2.0 in, Sep. Gas Line ID: 5.761 in, Beta ratio: 0.34, Sensor range: 0-1000 in H2O, UCT.: 131.8 F, DCT: 126 F, ANN P: 0 psi, Gas Specific Gravity: 0.803, H2S: 4500 ppm, CO2: 3, Surge tank press: 10 psi, Shrinkage factor: -
14-February -2024	19:20	Rate Measurement, Time: (19:20-19:35) Choke Size: 24/64” Fix UCP: 3618 PSI, DCP: 273 PSI, Oil Rate: 3346 STBD, Gas Rate: 3.3267 MMSCFD, GOR: 964 SCF/STB, Oil Specific Gravity: 0.835 @ 60 F, API @ 60 F: 37.79, Weight: 52 PCF, BS&W: 0 *Further Info: *Sep.P: 257 PSI, Sep.T: 121.7 F, Orifice Diff. P: 76 in H2O, Orifice Plate Size: 2.0 in, Sep. Gas Line ID: 5.761 in, Beta ratio: 0.34, Sensor range: 0-1000 in H2O, UCT.: 131.8 F, DCT: 126 F, ANN P: 0 psi, Gas Specific Gravity: 0.802, H2S: 4500 ppm, CO2: 3, Surge tank press: 10 psi, Shrinkage factor: -
15-February -2024	00:00	Closed the well fom C/M.
15-February -2024	09:45	closed the well from TMV.
15-February -2024	09:30	Started Rig up the CT BOP.
15-February -2024	09:46	Release pressure to storage tanks.
15-February -2024	10:40	function test All Rams.
15-February -2024	10:05	perform pull test of CT connector sub 10 k lb.
15-February -2024	12:00	CT BOP and blind rams pressure test (with 6500 psi) and leakage from TMV and the LPU line.
15-February -2024	13:05	Performed CT BOP and blind rams pressure test (with 6500 psi) "OK" at 13:05-13:15.
15-February -2024	13:45	The CT inj./up.
15-February -2024	14:10	Fill the CT reel with pumping 2bbl diesel.
15-February -2024	14:20	Performed The CT stripper low pressure test up to 600psi "OK".
15-February -2024	14:42	Performed The CT stripper high pressure test up to 6500psi "OK".
15-February -2024	15:10	Released the pressure from C/M to 1500 psi and then performed the double flapper check valve

		with differential pressure 1500psi"OK".
15-February -2024	15:55	CT pipe and pipe ram pressure tested up to 6400 for 15 min, OK.
16-February -2024	17:13	Open C/M by 16/64 adj choke and flowing through separator.
16-February -2024	18:34	Increase choke size gradually to 48/64 adj since 17:25 to 18:27 and switch to 48/64 fix choke.
17-February -2024	00:25	Rate Measurement, Time: (00:25-00:35) Choke Size: 48/64” Fix. UCP: 2025 PSI, DCP: 343 PSI, Oil Rate: 7200 STBD, Gas Rate: 7.156 MMSCFD, GOR: 993 SCF/STB, Oil Specific Gravity: 0.836 @ 60 F, API @ 60 F: 37.6, Weight: 52 PCF, BS&W: 0 *Further Info*: Sep.P: 310 PSI, Sep.T: 138 F, Orifice Diff. P: 82 in H2O, Orifice Plate Size: 2.750 in, Sep. Gas Line ID: 5.761 in, Beta ratio: 0.47, Sensor range: 0-1000 in H2O, UCT: 153 F, DCT: 150 F, ANN P: 284 psi, Gas Specific Gravity: 0.801, H2S: 6000 ppm, CO2: 3, Surge tank press: 30 psi, Shrinkage factor: -
17-February -2024	00:54	Rate Measurement, Time: (00:54-01:04) Choke Size: 48/64” Fix UCP: 2021 PSI, DCP: 342 PSI, Oil Rate: 7370 STBD, Gas Rate: 7.2247 MMSCFD, GOR: 981 SCF/STB, Oil Specific Gravity: 0.836 @ 60 F, API @ 60 F: 37.67, Weight: 52 PCF, BS&W: 0 *Further Info*: Sep.P: 310 PSI, Sep.T: 138 F, Orifice Diff. P: 83.7 in H2O, Orifice Plate Size: 2.750 in, Sep. Gas Line ID: 5.761 in, Beta ratio: 0.47, Sensor range: 0-1000 in H2O, UCT.: 154 F, DCT: 150 F, ANN P: 349 psi, Gas Specific Gravity: 0.802, H2S: 6000 ppm, CO2: 3, Surge tank press: 30 psi, Shrinkage factor: -
17-February -2024	01:16	Close the well from CM for the oil transportation operation
18-February -2024	06:00	Annulus section B pressure= 180 psi.
18-February -2024	14:50	Open C/M by 16/64" adj choke and flowing through separator.
18-February -2024	15:58	Choke size increased gradually to 44/64" adj.
18-February -2024	16:40	Job stopped due to the HSE concerns about gas spreading in the pad area; H2S @ camp= 9 ppm, LEL= 7 %.

18-February -2024	16:42	Close the well from CM.
19-February -2024	19:26	Open the well from CM on 12/64" adj. choke size.
19-February -2024	20:37	Choke size increased gradually to 44/64" adj.
19-February -2024	20:51	CT RIH started while pumping Diesel with min. rate.
19-February -2024	21:33	Choke size decreased gradually to 40/64" adj.
19-February -2024	22:04	Choke size decreased gradually to 36/64" adj.
19-February -2024	22:56	Choke size decreased gradually to 32/64" adj.
19-February -2024	23:59	Finished 55 bbl pumping.
20-February -2024	00:33	CT tag at 4088 m.
20-February -2024	00:35	Decrease C/S to 16/64" adj.
20-February -2024	00:45	Start displace the well by diesel (Rate:1 BPM, P:4850-5100psi).
20-February -2024	00:49	Decrease C/S to 12/64" adj. @ 00:49
20-February -2024	01:09	Increased C/S to 16/64" adj.
20-February -2024	01:10	Started POOH CT to 4000 m.
20-February -2024	01:17	CT at 4000 m.
20-February -2024	02:10	POOH CT to 3750 m.
20-February -2024	02:14	Increased C/S to 20/64" adj.
20-February -2024	02:38	POOH CT to 3350 m.
20-February -2024	03:20	CT at 3350 m.
20-February -2024	03:52	POOH CT to 3000 m.

20-February -2024	04:18	CT at 3000 m.
20-February -2024	04:55	POOH CT to 2500 m.
20-February -2024	05:32	CT at 2500 m.
20-February -2024	06:22	POOH CT to 2000 m.
20-February -2024	06:55	CT at 2000 m.
20-February -2024	08:02	Closed the well from C.M.
20-February -2024	09:33	Stop pumping diesel (total pumping 254 bbl).
20-February -2024	09:34	CT POOH Simultaneously pumping diesel
20-February -2024	15:46	Stop pumping diesel (total pumping 30 bbl).
21-February -2024	15:53	Open the well from C.M on 16/64" adj C/S and divert to separator.
21-February -2024	15:58	Increase C/S gradually to 20/64" adj.
21-February -2024	16:03	Increased C/S to 24/64" adj.
21-February -2024	16:08	Increased C/S to 28/64" adj.
21-February -2024	16:50	Start RIH CT on 40/64" C/S (WHP: 2800 psi, Pumping pressure:3200 psi, Rate:0.3 bpm).
21-February -2024	18:00	Stop RIH CT due to CT counter failure and stop pumping diesel.
21-February -2024	18:08	Continue RIH CT and continue pumping diesel.
21-February -2024	19:00	CT tag 4081m.
21-February -2024	19:01	Stop CT and stop pumping diesel (55 bbl pumping) due to CT digital counter failure.
21-February -2024	19:14	Close the well from C.M.
21-February -2024	19:18	Open the well from C.M on 12/64" adj C/S and divert to separator.

21-February -2024	19:55	Start displacement the well by pumping diesel.
21-February -2024	20:04	closed the C.M.
21-February -2024	20:15	Open the well on 4/64" adj C/S and POOH CT to 4000 m and continue pumping diesel.
21-February -2024	20:30	Stop CT at 4000 m and continue pumping diesel.
21-February -2024	20:46	POOH CT to 3750 m and continue pumping diesel.
21-February -2024	21:00	Stop CT at 3750 m and continue pumping diesel.
21-February -2024	21:25	POOH CT to 3350 m and continue pumping diesel.
21-February -2024	21:56	Increased C/S to 16/64" adj. C/S.
21-February -2024	22:00	Stop CT at 3350 m and continue pumping diesel.
21-February -2024	22:00	Check vent port and observe no outlet.
21-February -2024	22:15	POOH CT to 3000 m and continue pumping diesel.
21-February -2024	22:40	Stop CT at 3000 m and continue pumping diesel.
21-February -2024	23:00	Decrease C/S to 14/64" adj.
21-February -2024	23:00	POOH CT to 2500 m and continue pumping diesel.
21-February -2024	23:20	Stop CT at 2500 m and continue pumping diesel.
21-February -2024	23:25	POOH CT to 2000 m and continue pumping diesel.
21-February -2024	23:49	Stop CT at 2000 m and continue pumping diesel (188 bbl pumping).
22-February -2024	00:01	POOH CT to 3000 m and continue pumping diesel.
22-February -2024	00:10	Divert flow to storage tanks from separator bypass.
22-February -2024	00:15	Observe more than 50% diesel at return sample.

22-February -2024	00:43	Stop pumping (total pumping 25 bbl)
22-February -2024	00:44	CT at 1000 m and CT Displace with pumping water
22-February -2024	01:17	Stop pumping (total pumping 26 bbl)
22-February -2024	01:32	Decrease C/S to 8/64" adj.
22-February -2024	01:44	CT at surface and Close swab valve.
22-February -2024	01:45	Close the well from C.M.
22-February -2024	02:20	Start bullheading.
22-February -2024	02:33	End of bullheading 30 bbl diesel.
22-February -2024	02:40	Close TMV.
22-February -2024	02:50	Rig down CT.
22-February -2024	04:45	Rig up S/L.
22-February -2024	05:30	Function test of SL BOP (ok).
22-February -2024	05:50	Pull test wire of S/L by 1000 lb (ok).
22-February -2024	06:44	Rig up S/L lubricator.
22-February -2024	06:25	Pressure test SL BOP & lubricator by 4500 psi (ok).
22-February -2024	06:28	Open TMV and Swab valve.
22-February -2024	06:30	Start RIH S/L for retrieve protection sleeve.
22-February -2024	06:40	SL at 53 m and Start POOH to surface.
22-February -2024	06:45	S/L at surface, retrieve P.S successfully (Profile: AF, OD: 3.437", ID :2.9", company: WQX).
22-February -2024	06:50	Close the TMV & swab valve.

22-February -2024	06:55	Rig down S/L.
22-February -2024	06:47	Close TMV.
22-February -2024	08:20	Open the well from TMV to CM (Observed SIWHP: 4314 PSI).
22-February -2024	08:35	3SV function test.
22-February -2024	08:45	3SV closed, inflow test performed, WHP released from 4300 psi to 330 psi, and pressure was stable in 20 min @ 08:45 – 09:05.
22-February -2024	10:20	Rig up lubricator.
22-February -2024	10:55	Lubricator pressure tested up to 6000 psi for 5 min.
22-February -2024	11:10	1-way BPV installed.
22-February -2024	11:15	Close TMV.
22-February -2024	11:30	Rig down lubricator.
22-February -2024	13:10	Annulus section B pressure released from 80 psi to zero (Sample Weight= 73 pcf, Duration= 20 Sec, Tot. Vol.= 1.5 lit).
22-February -2024	16:40	Install WH fixed chock (S.N:0179232 A/O).
22-February -2024	16:55	Pump truck line pressure tested against Kill valve up to 6500 psi for 10 min, OK.
22-February -2024	17:10	Pressure test against Choke cap, TMV, and X-Tree's cap performed up to 6500 psi for 10 min (TMV close, Swab valve open, Production valve open), leakage observed from instrument flange.
22-February -2024	17:40	Pressure test against Choke cap, TMV, and X-Tree's cap performed up to 6500 psi for 10 min (TMV close, Swab valve open, Production valve open), OK.
22-February -2024	13:10	Annulus section B pressure released from 80 psi to zero (Sample Weight= 73 pcf, Duration= 20 Sec, Tot. Vol.= 1.5 lit).

22-February -2024	16:40	Install WH fixed chock (S.N:0179232 A/O).
22-February -2024	16:55	Pump truck line pressure tested against Kill valve up to 6500 psi for 10 min, OK.
22-February -2024	17:10	Pressure test against Choke cap, TMV, and X-Tree's cap performed up to 6500 psi for 10 min (TMV close, Swab valve open, Production valve open), leakage observed from instrument flange.
22-February -2024	17:40	Pressure test against Choke cap, TMV, and X-Tree's cap performed up to 6500 psi for 10 min (TMV close, Swab valve open, Production valve open), OK.
22-February -2024	18:05	Pressure test against Production valve, TMV, and Swab Valve performed up to 6500 psi for 10 min (TMV close, Swab valve close, Production valve close).
22-February -2024	18:20	All of the X-Tree's valves closed.
22-February -2024	18:25	Annulus pressures checked: Section C: 0 psi, No outlet fluid. Section B: Sample Weight= 73 pcf, Duration= 20 Sec, Tot. Vol.= 1.5 lit. Section A: No outlet fluid.
22-February -2024	18:30	Secure the well.
End Of Job		

Table 10-5: AZN-10 Sequence of Operation

Operation Results

10.4.4.1 Summary of Well Testing Results

Well Testing Rate Measurement (Before Acidizing)																			
Date (DD-MM-YYYY)	Time	Choke size (Inch)	UCP (psi)	DCP (psi)	SEP.P (psi)	S.T (F)	D.P (in.H2O)	Orifice Size (inch)	API (oil)	gas SP.GR	H2S (ppm)	CO2 (%)	BS&W (%)	gas outlet (ID)	beta ratio	Ann P(psi)	Weight (PCF)	Q- Oil (BBL/Day)	Q-Gas (SCF/Day)	GOR (SCF/BBL)
09-February -2024	4:51	16/64	3332	216	213	76	45.5	1.5	37.3	0.799	5000	3	0	5.761	0.26	0	52	1424	1.257717	883
09-February -2024	6:00	16/64	3327	196	213	76	45.46	1.5	37.8	0.772	5000	3	0	5.761	0.26	0	52	1446	1.278961	884
09-February -2024	9:36	16/64	3313	213	213	76	45.29	1.5	37.1	0.779	5000	3	0	5.761	0.26	0	52	1446	1.270831	879
09-February -2024	10:42	16/64	3310	214	216	77	45.66	1.5	37.1	0.779	5000	3	0	5.761	0.26	0	52	1435	1.283206	894
09-February -2024	12:39	16/64	3303	201	199	76	51.05	1.5	37.1	0.779	5000	3	0	5.761	0.26	0	52	1446	1.306491	904

Well Testing Rate Measurement (After Acidizing)

Date (DD-MM-YYYY)	Time	Choke size (Inch)	UCP (psi)	DCP (psi)	SEP.P (psi)	S.T (F)	D.P (in.H2O)	Orifice Size (inch)	API (oil)	gas SP.GR	H2S (ppm)	CO2 (%)	BS&W (%)	gas outlet (ID)	beta ratio	Ann P(psi)	Weight (PCF)	Q- Oil (BBL/Day)	Q-Gas (SCF/Day)	GOR (SCF/BBL)
13-February -2024	22:25	16/64	4351	215	204	103	76	1.5	37.5	0.792	5000	3	0	5.761	0.26	0	52	1808	1.558562	862
13-February -2024	22:45	16/64	4350	215	204	135	76	1.5	37.6	0.793	5000	3	0	5.761	0.26	0	52	1808	1.515092	838
14-February -2024	2:30	16/64	4340	214	204	110	76	1.5	37.3	0.793	5000	3	0	5.761	0.26	0	52	1779	1.54798	870
14-February -2024	3:25	16/64	4337	214	205	102	76	1.5	37.2	0.792	5000	3	0	5.761	0.26	0	52	1808	1.56354	864
14-February -2024	18:18	24/64	3631	275	258.1	122.8	72.65	2	37.5	0.802	4000	3	0	5.761	0.34	0	52	3368	2.971607	882
14-February -2024	18:35	24/64	3629	276.2	258	122.8	72.55	2	37.7	0.803	4500	3	0	5.761	0.34	0	52	3346	2.967725	887
14-February -2024	19:35	24/64	3618	273	257	121.7	76	2	37.79	0.802	4500	3	0	5.761	0.34	0	52	3346	3.03591	907
17-February -2024	0:35	48/64	2025	343	310	138.9	82.26	2.75	37.6	0.801	6000	3	0	5.761	0.47	0	52	7200	6.565922	912
17-February -2024	1:04	48/64	2021	342	310.7	138.5	83.77	2.75	37.67	0.802	6000	3	0	5.761	0.47	0	52	7370	6.631663	900

Surface testing data were analyzed to investigate the effect of stimulation operation. According to pre-stimulation clean up results well head pressure and oil rate were observed 3303 PSI and 1446 BPD at 16/64” choke size. On the other hand, data gathered from post stimulation clean up shows that well head pressure and oil rate have increased to 3631 PSI and 3368 BPD at 16/64 choke size. The above data illustrates that stimulation operation has led to a 328-PSI increase in wellhead pressure and a 1922-BPD increase in oil rate. Generally, the stimulation operation was carried out successfully.

10.4.4.2 Well Testing Main Results & Plots

Date (DD-MM-YYYY)	Time HH:MM	U.C.P (PSI)	U.C.T(°F)	D.C.P (PSI)	D.C.T (°F)	SEP.P (PSI)	SEP.T (°F)	Annulus pressure (PSI)	DP (in.H2O)	Orifice size (inch)	Q-Oil (STB/Day)	Q-Gas (SCF/Day)
04-February -2024	10:30	31	0.0	0	0.0	0	0.0	13	0	0	0	0
04-February -2024	10:50	5	0.0	0	0.0	0	0.0	13	0	0	0	0
04-February -2024	11:10	5	0.0	0	0.0	0	0.0	13	0	0	0	0
04-February -2024	11:30	74	0.0	0	0.0	0	0.0	13	0	0	0	0
04-February -2024	11:50	168	0.0	0	0.0	0	0.0	13	0	0	0	0
05-February -2024	0:10	5	0.0	0	0.0	0	0.0	13	0	0	0	0
05-February -2024	0:30	6378	0.0	0	0.0	0	0.0	13	0	0	0	0
05-February -2024	0:50	7	0.0	0	0.0	0	0.0	13	0	0	0	0

05-February -2024	1:10	774	0.0	0	0.0	0	0.0	19	0	0	0	0
05-February -2024	1:30	172	0.0	0	0.0	0	0.0	13	0	0	0	0
05-February -2024	1:50	143	0.0	0	0.0	0	0.0	11	0	0	0	0
05-February -2024	2:10	117	0.0	0	0.0	0	0.0	9	0	0	0	0
05-February -2024	2:30	188	0.0	0	0.0	0	0.0	8	0	0	0	0
05-February -2024	2:50	125	0.0	0	0.0	0	0.0	7	0	0	0	0
05-February -2024	3:10	88	0.0	0	0.0	0	0.0	7	0	0	0	0
05-February -2024	3:30	109	0.0	0	0.0	0	0.0	6	0	0	0	0
05-February -2024	3:50	93	0.0	0	0.0	0	0.0	6	0	0	0	0
05-February -2024	4:10	89	0.0	0	0.0	0	0.0	7	0	0	0	0
05-February -2024	4:30	86	0.0	0	0.0	0	0.0	7	0	0	0	0

05-February -2024	4:50	74	0.0	0	0.0	0	0.0	4	0	0	0	0
05-February -2024	5:10	143	0.0	0	0.0	0	0.0	4	0	0	0	0
05-February -2024	5:30	90	0.0	0	0.0	0	0.0	4	0	0	0	0
05-February -2024	5:50	97	0.0	0	0.0	0	0.0	3	0	0	0	0
05-February -2024	6:10	143	0.0	0	0.0	0	0.0	3	0	0	0	0
05-February -2024	6:30	90	0.0	0	0.0	0	0.0	3	0	0	0	0
05-February -2024	6:50	28	0.0	0	0.0	0	0.0	3	0	0	0	0
05-February -2024	7:10	19	0.0	0	0.0	0	0.0	4	0	0	0	0
05-February -2024	7:29	14	0.0	0	0.0	0	0.0	4	0	0	0	0
05-February -2024	7:49	9	0.0	0	0.0	0	0.0	4	0	0	0	0
05-February -2024	8:09	14	0.0	0	0.0	0	0.0	4	0	0	0	0

05-February -2024	8:29	25	0.0	0	0.0	0	0.0	5	0	0	0	0
05-February -2024	8:49	8	0.0	0	0.0	0	0.0	6	0	0	0	0
05-February -2024	9:09	8	0.0	0	0.0	0	0.0	6	0	0	0	0
05-February -2024	9:29	8	0.0	0	0.0	0	0.0	7	0	0	0	0
05-February -2024	9:49	9	0.0	0	0.0	0	0.0	8	0	0	0	0
05-February -2024	10:09	9	0.0	0	0.0	0	0.0	8	0	0	0	0
05-February -2024	10:29	9	0.0	0	0.0	0	0.0	8	0	0	0	0
05-February -2024	10:49	9	0.0	0	0.0	0	0.0	9	0	0	0	0
05-February -2024	11:09	9	0.0	0	0.0	0	0.0	9	0	0	0	0
05-February -2024	11:29	9	0.0	0	0.0	0	0.0	9	0	0	0	0
05-February -2024	11:49	9	0.0	0	0.0	0	0.0	9	0	0	0	0

05-February -2024	12:09	8	0.0	0	0.0	0	0.0	9	0	0	0	0
05-February -2024	12:29	2120	0.0	0	0.0	0	0.0	9	0	0	0	0
05-February -2024	12:49	7	0.0	0	0.0	0	0.0	10	0	0	0	0
05-February -2024	13:09	8	0.0	0	0.0	0	0.0	11	0	0	0	0
05-February -2024	13:29	56	0.0	0	0.0	0	0.0	12	0	0	0	0
05-February -2024	13:49	74	0.0	0	0.0	0	0.0	13	0	0	0	0
05-February -2024	14:09	3548	0.0	0	0.0	0	0.0	53	0	0	0	0
05-February -2024	14:29	3676	0.0	0	0.0	0	0.0	58	0	0	0	0
05-February -2024	14:49	6	0.0	0	0.0	0	0.0	57	0	0	0	0
05-February -2024	15:09	1420	0.0	0	0.0	0	0.0	55	0	0	0	0
05-February -2024	15:29	3695	0.0	0	0.0	0	0.0	54	0	0	0	0

05-February -2024	15:49	3701	0.0	0	0.0	0	0.0	53	0	0	0	0
05-February -2024	16:09	3698	0.0	0	0.0	0	0.0	52	0	0	0	0
05-February -2024	16:29	3681	0.0	0	0.0	0	0.0	51	0	0	0	0
05-February -2024	16:49	3675	0.0	0	0.0	0	0.0	50	0	0	0	0
05-February -2024	17:09	3675	0.0	0	0.0	0	0.0	50	0	0	0	0
05-February -2024	17:29	3657	0.0	0	0.0	0	0.0	50	0	0	0	0
05-February -2024	17:49	3665	0.0	0	0.0	0	0.0	50	0	0	0	0
05-February -2024	18:09	5	0.0	0	0.0	0	0.0	51	0	0	0	0
05-February -2024	18:29	7	0.0	0	0.0	0	0.0	51	0	0	0	0
05-February -2024	18:49	8	0.0	0	0.0	0	0.0	51	0	0	0	0
05-February -2024	19:09	689	0.0	0	0.0	0	0.0	50	0	0	0	0

06-February -2024	8:15	3504	67.1	6	52.1	10	0.0	32	0	0	0	0
06-February -2024	8:35	823	77.8	19	77.2	10	0.0	37	7.05	1.5	0	0
06-February -2024	8:55	265	82.2	27	83.8	9	0.0	180	18.36	1.5	0	0
06-February -2024	9:15	301	81.6	15	82.9	9	0.0	423	26.02	1.5	0	0
06-February -2024	9:35	1405	80.8	4	78.6	9	0.0	0	31.38	1.5	0	0
06-February -2024	9:55	3700	78.3	3	74.9	9	0.0	192	35.6	1.5	0	0
06-February -2024	10:15	4079	76.6	3	74.6	9	0.0	134	36.49	1.5	0	0
06-February -2024	10:35	4137	75.3	6	74.8	9	0.0	83	40.33	1.5	0	0
06-February -2024	10:55	4153	76.5	5	76.0	10	0.0	48	44.33	1.5	0	0
06-February -2024	11:15	4151	77.6	7	76.2	9	0.0	28	45.37	1.5	0	0
06-February -2024	11:35	1228	81.6	19	82.6	8	0.0	7	46.61	1.5	0	0

06-February -2024	11:55	614	84.0	20	87.8	8	0.0	39	38.95	1.5	0	0
06-February -2024	12:15	680	85.7	33	90.9	64	0.0	125	55.47	1.5	0	0
06-February -2024	12:35	673	85.1	30	83.2	9	0.0	218	50.81	1.5	0	0
06-February -2024	12:55	777	86.0	21	88.7	8	0.0	322	0	0	0	0
06-February -2024	13:15	874	88.7	35	92.1	33	0.0	385	0	0	0	0
06-February -2024	13:35	875	87.9	37	81.7	34	0.0	431	0	0	0	0
06-February -2024	13:55	934	86.9	70	76.8	67	0.0	466	0	0	0	0
06-February -2024	14:15	905	86.6	69	74.3	67	0.0	209	0	0	0	0
06-February -2024	14:35	902	87.6	72	72.9	67	0.0	255	0	0	0	0
06-February -2024	14:55	844	87.2	64	70.4	59	0.0	299	0	0	0	0
06-February -2024	15:15	735	87.9	64	76.1	59	0.0	351	0	0	0	0

06-February -2024	15:35	755	88.2	67	77.2	62	0.0	425	0	0	0	0
06-February -2024	15:55	679	89.1	101	80.5	95	0.0	446	0	0	0	0
06-February -2024	16:15	690	89.4	86	80.3	77	0.0	19	0	0	0	0
06-February -2024	16:35	694	90.0	89	79.8	82	0.0	282	0	0	0	0
06-February -2024	16:55	570	90.4	98	82.9	0	0.0	192	0.08	0	0	0
06-February -2024	17:15	529	90.8	92	82.8	79	0.0	299	0.05	0	0	0
06-February -2024	17:35	460	91.3	99	84.6	85	0.0	401	0	0	0	0
06-February -2024	17:55	431	92.2	94	85.7	79	0.0	359	0	0	0	0
06-February -2024	18:15	417	92.5	91	86.0	76	0.0	257	0	0	0	0
06-February -2024	18:35	367	92.4	98	87.4	85	0.0	331	0	0	0	0
06-February -2024	18:55	475	92.7	118	87.0	102	0.0	556	0	0	0	0

06-February -2024	19:15	557	93.9	131	90.1	114	0.0	331	0	0	0	0
06-February -2024	19:35	571	94.5	137	89.8	120	0.0	284	0	0	0	0
06-February -2024	19:55	612	94.6	103	87.5	90	0.0	322	0	0	0	0
06-February -2024	20:15	562	95.1	130	90.1	114	0.0	430	0	0	0	0
06-February -2024	20:35	523	95.4	121	90.8	107	0.0	209	0	0	0	0
06-February -2024	20:55	556	95.9	128	91.2	114	0.0	341	0	0	0	0
06-February -2024	21:15	567	96.2	130	91.9	114	0.0	483	0	0	0	0
06-February -2024	21:35	556	96.7	129	92.4	113	0.0	271	0	0	0	0
06-February -2024	21:55	537	97.1	125	93.2	109	0.0	353	0	0	0	0
06-February -2024	22:15	535	97.6	124	93.7	109	0.0	431	0	0	0	0
06-February -2024	22:35	528	98.0	122	94.2	109	0.0	462	0	0	0	0

06-February -2024	22:55	519	98.5	119	94.6	106	0.0	236	0	0	0	0
06-February -2024	23:15	509	98.5	118	94.6	104	0.0	288	0	0	0	0
06-February -2024	23:35	517	98.7	121	95.4	105	0.0	355	0	0	0	0
06-February -2024	23:55	508	99.0	118	95.9	102	0.0	411	0	0	0	0
07-February -2024	0:15	501	99.2	116	96.5	101	0.0	463	0	0	0	0
07-February -2024	0:35	512	99.8	119	96.6	105	0.0	206	0	0	0	0
07-February -2024	0:55	510	100.3	119	97.3	104	0.0	247	0	0	0	0
07-February -2024	1:15	538	100.8	124	97.9	108	0.0	327	0	0	0	0
07-February -2024	1:35	548	101.3	128	98.4	111	0.0	398	0	0	0	0
07-February -2024	1:55	543	101.7	125	99.0	109	0.0	458	0	0	0	0
07-February -2024	2:15	535	102.0	124	99.9	109	0.0	176	0	0	0	0

07-February -2024	2:35	526	101.8	122	99.5	107	0.0	215	0	0	0	0
07-February -2024	2:55	516	102.2	123	100.2	106	0.0	246	0	0	0	0
07-February -2024	3:15	514	102.4	123	100.7	106	0.0	277	0	0	0	0
07-February -2024	3:35	513	102.6	122	101.1	105	0.0	310	0	0	0	0
07-February -2024	3:55	501	102.6	118	101.1	104	0.0	332	0	0	0	0
07-February -2024	4:15	494	103.3	117	101.3	103	0.0	348	0	0	0	0
07-February -2024	4:35	494	103.4	118	101.5	102	0.0	370	0	0	0	0
07-February -2024	4:55	489	103.5	118	102.5	102	0.0	387	0	0	0	0
07-February -2024	5:15	484	103.2	115	102.6	101	0.0	402	0	0	0	0
07-February -2024	5:35	502	103.4	118	102.1	104	0.0	430	0	0	0	0
07-February -2024	5:55	549	104.4	130	102.7	114	0.0	354	0	0	0	0

07-February -2024	6:15	563	104.5	131	104.6	114	0.0	256	0	0	0	0
07-February -2024	6:35	557	105.1	132	104.8	115	0.0	298	0	0	0	0
07-February -2024	6:55	547	105.3	130	105.5	114	0.0	334	0	0	0	0
07-February -2024	7:15	537	105.6	128	105.3	113	0.0	360	0	0	0	0
07-February -2024	7:35	539	106.2	127	106.0	114	0.0	389	0	0	0	0
07-February -2024	7:55	540	106.3	127	107.0	114	0.0	421	0	0	0	0
07-February -2024	8:15	532	106.6	121	107.4	112	0.0	441	0	0	0	0
07-February -2024	8:35	462	107.1	145	110.7	132	0.0	463	0	0	0	0
07-February -2024	8:55	424	107.7	133	112.2	123	0.0	494	0	0	0	0
07-February -2024	9:15	404	107.5	132	111.5	119	0.0	172	0	0	0	0
07-February -2024	9:35	375	108.0	139	113.7	126	0.0	197	0	0	0	0

07-February -2024	9:55	337	109.1	143	116.7	131	0.0	249	0	0	0	0
07-February -2024	10:15	337	109.5	164	119.3	156	0.0	306	0	0	0	0
07-February -2024	10:35	339	109.2	167	120.4	158	0.0	352	0	0	0	0
07-February -2024	10:55	336	109.3	167	120.3	156	0.0	388	0	0	0	0
07-February -2024	11:15	331	109.7	163	120.8	154	0.0	412	0	0	0	0
07-February -2024	11:35	330	109.9	163	121.7	153	0.0	435	0	0	0	0
07-February -2024	11:55	369	109.6	146	118.8	138	0.0	461	0	0	0	0
07-February -2024	12:15	335	109.9	172	121.9	159	0.0	481	0	0	0	0
07-February -2024	12:35	469	111.7	224	123.9	211	0.0	271	0	0	0	0
07-February -2024	12:55	533	113.6	214	124.4	194	0.0	468	0	0	0	0
07-February -2024	13:15	527	115.1	215	126.4	196	0.0	262	0	0	0	0

07-February -2024	13:35	523	116.0	215	127.1	195	0.0	378	0	0	0	0
07-February -2024	13:55	518	116.3	212	128.7	194	0.0	420	0	0	0	0
07-February -2024	14:15	507	107.5	14	112.2	54	0.0	305	0	0	0	0
07-February -2024	14:35	6	95.3	11	100.2	51	0.0	280	0	0	0	0
07-February -2024	14:55	6	90.0	11	92.4	38	0.0	28	0	0	0	0
07-February -2024	15:15	6	87.6	9	89.1	26	0.0	0	0	0	0	0
07-February -2024	15:35	6	30.1	9	30.0	16	0.0	0	0	0	0	0
07-February -2024	15:55	6	29.5	9	28.7	8	0.0	0	0	0	0	0
07-February -2024	16:15	5	29.3	8	27.0	6	0.0	0	0	0	0	0
07-February -2024	16:35	6	28.9	9	25.7	7	0.0	0	0	0	0	0
07-February -2024	16:55	6	28.2	10	25.1	7	0.0	0	0	0	0	0

07-February -2024	17:15	850	24.7	14	21.7	6	0.0	0	0	0	0	0
07-February -2024	17:35	1058	24.9	15	22.7	7	0.0	0	0	0	0	0
07-February -2024	17:55	444	24.1	14	21.9	6	0.0	0	0	0	0	0
07-February -2024	18:15	624	23.8	14	21.0	6	0.0	0	0	0	0	0
07-February -2024	18:35	5422	23.6	13	20.2	6	0.0	0	0	0	0	0
07-February -2024	18:55	4547	23.5	13	19.6	6	0.0	0	0	0	0	0
07-February -2024	19:15	4600	72.4	13	65.9	6	0.0	0	0	0	0	0
07-February -2024	19:35	174	66.2	8	59.0	6	0.0	0	0	0	0	0
07-February -2024	19:55	179	66.5	9	58.3	6	0.0	0	0	0	0	0
07-February -2024	20:15	8	68.3	14	59.9	6	0.0	0	0	0	0	0
07-February -2024	20:35	27	68.0	14	59.4	6	0.0	0	0	0	0	0

07-February -2024	20:55	5	67.7	13	59.1	6	0.0	0	0	0	0	0
07-February -2024	21:15	10	67.4	13	58.6	5	0.0	0	0	0	0	0
07-February -2024	21:35	5854	67.4	13	58.2	5	0.0	0	0	0	0	0
07-February -2024	21:55	4871	67.0	9	57.7	5	0.0	0	0	0	0	0
07-February -2024	22:15	2743	96.5	70	76.6	66	0.0	0	0	0	0	0
07-February -2024	22:35	2784	99.2	99	84.2	90	0.0	0	0	0	0	0
07-February -2024	22:55	2426	100.2	124	87.7	114	0.0	0	0	0	0	0
07-February -2024	23:15	2399	100.7	140	90.1	131	0.0	0	0	0	0	0
07-February -2024	23:35	2378	101.7	139	91.1	130	0.0	0	0	0	0	0
07-February -2024	23:55	2365	102.6	198	93.7	194	0.0	0	0	0	0	0
08-February -2024	0:15	2359	103.0	209	96.1	205	0.0	0	0	0	0	0

08-February -2024	0:35	2353	103.6	209	96.7	204	0.0	0	0	0	0	0
08-February -2024	0:55	2345	103.8	209	97.5	205	0.0	0	0	0	0	0
08-February -2024	1:15	2338	103.1	208	97.9	204	0.0	0	0	0	0	0
08-February -2024	1:35	2331	103.5	209	98.5	204	0.0	0	0	0	0	0
08-February -2024	1:55	2329	104.2	212	99.1	206	0.0	0	0	0	0	0
08-February -2024	2:15	5	86.3	10	79.9	6	0.0	0	0	0	0	0
08-February -2024	2:35	2	74.6	10	70.9	5	0.0	0	0	0	0	0
08-February -2024	2:55	4	68.1	10	64.3	6	0.0	0	0	0	0	0
08-February -2024	3:15	4	65.0	10	59.5	6	0.0	0	0	0	0	0
08-February -2024	3:35	5	63.9	10	56.2	6	0.0	0	0	0	0	0
08-February -2024	3:55	5	63.2	9	53.9	6	0.0	0	0	0	0	0

08-February -2024	4:15	6	62.8	9	52.4	6	0.0	0	0	0	0	0
08-February -2024	4:35	5	62.4	9	51.3	6	0.0	0	0	0	0	0
08-February -2024	4:55	6	62.1	9	50.4	6	0.0	0	0	0	0	0
08-February -2024	5:15	7	61.9	9	49.7	6	0.0	0	0	0	0	0
08-February -2024	5:35	7	61.8	9	49.3	5	0.0	0	0	0	0	0
08-February -2024	5:55	8	61.6	9	48.9	6	0.0	0	0	0	0	0
08-February -2024	6:15	8	61.4	9	48.6	6	0.0	0	0	0	0	0
08-February -2024	6:35	9	61.3	9	48.2	6	0.0	0	0	0	0	0
08-February -2024	6:55	8	60.9	10	47.7	6	0.0	0	0	0	0	0
08-February -2024	7:15	9	60.6	10	47.3	5	0.0	0	0	0	0	0
08-February -2024	7:35	16	61.6	5	47.4	5	0.0	0	0	0	0	0

08-February -2024	7:55	4939	62.7	3	48.3	6	0.0	0	0	0	0	0
08-February -2024	8:15	4948	64.4	3	50.8	6	0.0	0	0	0	0	0
08-February -2024	8:35	4953	66.0	3	53.4	5	0.0	0	0	0	0	0
08-February -2024	8:55	4959	67.3	4	56.6	6	0.0	0	0	0	0	0
08-February -2024	9:15	4954	68.5	3	59.2	6	0.0	0	0	0	0	0
08-February -2024	9:35	4958	69.6	5	61.3	7	0.0	0	0	0	0	0
08-February -2024	9:55	4981	70.5	3	63.4	6	0.0	0	0	0	0	0
08-February -2024	10:15	4989	71.1	2	66.2	7	0.0	0	0	0	0	0
08-February -2024	10:35	4997	71.3	1	70.0	7	0.0	0	0	0	0	0
08-February -2024	10:55	5004	72.9	3	72.3	6	0.0	0	0	0	0	0
08-February -2024	11:15	5008	74.8	3	74.2	6	0.0	0	0	0	0	0

08-February -2024	11:35	5013	75.7	3	76.2	7	0.0	0	0	0	0	0
08-February -2024	11:55	5017	75.1	5	75.6	6	0.0	0	0	0	0	0
08-February -2024	12:15	5020	0.0	5	75.5	6	0.0	0	0	0	0	0
08-February -2024	12:35	5024	76.7	4	75.7	8	0.0	0	0	0	0	0
08-February -2024	12:55	5028	77.8	3	76.7	8	0.0	0	0	0	0	0
08-February -2024	13:15	22	73.0	5	76.8	7	0.0	0	0	0	0	0
08-February -2024	13:35	4933	87.9	5	69.2	8	0.0	0	0	0	0	0
08-February -2024	13:55	36	78.3	6	71.8	7	0.0	0	0	0	0	0
08-February -2024	14:15	37	81.3	5	76.9	7	0.0	0	0	0	0	0
08-February -2024	14:35	38	83.5	6	80.6	8	0.0	0	0	0	0	0
08-February -2024	14:55	14	84.0	9	82.7	7	0.0	0	0	0	0	0

08-February -2024	15:15	5855	83.3	9	83.7	6	0.0	0	0	0	0	0
08-February -2024	15:35	19	81.9	9	82.2	7	0.0	0	0	0	0	0
08-February -2024	15:55	5	82.1	10	79.4	6	0.0	0	0	0	0	0
08-February -2024	16:15	2	81.5	9	79.4	5	0.0	0	0	0	0	0
08-February -2024	16:35	3	81.9	11	77.5	5	0.0	0	0	0	0	0
08-February -2024	16:55	2	81.8	9	77.5	6	0.0	0	0	0	0	0
08-February -2024	17:15	3	80.9	9	78.0	6	0.0	0	0	0	0	0
08-February -2024	17:35	3	78.4	11	75.2	5	0.0	0	0	0	0	0
08-February -2024	17:55	2	77.1	13	72.3	5	0.0	0	0	0	0	0
08-February -2024	18:15	2	75.5	12	69.6	5	0.0	0	0	0	0	0
08-February -2024	18:35	2	74.4	11	67.3	6	0.0	0	0	0	0	0

08-February -2024	18:55	3	73.3	11	65.4	5	0.0	0	0	0	0	0
08-February -2024	19:15	557	72.9	10	63.8	5	0.0	0	0	0	0	0
08-February -2024	19:35	4	70.3	9	62.3	6	0.0	0	0	0	0	0
08-February -2024	19:55	5067	70.4	9	61.5	5	0.0	0	0	0	0	0
08-February -2024	20:15	4605	86.8	47	66.3	46	0.0	0	0	0	0	0
08-February -2024	20:35	3293	92.4	99	75.3	91	0.0	0	0	0	0	0
08-February -2024	20:55	3339	93.4	90	76.3	84	0.0	0	0	0	0	0
08-February -2024	21:15	3377	93.3	86	76.9	81	0.0	0	0	0	0	0
08-February -2024	21:35	3398	93.0	84	78.0	79	0.0	0	0	0	0	0
08-February -2024	21:55	3399	93.3	84	77.9	78	0.0	0	0	0	0	0
08-February -2024	22:15	3400	93.7	100	80.6	96	0.0	0	0	0	0	0

08-February -2024	22:35	3391	94.6	102	81.3	96	0.0	0	0	0	0	0
08-February -2024	22:55	3381	95.5	104	81.9	96	0.0	0	0	0	0	0
08-February -2024	23:15	3380	95.3	106	82.3	98	0.0	0	0	0	0	0
08-February -2024	23:35	3373	95.7	107	82.9	100	0.0	0	0	0	0	0
08-February -2024	23:55	3369	96.2	108	83.1	100	0.0	0	0	0	0	0
09-February -2024	0:15	3365	94.3	111	82.6	102	0.0	0	63.77	1.5	0	0
09-February -2024	0:35	3361	94.6	110	82.8	103	0.0	0	63.52	1.5	0	0
09-February -2024	0:55	3358	95.4	225	88.9	222	0.0	0	0	0	0	0
09-February -2024	1:15	3358	96.7	233	91.2	229	0.0	0	42.89	2	0	0
09-February -2024	1:35	3355	96.5	236	91.2	232	0.0	0	41.97	2	0	0
09-February -2024	1:55	3351	95.9	238	90.7	236	0.0	0	41.99	2	0	0

09-February -2024	2:15	3348	95.7	217	89.4	213	0.0	0	47.05	2	0	0
09-February -2024	2:35	3347	96.3	218	89.8	215	0.0	0	47.61	2	0	0
09-February -2024	2:55	3344	96.6	217	90.4	214	0.0	0	47.1	2	0	0
09-February -2024	3:15	3341	96.6	218	90.2	214	0.0	0	46.91	2	0	0
09-February -2024	3:35	3338	97.3	217	90.2	215	0.0	0	46.95	2	0	0
09-February -2024	3:55	3337	97.1	214	90.4	213	0.0	0	46.3	2	0	0
09-February -2024	4:15	3334	97.2	216	90.2	213	0.0	0	46.09	2	0	0
09-February -2024	4:35	3333	95.8	216	90.1	212	0.0	0	45.97	2	0	0
09-February -2024	4:51	3332	97	216	90.5	213	76	0	45.5	1.5	1424	1257717
09-February -2024	4:55	3333	97.5	216	90.6	213	0.0	0	45.5	2	0	0
09-February -2024	5:15	3332	96.4	217	91.7	215	0.0	0	45.6	2	0	0

09-February -2024	5:35	3329	95.9	216	91.4	214	0.0	0	45.72	2	0	0
09-February -2024	5:55	3326	97.6	217	91.9	213	0.0	0	45.26	2	0	0
09-February -2024	6:00	3327	98.1	196	92.4	213	76	0	45.46	1.5	1446	1278961
09-February -2024	6:15	3326	98.5	219	92.6	215	0.0	0	45.9	2	0	0
09-February -2024	6:35	3324	99.0	217	93.2	215	0.0	0	0	0	0	0
09-February -2024	6:55	3324	99.6	216	93.5	213	0.0	0	0	0	0	0
09-February -2024	7:15	3323	99.1	216	93.6	213	0.0	0	0	0	0	0
09-February -2024	7:35	3322	98.8	214	93.9	216	0.0	0	0	0	0	0
09-February -2024	7:55	3319	97.3	213	93.9	216	0.0	0	0	0	0	0
09-February -2024	8:15	3318	97.8	211	94.3	217	0.0	0	0	0	0	0
09-February -2024	8:35	3317	98.4	212	95.1	216	0.0	0	45.33	0	0	0

09-February -2024	8:55	3315	98.9	213	95.5	218	0.0	0	45.56	0	0	0
09-February -2024	9:15	3314	99.7	213	96.0	218	0.0	0	45.48	1.5	0	0
09-February -2024	9:36	3313	100.0	213	96.6	213	76	0	45.29	1.5	879	1270831
09-February -2024	9:55	3312	100.2	212	97.1	216	0.0	0	45.37	1.5	0	0
09-February -2024	10:15	3312	100.2	213	97.4	215	0.0	0	45.44	1.5	0	0
09-February -2024	10:35	3310	100.6	213	97.6	216	0.0	0	45.49	1.5	0	0
09-February -2024	10:42	3310	100.1	214	88.8	216	77	0	45.66	1.5	1435	1283206
09-February -2024	10:55	3310	101.1	68	88.9	57	0.0	0	0	0	0	0
09-February -2024	11:15	3308	101.5	212	97.1	215	0.0	0	39.84	1.5	0	0
09-February -2024	11:35	3306	101.1	198	98.1	200	0.0	0	51.36	1.5	0	0
09-February -2024	11:55	3306	100.7	197	98.3	198	0.0	0	51.17	1.5	0	0

09-February -2024	12:15	3305	100.9	198	98.4	199	0.0	0	51.31	1.5	0	0
09-February -2024	12:35	3304	101.6	198	99.4	200	0.0	0	50.72	1.5	0	0
09-February -2024	12:39	3303	101.5	201	99.6	199	76	0	51.05	1.5	1446	1306491
09-February -2024	12:55	3301	101.5	200	99.6	203	0.0	0	0	0	0	0
09-February -2024	13:15	678	97.6	20	90.0	18	0.0	0	0.04	0	0	0
09-February -2024	13:35	12	91.1	11	85.0	9	0.0	0	0.04	0	0	0
09-February -2024	13:55	13	87.4	9	88.5	8	0.0	0	0.03	0	0	0
09-February -2024	14:15	13	88.4	9	92.1	8	0.0	0	0.02	0	0	0
09-February -2024	14:35	14	89.7	11	94.2	7	0.0	0	0	0	0	0
09-February -2024	14:55	15	90.9	11	96.3	6	0.0	0	0	0	0	0
09-February -2024	15:15	15	90.9	14	97.0	6	0.0	0	0	0	0	0

09-February -2024	15:35	4	90.4	14	96.8	6	0.0	0	0	0	0	0
09-February -2024	15:55	5	89.1	15	94.6	7	0.0	0	0	0	0	0
09-February -2024	16:15	5	88.6	14	91.9	6	0.0	0	0	0	0	0
09-February -2024	16:35	5	88.1	13	90.0	5	0.0	0	0	0	0	0
09-February -2024	16:55	5	88.0	13	89.3	6	0.0	0	0	0	0	0
09-February -2024	17:15	3	86.2	14	88.3	5	0.0	0	0	0	0	0
09-February -2024	17:35	529	80.8	17	82.4	5	0.0	0	0	0	0	0
09-February -2024	17:55	690	80.7	18	80.1	5	0.0	0	0	0	0	0
09-February -2024	18:15	3257	80.5	17	78.5	5	0.0	0	0	0	0	0
09-February -2024	18:35	4918	79.7	15	76.5	5	0.0	0	0	0	0	0
09-February -2024	18:55	4934	78.7	14	74.9	5	0.0	0	0	0	0	0

09-February -2024	19:15	4960	77.6	12	73.4	5	0.0	0	0	0	0	0
09-February -2024	19:35	1872	75.2	11	72.0	5	0.0	0	0	0	0	0
09-February -2024	19:55	9	71.0	16	66.2	5	0.0	0	0	0	0	0
09-February -2024	20:15	8	69.8	15	64.4	5	0.0	0	0	0	0	0
09-February -2024	20:35	23	70.8	14	64.2	5	0.0	0	0	0	0	0
09-February -2024	20:55	6109	71.4	15	64.0	6	0.0	0	0	0	0	0
09-February -2024	21:15	1164	70.7	10	63.4	5	0.0	0	0	0	0	0
09-February -2024	21:35	6125	70.0	9	61.3	5	0.0	0	0	0	0	0
09-February -2024	21:55	3578	70.4	8	59.7	6	0.0	0	0	0	0	0
09-February -2024	22:15	9	68.0	9	59.9	5	0.0	0	0	0	0	0
09-February -2024	22:35	5006	68.4	9	58.1	5	0.0	0	0	0	0	0

09-February -2024	22:55	5009	68.0	9	57.0	6	0.0	0	0	0	0	0
09-February -2024	23:15	5013	67.7	9	56.1	6	0.0	0	0	0	0	0
09-February -2024	23:35	5016	67.5	9	55.3	5	0.0	0	0	0	0	0
09-February -2024	23:55	5020	67.2	9	54.6	6	0.0	0	0	0	0	0
10-February -2024	0:15	5022	67.0	9	0.0	6	0.0	0	0	0	0	0
10-February -2024	0:35	5026	66.7	9	0.0	5	0.0	0	0	0	0	0
10-February -2024	0:55	5028	66.4	9	0.0	5	0.0	0	0	0	0	0
10-February -2024	1:15	5031	66.1	9	0.0	5	0.0	0	0	0	0	0
10-February -2024	1:35	5034	65.8	9	0.0	6	0.0	0	0	0	0	0
10-February -2024	1:55	5037	65.6	9	0.0	6	0.0	0	0	0	0	0
10-February -2024	2:15	5039	65.4	9	0.0	5	0.0	0	0	0	0	0

10-February -2024	2:35	5041	65.1	9	0.0	6	0.0	0	0	0	0	0
10-February -2024	2:55	5044	64.9	9	0.0	5	0.0	0	0	0	0	0
10-February -2024	3:15	5047	64.7	7	0.0	5	0.0	0	0	0	0	0
10-February -2024	3:35	5049	64.5	8	0.0	6	0.0	0	0	0	0	0
10-February -2024	3:55	5052	64.2	10	0.0	5	0.0	0	0	0	0	0
10-February -2024	4:15	5054	63.9	10	0.0	5	0.0	0	0	0	0	0
10-February -2024	4:35	5057	63.5	11	0.0	5	0.0	0	0	0	0	0
10-February -2024	4:55	5059	63.2	10	0.0	6	0.0	0	0	0	0	0
10-February -2024	5:15	5060	63.0	10	0.0	5	0.0	0	0	0	0	0
10-February -2024	5:35	5062	62.6	11	0.0	6	0.0	0	0	0	0	0
10-February -2024	5:55	5065	62.0	12	0.0	6	0.0	0	0	0	0	0

10-February -2024	6:15	5066	61.7	10	0.0	6	0.0	0	0	0	0	0
10-February -2024	6:35	5069	61.6	8	0.0	6	0.0	0	0	0	0	0
10-February -2024	6:55	5071	61.3	9	0.0	5	0.0	0	0	0	0	0
10-February -2024	7:15	5072	61.2	9	0.0	6	0.0	0	0	0	0	0
10-February -2024	7:35	5075	61.2	8	0.0	6	0.0	0	0	0	0	0
10-February -2024	7:55	5077	61.4	6	0.0	6	0.0	0	0	0	0	0
10-February -2024	8:15	5079	62.1	4	0.0	6	0.0	0	0	0	0	0
10-February -2024	8:35	5080	62.9	2	0.0	6	0.0	0	0	0	0	0
10-February -2024	8:55	5082	63.8	0	0.0	6	0.0	0	0	0	0	0
10-February -2024	9:15	5084	65.5	1	0.0	6	0.0	0	0	0	0	0
10-February -2024	9:35	5085	67.3	7	0.0	6	0.0	0	0	0	0	0

10-February -2024	9:55	5087	68.4	5	0.0	6	0.0	0	0	0	0	0
10-February -2024	10:15	5049	77.6	1	0.0	6	0.0	0	0	0	0	0
10-February -2024	10:35	4387	87.7	260	0.0	6	0.0	0	0	0	0	0
10-February -2024	10:55	3423	91.2	480	0.0	6	0.0	0	0	0	0	0
10-February -2024	11:15	3472	92.5	371	0.0	6	0.0	0	0	0	0	0
10-February -2024	11:35	4758	91.4	3	0.0	6	0.0	0	0	0	0	0
10-February -2024	11:55	4949	87.7	6	0.0	6	0.0	0	0	0	0	0
10-February -2024	12:15	4999	86.3	7	0.0	6	0.0	0	0	0	0	0
10-February -2024	12:35	5023	85.8	6	0.0	7	0.0	0	0	0	0	0
10-February -2024	12:55	5039	86.2	8	0.0	9	0.0	0	0	0	0	0
10-February -2024	13:15	5049	87.6	11	0.0	7	0.0	0	0	0	0	0

10-February -2024	13:35	5057	87.5	15	0.0	7	0.0	0	0	0	0	0
10-February -2024	13:55	5062	86.94	15.82	0.0	5.9	0.0	0	0	0	0	0
10-February -2024	14:15	5068	86.0	14	0.0	5	0.0	0	0	0	0	0
10-February -2024	14:35	5073	85.3	12	0.0	5	0.0	0	0	0	0	0
10-February -2024	14:55	5078	85.0	10	0.0	6	0.0	0	0	0	0	0
10-February -2024	15:15	5081	85.2	7	0.0	5	0.0	0	0	0	0	0
10-February -2024	15:35	5085	85.0	9	0.0	6	0.0	0	0	0	0	0
10-February -2024	15:55	5087	85.2	8	0.0	6	0.0	0	0	0	0	0
10-February -2024	16:15	5089	84.0	13	0.0	5	0.0	0	0	0	0	0
10-February -2024	16:35	5091	82.6	12	0.0	6	0.0	0	0	0	0	0
10-February -2024	16:55	5094	81.3	10	0.0	5	0.0	0	0	0	0	0

10-February -2024	17:15	5096	80.3	9	0.0	5	0.0	0	0	0	0	0
10-February -2024	17:35	5099	79.4	8	0.0	5	0.0	0	0	0	0	0
10-February -2024	17:55	5100	78.3	9	0.0	5	0.0	0	0	0	0	0
10-February -2024	18:15	5103	77.4	9	0.0	5	0.0	0	0	0	0	0
10-February -2024	18:35	5105	76.6	9	0.0	5	0.0	0	0	0	0	0
10-February -2024	18:55	5107	75.8	9	0.0	5	0.0	0	0	0	0	0
10-February -2024	19:15	5108	75.2	8	0.0	5	0.0	0	0	0	0	0
10-February -2024	19:35	5109	74.5	8	0.0	6	67.0	0	0	0	0	0
10-February -2024	19:55	5112	74.0	7	0.0	0	66.4	0	0	0	0	0
10-February -2024	20:15	5113	73.4	8	0.0	0	65.7	0	0	0	0	0
10-February -2024	20:35	5114	72.9	8	0.0	0	65.0	0	0	0	0	0

10-February -2024	20:55	5115	72.3	8	0.0	0	64.4	0	0	0	0	0
10-February -2024	21:15	5117	71.9	9	0.0	0	63.6	0	0	0	0	0
10-February -2024	21:35	5119	71.3	9	0.0	1	63.1	0	0	0	0	0
10-February -2024	21:55	5119	70.8	9	0.0	0	62.2	0	0	0	0	0
10-February -2024	22:15	5121	70.4	9	0.0	0	61.5	0	0	0	0	0
10-February -2024	22:35	5123	69.9	9	0.0	0	60.9	0	0	0	0	0
10-February -2024	22:55	5123	69.5	9	0.0	0	60.6	0	0	0	0	0
10-February -2024	23:15	5125	69.2	8	0.0	0	60.4	0	0	0	0	0
10-February -2024	23:35	5126	68.7	9	0.0	1	59.7	0	0	0	0	0
10-February -2024	23:55	5127	68.5	9	0.0	1	59.4	0	0	0	0	0
11-February -2024	0:15	5128	68.5	7	0.0	0	59.5	0	0	0	0	0

11-February -2024	0:35	5129	68.3	6	0.0	1	59.5	0	0	0	0	0
11-February -2024	0:55	5131	68.3	5	0.0	1	59.6	0	0	0	0	0
11-February -2024	1:15	5132	68.2	5	0.0	1	59.8	0	0	0	0	0
11-February -2024	1:35	5132	68.3	5	0.0	0	60.3	0	0	0	0	0
11-February -2024	1:55	5133	68.2	5	0.0	0	60.2	0	0	0	0	0
11-February -2024	2:15	5134	68.2	5	0.0	0	60.2	0	0	0	0	0
11-February -2024	2:35	5136	68.1	5	0.0	1	60.1	0	0	0	0	0
11-February -2024	2:55	5137	68.2	5	0.0	1	60.4	0	0	0	0	0
11-February -2024	3:15	5137	68.1	5	0.0	1	60.3	0	0	0	0	0
11-February -2024	3:35	5139	67.9	6	0.0	0	60.1	0	0	0	0	0
11-February -2024	3:55	5139	67.5	7	0.0	1	59.0	0	0	0	0	0

11-February -2024	4:15	5140	67.4	7	0.0	0	58.4	0	0	0	0	0
11-February -2024	4:35	5142	67.3	7	0.0	0	58.3	0	0	0	0	0
11-February -2024	4:55	5142	67.2	7	0.0	0	58.2	0	0	0	0	0
11-February -2024	5:15	5143	67.2	6	0.0	0	58.2	0	0	0	0	0
11-February -2024	5:35	5144	67.0	7	0.0	1	58.1	0	0	0	0	0
11-February -2024	5:55	5145	66.9	7	0.0	0	58.0	0	0	0	0	0
11-February -2024	6:15	5146	66.9	6	0.0	0	58.0	0	0	0	0	0
11-February -2024	6:35	5147	66.9	6	0.0	0	58.0	0	0	0	0	0
11-February -2024	6:55	5147	66.8	6	0.0	0	58.1	0	0	0	0	0
11-February -2024	7:15	5148	66.8	63	0.0	0	58.1	0	0	0	0	0
11-February -2024	7:35	5149	66.8	11	0.0	0	58.1	0	0	0	0	0

11-February -2024	7:55	5150	67.0	16	0.0	0	58.3	0	0	0	0	0
11-February -2024	8:15	5151	67.5	7	0.0	1	59.0	0	0	0	0	0
11-February -2024	8:35	5152	68.3	5	0.0	1	59.9	0	0	0	0	0
11-February -2024	8:55	5152	68.7	5	0.0	1	60.7	0	0	0	0	0
11-February -2024	9:15	5154	68.8	6	0.0	1	60.9	0	0	0	0	0
11-February -2024	9:35	5155	68.8	6	0.0	1	61.0	0	0	0	0	0
11-February -2024	9:55	5155	68.9	6	0.0	1	61.3	0	0	0	0	0
11-February -2024	10:15	5156	69.5	5	0.0	0	61.5	0	0	0	0	0
11-February -2024	10:35	5156	69.9	5	0.0	1	61.8	0	0	0	0	0
11-February -2024	10:55	5157	70.5	5	0.0	1	62.6	0	0	0	0	0
11-February -2024	11:15	5158	70.8	6	0.0	1	63.0	0	0	0	0	0

11-February -2024	11:35	5158	71.0	8	0.0	1	63.2	0	0	0	0	0
11-February -2024	11:55	5159	70.2	13	0.0	1	61.6	0	0	0	0	0
11-February -2024	12:14	5160	69.8	12	0.0	0	61.1	0	0	0	0	0
11-February -2024	12:34	5161	70.5	9	0.0	0	62.1	0	0	0	0	0
11-February -2024	12:54	4342	82.5	331	0.0	1	63.0	0	0	0	0	0
11-February -2024	13:14	4156	86.8	329	0.0	1	63.5	0	0	0	0	0
11-February -2024	13:34	4986	85.3	12	0.0	1	63.5	0	0	0	0	0
11-February -2024	13:54	5086	81.1	11	0.0	1	63.6	0	0	0	0	0
11-February -2024	14:14	5113	79.0	10	0.0	1	63.9	0	0	0	0	0
11-February -2024	14:34	5126	77.3	10	0.0	1	64.1	0	0	0	0	0
11-February -2024	14:54	5135	75.8	11	0.0	1	64.3	0	0	0	0	0

11-February -2024	15:14	5140	74.5	11	0.0	1	64.6	0	0	0	0	0
11-February -2024	15:34	5144	73.9	10	0.0	1	65.0	0	0	0	0	0
11-February -2024	15:54	5147	73.3	10	0.0	1	64.8	0	0	0	0	0
11-February -2024	16:14	5149	72.6	9	0.0	1	64.3	0	0	0	0	0
11-February -2024	16:34	5151	72.1	9	0.0	1	63.9	0	0	0	0	0
11-February -2024	16:54	4694	81.1	130	0.0	128	67.7	0	0	0	0	0
11-February -2024	17:14	3560	86.5	376	0.0	136	68.3	0	0	0	0	0
11-February -2024	17:34	3934	89.0	52	0.0	137	65.8	0	0	0	0	0
11-February -2024	17:54	16	76.1	6	0.0	1	62.5	0	0	0	0	0
11-February -2024	18:14	11	73.7	6	0.0	1	62.2	0	0	0	0	0
11-February -2024	18:34	7	71.3	11	0.0	0	61.9	0	0	0	0	0

11-February -2024	18:54	8	71.2	13	0.0	0	61.7	0	0	0	0	0
11-February -2024	19:14	8	73.2	11	0.0	0	61.3	0	0	0	0	0
11-February -2024	19:34	3	66.1	323	0.0	0	60.8	0	0	0	0	0
11-February -2024	19:54	3	66.3	0	0.0	0	60.4	0	0	0	0	0
11-February -2024	20:14	2	66.1	694	0.0	0	59.3	0	0	0	0	0
11-February -2024	20:34	3	66.0	770	0.0	0	58.6	0	0	0	0	0
11-February -2024	20:54	3	66.2	0	0.0	0	58.0	0	0	0	0	0
11-February -2024	21:14	10	68.4	44	0.0	0	57.5	0	0	0	0	0
11-February -2024	21:34	11	67.9	0	0.0	1	57.4	0	0	0	0	0
11-February -2024	21:54	5143	68.2	0	0.0	0	57.1	0	0	0	0	0
11-February -2024	22:14	5146	67.7	0	0.0	1	56.9	0	0	0	0	0

11-February -2024	22:34	5148	67.3	0	0.0	1	56.7	0	0	0	0	0
11-February -2024	22:54	5149	66.8	0	0.0	1	56.5	0	0	0	0	0
11-February -2024	23:14	5151	66.3	0	0.0	0	56.1	0	0	0	0	0
11-February -2024	23:34	5153	65.7	0	0.0	0	55.5	0	0	0	0	0
11-February -2024	23:54	5154	65.4	0	0.0	1	55.1	0	0	0	0	0
12-February -2024	0:15	5156	65.1	0	0.0	1	54.7	0	0	0	0	0
12-February -2024	0:35	5157	64.8	0	0.0	1	54.4	0	0	0	0	0
12-February -2024	0:55	4648	82.6	65	0.0	59	57.2	0	0	0	0	0
12-February -2024	1:15	2392	88.4	236	0.0	205	67.4	0	0	0	0	0
12-February -2024	1:35	2007	90.5	239	0.0	202	70.3	0	0	0	0	0
12-February -2024	1:55	1856	92.2	242	0.0	213	72.1	0	0	0	0	0

12-February -2024	2:15	1819	93.4	226	0.0	205	73.3	0	0	0	0	0
12-February -2024	2:35	1998	93.1	203	0.0	183	72.6	0	0	0	0	0
12-February -2024	2:55	2036	93.4	206	0.0	190	72.7	0	0	0	0	0
12-February -2024	3:15	2101	91.7	200	0.0	181	72.0	0	0	0	0	0
12-February -2024	3:35	2099	92.9	199	0.0	182	71.4	0	0	0	0	0
12-February -2024	3:55	2110	93.1	197	0.0	181	72.0	0	0	0	0	0
12-February -2024	4:15	2091	94.0	196	0.0	180	72.7	0	0	0	0	0
12-February -2024	4:35	2067	93.9	199	0.0	181	72.6	0	0	0	0	0
12-February -2024	4:55	2040	95.2	343	0.0	178	74.7	0	0	0	0	0
12-February -2024	5:15	2040	95.3	177	0.0	176	74.6	0	0	0	0	0
12-February -2024	5:35	2011	95.8	184	0.0	173	74.5	0	0	0	0	0

12-February -2024	5:55	2085	92.5	194	0.0	180	72.7	0	0	0	0	0
12-February -2024	6:15	2083	92.2	180	0.0	180	71.1	0	0	0	0	0
12-February -2024	6:35	131	34.9	6	0.0	10	22.3	0	0	0	0	0
12-February -2024	6:55	140	33.7	2	0.0	10	23.1	0	0	0	0	0
12-February -2024	7:15	3887	30.9	117	0.0	87	21.0	0	0	0	0	0
12-February -2024	7:35	3922	86.1	113	0.0	98	68.6	0	0	0	0	0
12-February -2024	7:55	4962	79.7	54	0.0	1	61.1	0	0	0	0	0
12-February -2024	8:15	5600	75.6	119	0.0	1	58.5	0	0	0	0	0
12-February -2024	8:35	4632	72.7	153	0.0	1	57.7	0	0	0	0	0
12-February -2024	8:55	4632	71.9	58	0.0	1	57.8	0	0	0	0	0
12-February -2024	9:15	4635	71.6	42	0.0	1	58.5	0	0	0	0	0

12-February -2024	9:35	4659	72.1	12	0.0	1	60.2	0	0	0	0	0
12-February -2024	9:55	4663	72.8	0	0.0	1	62.0	0	0	0	0	0
12-February -2024	10:15	4675	74.1	0	0.0	1	63.6	0	0	0	0	0
12-February -2024	10:35	4724	75.0	0	0.0	1	64.2	0	0	0	0	0
12-February -2024	10:55	4698	75.2	0	0.0	1	65.7	0	0	0	0	0
12-February -2024	11:15	4745	75.6	0	0.0	1	66.8	0	0	0	0	0
12-February -2024	11:35	4730	75.7	0	0.0	1	67.7	0	0	0	0	0
12-February -2024	11:55	4932	76.3	0	0.0	1	68.0	0	0	0	0	0
12-February -2024	12:14	4851	77.5	0	0.0	1	68.9	0	0	0	0	0
12-February -2024	12:34	4707	78.5	0	0.0	1	70.0	0	0	0	0	0
12-February -2024	12:54	4768	79.7	0	0.0	2	72.5	0	0	0	0	0

12-February -2024	13:14	4756	81.9	0	0.0	1	76.0	0	0	0	0	0
12-February -2024	13:34	4745	83.2	0	0.0	1	79.3	0	0	0	0	0
12-February -2024	13:54	4755	83.4	0	0.0	1	82.5	0	0	0	0	0
12-February -2024	14:14	3976	86.2	0	0.0	4	83.9	0	0	0	0	0
12-February -2024	14:34	3015	89.2	196	0.0	186	83.5	0	0	0	0	0
12-February -2024	14:54	3374	90.1	175	0.0	164	83.6	0	0	0	0	0
12-February -2024	15:14	1966	90.8	159	0.0	139	83.8	0	0	0	0	0
12-February -2024	15:34	4851	91.8	27	0.0	14	85.4	0	0	0	0	0
12-February -2024	15:54	5416	90.0	15	0.0	1	86.6	0	0	0	0	0
12-February -2024	16:14	5592	89.1	15	0.0	0	82.4	0	0	0	0	0
12-February -2024	16:34	5207	88.3	14	0.0	0	79.6	0	0	0	0	0

12-February -2024	16:54	4825	87.1	14	0.0	0	78.1	0	0	0	0	0
12-February -2024	17:14	4997	85.5	15	0.0	0	76.6	0	0	0	0	0
12-February -2024	17:34	5014	83.7	15	0.0	0	75.0	0	0	0	0	0
12-February -2024	17:54	4991	81.5	16	0.0	0	73.3	0	0	0	0	0
12-February -2024	18:14	4859	79.6	15	0.0	0	71.4	0	0	0	0	0
12-February -2024	18:34	4890	78.2	13	0.0	0	69.8	0	0	0	0	0
12-February -2024	18:54	4928	77.5	11	0.0	1	68.6	0	0	0	0	0
12-February -2024	19:14	4940	76.6	11	0.0	1	67.7	0	0	0	0	0
12-February -2024	19:34	4957	75.8	11	0.0	1	67.0	0	0	0	0	0
12-February -2024	19:54	4849	74.5	10	0.0	0	66.1	0	0	0	0	0
12-February -2024	20:14	5798	74.4	9	0.0	1	65.4	0	0	0	0	0

12-February -2024	20:34	3972	86.9	23	0.0	0	64.7	0	0	0	0	0
12-February -2024	20:54	3609	92.5	204	0.0	170	82.0	0	0	0	0	0
12-February -2024	21:14	4196	93.4	76	0.0	63	84.4	0	0	0	0	0
12-February -2024	21:34	2916	100.4	318	0.0	267	92.5	0	0	0	0	0
12-February -2024	21:54	2967	103.9	290	0.0	240	91.8	0	0	0	0	0
12-February -2024	22:14	2604	107.0	378	0.0	328	98.9	0	0	0	0	0
12-February -2024	22:34	2266	111.2	437	0.0	387	106.2	0	0	0	0	0
12-February -2024	22:54	2243	114.5	421	0.0	369	110.7	0	0	0	0	0
12-February -2024	23:14	2067	117.9	477	0.0	423	116.4	0	0	0	0	0
12-February -2024	23:34	2063	121.2	456	0.0	399	121.2	0	0	0	0	0
12-February -2024	23:54	2042	124.3	472	0.0	414	125.6	0	0	0	0	0

13-February -2024	3:35	3801	124.1	217	0.0	208	113.7	0	0	0	0	0
13-February -2024	3:55	3931	122.9	215	0.0	206	112.0	0	0	0	0	0
13-February -2024	4:15	4009	122.0	219	0.0	209	114.2	0	0	0	0	0
13-February -2024	4:35	4063	121.0	220	0.0	209	110.1	0	0	0	0	0
13-February -2024	4:55	4104	119.7	221	0.0	210	108.4	0	0	0	0	0
13-February -2024	5:15	4134	119.4	225	0.0	216	109.0	0	0	0	0	0
13-February -2024	5:35	4157	118.8	224	0.0	214	109.6	0	0	0	0	0
13-February -2024	5:55	4176	116.6	224	0.0	216	110.2	0	0	0	0	0
13-February -2024	6:15	4192	116.9	225	0.0	216	110.4	0	0	0	0	0
13-February -2024	6:35	4206	117.8	225	0.0	215	109.4	0	0	0	0	0
13-February -2024	6:55	4218	116.7	226	0.0	216	108.8	0	0	0	0	0

13-February -2024	7:15	4227	115.1	225	0.0	217	108.6	0	0	0	0	0
13-February -2024	7:35	4235	116.3	224	0.0	217	107.9	0	0	0	0	0
13-February -2024	7:55	4242	116.9	223	0.0	217	107.9	0	0	0	0	0
13-February -2024	8:15	4247	115.9	221	0.0	219	109.7	0	0	0	0	0
13-February -2024	8:35	4358	115.1	127	0.0	128	110.0	0	0	0	0	0
13-February -2024	8:55	5	95.4	4	0.0	1	94.7	0	0	0	0	0
13-February -2024	9:15	6	84.2	5	0.0	1	85.0	0	0	0	0	0
13-February -2024	9:35	6	78.8	5	0.0	0	80.6	0	0	0	0	0
13-February -2024	9:55	6	76.8	5	0.0	1	78.9	0	0	0	0	0
13-February -2024	10:15	7	77.1	4	0.0	1	77.7	0	0	0	0	0
13-February -2024	10:35	7	78.4	6	0.0	1	77.2	0	0	0	0	0

13-February -2024	10:55	7	79.5	7	0.0	1	77.0	0	0	0	0	0
13-February -2024	11:15	8	80.9	10	0.0	0	76.7	0	0	0	0	0
13-February -2024	11:35	8	82.2	9	0.0	1	77.0	0	0	0	0	0
13-February -2024	11:55	7	81.6	10	0.0	1	76.7	0	0	0	0	0
13-February -2024	12:15	8	81.6	9	0.0	1	77.1	0	0	0	0	0
13-February -2024	12:35	8	82.3	10	0.0	1	78.1	0	0	0	0	0
13-February -2024	12:55	3	84.0	14	0.0	2	80.6	0	0	0	0	0
13-February -2024	13:15	4705	81.4	19	0.0	2	84.5	0	0	0	0	0
13-February -2024	13:35	6086	85.3	22	0.0	2	87.5	0	0	0	0	0
13-February -2024	13:55	6364	86.7	14	0.0	2	91.0	0	0	0	0	0
13-February -2024	14:15	4580	105.7	193	0.0	178	101.0	0	0	0	0	0

13-February -2024	14:35	4513	107.2	236	0.0	224	105.7	0	0	0	0	0
13-February -2024	14:55	4480	107.5	239	0.0	223	107.1	0	0	0	0	0
13-February -2024	15:15	4458	107.6	240	0.0	224	108.2	0	0	0	0	0
13-February -2024	15:35	4443	108.6	269	0.0	257	110.2	0	0	0	0	0
13-February -2024	15:55	4429	108.1	280	0.0	263	107.4	0	0	0	0	0
13-February -2024	16:15	4416	109.0	271	0.0	261	107.5	0	0	0	0	0
13-February -2024	16:35	4405	109.1	270	0.0	257	107.9	0	0	0	0	0
13-February -2024	16:55	4399	109.1	279	0.0	267	107.9	0	0	0	0	0
13-February -2024	17:15	4393	108.3	287	0.0	272	107.6	0	0	0	0	0
13-February -2024	17:35	4389	110.1	287	0.0	273	108.2	0	0	0	0	0
13-February -2024	17:55	4385	109.9	289	0.0	275	108.3	0	0	0	0	0

13-February -2024	18:15	4382	109.8	290	0.0	276	108.4	0	0	0	0	0
13-February -2024	18:35	4379	109.2	291	0.0	279	108.0	0	0	0	0	0
13-February -2024	18:55	4376	110.1	290	0.0	280	107.6	0	0	0	0	0
13-February -2024	19:15	4373	109.7	292	0.0	283	108.0	0	0	0	0	0
13-February -2024	19:35	4371	109.1	276	0.0	266	106.5	0	0	0	0	0
13-February -2024	19:55	4368	108.6	274	0.0	265	106.4	0	0	0	0	0
13-February -2024	20:15	4366	109.2	235	0.0	226	105.2	0	0	0	0	0
13-February -2024	20:35	4364	112.6	222	0.0	209	104.6	0	79.4	2	0	0
13-February -2024	20:55	4362	113.9	218	0.0	205	105.0	0	78.5	2	0	0
13-February -2024	21:15	4360	114.0	217	0.0	205	103.8	0	77.99	2	0	0
13-February -2024	21:35	4358	114.2	214	0.0	203	102.6	0	76.44	2	0	0

13-February -2024	21:55	4357	114.2	215	0.0	203	102.4	0	75.3	2	0	0
13-February -2024	22:15	4354	114.4	214	0.0	203	103.3	0	76.15	2	0	0
13-February -2024	22:25	4351	114.4	215	0.0	204	103	0	76	1.5	1808	1558562
13-February -2024	22:35	4353	114.4	214	0.0	202	103.2	0	75.85	2	0	0
13-February -2024	22:45	4350	114.5	215	0.0	204	103	0	76	1.5	1808	1515092
13-February -2024	22:55	4352	114.5	214	0.0	203	103.2	0	76.04	2	0	0
13-February -2024	23:15	4350	114.0	213	0.0	203	102.6	0	76.28	2	0	0
13-February -2024	23:35	4348	113.7	212	0.0	204	101.7	0	0.14	0	0	0
13-February -2024	23:55	4346	114.0	213	0.0	204	101.6	0	0.14	0	0	0
14-February -2024	0:15	4345	113.1	213	0.0	204	101.8	0	0.13	0	0	0
14-February -2024	0:35	4343	114.6	212	0.0	204	101.9	0	0.13	0	0	0

14-February -2024	0:55	4340	114.8	213	0.0	204	102.3	0	0.12	0	0	0
14-February -2024	1:15	4342	114.8	212	0.0	204	102.5	0	0.11	0	0	0
14-February -2024	1:35	4341	114.9	216	0.0	204	102.7	0	76.41	0	0	0
14-February -2024	1:55	4339	113.7	217	0.0	205	103.2	0	76.7	0	0	0
14-February -2024	2:15	4338	113.7	217	0.0	204	103.1	0	76.42	0	0	0
14-February -2024	2:30	4340	114.3	214	0.0	204	110	0	76	1.5	1779	1547985
14-February -2024	2:35	4338	114.4	214	0.0	205	102.6	0	0.1	0	0	0
14-February -2024	2:55	4338	114.1	213	0.0	205	102.4	0	0.12	0	0	0
14-February -2024	3:15	4336	114.3	216	0.0	205	103.0	0	78.63	0	0	0
14-February -2024	3.25	4337	11.3.0	214	0.0	205	102	0	76	1.5	1808	1563541
14-February -2024	3:35	4336	113.0	213	0.0	205	102.2	0	0	0	0	0

14-February -2024	3:55	4335	112.3	213	0.0	205	102.1	0	0	0	0	0
14-February -2024	4:15	4334	114.3	213	0.0	205	101.7	0	0	0	0	0
14-February -2024	4:35	4333	113.4	212	0.0	204	101.8	0	0	0	0	0
14-February -2024	4:55	4332	112.5	213	0.0	204	101.5	0	0	0	0	0
14-February -2024	5:15	4331	113.4	213	0.0	204	102.0	0	0	0	0	0
14-February -2024	5:35	4331	113.6	213	0.0	205	102.0	0	0	0	0	0
14-February -2024	5:55	4330	114.2	212	0.0	205	102.7	0	0	0	0	0
14-February -2024	6:15	4330	113.8	212	0.0	204	102.9	0	0	0	0	0
14-February -2024	6:35	4328	114.5	213	0.0	205	103.1	0	0	0	0	0
14-February -2024	6:55	4363	114.3	211	0.0	199	103.1	0	0	0	0	0
14-February -2024	7:15	13	93.3	10	0.0	0	83.5	0	0	0	0	0

14-February -2024	7:35	4749	84.7	7	0.0	0	75.6	0	0	0	0	0
14-February -2024	7:55	4788	79.7	4	0.0	0	70.8	0	0	0	0	0
14-February -2024	8:15	4080	115.8	190	0.0	175	96.5	0	0	0	0	0
14-February -2024	8:35	3918	119.0	221	0.0	208	107.2	0	0	0	0	0
14-February -2024	8:55	3856	119.8	225	0.0	207	107.6	0	0	0	0	0
14-February -2024	9:15	3819	122.2	220	0.0	206	109.3	1	0	0	0	0
14-February -2024	9:35	3793	122.4	221	0.0	204	110.0	2	0	0	0	0
14-February -2024	9:55	3774	123.7	221	0.0	205	112.5	4	0	0	0	0
14-February -2024	10:15	3758	123.5	224	0.0	204	112.8	5	0	0	0	0
14-February -2024	10:35	3745	124.2	225	0.0	205	113.4	6	0	0	0	0
14-February -2024	10:55	3736	126.0	267	0.0	250	115.0	0	0	0	0	0

14-February -2024	11:15	3727	126.2	270	0.0	250	116.4	0	77.73	2	0	0
14-February -2024	11:35	3720	127.2	269	0.0	250	118.4	1	77.1	2	0	0
14-February -2024	11:55	3713	127.0	270	0.0	249	118.7	2	76.51	2	0	0
14-February -2024	12:15	3706	127.9	271	0.0	252	119.7	3	76.59	2	0	0
14-February -2024	12:35	3701	128.0	272	0.0	249	120.1	3	76.16	2	0	0
14-February -2024	12:55	3695	129.0	269	0.0	250	121.5	4	76.39	2	0	0
14-February -2024	13:15	3691	129.7	270	0.0	249	123.0	4	75.94	2	0	0
14-February -2024	13:35	3685	129.6	272	0.0	248	122.9	5	76.18	2	0	0
14-February -2024	13:55	3680	129.7	272	0.0	247	122.7	6	76.05	2	0	0
14-February -2024	14:15	3676	130.4	268	0.0	249	123.6	7	76.01	2	0	0
14-February -2024	14:35	3671	130.9	271	0.0	248	122.5	7	75.99	2	0	0

14-February -2024	14:55	3667	131.1	271	0.0	247	122.5	8	75.91	2	0	0
14-February -2024	15:15	3664	130.8	270	0.0	249	122.2	8	76.56	2	0	0
14-February -2024	15:35	3659	131.1	261	0.0	241	121.1	9	74.32	2	0	0
14-February -2024	15:55	3656	131.1	274	0.0	258	122.2	0	72.76	2	0	0
14-February -2024	16:15	3652	131.4	276	0.0	258	121.4	0	72.43	2	0	0
14-February -2024	16:35	3648	130.6	274	0.0	257	120.8	1	72.36	2	0	0
14-February -2024	16:55	3645	130.5	275	0.0	258	120.4	1	72.16	2	0	0
14-February -2024	17:15	3640	131.4	275	0.0	257	120.5	2	72.05	2	0	0
14-February -2024	17:35	3638	131.8	275	0.0	258	120.8	2	72.09	2	0	0
14-February -2024	17:55	3635	131.8	274	0.0	257	122.3	3	72.51	2	0	0
14-February -2024	18:15	3631	132.4	275	0.0	258	122.7	3	72.54	2	0	0

14-February -2024	18:18	3631	132.5	275	0.0	258	122.8	0	72.65	2	3368	2971607
14-February -2024	18:35	3628	132.9	276	0.0	258	122.4	3	75.95	2	3346	2967725
14-February -2024	18:55	3624	132.4	277	0.0	258	122.5	3	75.96	2	0	0
14-February -2024	19:15	3621	132.5	270	0.0	258	120.9	3	0.1	0	0	0
14-February -2024	19:35	3618	133.0	273	0.0	257	121.0	4	76	2	3346	303591
14-February -2024	19:55	3615	133.0	272	0.0	257	122.1	4	76.22	0	0	0
14-February -2024	20:15	4299	112.3	8	0.0	1	93.3	0	0.21	0	0	0
14-February -2024	20:35	4447	97.5	7	0.0	1	79.4	0	0.19	0	0	0
14-February -2024	20:55	4524	89.7	7	0.0	1	73.8	0	0.18	0	0	0
14-February -2024	21:15	4577	85.6	8	0.0	1	71.0	0	0.17	0	0	0
14-February -2024	21:35	4616	82.8	6	0.0	1	69.9	0	0.15	0	0	0

14-February -2024	21:55	4648	80.7	7	0.0	1	69.2	0	0.14	0	0	0
14-February -2024	22:15	4673	79.8	6	0.0	1	69.1	0	0.14	0	0	0
14-February -2024	22:35	4695	77.3	20	0.0	1	67.0	0	0.14	0	0	0
14-February -2024	22:55	4715	76.9	19	0.0	0	66.6	0	0.14	0	0	0
14-February -2024	23:15	4731	76.4	6	0.0	1	66.8	0	0.12	0	0	0
14-February -2024	23:35	4746	75.8	6	0.0	1	66.6	0	0.13	0	0	0
14-February -2024	23:55	4760	75.2	6	0.0	0	66.4	0	0.12	0	0	0
15-February -2024	0:15	4773	74.8	6	0.0	0	66.1	0	0.11	0	0	0
15-February -2024	0:35	4784	74.5	5	0.0	1	66.1	0	0.1	0	0	0
15-February -2024	0:55	4795	74.3	5	0.0	1	65.9	0	0.11	0	0	0
15-February -2024	1:15	4806	73.8	6	0.0	1	65.3	0	0.1	0	0	0

15-February -2024	1:35	4815	73.5	6	0.0	1	64.9	0	0.11	0	0	0
15-February -2024	1:55	4824	73.3	6	0.0	1	64.7	0	0.09	0	0	0
15-February -2024	2:15	4832	73.1	5	0.0	0	64.6	0	0.1	0	0	0
15-February -2024	2:35	4840	73.0	5	0.0	1	64.4	0	0.09	0	0	0
15-February -2024	2:55	4847	72.8	6	0.0	1	64.1	0	0.09	0	0	0
15-February -2024	3:15	4854	72.6	6	0.0	0	63.9	0	0.08	0	0	0
15-February -2024	3:35	4862	72.5	6	0.0	1	63.7	0	0.08	0	0	0
15-February -2024	3:55	4867	72.4	6	0.0	0	63.6	0	0.08	0	0	0
15-February -2024	4:15	4873	72.3	6	0.0	0	63.5	0	0.08	0	0	0
15-February -2024	4:35	4880	72.2	6	0.0	1	63.3	0	0.07	0	0	0
15-February -2024	4:55	4885	72.1	6	0.0	0	63.3	0	0.07	0	0	0

15-February -2024	5:15	4891	72.1	6	0.0	1	63.3	0	0.06	0	0	0
15-February -2024	5:35	4896	72.2	0	0.0	1	63.4	0	0.07	0	0	0
15-February -2024	5:55	4901	72.4	0	0.0	0	63.6	0	0.06	0	0	0
15-February -2024	6:15	4906	72.6	0	0.0	0	63.9	0	0.07	0	0	0
15-February -2024	6:35	4910	72.8	0	0.0	1	64.1	0	0.07	0	0	0
15-February -2024	6:55	4915	73.1	0	0.0	0	64.6	0	0.07	0	0	0
15-February -2024	7:15	4920	73.6	0	0.0	0	65.4	0	0.08	0	0	0
15-February -2024	7:35	4924	73.9	0	0.0	1	65.9	0	0.09	0	0	0
15-February -2024	7:55	4928	74.2	0	0.0	0	66.4	0	0.09	0	0	0
15-February -2024	8:15	4932	75.0	0	0.0	1	67.4	0	0.1	0	0	0
15-February -2024	8:35	4935	75.3	0	0.0	1	68.0	0	0.11	0	0	0

15-February -2024	8:55	4939	76.2	0	0.0	1	69.1	0	0.1	0	0	0
15-February -2024	9:15	4943	77.1	0	0.0	1	70.0	0	0.1	0	0	0
15-February -2024	9:35	4947	78.5	0	0.0	1	71.5	0	0.11	0	0	0
15-February -2024	9:55	10	75.4	1	0.0	1	71.4	0	0.13	0	0	0
15-February -2024	10:15	15	77.3	2	0.0	1	70.5	0	0.14	0	0	0
15-February -2024	10:35	16	78.6	0	0.0	1	71.9	0	0.12	0	0	0
15-February -2024	10:55	17	79.2	0	0.0	1	72.5	0	0.14	0	0	0
15-February -2024	11:15	18	78.5	2	0.0	1	72.1	0	0.14	0	0	0
15-February -2024	11:35	17	78.9	0	0.0	1	71.8	0	0.14	0	0	0
15-February -2024	11:55	420	79.3	0	0.0	1	71.9	0	0.14	0	0	0
15-February -2024	12:15	6231	78.9	1	0.0	1	72.0	0	0.14	0	0	0

15-February -2024	12:35	4245	79.7	0	0.0	1	73.8	0	0.12	0	0	0
15-February -2024	12:55	6265	81.1	0	0.0	3	75.8	0	0.12	0	0	0
15-February -2024	13:15	2488	78.6	5	0.0	1	74.1	0	0.16	0	0	0
15-February -2024	13:35	11	80.3	3	0.0	1	74.5	0	0.14	0	0	0
15-February -2024	13:55	11	78.9	8	0.0	1	73.0	0	0.14	0	0	0
15-February -2024	14:15	587	77.6	9	0.0	1	70.6	0	0.13	0	0	0
15-February -2024	14:35	6359	79.2	7	0.0	1	70.0	0	0.14	0	0	0
15-February -2024	14:55	1058	76.1	6	0.0	1	69.8	0	0.12	0	0	0
15-February -2024	15:15	1618	76.9	6	0.0	1	69.2	0	0.12	0	0	0
15-February -2024	15:35	5	77.9	2	0.0	1	70.9	0	0.11	0	0	0
15-February -2024	15:55	6361	80.4	1	0.0	1	72.1	0	0.12	0	0	0

15-February -2024	16:15	4444	78.4	5	0.0	1	71.6	0	0.13	0	0	0
15-February -2024	16:35	5003	78.1	8	0.0	0	70.8	0	0.12	0	0	0
15-February -2024	16:55	5006	77.2	7	0.0	1	69.4	0	0.13	0	0	0
15-February -2024	17:15	5008	76.9	7	0.0	1	68.8	0	0.14	0	0	0
15-February -2024	17:35	5010	76.7	6	0.0	1	68.9	0	0.12	0	0	0
15-February -2024	17:55	5012	76.5	6	0.0	0	68.9	0	0.11	0	0	0
15-February -2024	18:15	5014	76.0	8	0.0	0	68.7	0	0	0	0	0
15-February -2024	18:35	5017	75.3	9	0.0	0	67.4	0	0	0	0	0
15-February -2024	18:55	5018	74.9	8	0.0	0	67.1	0	0.01	0	0	0
15-February -2024	19:15	5020	75.0	0	0.0	1	67.3	0	0.06	0	0	0
15-February -2024	19:35	5022	74.7	1	0.0	0	67.0	0	0.1	0	0	0

15-February -2024	19:55	5024	74.5	1	0.0	1	66.8	0	0.09	0	0	0
15-February -2024	20:15	5027	74.4	0	0.0	0	66.7	0	0.09	0	0	0
15-February -2024	20:35	5028	74.5	0	0.0	1	66.9	0	0.08	0	0	0
15-February -2024	20:55	5029	74.4	0	0.0	1	66.7	0	0.08	0	0	0
15-February -2024	21:15	5032	74.4	0	0.0	0	66.7	0	0.08	0	0	0
15-February -2024	21:35	5033	74.5	0	0.0	1	66.8	0	0.08	0	0	0
15-February -2024	21:55	5035	74.5	0	0.0	1	66.8	0	0.08	0	0	0
15-February -2024	22:15	5037	74.5	0	0.0	0	66.8	0	0	0	0	0
15-February -2024	22:35	5038	74.2	0	0.0	1	65.9	0	0.08	0	0	0
15-February -2024	22:55	5040	74.0	0	0.0	0	65.6	0	0.08	0	0	0
15-February -2024	23:15	5042	73.8	0	0.0	0	65.2	0	0.08	0	0	0

15-February -2024	23:35	5044	73.7	0	0.0	1	65.2	0	0.09	0	0	0
15-February -2024	23:55	5045	73.5	4	0.0	1	64.8	0	0.07	0	0	0
16-February -2024	0:15	5047	73.1	0	0.0	0	64.1	0	0.07	0	0	0
16-February -2024	0:35	5048	72.8	0	0.0	0	63.5	0	0.06	0	0	0
16-February -2024	0:55	5050	72.7	0	0.0	1	63.4	0	0.06	0	0	0
16-February -2024	1:15	5051	72.8	0	0.0	1	63.5	0	0.07	0	0	0
16-February -2024	1:35	5053	72.8	0	0.0	1	63.6	0	0.06	0	0	0
16-February -2024	1:55	5054	72.8	0	0.0	1	63.6	0	0.06	0	0	0
16-February -2024	2:15	5056	72.8	0	0.0	0	63.7	0	0.08	0	0	0
16-February -2024	2:35	5057	72.9	0	0.0	0	63.7	0	0.15	0	0	0
16-February -2024	2:55	5058	72.8	0	0.0	0	63.7	0	0.11	0	0	0

16-February -2024	3:15	5060	72.9	0	0.0	0	63.7	0	0.09	0	0	0
16-February -2024	3:35	5061	72.7	0	0.0	1	63.6	0	0.08	0	0	0
16-February -2024	3:55	5062	72.7	0	0.0	0	63.3	0	0.09	0	0	0
16-February -2024	4:15	5064	72.6	0	0.0	0	63.2	0	0.08	0	0	0
16-February -2024	4:35	5065	72.3	0	0.0	0	62.8	0	0.08	0	0	0
16-February -2024	4:55	5066	72.1	0	0.0	0	62.5	0	0.08	0	0	0
16-February -2024	5:15	5067	72.0	0	0.0	1	62.4	0	0.06	0	0	0
16-February -2024	5:35	5069	72.0	0	0.0	0	62.4	0	0.05	0	0	0
16-February -2024	5:55	5070	71.8	0	0.0	1	62.1	0	0.05	0	0	0
16-February -2024	6:15	5071	71.6	0	0.0	1	61.9	0	0.05	0	0	0
16-February -2024	6:35	5072	71.5	0	0.0	1	61.9	0	0.05	0	0	0

16-February -2024	6:55	5073	71.5	0	0.0	0	61.9	0	0.04	0	0	0
16-February -2024	7:15	5074	71.5	0	0.0	0	61.9	0	0.04	0	0	0
16-February -2024	7:35	5076	71.7	0	0.0	0	62.3	0	0.05	0	0	0
16-February -2024	7:55	5077	71.9	0	0.0	1	62.7	0	0.04	0	0	0
16-February -2024	8:15	5079	72.7	0	0.0	0	64.0	0	0.04	0	0	0
16-February -2024	8:35	5079	73.4	0	0.0	1	64.9	0	0.05	0	0	0
16-February -2024	8:55	5080	73.9	0	0.0	1	65.8	0	0.04	0	0	0
16-February -2024	9:15	5081	74.9	0	0.0	1	67.0	0	0.05	0	0	0
16-February -2024	9:35	5082	75.9	0	0.0	1	68.3	0	0.06	0	0	0
16-February -2024	9:55	5084	77.1	0	0.0	1	69.8	0	0.08	0	0	0
16-February -2024	10:15	5084	78.7	0	0.0	1	71.4	0	0.08	0	0	0

16-February -2024	10:35	5086	79.4	0	0.0	1	72.9	0	0.08	0	0	0
16-February -2024	10:55	5086	80.3	1	0.0	1	74.8	0	0.09	0	0	0
16-February -2024	11:15	5088	80.9	0	0.0	1	75.4	0	0.11	0	0	0
16-February -2024	11:35	5089	82.1	0	0.0	1	76.3	0	0.12	0	0	0
16-February -2024	11:55	5090	82.1	1	0.0	1	76.2	0	0.13	0	0	0
16-February -2024	12:15	5091	83.4	0	0.0	1	77.4	0	0.13	0	0	0
16-February -2024	12:35	5092	84.2	0	0.0	2	78.3	0	0.15	0	0	0
16-February -2024	12:55	5093	84.3	5	0.0	2	78.1	0	0.12	0	0	0
16-February -2024	13:15	5094	86.2	0	0.0	3	80.9	0	0.14	0	0	0
16-February -2024	13:35	5094	87.2	3	0.0	3	82.9	0	0.14	0	0	0
16-February -2024	13:55	5096	86.9	6	0.0	3	83.2	0	0.12	0	0	0

16-February -2024	14:15	5095	87.7	7	0.0	2	84.0	0	0.14	0	0	0
16-February -2024	14:35	5097	88.4	6	0.0	2	85.2	0	0.13	0	0	0
16-February -2024	14:55	5097	86.8	11	0.0	2	82.6	0	0.16	0	0	0
16-February -2024	15:15	5098	86.5	11	0.0	1	82.3	0	0.15	0	0	0
16-February -2024	15:35	5099	87.4	3	0.0	1	84.0	0	0.13	0	0	0
16-February -2024	15:55	5100	85.9	7	0.0	1	81.9	0	0.15	0	0	0
16-February -2024	16:15	5101	86.0	7	0.0	1	81.5	0	0.16	0	0	0
16-February -2024	16:35	5103	86.9	3	0.0	1	80.9	0	0.14	0	0	0
16-February -2024	16:55	5103	86.7	3	0.0	1	80.6	0	0.15	0	0	0
16-February -2024	17:15	5021	89.9	79	0.0	81	80.9	0	0.03	0	0	0
16-February -2024	17:35	4518	100.0	228	0.0	217	84.2	0	0	0	0	0

16-February -2024	17:55	3119	114.8	304	0.0	271	101.8	0	0.02	0	0	0
16-February -2024	18:15	2879	120.8	318	0.0	286	110.0	0	0	0	0	0
16-February -2024	18:35	2229	126.4	390	0.0	350	118.1	0	0.02	0	0	0
16-February -2024	18:55	2219	128.4	371	0.0	332	121.1	0	0	0	0	0
16-February -2024	19:15	2190	130.8	358	0.0	319	125.4	0	0	0	0	0
16-February -2024	19:35	2166	132.7	356	0.0	318	127.2	0	0	0	0	0
16-February -2024	19:55	2145	134.2	353	0.0	314	129.5	0	0	0	0	0
16-February -2024	20:15	2126	135.2	352	0.0	312	130.0	0	87.06	2.75	0	0
16-February -2024	20:35	2111	136.0	352	0.0	313	131.4	0	85.58	2.75	0	0
16-February -2024	20:55	2098	136.9	350	0.0	310	131.9	0	85.28	2.75	0	0
16-February -2024	21:15	2087	137.8	349	0.0	310	134.0	0	85.36	2.75	0	0

16-February -2024	21:35	2077	138.8	347	0.0	309	136.5	0	85.95	2.75	0	0
16-February -2024	21:55	2068	139.6	346	0.0	309	138.0	0	84.94	2.75	0	0
16-February -2024	22:15	2061	140.6	345	0.0	308	139.4	0	84.94	2.75	0	0
16-February -2024	22:35	2054	141.3	343	0.0	305	139.9	0	84.93	2.75	0	0
16-February -2024	22:55	2048	141.9	344	0.0	308	139.7	0	84.07	2.75	0	0
16-February -2024	23:15	2042	142.3	346	0.0	310	140.4	0	81.94	2.75	0	0
16-February -2024	23:35	2036	142.8	346	0.0	313	139.3	0	82.03	2.75	0	0
16-February -2024	23:55	2032	143.6	346	0.0	313	138.6	0	81.21	2.75	0	0
17-February -2024	0:15	2029	144.4	345	0.0	312	139.3	0	81.94	2.75	0	0
17-February -2024	0:35	2025	145.4	343	0.0	309	137.9	0	81.97	2.75	0	0
17-February -2024	0:55	2021	145.9	342	0.0	310	138.5	0	83.38	2.75	7200	6565922

17-February -2024	1:04	2021	144.0	342	0.0	310	138.5	0	83.7	2.75	7370	6631663
17-February -2024	1:15	2807	143.6	260	138.6	0	0.0	380	0	0	0	0
17-February -2024	1:35	3777	119.5	0	114.5	0	0.0	150	0	0	0	0
17-February -2024	1:55	4037	109.3	0	104.3	0	0.0	60	0	0	0	0
17-February -2024	2:15	4184	104.5	0	99.5	0	0.0	33	0	0	0	0
17-February -2024	2:35	4283	100.5	0	95.5	0	0.0	21	0	0	0	0
17-February -2024	2:55	4357	95.4	0	90.4	0	0.0	14	0	0	0	0
17-February -2024	3:15	4416	90.6	0	85.6	0	0.0	9	0	0	0	0
17-February -2024	3:35	4465	87.1	0	82.1	0	0.0	6	0	0	0	0
17-February -2024	3:55	4504	83.8	0	78.8	0	0.0	4	0	0	0	0
17-February -2024	4:15	4539	80.7	0	75.7	0	0.0	2	0	0	0	0

17-February -2024	4:35	4570	78.7	0	73.7	0	0.0	1	0	0	0	0
17-February -2024	4:55	4597	77.6	0	72.6	0	0.0	0	0	0	0	0
17-February -2024	5:15	4621	76.2	0	71.2	0	0.0	0	0	0	0	0
17-February -2024	5:35	4642	74.6	0	69.6	0	0.0	0	0	0	0	0
17-February -2024	5:55	4661	73.0	0	68.0	0	0.0	0	0	0	0	0
17-February -2024	6:15	4679	72.3	0	67.3	0	0.0	0	0	0	0	0
17-February -2024	6:35	4695	71.3	0	66.3	0	0.0	0	0	0	0	0
17-February -2024	6:55	4710	70.6	0	65.6	0	0.0	0	0	0	0	0
17-February -2024	7:15	4724	70.0	0	65.0	0	0.0	0	0	0	0	0
17-February -2024	7:35	4737	69.5	0	64.5	0	0.0	0	0	0	0	0
17-February -2024	7:55	4750	69.3	0	64.3	0	0.0	0	0	0	0	0

17-February -2024	8:15	4761	69.2	0	64.2	0	0.0	0	0	0	0	0
17-February -2024	8:35	4772	69.4	0	64.4	0	0.0	0	0	0	0	0
17-February -2024	8:55	4782	70.3	0	65.3	0	0.0	0	0	0	0	0
17-February -2024	9:15	4791	71.3	0	66.3	0	0.0	0	0	0	0	0
17-February -2024	9:35	4800	72.3	0	67.3	0	0.0	0	0	0	0	0
17-February -2024	9:55	4809	73.6	0	68.6	0	0.0	0	0	0	0	0
17-February -2024	10:15	4817	75.0	0	70.0	0	0.0	0	0	0	0	0
17-February -2024	10:35	4824	76.1	0	71.1	0	0.0	0	0	0	0	0
17-February -2024	10:55	4831	76.7	0	71.7	0	0.0	0	0	0	0	0
17-February -2024	11:15	4839	77.0	0	72.0	0	0.0	0	0	0	0	0
17-February -2024	11:35	4845	77.9	0	72.9	0	0.0	0	0	0	0	0

17-February -2024	11:55	4852	78.4	0	73.4	0	0.0	0	0	0	0	0
17-February -2024	12:15	4858	79.1	0	74.1	0	0.0	0	0	0	0	0
17-February -2024	12:35	4864	79.9	0	74.9	0	0.0	0	0	0	0	0
17-February -2024	12:55	4870	81.2	0	76.2	0	0.0	0	0	0	0	0
17-February -2024	13:15	4876	83.0	0	78.0	0	0.0	0	0	0	0	0
17-February -2024	13:35	4881	83.9	0	78.9	0	0.0	0	0	0	0	0
17-February -2024	13:55	4886	84.0	0	79.0	0	0.0	0	0	0	0	0
17-February -2024	14:15	4890	85.2	0	80.2	0	0.0	0	0	0	0	0
17-February -2024	14:35	4896	86.5	0	81.5	0	0.0	0	0	0	0	0
17-February -2024	14:55	4900	87.1	0	82.1	0	0.0	0	0	0	0	0
17-February -2024	15:15	4904	87.8	0	82.8	0	0.0	0	0	0	0	0

17-February -2024	15:35	4909	87.8	0	82.8	0	0.0	0	0	0	0	0
17-February -2024	15:55	4913	88.1	0	83.1	0	0.0	0	0	0	0	0
17-February -2024	16:15	4917	87.5	0	82.5	0	0.0	0	0	0	0	0
17-February -2024	16:35	4920	86.4	0	81.4	0	0.0	0	0	0	0	0
17-February -2024	16:55	4924	84.9	0	79.9	0	0.0	0	0	0	0	0
17-February -2024	17:15	4928	84.2	0	79.2	0	0.0	0	0	0	0	0
17-February -2024	17:35	4931	83.3	2	78.3	0	0.0	0	0	0	0	0
17-February -2024	17:55	4934	81.7	3	76.7	0	0.0	0	0	0	0	0
17-February -2024	18:15	4938	79.4	4	74.4	0	0.0	0	0	0	0	0
17-February -2024	18:35	4941	77.7	2	72.7	0	0.0	0	0	0	0	0
17-February -2024	18:55	4945	74.1	1	69.1	0	0.0	0	0	0	0	0

17-February -2024	19:15	4948	73.3	0	68.3	0	0.0	0	0	0	0	0
17-February -2024	19:35	4951	72.5	0	67.5	0	0.0	0	0	0	0	0
17-February -2024	19:55	4954	71.9	0	66.9	0	0.0	0	0	0	0	0
17-February -2024	20:15	4957	71.1	0	66.1	0	0.0	0	0	0	0	0
17-February -2024	20:35	4960	70.6	0	65.6	0	0.0	0	0	0	0	0
17-February -2024	20:55	4963	70.1	0	65.1	0	0.0	0	0	0	0	0
17-February -2024	21:15	4966	69.5	0	64.5	0	0.0	0	0	0	0	0
17-February -2024	21:35	4968	69.0	0	64.0	0	0.0	0	0	0	0	0
17-February -2024	21:55	4972	68.6	0	63.6	0	0.0	0	0	0	0	0
17-February -2024	22:15	4974	68.3	0	63.3	0	0.0	0	0	0	0	0
17-February -2024	22:35	4976	67.9	0	62.9	0	0.0	0	0	0	0	0

17-February -2024	22:55	4979	67.7	0	62.7	0	0.0	0	0	0	0	0
17-February -2024	23:15	4981	67.4	0	62.4	0	0.0	0	0	0	0	0
17-February -2024	23:35	4984	67.2	0	62.2	0	0.0	0	0	0	0	0
17-February -2024	23:55	4986	67.0	0	62.0	0	0.0	0	0	0	0	0
18-February -2024	0:15	4988	66.7	0	61.7	0	0.0	0	0	0	0	0
18-February -2024	0:35	4991	66.3	0	61.3	0	0.0	0	0	0	0	0
18-February -2024	0:55	4993	66.2	0	61.2	0	0.0	0	0	0	0	0
18-February -2024	1:15	4995	65.9	0	60.9	0	0.0	0	0	0	0	0
18-February -2024	1:35	4997	65.7	0	60.7	0	0.0	0	0	0	0	0
18-February -2024	1:55	4999	65.4	0	60.4	0	0.0	0	0	0	0	0
18-February -2024	2:15	5001	65.0	0	60.0	0	0.0	0	0	0	0	0

18-February -2024	2:35	5003	64.8	0	59.8	0	0.0	0	0	0	0	0
18-February -2024	2:55	5005	64.5	0	59.5	0	0.0	0	0	0	0	0
18-February -2024	3:15	5007	64.3	0	59.3	0	0.0	0	0	0	0	0
18-February -2024	3:35	5009	64.2	0	59.2	0	0.0	0	0	0	0	0
18-February -2024	3:55	5011	64.0	0	59.0	0	0.0	0	0	0	0	0
18-February -2024	4:15	5013	63.8	0	58.8	0	0.0	0	0	0	0	0
18-February -2024	4:35	5014	63.5	0	58.5	0	0.0	0	0	0	0	0
18-February -2024	4:55	5016	63.3	0	58.3	0	0.0	0	0	0	0	0
18-February -2024	5:15	5018	63.0	0	58.0	0	0.0	0	0	0	0	0
18-February -2024	5:35	5020	62.7	0	57.7	0	0.0	0	0	0	0	0
18-February -2024	5:55	5022	62.5	0	57.5	0	0.0	0	0	0	0	0

18-February -2024	6:15	5024	62.3	0	57.3	0	0.0	0	0	0	0	0
18-February -2024	6:35	5024	62.1	0	57.1	0	0.0	0	0	0	0	0
18-February -2024	6:55	5027	62.0	0	57.0	0	0.0	0	0	0	0	0
18-February -2024	7:15	5028	61.8	0	56.8	0	0.0	0	0	0	0	0
18-February -2024	7:35	5030	62.1	0	57.1	0	0.0	0	0	0	0	0
18-February -2024	7:55	5032	62.6	0	57.6	0	0.0	0	0	0	0	0
18-February -2024	8:15	5033	63.6	0	58.6	0	0.0	0	0	0	0	0
18-February -2024	8:35	5035	65.0	0	60.0	0	0.0	0	0	0	0	0
18-February -2024	8:55	5036	66.5	0	61.5	0	0.0	0	0	0	0	0
18-February -2024	9:15	5037	68.1	0	63.1	0	0.0	0	0	0	0	0
18-February -2024	9:35	5038	69.6	0	64.6	0	0.0	0	0	0	0	0

18-February -2024	9:55	5040	71.1	0	66.1	0	0.0	0	0	0	0	0
18-February -2024	10:15	5041	72.6	0	67.6	0	0.0	0	0	0	0	0
18-February -2024	10:35	5043	73.7	0	68.7	0	0.0	0	0	0	0	0
18-February -2024	10:55	5043	74.8	0	69.8	0	0.0	0	0	0	0	0
18-February -2024	11:15	5046	75.3	0	70.3	0	0.0	0	0	0	0	0
18-February -2024	11:35	5047	76.3	0	71.3	0	0.0	0	0	0	0	0
18-February -2024	11:55	5048	77.0	0	72.0	0	0.0	0	0	0	0	0
18-February -2024	12:15	5050	77.8	0	72.8	0	0.0	0	0	0	0	0
18-February -2024	12:35	5052	78.6	0	73.6	0	0.0	0	0	0	0	0
18-February -2024	12:55	5052	80.1	0	75.1	0	0.0	0	0	0	0	0
18-February -2024	13:15	5054	81.9	0	76.9	0	0.0	0	0	0	0	0

18-February -2024	13:35	5055	82.8	0	77.8	0	0.0	0	0	0	0	0
18-February -2024	13:55	5057	83.3	0	78.3	0	0.0	0	0	0	0	0
18-February -2024	14:15	5057	84.7	0	79.7	0	0.0	0	0	0	0	0
18-February -2024	14:35	5058	84.7	4	79.7	0	0.0	0	0	0	0	0
18-February -2024	14:55	4766	93.8	129	88.8	0	0.0	0	0	0	0	0
18-February -2024	15:15	3685	103.6	290	98.6	0	0.0	0	0	0	0	0
18-February -2024	15:35	2997	108.3	395	103.3	0	0.0	0	0	0	0	0
18-February -2024	15:55	2589	113.0	473	108.0	0	0.0	0	0	0	0	0
18-February -2024	16:15	2357	116.5	445	111.5	0	0.0	0	0	0	0	0
18-February -2024	16:35	2306	119.9	426	114.9	0	0.0	0	0	0	0	0
18-February -2024	16:55	4050	113.7	4	108.7	0	0.0	0	0	0	0	0

18-February -2024	17:15	4429	104.8	4	99.8	0	0.0	0	0	0	0	0
18-February -2024	17:35	4589	98.6	5	93.6	0	0.0	0	0	0	0	0
18-February -2024	17:55	4681	94.3	5	89.3	0	0.0	0	0	0	0	0
18-February -2024	18:15	4741	90.8	5	85.8	0	0.0	0	0	0	0	0
18-February -2024	18:35	4785	88.1	4	83.1	0	0.0	0	0	0	0	0
18-February -2024	18:55	4818	85.9	2	80.9	0	0.0	0	0	0	0	0
18-February -2024	19:15	4844	83.9	2	78.9	0	0.0	0	0	0	0	0
18-February -2024	19:35	4864	82.4	1	77.4	0	0.0	0	0	0	0	0
18-February -2024	19:55	4882	80.7	0	75.7	0	0.0	0	0	0	0	0
18-February -2024	20:15	4897	79.2	0	74.2	0	0.0	0	0	0	0	0
18-February -2024	20:35	4910	77.1	0	72.1	0	0.0	0	0	0	0	0

18-February -2024	20:55	4921	75.5	0	70.5	0	0.0	0	0	0	0	0
18-February -2024	21:15	4931	74.0	0	69.0	0	0.0	0	0	0	0	0
18-February -2024	21:35	4939	73.0	0	68.0	0	0.0	0	0	0	0	0
18-February -2024	21:55	4947	72.1	0	67.1	0	0.0	0	0	0	0	0
18-February -2024	22:15	4955	71.2	0	66.2	0	0.0	0	0	0	0	0
18-February -2024	22:35	4961	70.5	0	65.5	0	0.0	0	0	0	0	0
18-February -2024	22:55	4967	69.8	0	64.8	0	0.0	0	0	0	0	0
18-February -2024	23:15	4972	69.0	0	64.0	0	0.0	0	0	0	0	0
18-February -2024	23:35	4977	68.0	0	63.0	0	0.0	0	0	0	0	0
18-February -2024	23:55	4983	67.4	0	62.4	0	0.0	0	0	0	0	0
19-February -2024	0:15	4988	66.9	0	61.9	0	0.0	0	0	0	0	0

19-February -2024	0:35	4991	66.6	0	61.6	0	0.0	0	0	0	0	0
19-February -2024	0:55	4995	66.4	0	61.4	0	0.0	0	0	0	0	0
19-February -2024	1:15	5000	66.0	0	61.0	0	0.0	0	0	0	0	0
19-February -2024	1:35	5003	65.6	0	60.6	0	0.0	0	0	0	0	0
19-February -2024	1:55	5006	65.3	0	60.3	0	0.0	0	0	0	0	0
19-February -2024	2:15	5009	64.9	0	59.9	0	0.0	0	0	0	0	0
19-February -2024	2:35	5012	64.7	0	59.7	0	0.0	0	0	0	0	0
19-February -2024	2:55	5015	64.5	0	59.5	0	0.0	0	0	0	0	0
19-February -2024	3:15	5018	64.4	0	59.4	0	0.0	0	0	0	0	0
19-February -2024	3:35	5020	64.2	0	59.2	0	0.0	0	0	0	0	0
19-February -2024	3:55	5023	63.9	0	58.9	0	0.0	0	0	0	0	0

19-February -2024	4:15	5025	63.7	0	58.7	0	0.0	0	0	0	0	0
19-February -2024	4:35	5028	63.4	0	58.4	0	0.0	0	0	0	0	0
19-February -2024	4:55	5030	63.1	0	58.1	0	0.0	0	0	0	0	0
19-February -2024	5:15	5033	62.9	0	57.9	0	0.0	0	0	0	0	0
19-February -2024	5:35	5034	62.7	0	57.7	0	0.0	0	0	0	0	0
19-February -2024	5:55	5036	62.6	0	57.6	0	0.0	0	0	0	0	0
19-February -2024	6:15	5038	62.5	0	57.5	0	0.0	0	0	0	0	0
19-February -2024	6:35	5040	62.2	0	57.2	0	0.0	0	0	0	0	0
19-February -2024	6:55	5042	62.1	0	57.1	0	0.0	0	0	0	0	0
19-February -2024	7:15	5044	61.9	0	56.9	0	0.0	0	0	0	0	0
19-February -2024	7:35	5046	62.0	0	57.0	0	0.0	0	0	0	0	0

19-February -2024	7:55	5047	62.7	0	57.7	0	0.0	0	0	0	0	0
19-February -2024	8:15	5049	63.8	0	58.8	0	0.0	0	0	0	0	0
19-February -2024	8:35	5051	65.3	0	60.3	0	0.0	0	0	0	0	0
19-February -2024	8:55	5052	66.9	0	61.9	0	0.0	0	0	0	0	0
19-February -2024	9:15	5053	68.7	0	63.7	0	0.0	0	0	0	0	0
19-February -2024	9:35	5055	70.7	0	65.7	0	0.0	0	0	0	0	0
19-February -2024	9:55	5056	72.5	0	67.5	0	0.0	0	0	0	0	0
19-February -2024	10:15	5057	73.4	0	68.4	0	0.0	0	0	0	0	0
19-February -2024	10:35	5060	73.8	0	68.8	0	0.0	0	0	0	0	0
19-February -2024	10:55	5060	74.1	0	69.1	0	0.0	0	0	0	0	0
19-February -2024	11:15	5061	74.5	0	69.5	0	0.0	0	0	0	0	0

19-February -2024	11:35	5063	75.2	0	70.2	0	0.0	0	0	0	0	0
19-February -2024	11:55	5065	75.9	0	70.9	0	0.0	0	0	0	0	0
19-February -2024	12:15	5066	76.9	0	71.9	0	0.0	0	0	0	0	0
19-February -2024	12:35	5067	78.0	0	73.0	0	0.0	0	0	0	0	0
19-February -2024	12:55	5069	78.7	0	73.7	0	0.0	0	0	0	0	0
19-February -2024	13:15	5069	79.1	0	76.1	0	0.0	0	0	0	0	0
19-February -2024	13:35	5071	80.3	0	77.3	0	0.0	0	0	0	0	0
19-February -2024	13:55	5071	80.2	0	77.2	0	0.0	0	0	0	0	0
19-February -2024	14:15	5073	80.9	0	77.9	0	0.0	0	0	0	0	0
19-February -2024	14:35	5075	82.7	0	79.7	0	0.0	0	0	0	0	0
19-February -2024	14:55	5076	84.3	0	81.3	0	0.0	0	0	0	0	0

19-February -2024	15:15	5076	83.6	3	80.6	0	0.0	0	0	0	0	0
19-February -2024	15:35	5077	84.6	0	81.6	0	0.0	0	0	0	0	0
19-February -2024	15:55	5078	84.0	1	81.0	0	0.0	0	0	0	0	0
19-February -2024	16:15	5079	83.8	2	80.8	0	0.0	0	0	0	0	0
19-February -2024	16:35	5079	81.5	7	78.5	0	0.0	0	0	0	0	0
19-February -2024	16:55	5082	80.5	2	77.5	0	0.0	0	0	0	0	0
19-February -2024	17:15	5082	80.2	0	77.2	0	0.0	0	0	0	0	0
19-February -2024	17:35	5083	78.8	2	75.8	0	0.0	0	0	0	0	0
19-February -2024	17:55	5083	77.4	1	74.4	0	0.0	0	0	0	0	0
19-February -2024	18:15	5085	76.0	2	73.0	0	0.0	0	0	0	0	0
19-February -2024	18:35	5085	75.0	1	72.0	0	0.0	0	0	0	0	0

19-February -2024	18:55	5087	74.1	1	71.1	0	0.0	0	0	0	0	0
19-February -2024	19:15	5087	73.4	1	70.4	0	0.0	0	0	0	0	0
19-February -2024	19:35	4940	87.4	57	84.4	50	65.3	0	0	0	0	0
19-February -2024	19:55	4016	98.0	215	95.0	192	83.5	0	0	0	0	0
19-February -2024	20:15	3209	105.6	301	102.6	268	94.1	0	0	0	0	0
19-February -2024	20:35	2668	111.0	355	108.0	319	102.1	3	0	0	0	0
19-February -2024	20:55	2384	114.2	376	111.2	334	109.2	8	0	0	0	0
19-February -2024	21:15	2296	115.9	359	112.9	318	111.1	12	0	0	0	0
19-February -2024	21:35	2659	113.2	254	110.2	225	109.3	17	0	0	0	0
19-February -2024	21:55	2446	116.5	299	113.5	264	112.3	21	0	0	0	0
19-February -2024	22:15	2639	115.9	254	112.9	226	110.3	25	0	0	0	0

19-February -2024	22:35	2578	116.9	255	113.9	227	112.3	29	0	0	0	0
19-February -2024	22:55	2482	117.2	248	114.2	222	112.1	33	0	0	0	0
19-February -2024	23:15	2766	115.7	208	112.7	187	107.5	35	0	0	0	0
19-February -2024	23:35	2743	116.6	210	113.6	189	108.1	37	0	0	0	0
19-February -2024	23:55	2677	118.1	210	115.1	190	111.7	39	0	0	0	0
20-February -2024	0:15	2658	117.5	209	114.5	188	110.8	42	0	0	0	0
20-February -2024	0:35	3483	116.7	99	113.7	91	110.0	45	0	0	0	0
20-February -2024	0:55	4186	100.4	10	97.4	9	88.8	34	0	0	0	0
20-February -2024	1:15	4287	97.0	43	94.0	38	81.2	25	0	0	0	0
20-February -2024	1:35	4275	95.1	43	92.1	38	78.7	20	0	0	0	0
20-February -2024	1:55	4210	95.2	41	92.2	36	77.1	17	0	0	0	0

20-February -2024	2:15	3984	94.7	80	91.7	69	76.8	15	0	0	0	0
20-February -2024	2:35	3750	99.1	84	96.1	75	82.1	15	0	0	0	0
20-February -2024	2:55	3702	106.7	16	103.7	4	92.1	16	0	0	0	0
20-February -2024	3:15	3929	109.6	53	106.6	48	99.7	17	0	0	0	0
20-February -2024	3:35	4048	103.7	91	100.7	72	100.7	17	0	0	0	0
20-February -2024	3:55	4051	104.9	86	101.9	71	91.5	17	0	0	0	0
20-February -2024	4:15	4051	108.3	80	105.3	68	95.2	17	0	0	0	0
20-February -2024	4:35	4098	103.6	81	100.6	71	94.4	17	0	0	0	0
20-February -2024	4:55	4097	103.4	86	100.4	76	94.1	16	0	0	0	0
20-February -2024	5:15	4076	108.0	83	105.0	73	99.2	17	0	0	0	0
20-February -2024	5:35	4100	107.8	84	104.8	74	102.6	17	0	0	0	0

20-February -2024	5:55	4131	105.5	85	102.5	75	98.4	17	0	0	0	0
20-February -2024	6:15	4126	105.4	86	102.4	76	97.8	17	0	0	0	0
20-February -2024	6:35	4119	109.1	87	106.1	76	99.8	17	0	0	0	0
20-February -2024	6:55	4134	109.4	88	106.4	78	101.4	17	0	0	0	0
20-February -2024	7:15	4158	106.3	84	103.3	75	98.0	17	0	0	0	0
20-February -2024	7:35	4163	106.5	85	103.5	76	99.1	17	0	0	0	0
20-February -2024	7:55	4144	112.5	85	109.5	78	102.1	18	0	0	0	0
20-February -2024	8:15	4525	105.5	0	102.5	1	93.1	17	0	0	0	0
20-February -2024	8:35	4631	97.2	0	94.2	1	84.9	14	0	0	0	0
20-February -2024	8:55	4682	91.6	0	88.6	1	78.3	12	0	0	0	0
20-February -2024	9:15	4717	89.6	0	86.6	1	73.3	11	0	0	0	0

20-February -2024	9:35	4741	88.3	0	85.3	1	71.4	9	0	0	0	0
20-February -2024	9:55	4760	86.8	0	83.8	1	70.0	8	0	0	0	0
20-February -2024	10:15	4776	85.7	0	82.7	1	69.7	8	0	0	0	0
20-February -2024	10:35	4792	85.0	0	82.0	2	70.8	7	0	0	0	0
20-February -2024	10:55	4804	84.7	0	81.7	1	72.9	7	0	0	0	0
20-February -2024	11:15	4816	84.1	0	81.1	1	72.8	6	0	0	0	0
20-February -2024	11:35	4828	83.4	0	80.4	1	72.6	5	0	0	0	0
20-February -2024	11:55	4835	83.4	0	80.4	1	72.2	5	0	0	0	0
20-February -2024	12:15	4844	83.8	0	80.8	1	72.4	4	0	0	0	0
20-February -2024	12:35	4850	84.2	0	81.2	1	73.1	4	0	0	0	0
20-February -2024	12:55	4858	84.9	0	81.9	4	75.1	4	0	0	0	0

20-February -2024	13:15	4865	85.3	0	82.3	4	78.3	4	0	0	0	0
20-February -2024	13:35	4870	86.0	0	83.0	3	79.0	3	0	0	0	0
20-February -2024	13:55	4876	85.7	1	82.7	4	80.1	3	0	0	0	0
20-February -2024	14:15	4882	87.2	0	84.2	4	83.1	2	0	0	0	0
20-February -2024	14:35	4888	88.1	1	85.1	4	87.0	2	0	0	0	0
20-February -2024	14:55	4889	86.4	9	83.4	1	85.3	2	0	0	0	0
20-February -2024	15:15	4897	86.1	0	0.0	0	0.0	1	0	0	0	0
20-February -2024	15:35	4901	87.2	0	0.0	0	0.0	1	0	0	0	0
20-February -2024	15:55	4906	86.2	0	0.0	0	0.0	1	0	0	0	0
20-February -2024	16:15	4910	85.9	0	0.0	0	0.0	1	0	0	0	0
20-February -2024	16:35	4913	86.3	0	0.0	0	0.0	1	0	0	0	0

20-February -2024	16:55	4912	84.5	0	0.0	0	0.0	1	0	0	0	0
20-February -2024	17:15	4915	83.6	0	0.0	0	0.0	0	0	0	0	0
20-February -2024	17:35	4918	81.9	0	0.0	0	0.0	0	0	0	0	0
20-February -2024	17:55	4922	80.8	0	0.0	0	0.0	0	0	0	0	0
20-February -2024	18:15	4923	79.7	0	0.0	0	0.0	0	0	0	0	0
20-February -2024	18:35	4925	78.6	0	0.0	0	0.0	0	0	0	0	0
20-February -2024	18:55	4928	77.6	0	0.0	0	0.0	0	0	0	0	0
20-February -2024	19:15	4944	76.8	0	0.0	0	0.0	0	0	0	0	0
20-February -2024	19:35	4930	76.1	0	0.0	0	0.0	0	0	0	0	0
20-February -2024	19:55	4933	75.3	0	0.0	0	0.0	0	0	0	0	0
20-February -2024	20:15	4935	74.8	0	0.0	0	0.0	0	0	0	0	0

20-February -2024	20:35	4934	73.7	0	0.0	0	0.0	0	0	0	0	0
20-February -2024	20:55	4936	72.8	0	0.0	0	0.0	0	0	0	0	0
20-February -2024	21:15	4936	71.8	0	0.0	0	0.0	0	0	0	0	0
20-February -2024	21:35	4936	71.3	0	0.0	0	0.0	0	0	0	0	0
20-February -2024	21:55	4939	70.7	0	0.0	0	0.0	0	0	0	0	0
20-February -2024	22:15	4940	70.1	0	0.0	0	0.0	0	0	0	0	0
20-February -2024	22:35	4939	69.6	0	0.0	0	0.0	0	0	0	0	0
20-February -2024	22:55	4941	69.1	0	0.0	0	0.0	0	0	0	0	0
20-February -2024	23:15	4939	68.6	0	0.0	0	0.0	0	0	0	0	0
20-February -2024	23:35	4940	68.2	0	0.0	0	0.0	0	0	0	0	0
20-February -2024	23:55	4942	67.8	0	0.0	0	0.0	0	0	0	0	0

21-February -2024	0:15	4949	67.4	0	0.0	0	0.0	0	0	0	0	0
21-February -2024	0:35	4943	67.1	0	0.0	0	0.0	0	0	0	0	0
21-February -2024	1:00	4944	66.6	0	0.0	0	0.0	0	0	0	0	0
21-February -2024	1:20	4946	66.3	0	0.0	0	0.0	0	0	0	0	0
21-February -2024	1:40	4944	65.9	0	0.0	0	0.0	0	0	0	0	0
21-February -2024	2:00	4946	65.6	0	0.0	0	0.0	0	0	0	0	0
21-February -2024	2:20	4947	65.3	0	0.0	0	0.0	0	0	0	0	0
21-February -2024	2:40	4949	65.1	0	0.0	0	0.0	0	0	0	0	0
21-February -2024	3:00	4951	64.8	0	0.0	0	0.0	0	0	0	0	0
21-February -2024	3:20	4952	64.5	0	0.0	0	0.0	0	0	0	0	0
21-February -2024	3:40	4970	64.3	0	0.0	0	0.0	0	0	0	0	0

21-February -2024	4:00	4952	64.2	0	0.0	0	0.0	0	0	0	0	0
21-February -2024	4:20	4952	64.0	0	0.0	0	0.0	0	0	0	0	0
21-February -2024	4:40	4953	63.9	0	0.0	0	0.0	0	0	0	0	0
21-February -2024	5:00	4954	63.7	0	0.0	0	0.0	0	0	0	0	0
21-February -2024	5:20	4955	63.4	0	0.0	0	0.0	0	0	0	0	0
21-February -2024	5:40	4958	63.2	0	0.0	0	0.0	0	0	0	0	0
21-February -2024	6:00	4955	62.9	0	0.0	0	0.0	0	0	0	0	0
21-February -2024	6:20	4956	62.8	0	0.0	0	0.0	0	0	0	0	0
21-February -2024	6:40	4961	62.6	0	0.0	0	0.0	0	0	0	0	0
21-February -2024	7:00	4955	62.8	0	0.0	0	0.0	0	0	0	0	0
21-February -2024	7:20	4954	62.8	0	0.0	0	0.0	0	0	0	0	0

21-February -2024	7:40	4957	63.2	0	0.0	0	0.0	0	0	0	0	0
21-February -2024	8:00	4954	63.8	0	0.0	0	0.0	0	0	0	0	0
21-February -2024	8:20	4954	64.8	0	0.0	0	0.0	0	0	0	0	0
21-February -2024	8:40	4958	66.4	0	0.0	0	0.0	0	0	0	0	0
21-February -2024	9:00	4953	68.0	0	0.0	0	0.0	0	0	0	0	0
21-February -2024	9:20	4953	69.4	0	0.0	0	0.0	0	0	0	0	0
21-February -2024	9:39	4954	71.0	0	0.0	0	0.0	0	0	0	0	0
21-February -2024	9:59	4953	72.7	0	0.0	0	0.0	0	0	0	0	0
21-February -2024	10:19	4952	74.6	0	0.0	0	0.0	0	0	0	0	0
21-February -2024	10:39	4958	75.6	0	0.0	0	0.0	0	0	0	0	0
21-February -2024	10:59	4949	75.9	0	0.0	0	0.0	0	0	0	0	0

21-February -2024	11:19	4950	75.7	0	0.0	0	0.0	0	0	0	0	0
21-February -2024	11:39	4954	76.7	0	0.0	0	0.0	0	0	0	0	0
21-February -2024	11:59	4949	77.9	0	0.0	0	0.0	0	0	0	0	0
21-February -2024	12:19	4949	79.0	0	0.0	0	0.0	0	0	0	0	0
21-February -2024	12:39	4949	79.7	0	0.0	0	0.0	0	0	0	0	0
21-February -2024	12:59	4969	80.9	0	0.0	0	0.0	0	0	0	0	0
21-February -2024	13:19	4948	82.4	0	0.0	0	0.0	0	0	0	0	0
21-February -2024	13:39	4948	83.2	0	0.0	0	0.0	0	0	0	0	0
21-February -2024	13:59	4948	83.4	0	0.0	0	0.0	0	0	0	0	0
21-February -2024	14:19	4951	84.5	0	0.0	0	0.0	0	0	0	0	0
21-February -2024	14:39	4948	85.4	0	0.0	0	0.0	0	0	0	0	0

21-February -2024	14:59	4947	86.1	0	0.0	0	0.0	0	0	0	0	0
21-February -2024	15:19	4948	86.4	0	0.0	0	0.0	0	0	0	0	0
21-February -2024	15:39	4949	87.0	0	0.0	0	0.0	0	0	0	0	0
21-February -2024	15:59	4682	89.7	91	86.7	78	85.7	0	0	0	0	0
21-February -2024	16:19	3894	99.4	227	93.4	204	94.7	0	0	0	0	0
21-February -2024	16:39	3227	103.8	293	97.8	254	95.3	1	0	0	0	0
21-February -2024	16:59	2720	106.5	344	100.5	302	99.6	4	0	0	0	0
21-February -2024	17:19	2538	109.0	312	103.0	271	101.7	8	0	0	0	0
21-February -2024	17:39	2449	111.8	311	105.8	268	104.9	13	0	0	0	0
21-February -2024	17:59	2394	115.0	310	109.0	271	107.6	18	0	0	0	0
21-February -2024	18:19	2334	116.1	311	110.1	271	109.6	25	0	0	0	0

21-February -2024	18:39	2223	116.7	297	110.7	260	111.8	31	0	0	0	0
21-February -2024	18:59	2244	119.4	301	113.4	266	114.2	40	0	0	0	0
21-February -2024	19:19	3700	118.1	10	112.1	9	97.9	46	0	0	0	0
21-February -2024	19:39	4188	109.0	11	103.0	9	85.1	32	0	0	0	0
21-February -2024	19:59	4490	100.8	3	94.8	1	78.9	24	0	0	0	0
21-February -2024	20:19	4615	92.7	2	86.7	1	72.4	17	0	0	0	0
21-February -2024	20:39	4703	85.9	70	79.9	72	70.7	13	0	0	0	0
21-February -2024	20:59	4715	87.3	12	81.3	10	69.7	10	0	0	0	0
21-February -2024	21:19	4759	85.5	10	79.5	8	67.5	8	0	0	0	0
21-February -2024	21:39	4719	84.7	10	78.7	8	65.6	6	0	0	0	0
21-February -2024	21:59	4597	86.4	51	80.4	47	65.8	5	0	0	0	0

21-February -2024	22:19	4396	89.7	47	83.7	42	67.8	5	0	0	0	0
21-February -2024	22:39	4273	91.2	42	85.2	38	68.8	5	0	0	0	0
21-February -2024	22:59	4224	89.9	27	83.9	25	66.7	4	0	0	0	0
21-February -2024	23:19	4228	87.7	20	81.7	18	63.4	4	0	0	0	0
21-February -2024	23:39	4201	86.1	19	80.1	17	61.4	3	0	0	0	0
21-February -2024	23:59	4144	85.6	16	79.6	13	60.2	3	0	0	0	0
22-February -2024	0:19	4138	88.6	5	82.6	5	58.7	3	0	0	0	0
22-February -2024	0:39	4158	89.5	3	83.5	3	57.7	3	0	0	0	0
22-February -2024	0:59	4176	91.1	2	85.1	2	57.0	2	0	0	0	0
22-February -2024	1:19	4196	92.0	3	86.0	1	56.2	2	0	0	0	0
22-February -2024	1:39	4250	91.5	3	85.5	1	55.5	2	0	0	0	0

22-February -2024	1:59	4292	84.4	3	78.4	1	55.2	2	0	0	0	0
22-February -2024	2:19	4310	80.1	3	74.1	0	54.8	1	0	0	0	0
22-February -2024	2:39	3	72.6	1	66.6	1	54.7	0	0	0	0	0
22-February -2024	2:59	9	71.1	3	65.1	1	54.1	0	0	0	0	0
22-February -2024	3:19	17	70.5	4	64.5	1	53.7	0	0	0	0	0
22-February -2024	3:39	18	70.2	4	64.2	0	53.5	0	0	0	0	0
22-February -2024	3:59	20	69.6	3	63.6	0	53.4	0	0	0	0	0
22-February -2024	4:19	19	68.9	3	62.9	1	53.0	0	0	0	0	0
22-February -2024	4:39	19	67.8	4	61.8	0	52.4	0	0	0	0	0
22-February -2024	4:59	20	67.1	4	61.1	1	52.0	0	0	0	0	0
22-February -2024	5:19	20	66.7	4	60.7	1	52.0	0	0	0	0	0

22-February -2024	5:39	20	66.2	3	60.2	1	51.5	0	0	0	0	0
22-February -2024	5:59	20	65.6	4	59.6	1	51.3	0	0	0	0	0
22-February -2024	6:19	21	64.9	4	58.9	1	50.8	0	0	0	0	0
22-February -2024	6:39	4298	65.3	3	59.3	0	50.8	0	0	0	0	0
22-February -2024	6:59	9	63.2	3	57.2	1	50.7	0	0	0	0	0
22-February -2024	7:19	12	62.7	0	56.7	1	50.8	0	0	0	0	0
22-February -2024	7:39	14	62.9	0	56.9	1	51.4	0	0	0	0	0
22-February -2024	7:59	14	63.6	0	57.6	1	54.2	0	0	0	0	0
22-February -2024	8:19	4140	66.7	0	60.7	1	55.8	0	0	0	0	0
22-February -2024	8:39	785	66.3	0	60.3	1	57.3	0	0	0	0	0
22-February -2024	8:59	331	71.4	0	65.4	1	58.5	0	0	0	0	0

22-February -2024	9:19	4	70.9	0	64.9	1	59.9	0	0	0	0	0
22-February -2024	9:39	4	72.1	0	66.1	1	61.6	0	0	0	0	0
22-February -2024	9:59	4	73.9	0	67.9	7	63.2	0	0	0	0	0
22-February -2024	10:19	5	75.4	0	69.4	17	64.7	0	0	0	0	0
22-February -2024	10:39	3	75.5	0	69.5	1	65.9	0	0	0	0	0
22-February -2024	10:59	4	76.6	0	70.6	1	67.3	0	0	0	0	0
22-February -2024	11:19	13	78.1	0	72.1	1	67.9	0	0	0	0	0
22-February -2024	11:39	5	78.6	0	72.6	1	68.7	0	0	0	0	0
22-February -2024	11:59	6	79.0	0	73.0	1	69.0	0	0	0	0	0

Figure 10-2: AZN-10 Wellhead Sketch

Figure 10-3: AZN-10 Wellhead Sketch

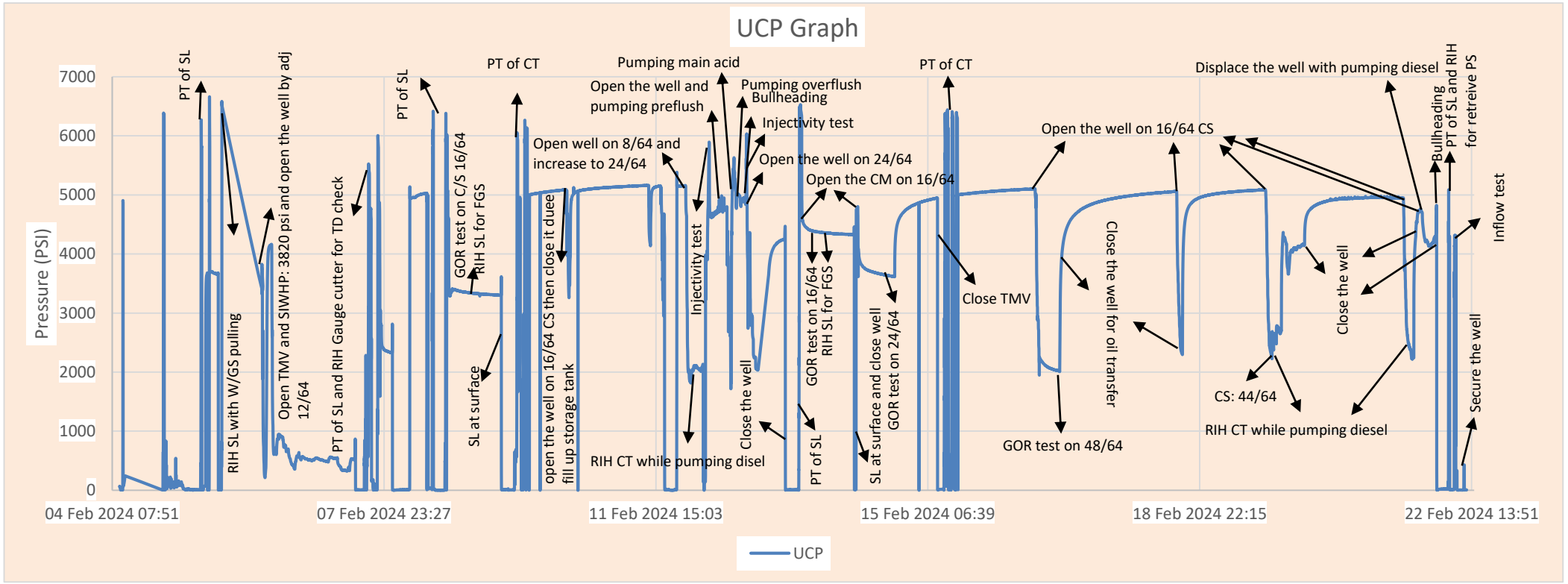


Figure 10-4: AZN-10 Upstream Pressure Graph

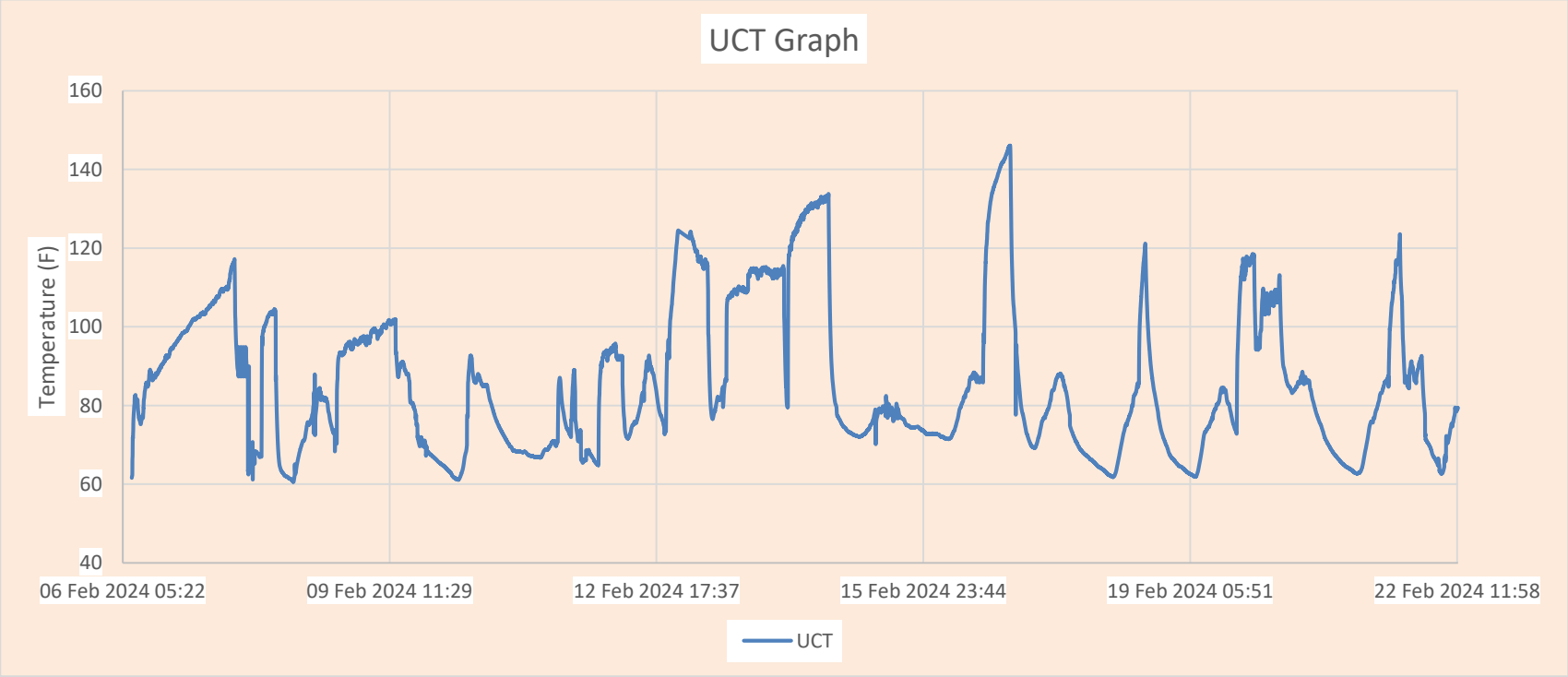


Figure 10-5: AZN-10 Upstream Temperature Graph

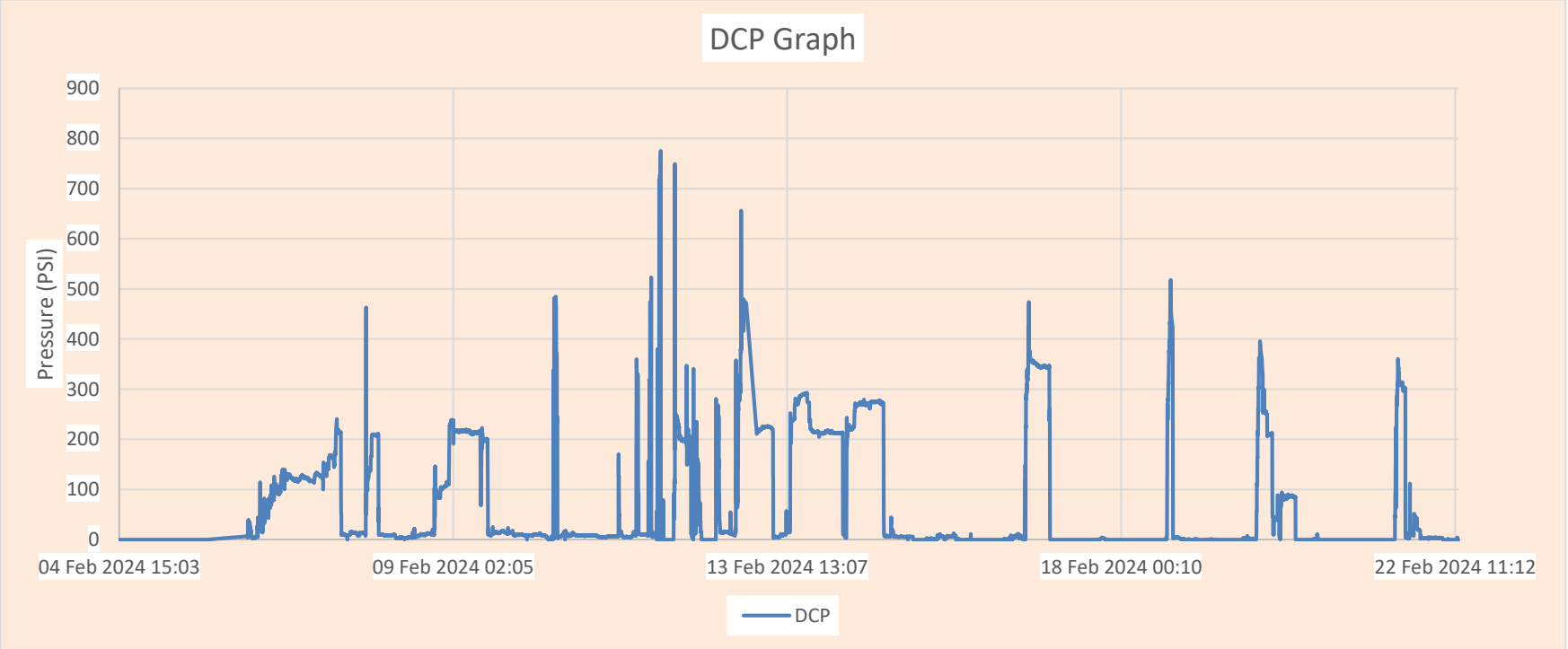


Figure 10-6: AZN-10 Downstream Pressure Graph

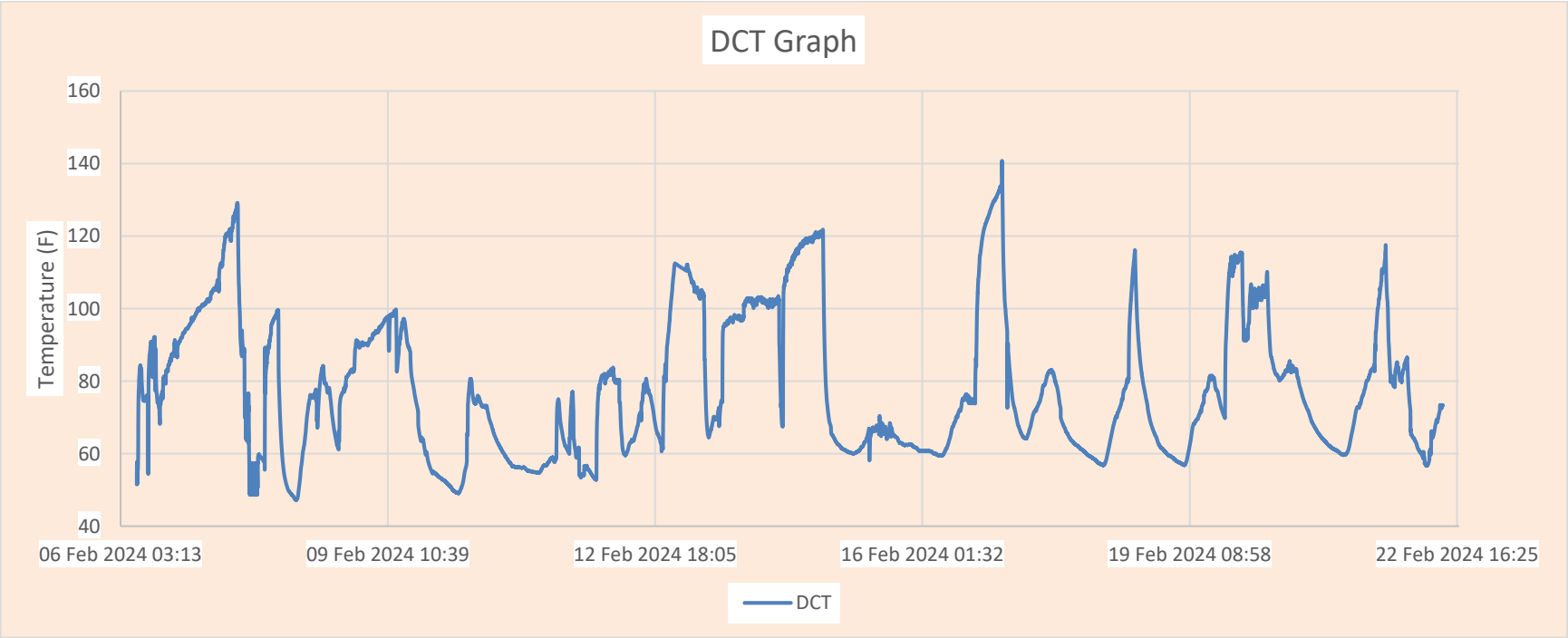


Figure 10-7: AZN-10 Downstream Temperature Graph

Summary of Pumping Data

ITEM	Pumped material	Planned	Total used	Unit
1	Diesel	542	1137	BBL
2	HCL 32%	71	71	BBL
3	Fresh Water	572	256	BBL
4	Mutual Solvent	5	5	Drum
5	Iron Reducing Agent	4	4	Drum
6	Iron Control Agent	2	2	Drum
7	Corrosion Inhibitor Aid	4	3.97	Drum
8	Anti-sludge Agent	4	4	Drum
9	Non-Emulsifier Agent	4	4	Drum
10	Corrosion Inhibitor	4	4	Drum
11	H2S Scavenger	4	4	Drum
12	Asphaltene Solvent (ADR)	30	0	BBL
13	Caustic Soda	5	4	Sacks
14	Liquid nitrogen	……	…….	USG
15	Anti-foam	55	…….	USG

Summary of Pumping Operation Results

Summary Of Pumping Operation Results					
Operation	Start Date and time DD/MM/YYYY, HH:MM	End Date and time DD/MM/YYYY, HH:MM	Volume		Unit
			USED	PLANNED	
Fill Up ant Pressure test Surface Lines with Water	03-February-2024, 08:25	06-February-2023, 19:22	100	100	BBL
acid pickling	11-February-2024, 17:45	11-February-2023, 17:55	5	5	BBL
Displace acid pickling by water	11-February-2024, 17:55	11-February-2024, 18:30	28	28	BBL
CT RIH Simultaneously Pumping diesel	12-February-2024, 01:30	12-February-2024, 07:48	190	190	BBL
injectivity test	12-February-2024, 07:49	12-February-2024, 08:21	26	10	BBL
Pumping pre flush	12-February-2024, 14:50	12-February-2024, 15:24	28	28	BBL
Pumping main acid	12-February-2024, 15:26	12-February-2024, 18:06	143	143	BBL
Pumping over flush	12-February-2024, 18:21	12-February-2024, 18:53	28	28	BBL
CT RIH simultaneously pumping diesel (to spot over flush in perforation)	12-February-2024, 19:00	12-February-2024, 19:35	26	26	BBL
post acid injectivity test	12-February-2024, 19:48	12-February-2024, 20:04	69	69	BBL
CT POOH Simultaneously pumping diesel	12-February-2024, 20:07	13-February-2024, 00:02	62	62	BBL
CT RIH simultaneously pumping diesel	19-February-2024, 21:51	19-February-2024, 24:00	55	55	BBL
Displace well by pumping diesel	20-February-2024, 00:01	20-February-2024, 09:33	254	140	BBL

CT POOH Simultaneously pumping diesel	20-February-2024, 09:34	21-February-2024, 15:46	30	….	BBL
CT RIH simultaneously pumping diesel	21-February-2024, 16:00	21-February-2024, 19:00	55	….	BBL
Displace well by pumping diesel	21-February-2024, 19:55	21-February-2024, 23:59	188	…..	BBL
CT POOH Simultaneously pumping diesel	22-February-2024, 00:00	22-February-2024, 00:43	25	….	BBL
CT Displace with pumping water	22-February-2024, 00:44	22-February-2024, 01:17	26	….	BBL
Bullheading with diesel	22-February-2024, 02:20	22-February-2024, 02:34	30	….	BBL

Pumping Operation Results & Plots

Pre-Acid Bullhead Operation Report				
Date (DD-MM-YYYY)	Time (HH:MM)	Pumping pressure (PSI)	Pump Rate (BPM)	Volume total (BBL)
12-February -2024	7:50	4950	0.69	0.41
12-February -2024	7:51	4986	0.56	0.89
12-February -2024	7:52	5024	0.53	1.35
12-February -2024	7:53	5056	0.5	1.81
12-February -2024	7:53	5102	0.61	2.27
12-February -2024	7:54	5094	0.53	2.73
12-February -2024	7:55	5126	0.66	3.18
12-February -2024	7:56	5133	0.5	3.64
12-February -2024	7:57	5148	0.48	4.09
12-February -2024	7:58	5156	0.56	4.54
12-February -2024	7:58	5175	0.56	4.99
12-February -2024	7:59	5245	0.87	5.59
12-February -2024	8:00	5331	0.82	6.29
12-February -2024	8:01	5424	1.03	7.07
12-February -2024	8:02	5459	1.03	7.9
12-February -2024	8:03	5556	1.01	8.74
12-February -2024	8:03	5588	0.99	9.57
12-February -2024	8:04	5658	0.98	10.41
12-February -2024	8:05	5688	1.06	11.23
12-February -2024	8:06	5735	0.95	12.06
12-February -2024	8:07	5796	0.98	12.87
12-February -2024	8:08	5781	0.95	13.69
12-February -2024	8:08	5917	1.03	14.49
12-February -2024	8:09	5983	0.98	15.29
12-February -2024	8:10	6006	0.93	16.08
12-February -2024	8:11	6060	0.98	16.87
12-February -2024	8:12	6150	1.01	17.7
12-February -2024	8:13	6173	1.08	18.54
12-February -2024	8:13	6216	1.03	19.36
12-February -2024	8:14	6263	1.03	20.22
12-February -2024	8:15	6290	0.95	21.05
12-February -2024	8:16	6309	0.98	21.87
12-February -2024	8:17	6243	0.98	22.7
12-February -2024	8:18	6383	1.01	23.52
12-February -2024	8:18	6297	0.95	24.34
12-February -2024	8:19	6209	0.44	24.93

12-February -2024	8:20	6045	0.46	25.29

Table 10-6: AZN-10 Pre-Acid Bullhead Operation Report

Pumping Pre-Flush Operation Report				
Date (DD-MM-YYYY)	Time (HH:MM)	Pumping pressure (PSI)	Pump Rate (BPM)	Volume total (BBL)
12-February -2024	14:50	4485	0.66	0.79
12-February -2024	14:51	4795	0.69	1.33
12-February -2024	14:52	4744	0.77	1.93
12-February -2024	14:53	4741	0.85	2.6
12-February -2024	14:54	4664	0.82	3.31
12-February -2024	14:54	4573	0.87	4.03
12-February -2024	14:55	4585	0.82	4.76
12-February -2024	14:56	4551	0.95	5.49
12-February -2024	14:57	4694	0.87	6.25
12-February -2024	14:58	4659	0.95	7.02
12-February -2024	14:59	4605	0.9	7.8
12-February -2024	14:59	4512	0.93	8.59
12-February -2024	15:00	4476	0.95	9.37
12-February -2024	15:01	4333	0.9	10.13
12-February -2024	15:02	4360	0.9	10.89
12-February -2024	15:03	4372	0.9	11.65
12-February -2024	15:04	4438	0.87	12.39
12-February -2024	15:04	4251	0.98	13.17
12-February -2024	15:05	4159	0.95	13.95
12-February -2024	15:06	3980	0.95	14.74
12-February -2024	15:07	3956	0.98	15.54
12-February -2024	15:08	3910	0.98	16.35
12-February -2024	15:09	4004	0.93	17.16
12-February -2024	15:09	3902	0.98	17.96
12-February -2024	15:10	3871	0.98	18.77
12-February -2024	15:11	3944	0.95	19.6
12-February -2024	15:12	3518	0.9	20.35
12-February -2024	15:13	3405	0.82	21.05
12-February -2024	15:14	2994	0.74	21.7
12-February -2024	15:14	2951	0.74	22.33
12-February -2024	15:15	3006	0.74	22.95
12-February -2024	15:16	3165	0.82	23.62
12-February -2024	15:17	3180	0.79	24.3
12-February -2024	15:18	3165	0.79	24.97
12-February -2024	15:19	3366	0.79	25.63

12-February -2024	15:19	2563	0	26.02
12-February -2024	15:20	3102	0	26.02
12-February -2024	15:21	4159	0.61	26.22
12-February -2024	15:22	4679	0.63	26.67
12-February -2024	15:23	4938	0.63	27.2
12-February -2024	15:24	5226	0.63	27.74
12-February -2024	15:24	4166	0	27.99

Table 10-7: AZN-10 Pumping Pre-Flush Operation Report

Pumping Main Acid Operation Report				
Date (DD-MM-YYYY)	**Time (HH:MM)**	**Pumping pressure (PSI)**	**Pump Rate (BPM)**	**Volume total (BBL)**
12-February -2024	15:26	5412	0.71	0.58
12-February -2024	15:27	5823	0.71	1.17
12-February -2024	15:28	5914	0.71	1.76
12-February -2024	15:29	5840	0.77	2.34
12-February -2024	15:30	5871	0.71	2.92
12-February -2024	15:30	5925	0.69	3.49
12-February -2024	15:31	5936	0.66	4.06
12-February -2024	15:32	6053	0.66	4.64
12-February -2024	15:33	6006	0.71	5.21
12-February -2024	15:34	6100	0.79	5.79
12-February -2024	15:35	6135	0.74	6.35
12-February -2024	15:35	6263	0.79	6.94
12-February -2024	15:36	5991	0.61	7.52
12-February -2024	15:37	5991	0.66	8.07
12-February -2024	15:38	5917	0.58	8.6

12-February -2024	15:39	5893	0.58	9.13
12-February -2024	15:40	5886	0.61	9.65
12-February -2024	15:40	5808	0.63	10.18
12-February -2024	15:41	5793	0.66	10.71
12-February -2024	15:42	5781	0.63	11.23
12-February -2024	15:43	5886	0.63	11.76
12-February -2024	15:44	5781	0.63	12.3
12-February -2024	15:45	5781	0.58	12.84
12-February -2024	15:45	5750	0.58	13.36
12-February -2024	15:46	5731	0.74	13.91
12-February -2024	15:47	5813	0.71	14.45
12-February -2024	15:48	5684	0.66	14.99
12-February -2024	15:49	5754	0.77	15.53
12-February -2024	15:50	5665	0.71	16.07
12-February -2024	15:50	5777	0.69	16.62
12-February -2024	15:51	5571	0.61	17.15
12-February -2024	15:52	5475	0.61	17.68
12-February -2024	15:53	5382	0.56	18.18
12-February -2024	15:54	5214	0.56	18.65
12-February -2024	15:55	5175	0.5	19.1
12-February -2024	15:55	5245	0.63	19.56
12-February -2024	15:56	5160	0.5	20.02

12-February -2024	15:57	5168	0.53	20.49
12-February -2024	15:58	5184	0.5	20.97
12-February -2024	15:59	5245	0.56	21.47
12-February -2024	16:00	5269	0.61	21.96
12-February -2024	16:00	5184	0.58	22.45
12-February -2024	16:01	5172	0.69	22.95
12-February -2024	16:02	5187	0.56	23.46
12-February -2024	16:03	5226	0.69	23.99
12-February -2024	16:04	5249	0.61	24.51
12-February -2024	16:05	5203	0.56	25.03
12-February -2024	16:05	5148	0.56	25.54
12-February -2024	16:06	5109	0.53	26.04
12-February -2024	16:07	5056	0.66	26.54
12-February -2024	16:08	4977	0.56	27.05
12-February -2024	16:09	5017	0.53	27.54
12-February -2024	16:10	5059	0.69	28.04
12-February -2024	16:10	4974	0.53	28.53
12-February -2024	16:11	4861	0.66	29.04
12-February -2024	16:12	4846	0.66	29.56
12-February -2024	16:13	4919	0.63	30.1
12-February -2024	16:14	4911	0.66	30.67
12-February -2024	16:15	4938	0.71	31.23

12-February -2024	16:15	5148	0.74	31.87
12-February -2024	16:16	5641	0.93	32.56
12-February -2024	16:17	6091	1.08	33.36
12-February -2024	16:18	6130	0.95	34.2
12-February -2024	16:19	6154	1.08	35.01
12-February -2024	16:20	6178	1.03	35.84
12-February -2024	16:20	5979	1.01	36.68
12-February -2024	16:21	6065	1.08	37.52
12-February -2024	16:22	5972	1.06	38.35
12-February -2024	16:23	6084	1.03	39.19
12-February -2024	16:24	5972	1.01	40.04
12-February -2024	16:25	6030	0.95	40.88
12-February -2024	16:25	5948	1.01	41.72
12-February -2024	16:26	5953	1.01	42.57
12-February -2024	16:27	6045	0.98	43.42
12-February -2024	16:28	6041	1.01	44.27
12-February -2024	16:29	5917	1.01	45.13
12-February -2024	16:30	5874	1.06	45.97
12-February -2024	16:30	6014	0.98	46.82
12-February -2024	16:31	5925	0.98	47.67
12-February -2024	16:32	6076	0.98	48.54
12-February -2024	16:33	6030	1.01	49.37

12-February -2024	16:34	6069	1.01	50.22
12-February -2024	16:35	6205	0.98	51.04
12-February -2024	16:35	6333	0.93	51.86
12-February -2024	16:36	6414	1.01	52.68
12-February -2024	16:37	6464	0.88	53.48
12-February -2024	16:38	6333	1.03	54.3
12-February -2024	16:39	6387	0.95	55.11
12-February -2024	16:40	6395	0.95	55.91
12-February -2024	16:40	6441	0.93	56.7
12-February -2024	16:41	6434	0.93	57.52
12-February -2024	16:42	6376	0.95	58.31
12-February -2024	16:43	6352	1.03	59.12
12-February -2024	16:44	6495	0.95	59.94
12-February -2024	16:45	6473	0.98	60.74
12-February -2024	16:45	6453	0.95	61.55
12-February -2024	16:46	3499	1.06	61.89
12-February -2024	16:47	3347	0.18	61.97
12-February -2024	16:48	3925	0	62.22
12-February -2024	16:49	3366	0	62.36
12-February -2024	16:50	3335	0	62.36
12-February -2024	16:50	3320	0	62.36
12-February -2024	16:51	3305	0	62.36

12-February -2024	16:52	3296	0	62.36
12-February -2024	16:53	3281	0	62.36
12-February -2024	16:54	3274	0	62.36
12-February -2024	16:55	3265	0	62.36
12-February -2024	16:55	3254	0	62.36
12-February -2024	16:56	5649	0.9	62.68
12-February -2024	16:57	5832	0.85	63.43
12-February -2024	16:58	5847	0.87	64.17
12-February -2024	16:59	5723	0.9	64.93
12-February -2024	17:00	5661	0.98	65.7
12-February -2024	17:00	5743	0.87	66.46
12-February -2024	17:01	5634	0.98	67.23
12-February -2024	17:02	5793	0.98	68.03
12-February -2024	17:03	5948	1.06	68.86
12-February -2024	17:04	5948	1.01	69.68
12-February -2024	17:05	6018	1.08	70.54
12-February -2024	17:05	5999	1.06	71.38
12-February -2024	17:06	6057	0.98	72.21
12-February -2024	17:07	6188	0.95	73.05
12-February -2024	17:08	6100	1.08	73.9
12-February -2024	17:09	6057	1.08	74.72
12-February -2024	17:10	6014	0.98	75.56

12-February -2024	17:10	5987	0.98	76.41
12-February -2024	17:11	5932	1.11	77.24
12-February -2024	17:12	5855	1.04	78.11
12-February -2024	17:13	5789	1.03	78.96
12-February -2024	17:14	5793	1.11	79.82
12-February -2024	17:15	5746	1.01	80.7
12-February -2024	17:15	5618	1.06	81.58
12-February -2024	17:16	5762	1.06	82.45
12-February -2024	17:17	5707	1.08	83.32
12-February -2024	17:18	5746	1.03	84.22
12-February -2024	17:19	5637	1.01	85.1
12-February -2024	17:20	5528	1.08	85.98
12-February -2024	17:20	5692	1.06	86.86
12-February -2024	17:21	5630	1.01	87.76
12-February -2024	17:22	5680	1.03	88.64
12-February -2024	17:23	5653	1.08	89.53
12-February -2024	17:24	5556	1.03	90.41
12-February -2024	17:25	5677	1.03	91.3
12-February -2024	17:25	5544	1.06	92.19
12-February -2024	17:26	5630	1.01	93.08
12-February -2024	17:27	5622	1.03	93.96
12-February -2024	17:28	5576	1.03	94.86

12-February -2024	17:29	5552	1.03	95.75
12-February -2024	17:30	5528	1.08	96.64
12-February -2024	17:30	5665	1.03	97.53
12-February -2024	17:31	5653	1.03	98.42
12-February -2024	17:32	5634	1.03	99.3
12-February -2024	17:33	5634	1.08	100.19
12-February -2024	17:34	5501	1.06	101.08
12-February -2024	17:35	5509	1.14	101.97
12-February -2024	17:35	5626	1.08	102.87
12-February -2024	17:36	5634	1.03	103.76
12-February -2024	17:37	5494	1.06	104.65
12-February -2024	17:38	5576	1.14	105.54
12-February -2024	17:39	5630	1.07	106.43
12-February -2024	17:40	5646	1.03	107.32
12-February -2024	17:40	5521	1.06	108.22
12-February -2024	17:41	5513	1.06	109.11
12-February -2024	17:42	5479	1.08	110
12-February -2024	17:43	5544	1.14	110.89
12-February -2024	17:44	5579	1.14	111.78
12-February -2024	17:45	5544	1.06	112.67
12-February -2024	17:45	5521	1.08	113.56
12-February -2024	17:46	5658	1.03	114.46

12-February -2024	17:47	5665	1.03	115.34
12-February -2024	17:48	5646	1.01	116.23
12-February -2024	17:49	5549	1.14	117.12
12-February -2024	17:50	5668	1.03	118.01
12-February -2024	17:50	5598	1.14	118.9
12-February -2024	17:51	5533	1.08	119.79
12-February -2024	17:52	5549	1.06	120.68
12-February -2024	17:53	5552	1.08	121.57
12-February -2024	17:54	5412	1.08	122.43
12-February -2024	17:55	5373	0.98	123.29
12-February -2024	17:55	5389	0.98	124.13
12-February -2024	17:56	5409	0.98	124.95
12-February -2024	17:57	5451	0.98	125.79
12-February -2024	17:58	5400	1.01	126.63
12-February -2024	17:59	5366	1.01	127.47
12-February -2024	18:00	5455	0.98	128.3
12-February -2024	18:00	5482	1.08	129.14
12-February -2024	18:01	5424	1.01	129.98
12-February -2024	18:02	5455	1.01	130.8
12-February -2024	18:03	5552	1.03	131.64
12-February -2024	18:04	5583	0.95	132.48
12-February -2024	18:05	5610	0.95	133.31

12-February -2024	18:05	5568	1.06	134.15
12-February -2024	17:00	5661	0.98	65.7
12-February -2024	17:00	5743	0.87	66.46
12-February -2024	17:01	5634	0.98	67.23
12-February -2024	17:02	5793	0.98	68.03
12-February -2024	17:03	5948	1.06	68.86
12-February -2024	17:04	5948	1.01	69.68
12-February -2024	17:05	6018	1.08	70.54
12-February -2024	17:05	5999	1.06	71.38
12-February -2024	17:06	6057	0.98	72.21
12-February -2024	17:07	6188	0.95	73.05
12-February -2024	17:08	6100	1.08	73.9
12-February -2024	17:09	6057	1.08	74.72
12-February -2024	17:10	6014	0.98	75.56
12-February -2024	17:10	5987	0.98	76.41
12-February -2024	17:11	5932	1.11	77.24
12-February -2024	17:12	5855	1.04	78.11
12-February -2024	17:13	5789	1.03	78.96
12-February -2024	17:14	5793	1.11	79.82
12-February -2024	17:15	5746	1.01	80.7
12-February -2024	17:15	5618	1.06	81.58
12-February -2024	17:16	5762	1.06	82.45

12-February -2024	17:17	5707	1.08	83.32
12-February -2024	17:18	5746	1.03	84.22
12-February -2024	17:19	5637	1.01	85.1
12-February -2024	17:20	5528	1.08	85.98
12-February -2024	17:20	5692	1.06	86.86
12-February -2024	17:21	5630	1.01	87.76
12-February -2024	17:22	5680	1.03	88.64
12-February -2024	17:23	5653	1.08	89.53
12-February -2024	17:24	5556	1.03	90.41
12-February -2024	17:25	5677	1.03	91.3
12-February -2024	17:25	5544	1.06	92.19
12-February -2024	17:26	5630	1.01	93.08
12-February -2024	17:27	5622	1.03	93.96
12-February -2024	17:28	5576	1.03	94.86
12-February -2024	17:29	5552	1.03	95.75
12-February -2024	17:30	5528	1.08	96.64
12-February -2024	17:30	5665	1.03	97.53
12-February -2024	17:31	5653	1.03	98.42
12-February -2024	17:32	5634	1.03	99.3
12-February -2024	17:33	5634	1.08	100.19
12-February -2024	17:34	5501	1.06	101.08
12-February -2024	17:35	5509	1.14	101.97

12-February -2024	17:35	5626	1.08	102.87
12-February -2024	17:36	5634	1.03	103.76
12-February -2024	17:37	5494	1.06	104.65
12-February -2024	17:38	5576	1.14	105.54
12-February -2024	17:39	5630	1.07	106.43
12-February -2024	17:40	5646	1.03	107.32
12-February -2024	17:40	5521	1.06	108.22
12-February -2024	17:41	5513	1.06	109.11
12-February -2024	17:42	5479	1.08	110
12-February -2024	17:43	5544	1.14	110.89
12-February -2024	17:44	5579	1.14	111.78
12-February -2024	17:45	5544	1.06	112.67
12-February -2024	17:45	5521	1.08	113.56
12-February -2024	17:46	5658	1.03	114.46
12-February -2024	17:47	5665	1.03	115.34
12-February -2024	17:48	5646	1.01	116.23
12-February -2024	17:49	5549	1.14	117.12
12-February -2024	17:50	5668	1.03	118.01
12-February -2024	17:50	5598	1.14	118.9
12-February -2024	17:51	5533	1.08	119.79
12-February -2024	17:52	5549	1.06	120.68
12-February -2024	17:53	5552	1.08	121.57

12-February -2024	17:54	5412	1.08	122.43
12-February -2024	17:55	5373	0.98	123.29
12-February -2024	17:55	5389	0.98	124.13
12-February -2024	17:56	5409	0.98	124.95
12-February -2024	17:57	5451	0.98	125.79
12-February -2024	17:58	5400	1.01	126.63
12-February -2024	17:59	5366	1.01	127.47
12-February -2024	18:00	5455	0.98	128.3
12-February -2024	18:00	5482	1.08	129.14
12-February -2024	18:01	5424	1.01	129.98
12-February -2024	18:02	5455	1.01	130.8
12-February -2024	18:03	5552	1.03	131.64
12-February -2024	18:04	5583	0.95	132.48
12-February -2024	18:05	5610	0.95	133.31
12-February -2024	18:05	5568	1.06	134.15
12-February -2024	17:45	5544	1.06	112.67
12-February -2024	17:45	5521	1.08	113.56
12-February -2024	17:46	5658	1.03	114.46
12-February -2024	17:47	5665	1.03	115.34
12-February -2024	17:48	5646	1.01	116.23
12-February -2024	17:49	5549	1.14	117.12
12-February -2024	17:50	5668	1.03	118.01

12-February -2024	17:50	5598	1.14	118.9
12-February -2024	17:51	5533	1.08	119.79
12-February -2024	17:52	5549	1.06	120.68
12-February -2024	17:53	5552	1.08	121.57
12-February -2024	17:54	5412	1.08	122.43
12-February -2024	17:55	5373	0.98	123.29
12-February -2024	17:55	5389	0.98	124.13
12-February -2024	17:56	5409	0.98	124.95
12-February -2024	17:57	5451	0.98	125.79
12-February -2024	17:58	5400	1.01	126.63
12-February -2024	17:59	5366	1.01	127.47
12-February -2024	18:00	5455	0.98	128.3
12-February -2024	18:00	5482	1.08	129.14
12-February -2024	18:01	5424	1.01	129.98
12-February -2024	18:02	5455	1.01	130.8
12-February -2024	18:03	5552	1.03	131.64
12-February -2024	18:04	5583	0.95	132.48
12-February -2024	18:05	5610	0.95	133.31
12-February -2024	18:05	5568	1.06	134.15

Table 10-8: AZN-10 Pumping Main Acid Operation Report

Pumping Over Flush Operation Report

Date (DD-MM-YYYY)	Time (HH:MM)	Pumping pressure (PSI)	Pump Rate (BPM)	Volume total (BBL)
12-February -2024	18:22	5459	0.82	0.65
12-February -2024	18:23	5486	0.98	1.38
12-February -2024	18:24	5424	0.95	2.12
12-February -2024	18:25	5494	0.82	2.85
12-February -2024	18:25	5400	0.87	3.58
12-February -2024	18:26	5498	0.87	4.32
12-February -2024	18:27	5595	0.98	5.1
12-February -2024	18:28	5470	0.93	5.87
12-February -2024	18:29	5614	0.87	6.62
12-February -2024	18:30	5680	0.95	7.39
12-February -2024	18:30	5886	0.95	8.16
12-February -2024	18:31	5991	1.06	8.96
12-February -2024	18:32	6065	0.87	9.75
12-February -2024	18:33	5987	1.01	10.55
12-February -2024	18:34	6038	0.93	11.33
12-February -2024	18:35	6057	0.98	12.1
12-February -2024	18:35	6135	0.87	12.89
12-February -2024	18:36	6209	0.87	13.67
12-February -2024	18:37	6231	0.93	14.44
12-February -2024	18:38	6188	1.01	15.22

12-February -2024	18:39	6205	0.93	15.99
12-February -2024	18:40	6173	0.93	16.74
12-February -2024	18:40	6328	1.01	17.5
12-February -2024	18:41	6383	0.85	18.28
12-February -2024	18:42	6336	1.01	19.03
12-February -2024	18:43	6403	0.85	19.78
12-February -2024	18:44	6461	0.9	20.52
12-February -2024	18:45	6488	0.9	21.27
12-February -2024	18:45	6414	0.9	22.01
12-February -2024	18:46	6523	0.82	22.77
12-February -2024	18:47	6468	0.87	23.49
12-February -2024	18:48	6483	0.87	24.24
12-February -2024	18:49	6558	0.98	24.97
12-February -2024	18:50	6534	0.87	25.69
12-February -2024	18:50	6662	0.87	26.41
12-February -2024	18:51	6671	0.87	27.14
12-February -2024	18:52	6558	0.87	27.86
12-February -2024	18:22	5459	0.82	0.65
12-February -2024	18:23	5486	0.98	1.38
12-February -2024	18:24	5424	0.95	2.12
12-February -2024	18:25	5494	0.82	2.85
12-February -2024	18:25	5400	0.87	3.58

12-February -2024	18:26	5498	0.87	4.32
12-February -2024	18:27	5595	0.98	5.1
12-February -2024	18:28	5470	0.93	5.87
12-February -2024	18:29	5614	0.87	6.62
12-February -2024	18:30	5680	0.95	7.39
12-February -2024	18:30	5886	0.95	8.16
12-February -2024	18:31	5991	1.06	8.96
12-February -2024	18:32	6065	0.87	9.75
12-February -2024	18:33	5987	1.01	10.55
12-February -2024	18:34	6038	0.93	11.33
12-February -2024	18:35	6057	0.98	12.1
12-February -2024	18:35	6135	0.87	12.89
12-February -2024	18:36	6209	0.87	13.67
12-February -2024	18:37	6231	0.93	14.44
12-February -2024	18:38	6188	1.01	15.22
12-February -2024	18:39	6205	0.93	15.99
12-February -2024	18:40	6173	0.93	16.74
12-February -2024	18:40	6328	1.01	17.5
12-February -2024	18:41	6383	0.85	18.28
12-February -2024	18:42	6336	1.01	19.03
12-February -2024	18:43	6403	0.85	19.78
12-February -2024	18:44	6461	0.9	20.52

12-February -2024	18:45	6488	0.9	21.27
12-February -2024	18:45	6414	0.9	22.01
12-February -2024	18:46	6523	0.82	22.77
12-February -2024	18:47	6468	0.87	23.49
12-February -2024	18:48	6483	0.87	24.24
12-February -2024	18:49	6558	0.98	24.97
12-February -2024	18:50	6534	0.87	25.69
12-February -2024	18:50	6662	0.87	26.41
12-February -2024	18:51	6671	0.87	27.14

Table 10-9: AZN-10 Pumping Over Flush Operation Report

Post Acid Bullhead Operation Report				
Date (DD-MM-YYYY)	**Time (HH:MM)**	**Pumping pressure (PSI)**	**Pump Rate (BPM)**	**Volume total (BBL)**
12-February -2024	19:49	5303	1.14	0.73
12-February -2024	19:49	5254	1.67	1.93
12-February -2024	19:50	5339	1.56	2.8
12-February -2024	19:51	5424	2.28	4.63
12-February -2024	19:52	5770	3.49	7.21
12-February -2024	19:53	5808	3.6	10.6
12-February -2024	19:54	5859	3.52	14.65
12-February -2024	19:54	5925	3.49	18.87
12-February -2024	19:55	6294	3.78	23.36
12-February -2024	19:56	5777	3.17	27.53

12-February -2024	19:57	6205	2.72	31.28
12-February -2024	19:58	6231	2.72	36.43
12-February -2024	19:59	6173	2.83	41.17
12-February -2024	19:59	6135	2.75	45.14
12-February -2024	20:00	6049	0	48.54
12-February -2024	20:01	6275	2.7	52.09
12-February -2024	20:02	6391	2.7	57.13
12-February -2024	20:03	6845	3.52	62.38
12-February -2024	20:04	5874	0	65.54

Table 10-10: AZN-10 Pumping Post-Acid Bullhead 1 Operation Report

Pumping Post Acid Bullhead (2) Operation Report				
Date (DD-MM-YYYY)	**Time (HH:MM)**	**Pumping pressure (PSI)**	**Pump Rate (BPM)**	**Volume total (BBL)**
22-February -2024	2:21	6379	1.04	0.79
22-February -2024	2:22	6473	1.5	1.92
22-February -2024	2:23	6468	1.5	3.15
22-February -2024	2:23	6437	1.5	4.41
22-February -2024	2:24	6391	1.8	5.83
22-February -2024	2:25	6348	1.8	7.31
22-February -2024	2:26	6318	2.01	8.98
22-February -2024	2:27	6267	2.01	10.63
22-February -2024	2:28	6231	2.01	12.31
22-February -2024	2:28	6193	2.41	14.16

22-February -2024	2:29	6150	2.41	16.12
22-February -2024	2:30	6115	2.76	18.28
22-February -2024	2:31	6282	2.76	20.62
22-February -2024	2:32	6473	3.52	23.15
22-February -2024	2:33	6473	3.53	26.1
22-February -2024	2:33	6445	3.47	29.01

Table 10-11: AZN-10 Pumping Post-Acid Bullhead 2 Operation Report

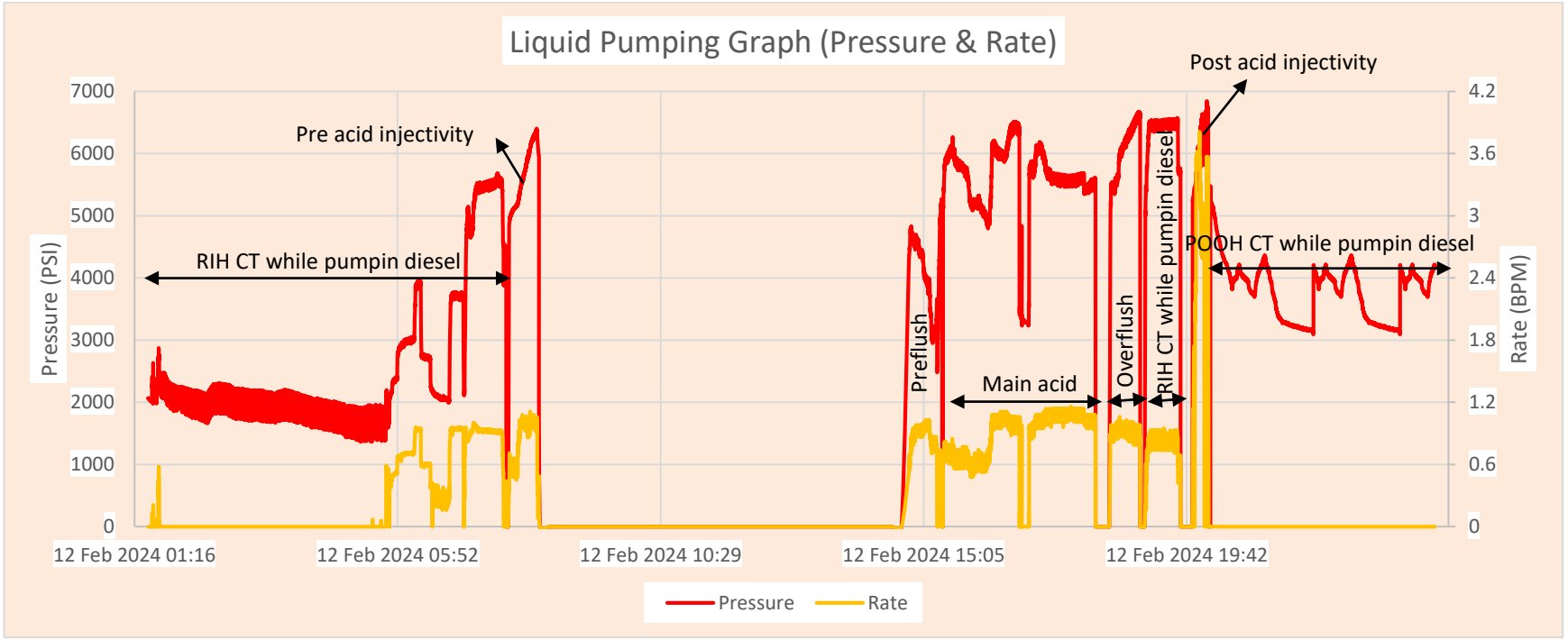


Figure 10-8: AZN-10 Liquid Pumping Graph (Pr. & Rate)

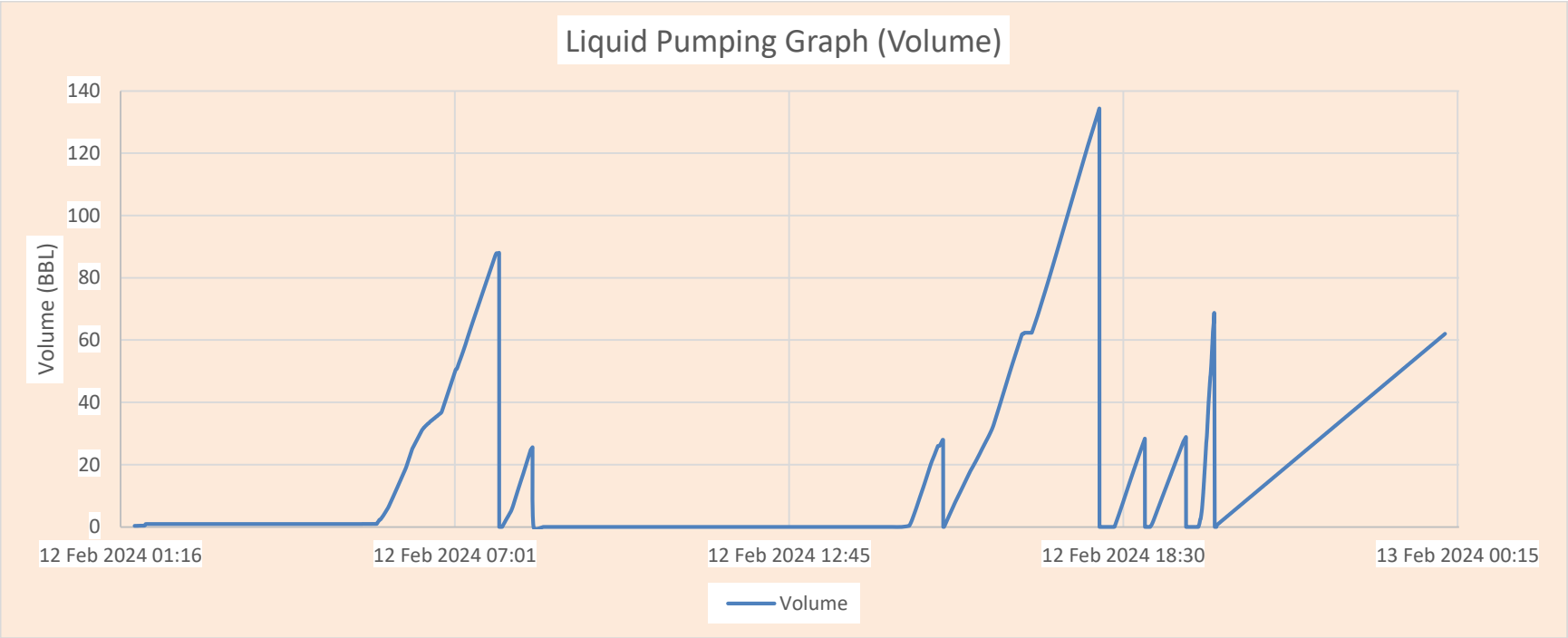


Figure 10-9: AZN-10 Liquid Pumping Graph (Volume)

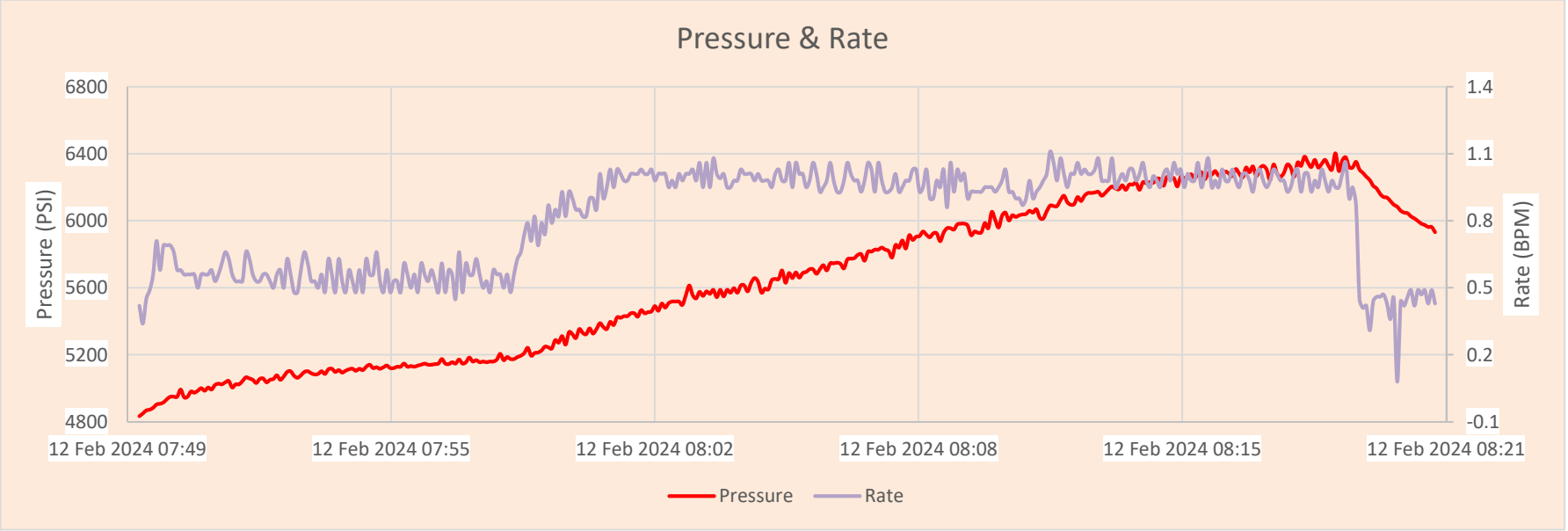


Figure 10-10: AZN-10 Pre-Acid Bullhead Graph (Pr. & Rate)

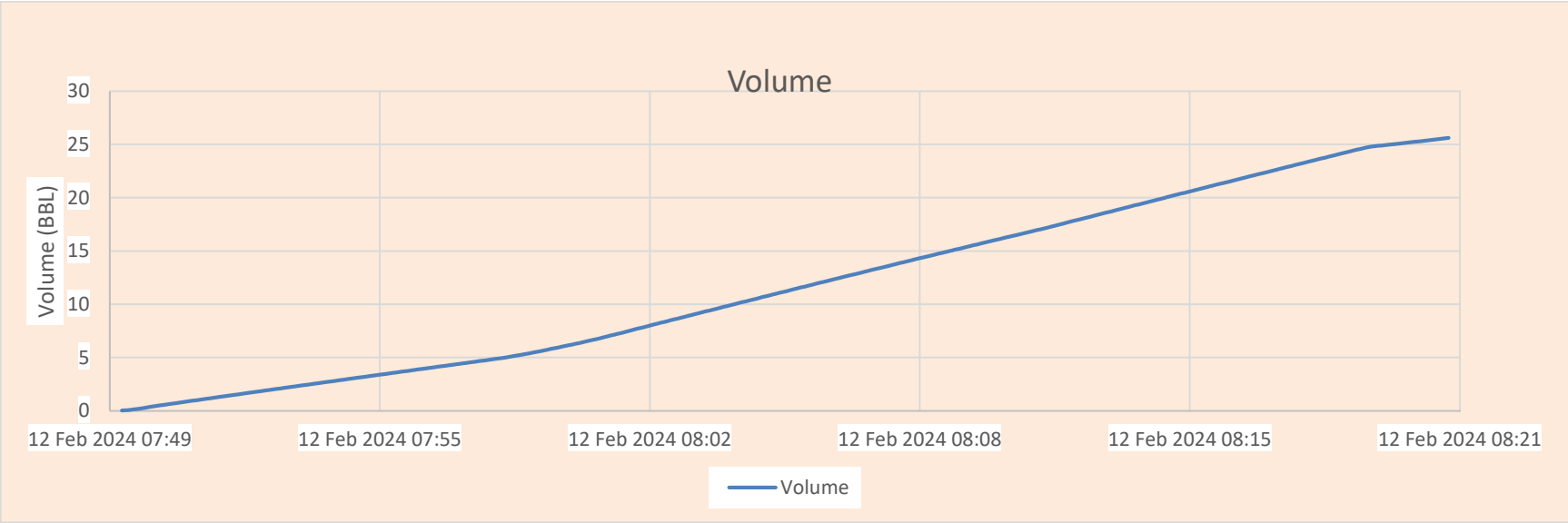


Figure 10-11: AZN-10 Pre-Acid Bullhead Graph (Volume)

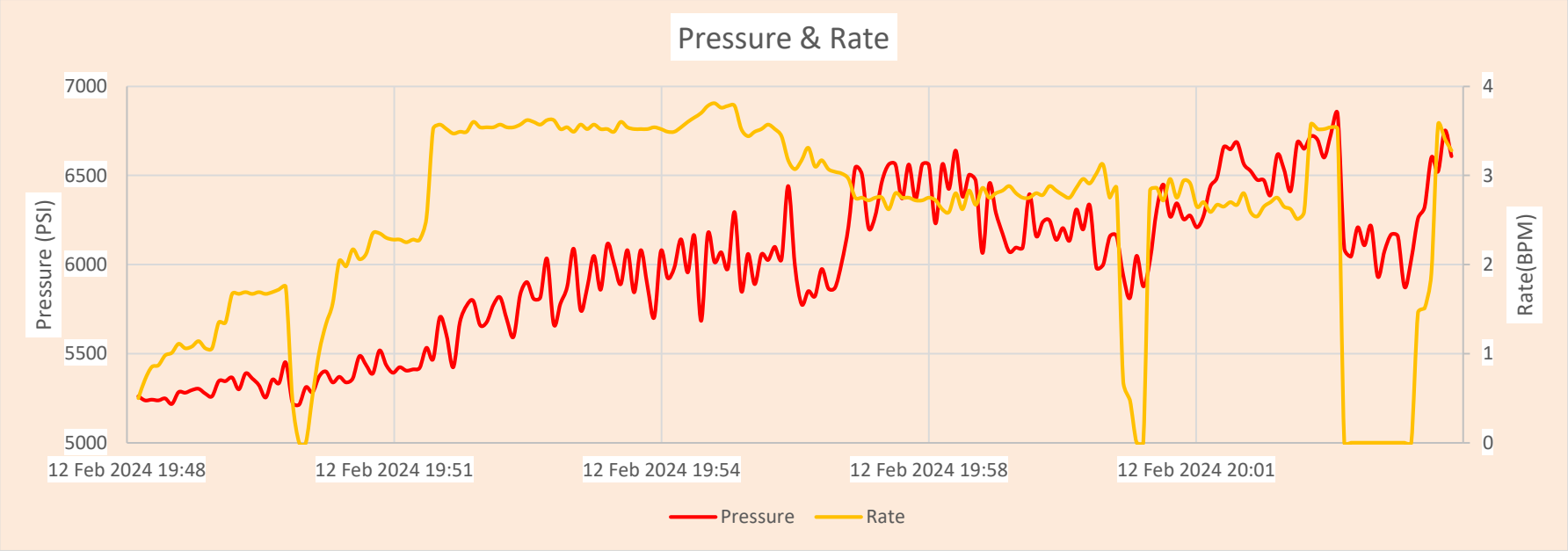


Figure 10-12: AZN-10 Post-Acid Bullhead 1 Graph (Pr. & Rate)

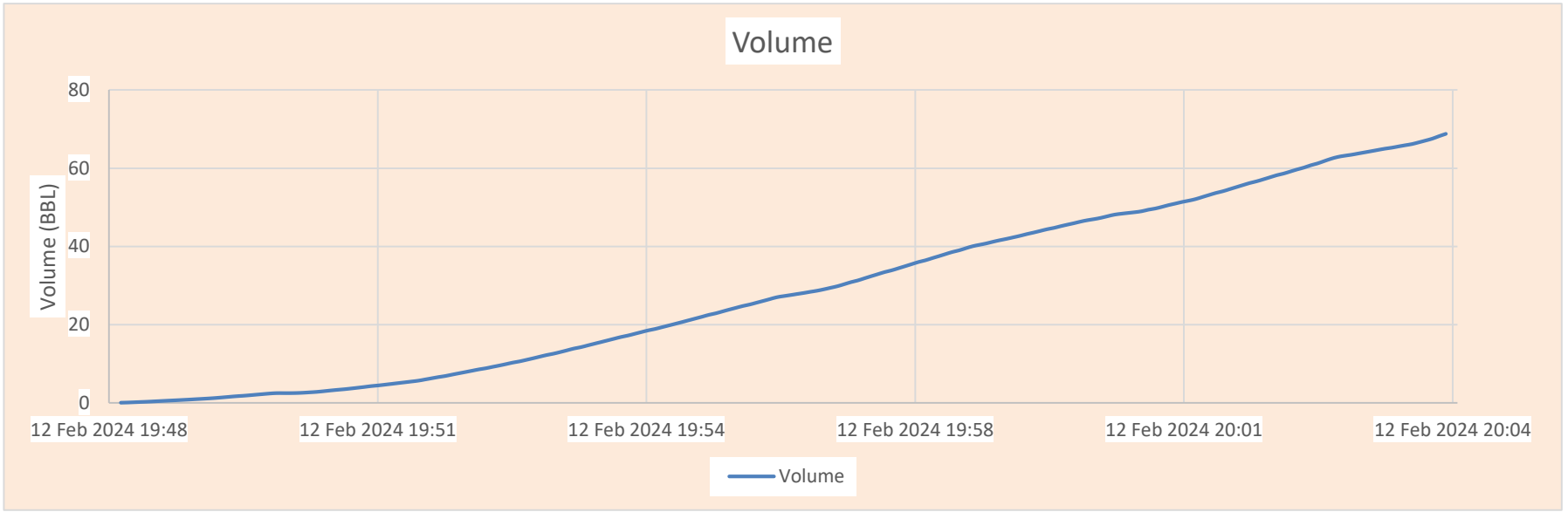


Figure 10-13: AZN-10 Post-Acid Bullhead 1 Graph (Volume)

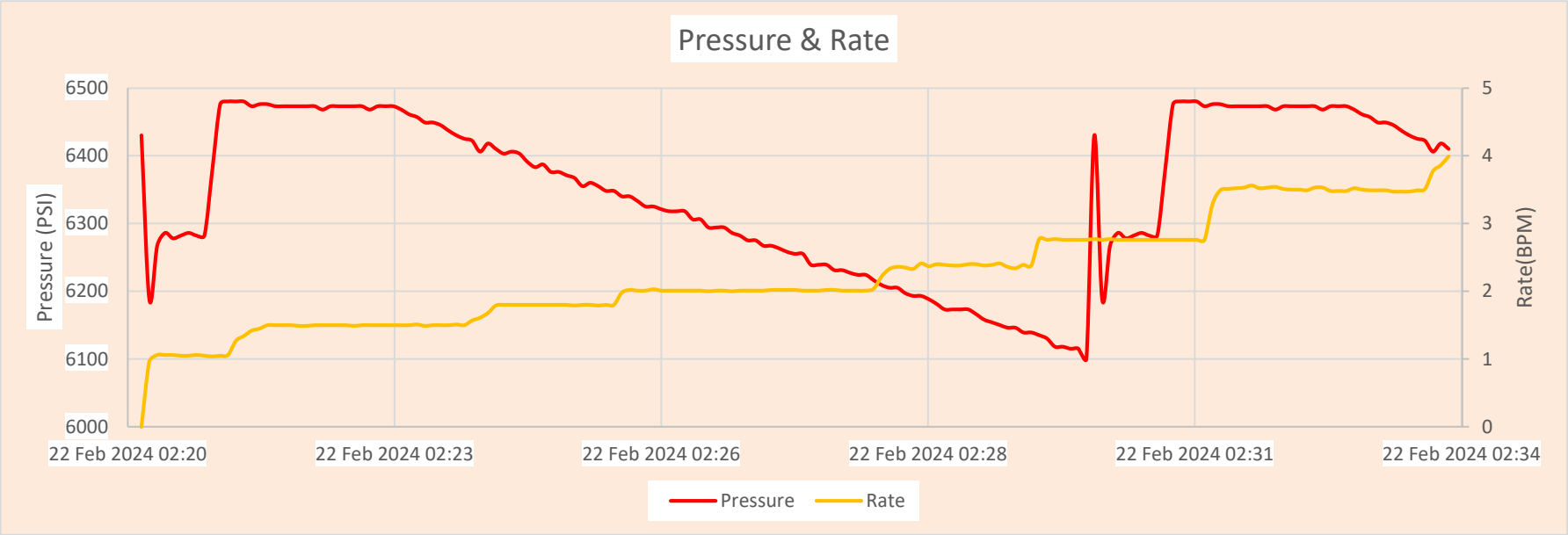


Figure 10-14: AZN-10 Post-Acid Bullhead 2 Graph (Pr. & Rate)

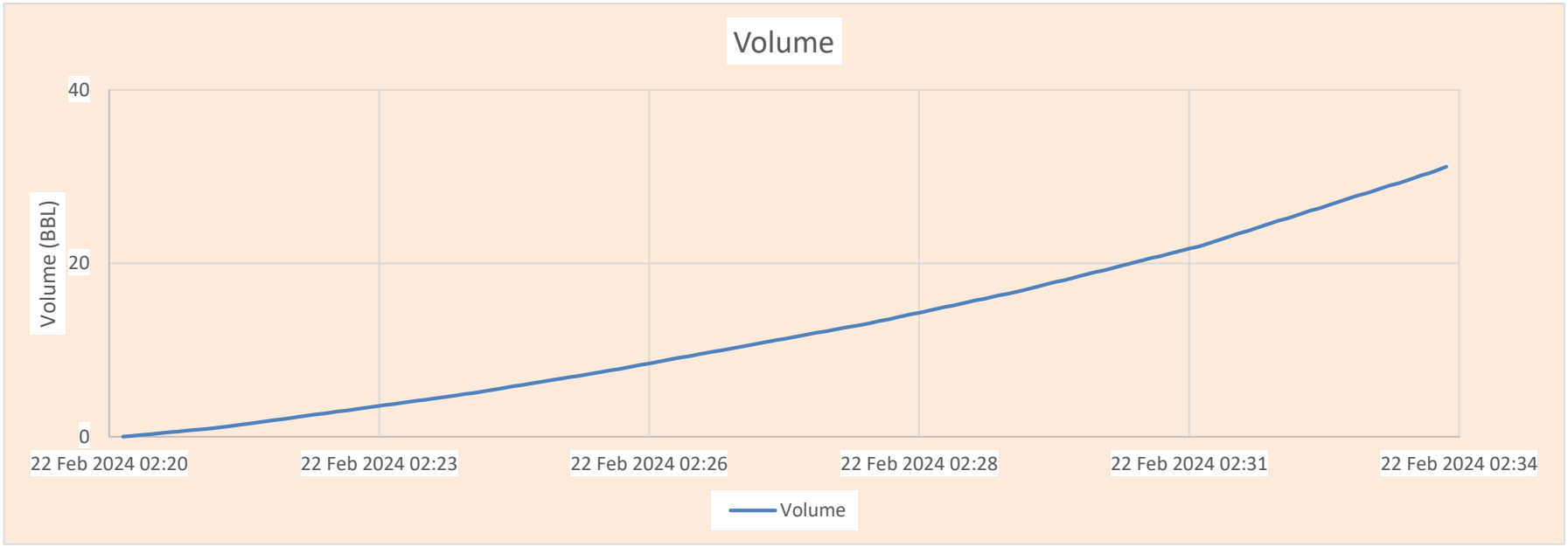


Figure 10-15: AZN-10 Post-Acid Bullhead 2 Graph (Volume)

Pumping Parameter in Pre-stimulation	
Pumping Pressure (PSI)	Pumping rate (bpm)
6383	1.01
Pumping Parameter in Post-stimulation	
Pumping Pressure (PSI)	Pumping rate (bpm)
6845	3.52

Table 10-12: AZN-10 Pumping Parameter

Flowing / Static Survey Data & Interpretation

10.4.8.1 Pre – Stimulation Downhole Memory Gauge Data, Results & Plots

AZNS # 10 - Before Stimulation										
Row	Date	Time(hh:mm:ss)	Event	Stop POOH Duration (min)	Depth MD (FT) Dipping Valve	Depth TVD (FT) Dipping Valve	Pressure (PSIA)	Temperature (°F)	Pressure Gradient (PSIA/FT)	Temperature Gradient (°F/FT)
1	09-February-2024		Connect		0	0				
2			Open the well		0	0				
3			Start POOH		0	0				
4		8:34:10 - 8:44:10	POOH Step 00	10	13114	13114	7095	274	0.29592	0.00000
5		8:46:00 - 8:56:50	POOH Step 02	10	13016	13016	7066	274	0.28000	0.00000
6		9:00:30 - 9:05:30	POOH Step 03	05	12816	12816	7010	274	0.27861	0.00498
7		9:08:23 - 9:13:23	POOH Step 04	05	12615	12615	6954	273	0.29000	0.00000
8		9:16:30 - 9:21:30	POOH Step 05	05	12415	12415	6896	273	0.27500	0.01500
9		9:24:00 - 9:31:00	POOH Step 06	07	12215	12215	6841	270	0.29000	0.00500
10		9:34:30 - 9:39:30	POOH Step 07	05	12015	12015	6783	269	0.28272	0.00500
11		9:47:10 - 9:52:10	POOH Step 08	05	11014	11014	6500	264	0.28000	0.00900
12		10:12:08 - 10:17:08	POOH Step 09	05	10014	10014	6220	255	0.28257	0.01002
13		10:23:43 - 10:28:43	POOH Step 10	05	9515	9515	6079	250	0.28486	0.00996

AZNS # 10 - Before Stimulation

Row	Date	Time(hh:mm:ss)	Event	Stop POOH Duration (min)	Depth MD (FT) Dipping Valve	Depth TVD (FT) Dipping Valve	Pressure (PSIA)	Temperature (°F)	Pressure Gradient (PSIA/FT)	Temperature Gradient (°F/FT)
14		10:33:10 - 10:40:10	POOH Step 11	07	9013	9013	5936	245	0.28257	0.01002
15		10:44:35 - 10:49:35	POOH Step 12	05	8514	8514	5795	240	0.28657	0.01403
16		10:54:20 - 10:59:20	POOH Step 13	05	8015	8015	5652	233	0.28287	0.00996
17		11:02:45 - 11:07:45	POOH Step 14	05	7513	7513	5510	228	0.28715	0.01406
18		11:10:00 - 11:15:00	POOH Step 15	05	7015	7015	5367	221	0.28971	0.01399
19		11:22:25 - 11:29:25	POOH Step 16	07	6014	6014	5077	207	0.28971	0.01199
20		11:39:00 - 11:44:00	POOH Step 17	05	5013	5013	4787	195	0.28900	0.01200
21		11:53:20 - 11:58:20	POOH Step 18	05	4013	4013	4498	183	0.29259	0.01503
22		12:06:25 - 12:11:25	POOH Step 19	05	3015	3015	4206	168	0.29100	0.01400
23		12:18:35 - 12:23:35	POOH Step 20	05	2015	2015	3915	154	0.29681	0.01394
24		12:27:50 - 12:34:50	POOH Step 21	07	1513	1513	3766	147	0.29659	0.01403
25		12:39:00 - 12:44:00	POOH Step 22	05	1014	1014	3618	140	0.29459	0.01403
26		12:48:40 - 12:53:40	POOH Step 23	05	515	515	3471	133	0.29249	0.01581
27		12:56:20 - 13:01:20	POOH Step 24	05	262	262	3397	129	0.30120	0.02008
28		13:04:00 - 13:09:00	POOH Step 25	05	13	13	3322	124		

Table 10-13 Pre-Stimulation Flowing Survey Report (Gauge No: 90504)

AZNS # 10 - Before Stimulation

Row	Date	Time(hh:mm:ss)	Event	Stop POOH Duration (min)	Depth MD (FT) Dipping Valve	Depth TVD (FT) Dipping Valve	Pressure (PSIA)	Temperature (°F)	Pressure Gradient (PSIA/FT)	Temperature Gradient (°F/FT)
1	09-February-2024		Connect		0	0				
2			Open the well		0	0				
3			Start POOH		0	0				
4		8:34:10 - 8:44:10	POOH Step 00	10	13114	13114	7095	274	0.29592	0.00000
5		8:46:00 - 8:56:50	POOH Step 02	10	13016	13016	7066	274	0.28000	0.00000
6		9:00:30 - 9:05:30	POOH Step 03	05	12816	12816	7010	274	0.27861	0.00498
7		9:08:23 - 9:13:23	POOH Step 04	05	12615	12615	6954	273	0.29000	0.00000
8		9:16:30 - 9:21:30	POOH Step 05	05	12415	12415	6896	273	0.27500	0.01500
9		9:24:00 - 9:31:00	POOH Step 06	07	12215	12215	6841	270	0.29000	0.00500
10		9:34:30 - 9:39:30	POOH Step 07	05	12015	12015	6783	269	0.28272	0.00500
11		9:47:10 - 9:52:10	POOH Step 08	05	11014	11014	6500	264	0.28000	0.00900
12		10:12:08 - 10:17:08	POOH Step 09	05	10014	10014	6220	255	0.28257	0.01002
13		10:23:43 - 10:28:43	POOH Step 10	05	9515	9515	6079	250	0.28486	0.00996
14		10:33:10 - 10:40:10	POOH Step 11	07	9013	9013	5936	245	0.28257	0.01002
15		10:44:35 - 10:49:35	POOH Step 12	05	8514	8514	5795	240	0.28657	0.01403
16		10:54:20 - 10:59:20	POOH Step 13	05	8015	8015	5652	233	0.28287	0.00996

AZNS # 10 - Before Stimulation

Row	Date	Time(hh:mm:ss)	Event	Stop POOH Duration (min)	Depth MD (FT) Dipping Valve	Depth TVD (FT) Dipping Valve	Pressure (PSIA)	Temperature (°F)	Pressure Gradient (PSIA/FT)	Temperature Gradient (°F/FT)
17		11:02:45 - 11:07:45	POOH Step 14	05	7513	7513	5510	228	0.28715	0.01406
18		11:10:00 - 11:15:00	POOH Step 15	05	7015	7015	5367	221	0.28971	0.01399
19		11:22:25 - 11:29:25	POOH Step 16	07	6014	6014	5077	207	0.28971	0.01199
20		11:39:00 - 11:44:00	POOH Step 17	05	5013	5013	4787	195	0.28900	0.01200
21		11:53:20 - 11:58:20	POOH Step 18	05	4013	4013	4498	183	0.29259	0.01503
22		12:06:25 - 12:11:25	POOH Step 19	05	3015	3015	4206	168	0.29100	0.01400
23		12:18:35 - 12:23:35	POOH Step 20	05	2015	2015	3915	154	0.29681	0.01394
24		12:27:50 - 12:34:50	POOH Step 21	07	1513	1513	3766	147	0.29659	0.01403
25		12:39:00 - 12:44:00	POOH Step 22	05	1014	1014	3618	140	0.29459	0.01403
26		12:48:40 - 12:53:40	POOH Step 23	05	515	515	3471	133	0.29249	0.01581
27		12:56:20 - 13:01:20	POOH Step 24	05	262	262	3397	129	0.30120	0.02008
28		13:04:00 - 13:09:00	POOH Step 25	05	13	13	3322	124		

Table 10-14 Pre-Stimulation Flowing Survey Report (Gauge No: 90506)

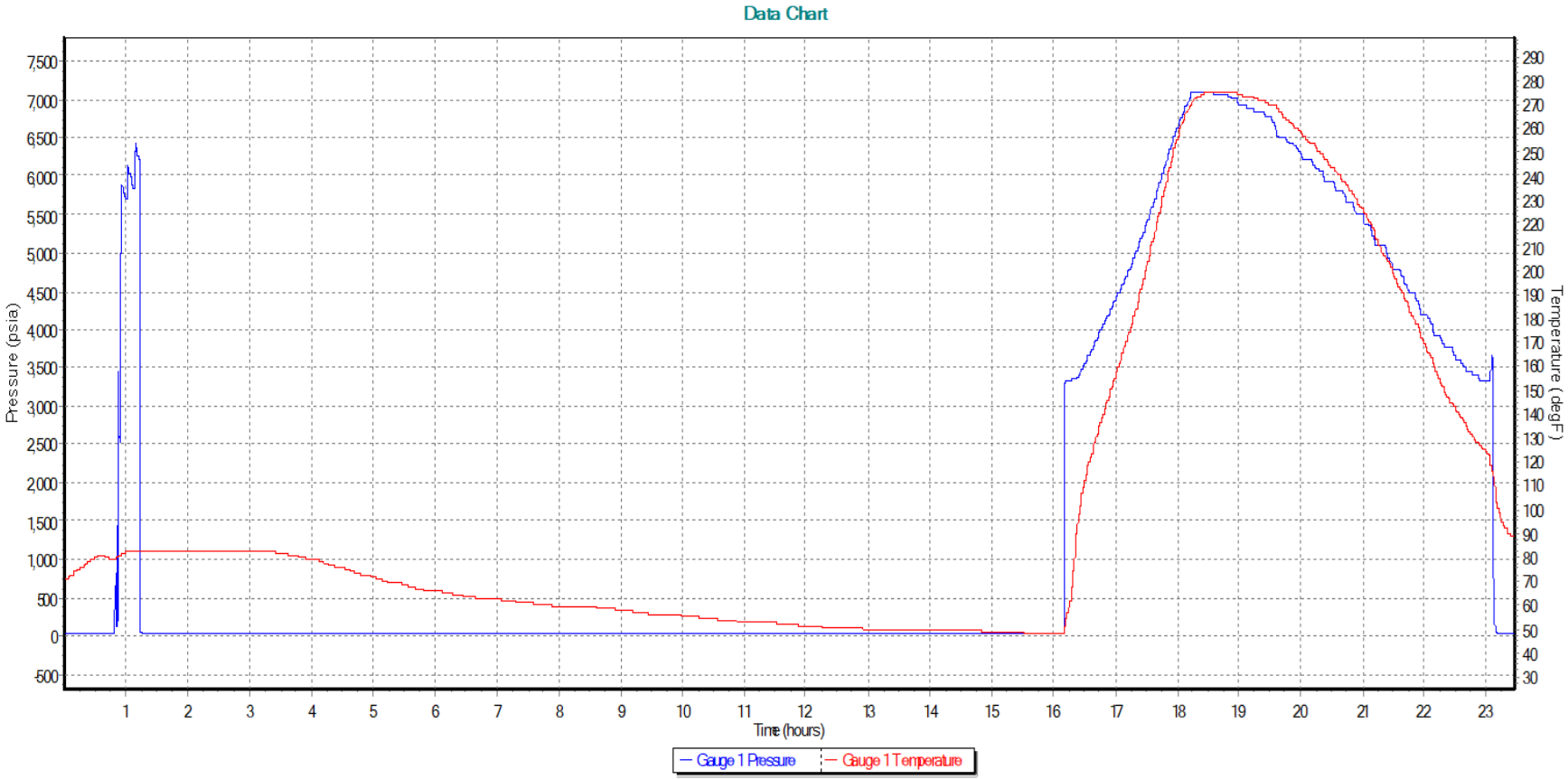
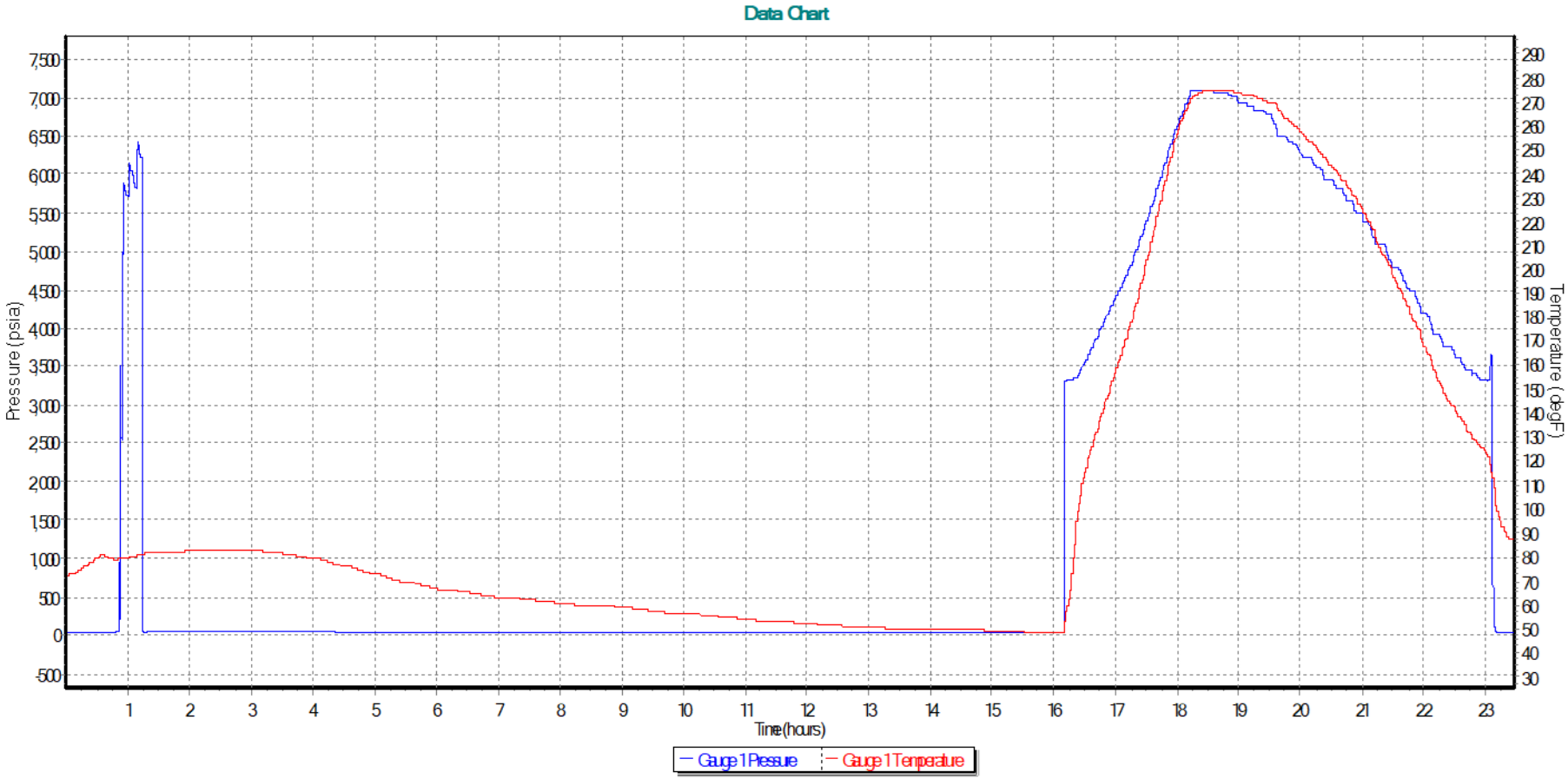


Figure 10-16: AZN-10 Pre-Stimulation Down Hole Memory Gauge (P&T) Plot (Gauge No: 90504)


Data Chart
Pressure (psia)
Temperature (degF)
Time (hours)
Gauge 1 Pressure
Gauge 1 Temperature

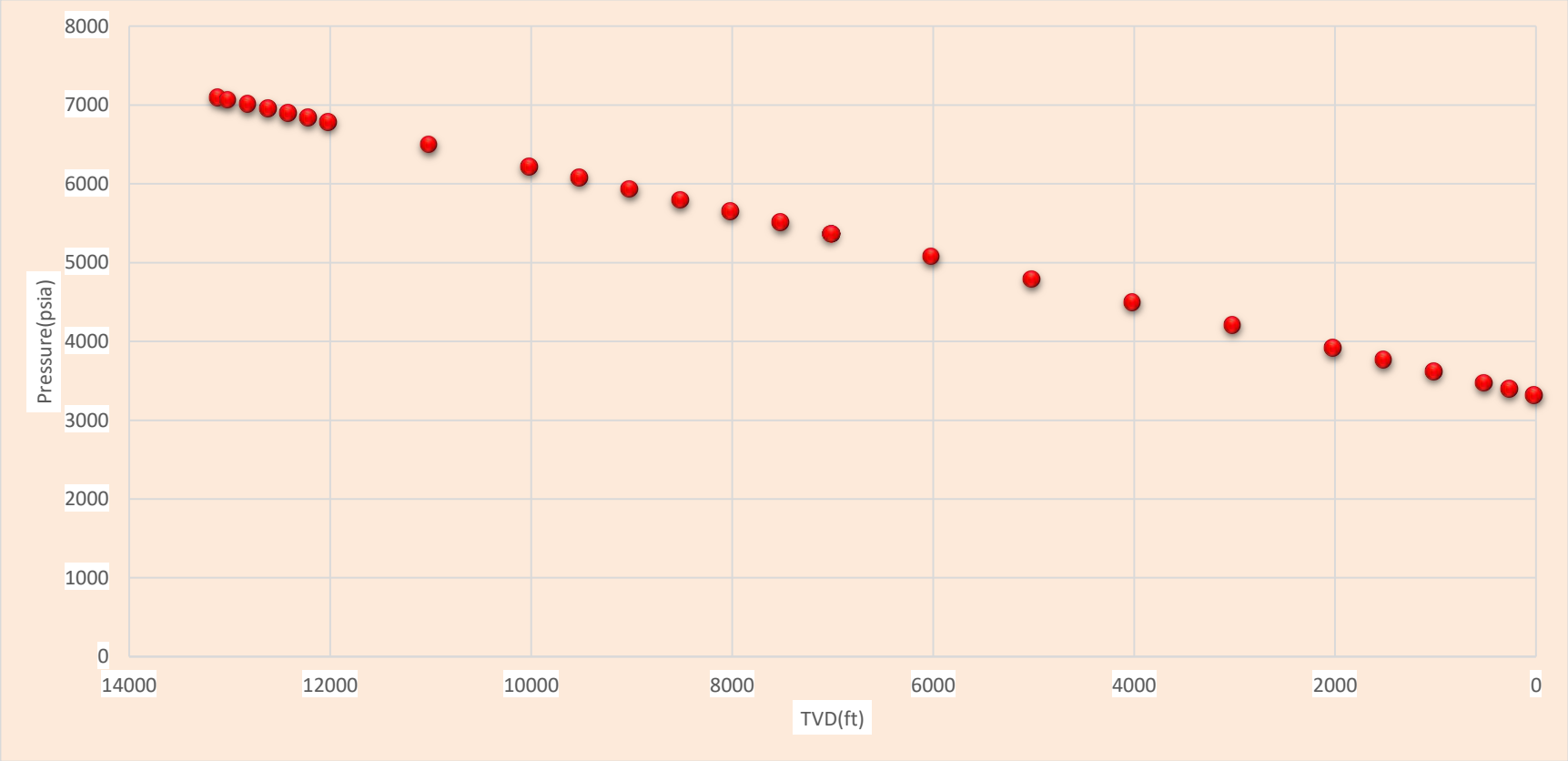


Figure 10-17: AZN-10 Pre-Stimulation Pressure Vs TVD Plot (Gauge No: 90504)

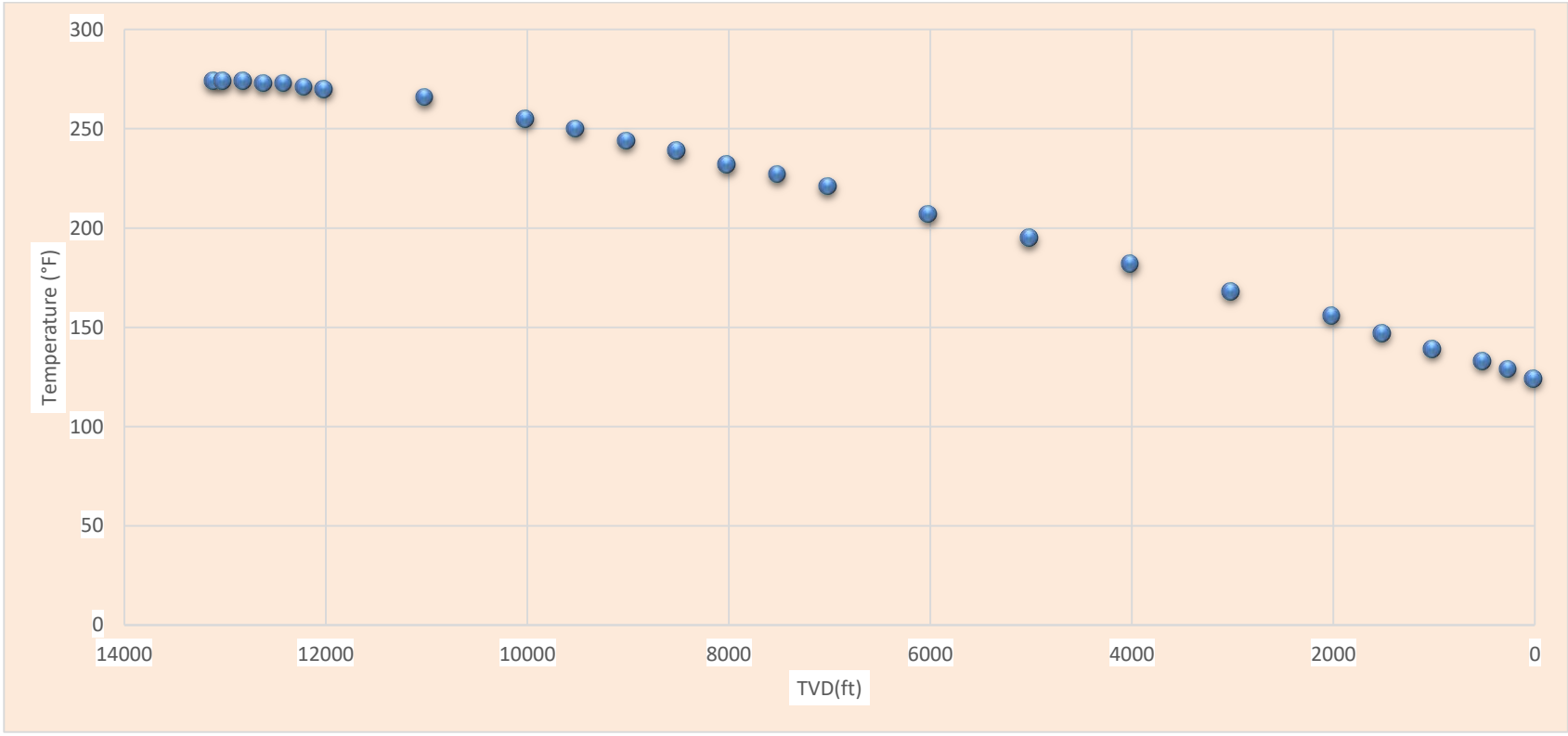


Figure 10-18: AZN-10 Pre-Stimulation Temperature Vs TVD Plot (Gauge No: 90504)

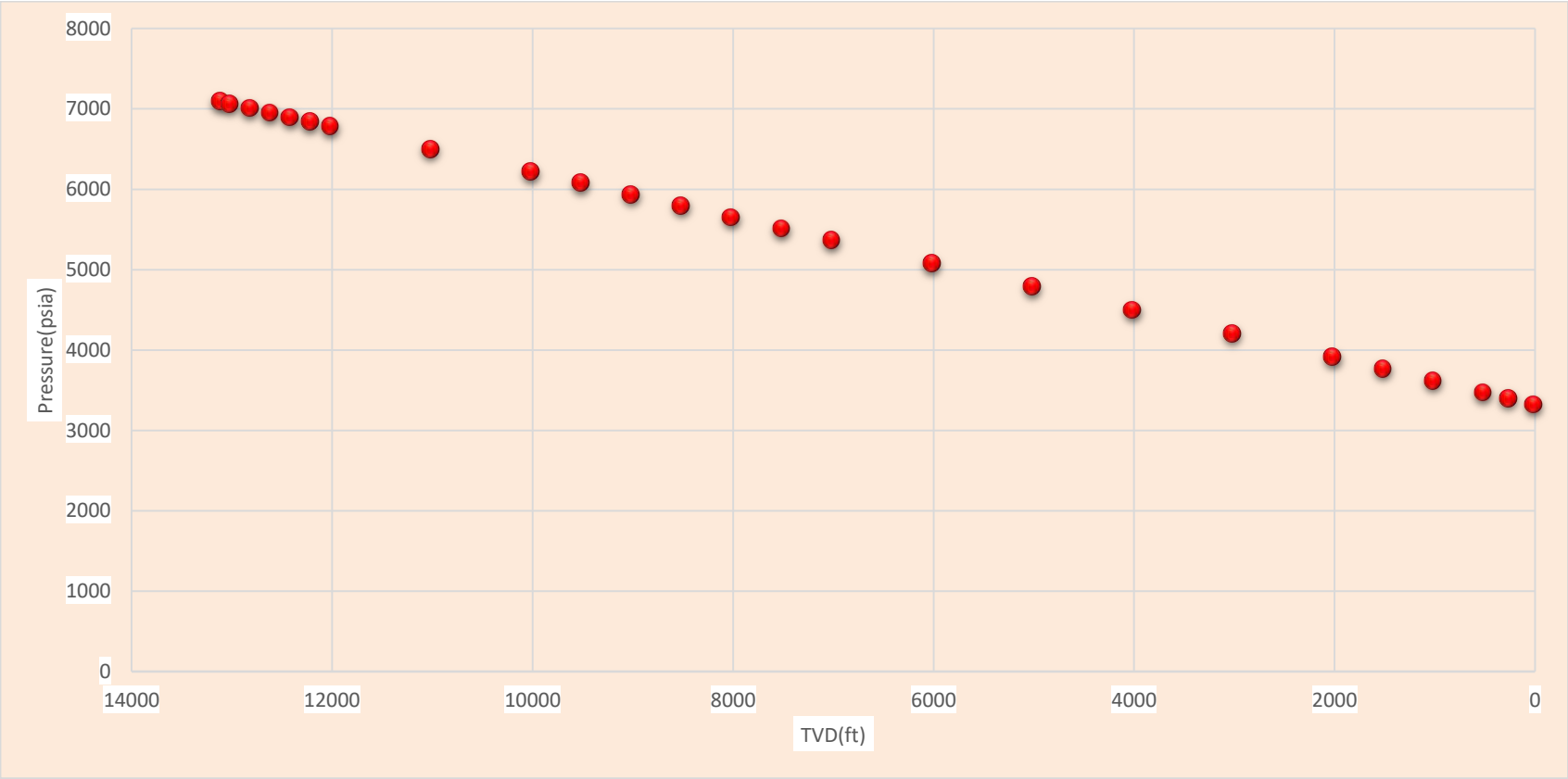


Figure 10-19: AZN-10 Pre-Stimulation Pressure Vs TVD Plot (Gauge No: 90506)

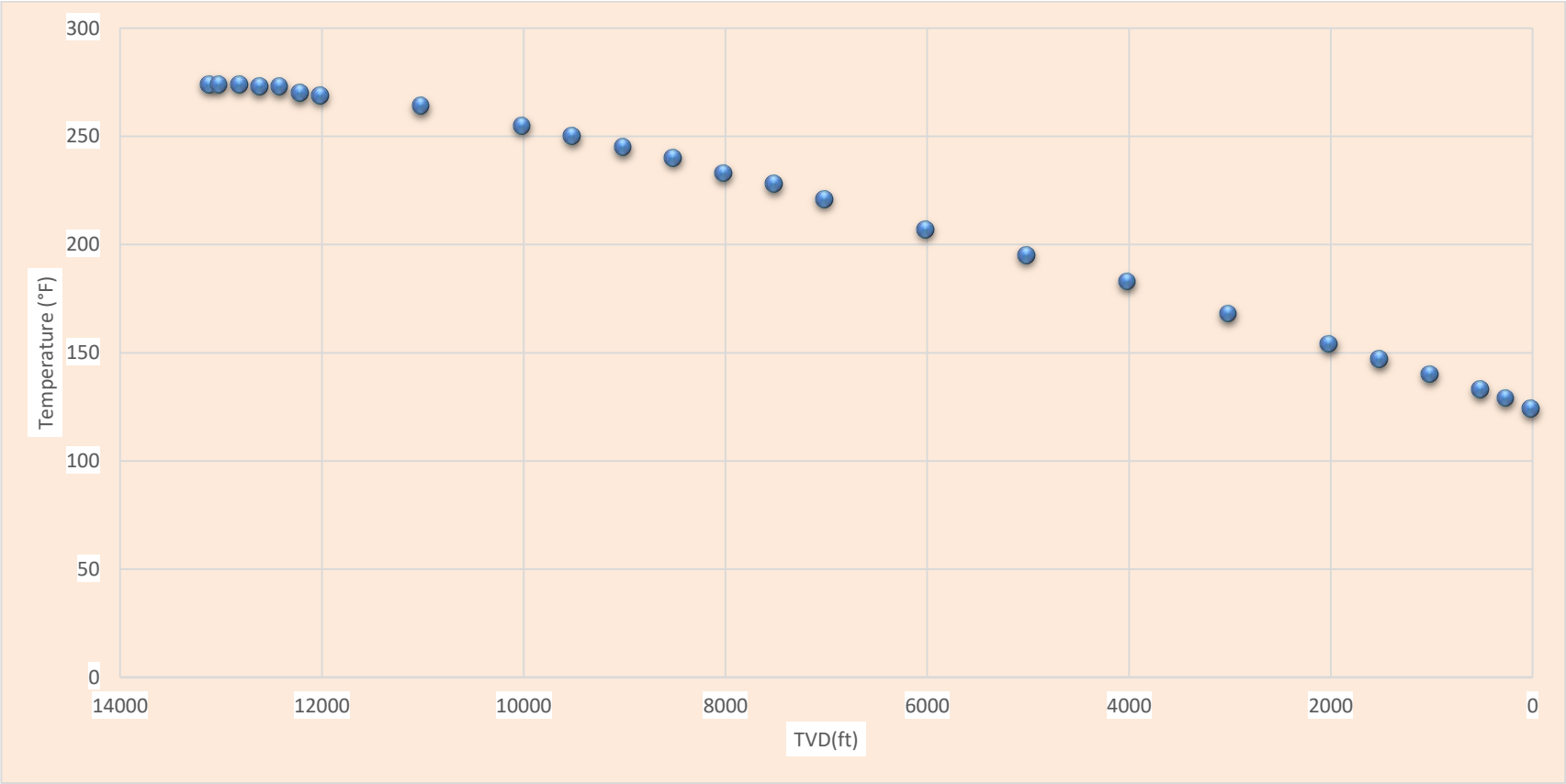


Figure 10-20: AZN-10 Pre-Stimulation Temperature Vs TVD Plot (Gauge No: 90506)

10.4.8.2 Post – Stimulation Downhole Memory Gauge Data, Results & Plots

AZNS # 10 - After Stimulation										
Row	Date	Time(hh:mm:ss)	Event	Stop POOH Duration (min)	Depth MD (FT) Dipping Valve	Depth TVD (FT) Dipping Valve	Pressure (PSIA)	Temperature (°F)	Pressure Gradient (PSIA/FT)	Temperature Gradient (°F/FT)
1	14-February-2024		Connect		0	0				
2			Open the well		0	0				
3			Start POOH		0	0				
4		2:30:00 - 2:40:00	POOH Step 00	10	13114	13114	8153	270	0.28571	0.00000
5		2:41:35 - 2:51:35	POOH Step 02	10	13016	13016	8125	270	0.28500	0.00500
6		2:54:20 - 2:59:20	POOH Step 03	05	12816	12816	8068	269	0.28358	0.00000
7		3:01:30 - 3:06:30	POOH Step 04	05	12615	12615	8011	269	0.28500	0.00500
8		3:08:40 - 3:13:40	POOH Step 05	05	12415	12415	7954	268	0.28000	0.00000
9		3:15:30 - 3:22:30	POOH Step 06	07	12215	12215	7898	268	0.28500	0.00500
10		3:24:50 - 3:29:50	POOH Step 07	05	12015	12015	7841	267	0.28571	0.00400
11		3:37:00 - 3:42:00	POOH Step 08	05	11014	11014	7555	263	0.28300	0.00400
12		3:48:15 - 3:53:15	POOH Step 09	05	10014	10014	7272	259	0.28657	0.01403
13		3:57:00 - 4:02:00	POOH Step 10	05	9515	9515	7129	252	0.28685	0.00996
14		4:05:45 - 4:12:45	POOH Step 11	07	9013	9013	6985	247	0.28657	0.00802
15		4:16:25 - 4:21:25	POOH Step 12	05	8514	8514	6842	243	0.28858	0.01002

AZNS # 10 - After Stimulation

Row	Date	Time(hh:mm:ss)	Event	Stop POOH Duration (min)	Depth MD (FT) Dipping Valve	Depth TVD (FT) Dipping Valve	Pressure (PSIA)	Temperature (°F)	Pressure Gradient (PSIA/FT)	Temperature Gradient (°F/FT)
16		4:25:55 - 4:30:55	POOH Step 13	05	8015	8015	6698	238	0.28685	0.01195
17		4:36:30 - 4:41:30	POOH Step 14	05	7513	7513	6554	232	0.28715	0.00803
18		4:46:30 - 4:51:30	POOH Step 15	05	7015	7015	6411	228	0.28971	0.01299
19		5:00:20 - 5:07:20	POOH Step 16	07	6014	6014	6121	215	0.28971	0.01099
20		5:15:00 - 5:20:00	POOH Step 17	05	5013	5013	5831	204	0.29100	0.01200
21		5:28:10 - 5:33:10	POOH Step 18	05	4013	4013	5540	192	0.29559	0.01202
22		5:41:25 - 5:46:25	POOH Step 19	05	3015	3015	5245	180	0.30200	0.01200
23		5:55:50 - 6:00:50	POOH Step 20	05	2015	2015	4943	168	0.38645	0.01594
24		6:05:00 - 6:12:00	POOH Step 21	07	1513	1513	4749	160	0.21242	0.01002
25		6:16:45 - 6:21:45	POOH Step 22	05	1014	1014	4643	155	0.29659	0.01202
26		6:26:20 - 6:31:20	POOH Step 23	05	515	515	4495	149	0.30435	0.01976
27		6:36:00 - 6:41:00	POOH Step 24	05	262	262	4418	144	0.30120	0.01606
28		6:46:15 - 6:51:15	POOH Step 25	05	13	13	4343	140		

Table 10-15 Post Stimulation Flowing Survey Report (Gauge No: 90504)

AZNS # 10 - After Stimulation

Row	Date	Time(hh:mm:ss)	Event	Stop POOH Duration (min)	Depth MD (FT) Dipping Valve	Depth TVD (FT) Dipping Valve	Pressure (PSIA)	Temperature (°F)	Pressure Gradient (PSIA/FT)	Temperature Gradient (°F/FT)
1	14-February-2024		Connect		0	0				
2			Open the well		0	0				
3			Start POOH		0	0				
4		2:30:00 - 2:40:00	POOH Step 00	10	13114	13114	8153	270	0.28571	0.00000
5		2:41:35 - 2:51:35	POOH Step 02	10	13016	13016	8125	270	0.28500	0.00500
6		2:54:20 - 2:59:20	POOH Step 03	05	12816	12816	8068	269	0.28358	0.00000
7		3:01:30 - 3:06:30	POOH Step 04	05	12615	12615	8011	269	0.28500	0.00500
8		3:08:40 - 3:13:40	POOH Step 05	05	12415	12415	7954	268	0.28000	0.00000
9		3:15:30 - 3:22:30	POOH Step 06	07	12215	12215	7898	268	0.28500	0.00500
10		3:24:50 - 3:29:50	POOH Step 07	05	12015	12015	7841	267	0.28571	0.00400
11		3:37:00 - 3:42:00	POOH Step 08	05	11014	11014	7555	263	0.28300	0.00400
12		3:48:15 - 3:53:15	POOH Step 09	05	10014	10014	7272	259	0.28657	0.01403
13		3:57:00 - 4:02:00	POOH Step 10	05	9515	9515	7129	252	0.28685	0.00996
14		4:05:45 - 4:12:45	POOH Step 11	07	9013	9013	6985	247	0.28657	0.00802
15		4:16:25 - 4:21:25	POOH Step 12	05	8514	8514	6842	243	0.28858	0.01002
16		4:25:55 - 4:30:55	POOH Step 13	05	8015	8015	6698	238	0.28685	0.01195

AZNS # 10 - After Stimulation

Row	Date	Time(hh:mm:ss)			Event	Stop POOH Duration (min)	Depth MD (FT) Dipping Valve	Depth TVD (FT) Dipping Valve	Pressure (PSIA)	Temperature (°F)	Pressure Gradient (PSIA/FT)	Temperature Gradient (°F/FT)
17		4:36:30	-	4:41:30	POOH Step 14	05	7513	7513	6554	232	0.28715	0.00803
18		4:46:30	-	4:51:30	POOH Step 15	05	7015	7015	6411	228	0.28971	0.01299
19		5:00:20	-	5:07:20	POOH Step 16	07	6014	6014	6121	215	0.28971	0.01099
20		5:15:00	-	5:20:00	POOH Step 17	05	5013	5013	5831	204	0.29100	0.01200
21		5:28:10	-	5:33:10	POOH Step 18	05	4013	4013	5540	192	0.29559	0.01202
22		5:41:25	-	5:46:25	POOH Step 19	05	3015	3015	5245	180	0.30200	0.01200
23		5:55:50	-	6:00:50	POOH Step 20	05	2015	2015	4943	168	0.38645	0.01594
24		6:05:00	-	6:12:00	POOH Step 21	07	1513	1513	4749	160	0.21242	0.01002
25		6:16:45	-	6:21:45	POOH Step 22	05	1014	1014	4643	155	0.29659	0.01202
26		6:26:20	-	6:31:20	POOH Step 23	05	515	515	4495	149	0.30435	0.01976
27		6:36:00	-	6:41:00	POOH Step 24	05	262	262	4418	144	0.30120	0.01606
28		6:46:15	-	6:51:15	POOH Step 25	05	13	13	4343	140		

Table 10-16 Post Stimulation Flowing Survey Report (Gauge No: 90506)

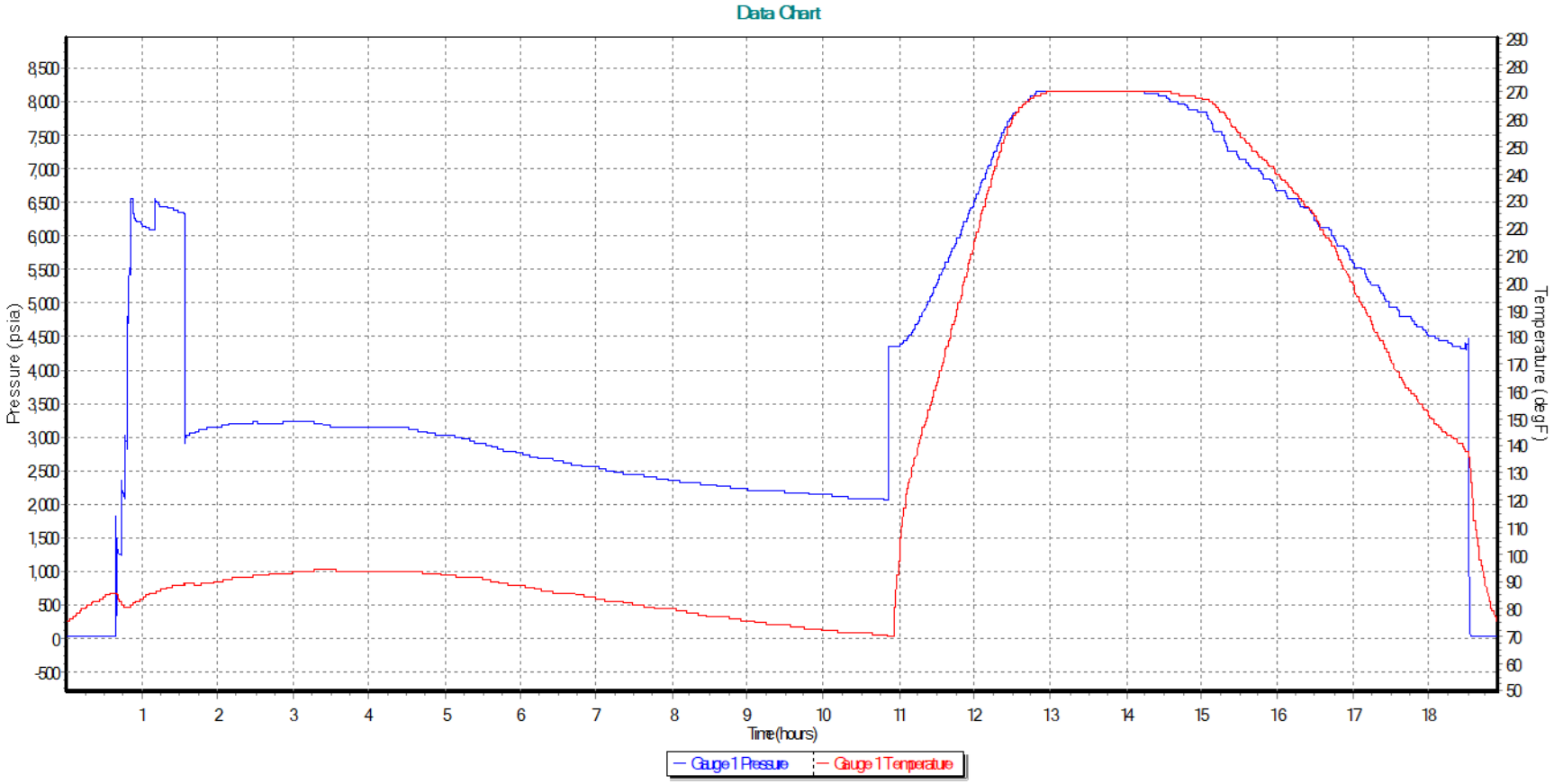


Figure 10-21 Post Stimulation Flowing Survey Down Hole Memory Gauge (P&T) Plot (Gauge No: 90504)

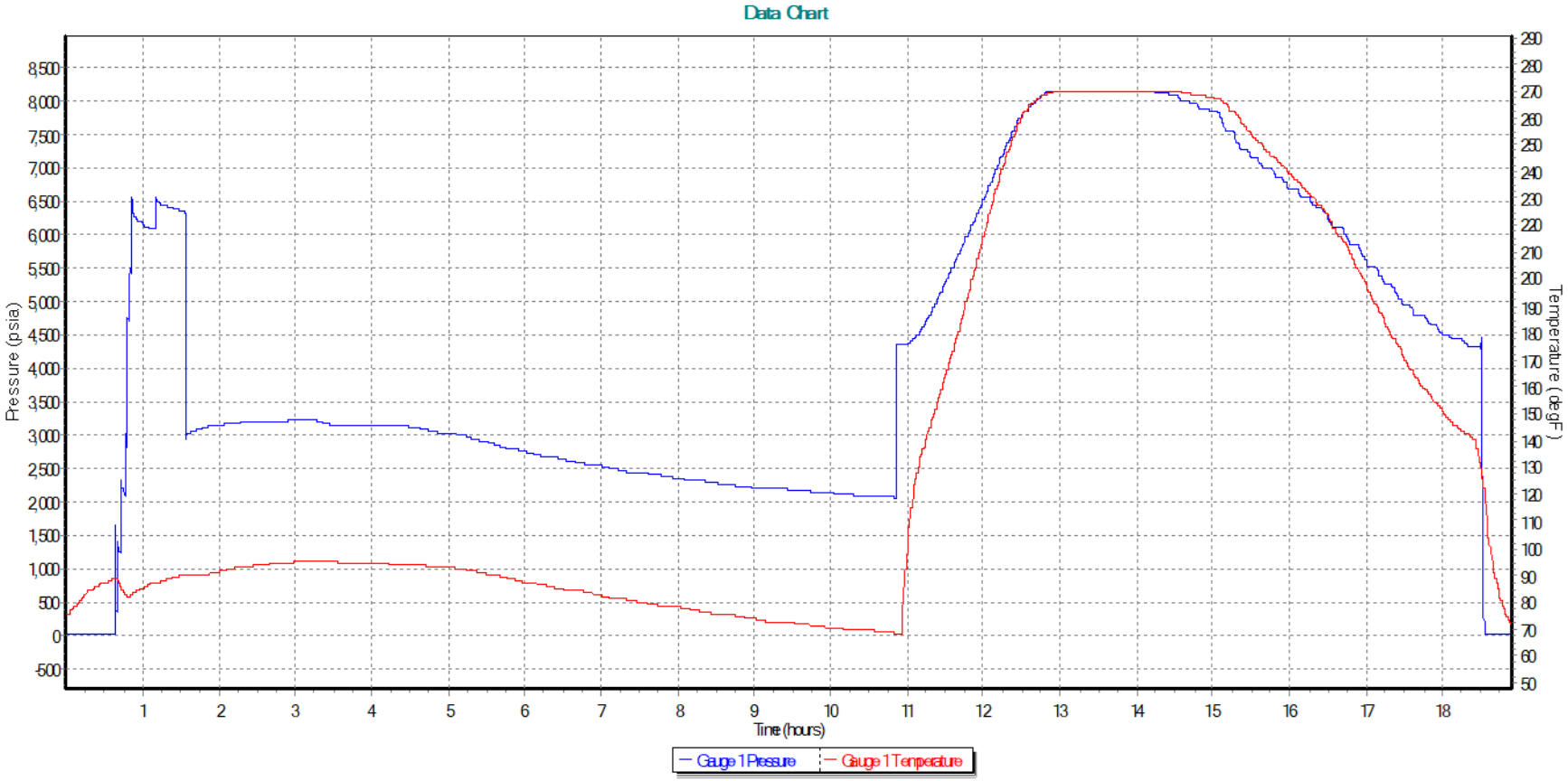


Figure 10-22 Post Stimulation Flowing Survey Down Hole Memory Gauge (P&T) Plot (Gauge No: 90506)



Figure 10-23 Post Stimulation Flowing Survey Pressure Vs TVD Plot (Gauge No: 90504)

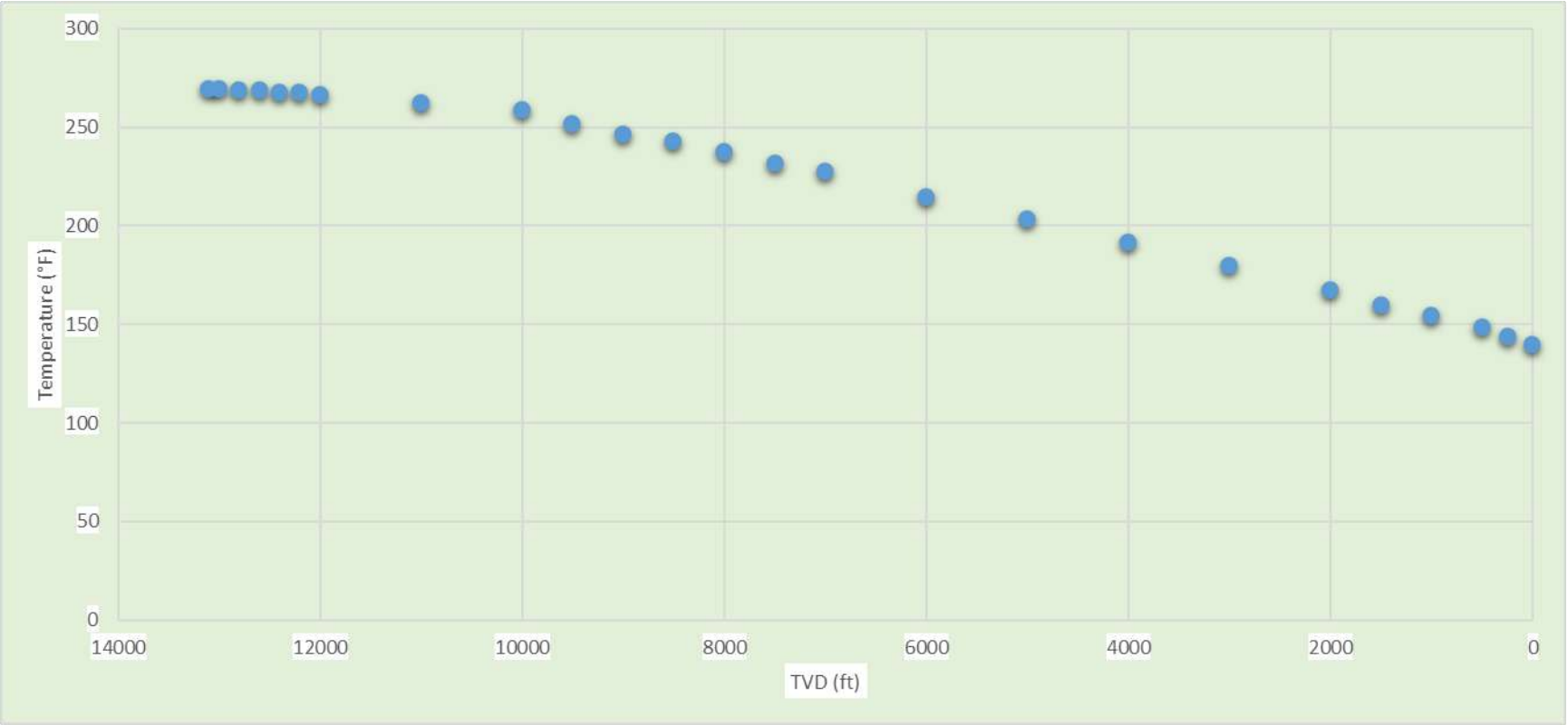


Figure 10-24 Post Stimulation Flowing Survey Temperature Vs TVD Plot (Gauge No: 90504)

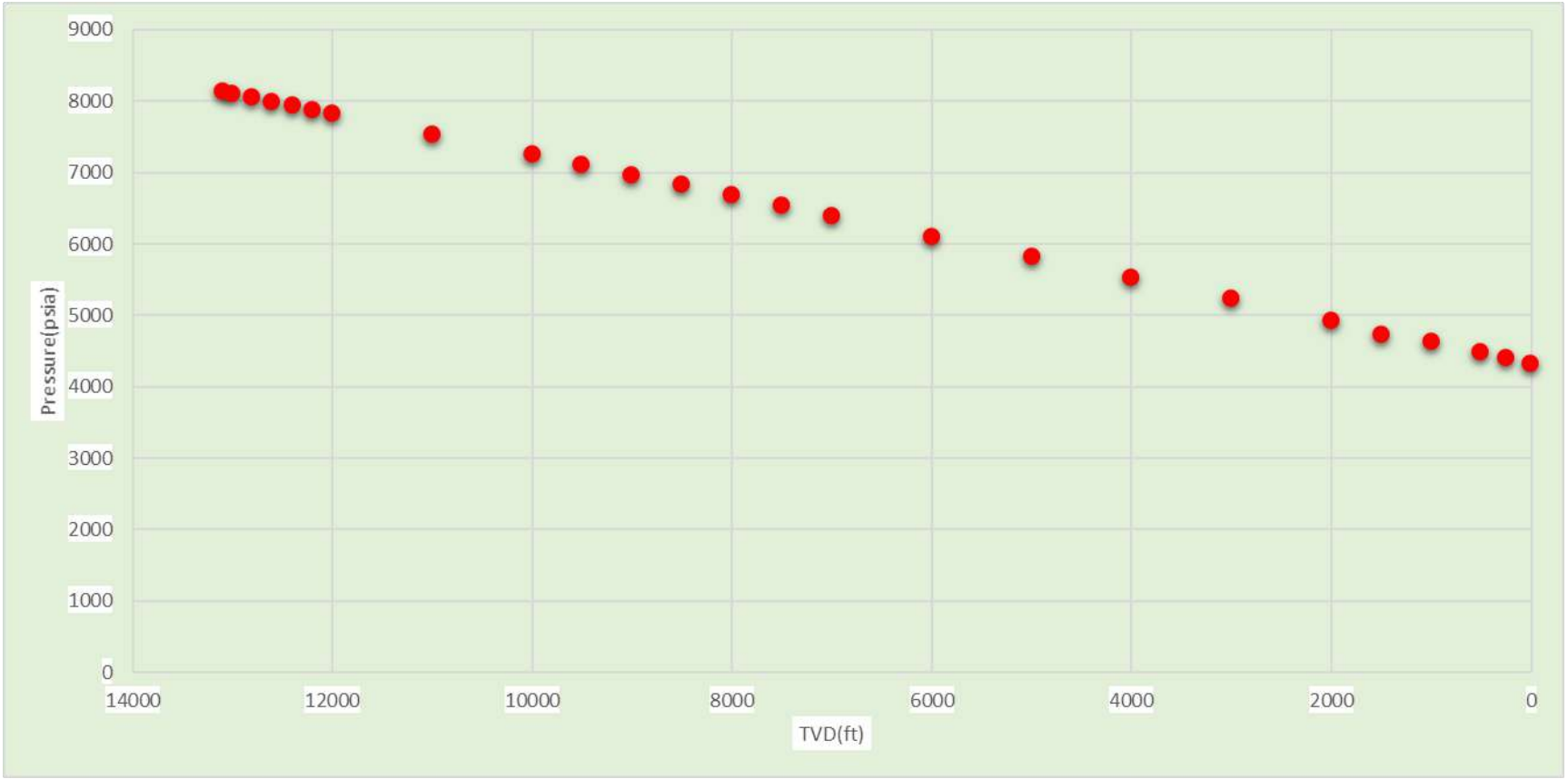


Figure 10-25 Post Stimulation Flowing Survey Pressure Vs TVD Plot (Gauge No: 90506)

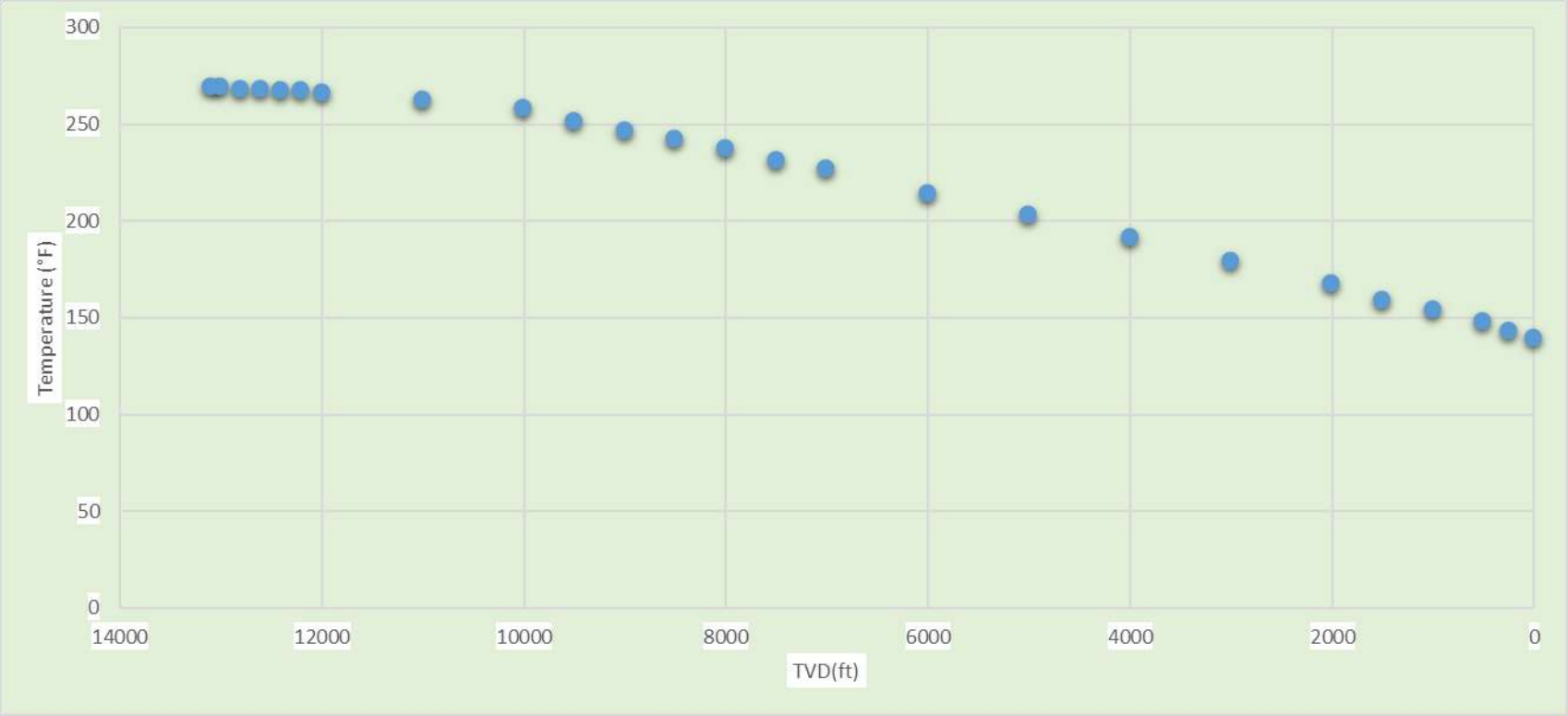


Figure 10-26 Post Stimulation Flowing Survey Temperature Vs TVD Plot (Gauge No: 90506)

The pre-stimulation PI of the well at the datum depth is calculated utilizing average reservoir pressure of the well extracted from the data provided by client, datum flowing pressure from flowing survey, and measured flow rate during flowing survey operation according to table below:

Pre-stimulation PI (@ 13124 FtTVDss): 0.66 (STB/D.PSI)	
Choke size (1/64" Fix)	16
Average Reservoir Pressure (PSIA)	9276
Bottom-hole Flowing Pressure (PSIA)	7105
Est. Rate (BBL/day)	1446

Table 10-17 Pre-Acid Productivity Index

The post-stimulation PI of the well at the datum depth is calculated utilizing average reservoir pressure of the well extracted from the data provided by client, datum flowing pressure from flowing survey, and measured flow rate during flowing survey operation according to table below:

Post-stimulation PI (@ 13124 ftTVDss): 3.261 (STB/D.PSI)	
Choke size (1/64" Fix)	16
Average Reservoir Pressure (PSIA)	9276
Bottom-hole Flowing Pressure (PSIA)	8162.85
Est. Rate (BBL/day)	3631

Table 10-18 Post Acid Productivity Index

10.5 Failure & Problems

N/A

10.6 Lesson Learned

N/A

10.7 Appendix

Gauges Information

DHMG raw data is attached as memory gauge's appendix.

Well Testing Raw Data

Well testing raw data is attached as well testing appendix.

Pumping Raw Data

Pmping raw data is attached as pumping appendix.

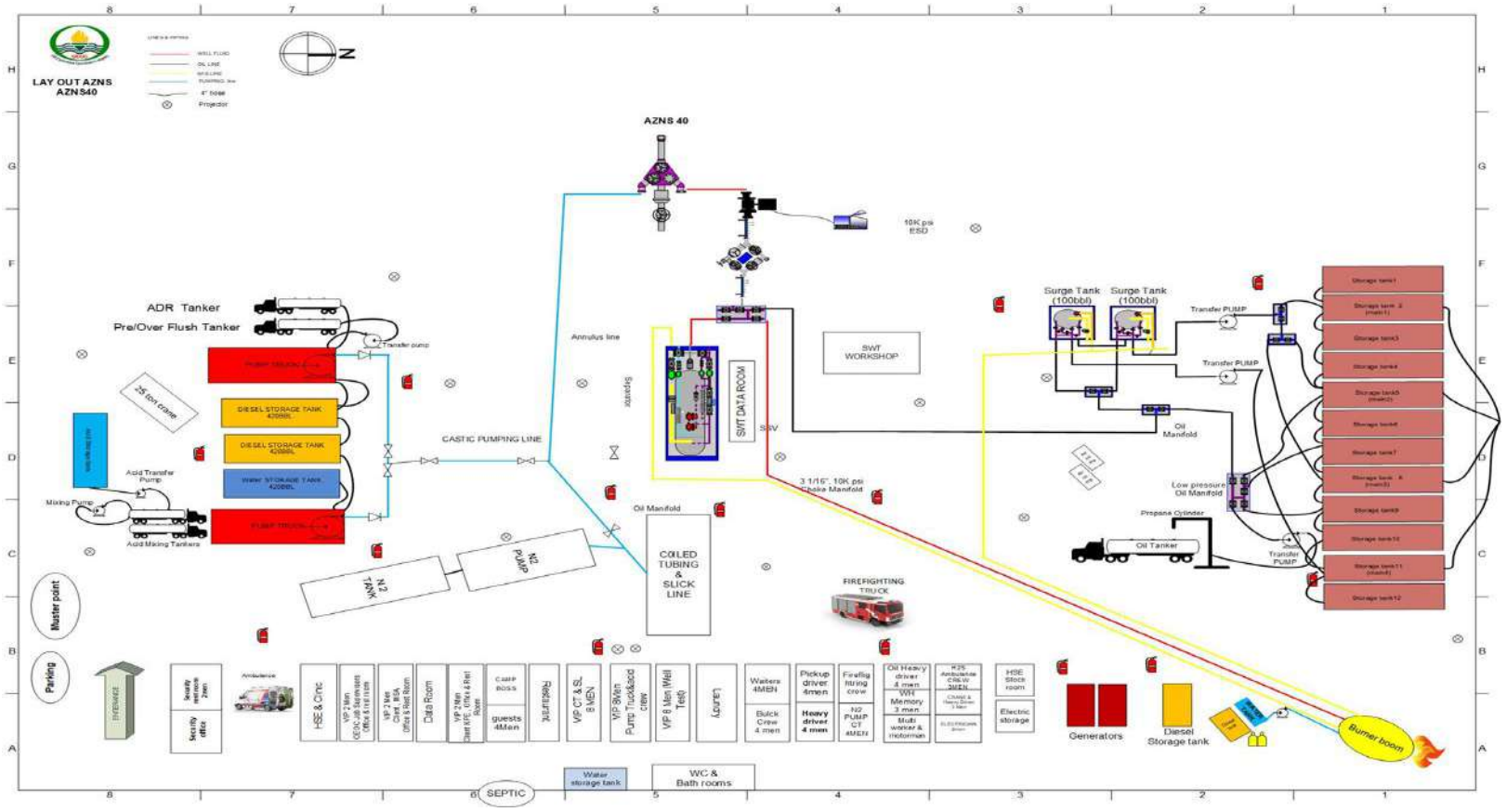
N_2 Lifting Raw Data

N2 lifting raw data is attached as N2 lifting appendix.

CT Raw Data

CT raw data is attached as CT appendix.

Surface Equipment Layout



CT & SL BHA Diagram

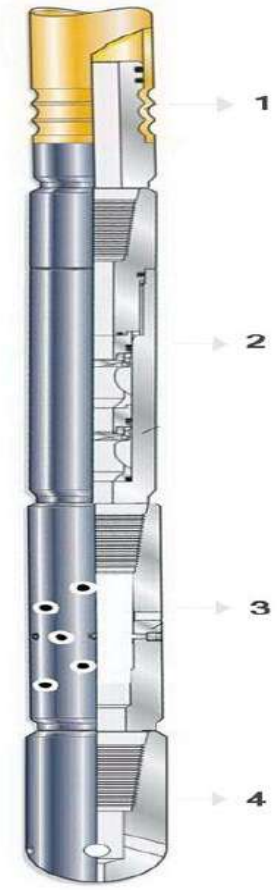
Row	BHA	Length (in)	OD size (in)	Schematic
1	Roll On Connector	3.3	1 3/4	
2	Two Check valves (Flapper Type)	8.3		
3	Perforation Sub (10 mm holes)	6.3		
4	Wash Nozzle (8 mm holes)	5		

Table 10-19 Coiled Tubing BHA Design

OD size (in)	Tools	Length (ft)	schematic
1 7/8	Rope socket	0.5	
1 7/8	Stem	10	
1 7/8	Mechanical Jar	4 (closed) & 6 (fully open)	
1 7/8	Hydraulic Jar	2	
3.4near 50	Gauge Cutter or GS Pulling or Memory Gauge	1	 

Table 10-20 S/L BHA Design

10.8 Onsite QC Test Report of Stimulation Additives & Fluids

QC tests and checklist fill up process were performed by Client, OEOC, and Energy Semnan Co. representatives prior to AZNS-10 well's stimulation operation and its report is mentioned in detail in the AZNS-10 operation program. It should be noted that samples of fresh water, raw acid, diesel, and additives were taken from onsite materials under client's supervision, and it was approved by client's Rep.

11. Drilling Fluid Information

11.1 Operation Remark

The OEOC201 was spud at AZN10, BOP was installed and tested then made up BHA and RIH to 1120^{m} after that wash and reamed down with 10^{pcf} OBM from hole to 1381^{m} (top of first cement plug) then POOH to surface.

Made 8 ½'' drill out BHA and RIH to 1381^{m} then drilled out cement from 1381^{m} to 1500^{m} then washed down from 1500^{m} to 3629^{m} (top of second cement plug) after that drilled out cement from 3629^{m} to 3707^{m} then POOH to surface.

Made up 5-7/8'' BHA and RIH to 3529^{m} then Displaced the hole with 102^{pcf} SALT/POL mud after that RIH to 4591^{m} and performed cement plug at this point then POOH to 3676^{m} and observed string plugged then commenced POOH to surface.

Made 5-7/8'' center cut bit and RIH to 4158^{m} then washed down from 4158^{m} to 4188^{m} after that drilled out cement from 4188^{m} to 4250^{m} then POOH to surface, made up 7'' and 9-5/8'' tandem scrapper to 3673^{m} and performed scrapping the POOH to surface.

Made BHA and RIH to 4250^{m} then spot 70^{bbl} $CaCl_2$ brine after that POOH to surface, performed logging at several stages after that made 9-5/8'' dry test and RIH to 3058^{m} then performed dry test after that POOH to surface.

Made 7'' polish mill and RIH to 3707^{m} then performed milling after that POOH to surface, RIH with SN packer to 3707^{m} and set the packer then POOH to surface, RIH with dry test tools to 3687^{m} and performed dry test.

Made up 9-5/8'' scrapper and RIH to top SN packer then performed scrapping inside 9-5/8'' casing after that POOH to surface, RIH to 315m and performed slow mud circulation then commenced RIH to 3699^{m} after that POOH to surface.

Made up 7'' tie back assembly and RIH to 3527^{m} then set packer after that performed cement job and POOH to surface, RIH with 8 ½'' BHA to 3442^{m} and drilled out soft to hard cement from 3442^{m} to 3527^{m} (7'' liner lap) then POOH to surface.

Made 6'' BHA and RIH to 3527^{m} then drilled cement and POB from 3527^{m} to 3676^{m} after that drilled out hard cement from 3676^{m} to 3704^{m} then POOH to surface, RIH with 5 7/8'' BHA to 3703^{m} and drilled out cement from 3703^{m} to 3711^{m} then RIH to 4242^{m} after that POOH to surface.

Made up 7'' and 9 5/8'' tandem scrapper BHA and RIH to 3700^{m} then performed scrapping after that POOH to surface, performed 9 5/8'' dry test.

Made up perforation gun and RIH to 3527^{m} /Not pass then POOH to surface, RIH with magnet and clean out BHA to 3700^{m} then POOH to surface, RIH as dummy run to 4120^{m} then POOH to surface, performed logging.

Performed perforation at several stages from 4082^{m} to 4077^{m} & from 4077^{m} to 4074^{m} & from 4057^{m} to 4051^{m} & from 4051^{m} to 4048^{m} & from 4048^{m} to 4042^{m} & from 4040.6^{m} to 4036.6^{m} and from 4036^{m} to 4036^{m} after that POOH to surface.

Made 5 7/8'' mill and RIH to 4244^{m} then performed mud circulation and POOH to surface, performed logging.

Made 7'' scrapper and RIH to 4087^{m} then pumped high viscosity pill and displaced the hole with 104^{pcf} inhibited $CaBr_2$ brine as completion fluid then POOH to surface, RIH with completion string to 3670^{m} then performed completion string tests/OK then RIH with slick line and installed protection sleeve of SSSV after that tried several times to pass SSD with check valve/No success then displaced the hole with 104^{pcf} Salt/POL mud, Performed slick line job successfully then displaced with 104^{pcf} inhibited $CaBr_2$ brine and installed XMAS tree.

11.2 Operation Remark

Based on client decision first cement plug was drilled with 100PCF OBM inside the hole, the 104PCF KCL/Polymer mud was used for commencing this interval.

The MW increased up to by adding Limestone Powder up to 104 PCF at end of this interval.

Treated frequently with essential chemical such as lime as Alkalinity controller to adjust pH, Starch LV for control FL, Soda Ash for control hardness & Xanthan gum as Viscosifier.

11.3 Drilling Fluid type

FLUID PROPERTIES RECAP					
KCL Pol Mud		Drilling start	Range		Drilling End
MW (min)	pcf	102	91	104	104
Funnel Viscosity	sec/qt	44	44	55	44
PV	cps	21	18	25	21
YP	lb/100sq.ft	10	8	30	10
API Fluid loss	cc	NC	NC		NC
pH		8.5	8.5	10.5	9.3
Solid	%	44	44	45	45
Cl (NaCl)	mg/L	180,000	160000	180000	160,000
Total Hardness	mg/L	2,800	2800	12800	12,800
Ca Hardness	mg/L	880	800	12000	11,000

Table 11-1: AZN-10 Drilling Fluid Properties

11.4 Drilling Fluid type

Products	Units Used	Unit wt lbs	Total wt		Effective Concentration	
CaCl2 84% *	11	2,200	24,200	lbs	220.00	ppb
CaCl2 97%	13	2,200	28,600	lbs	32.87	ppb
Caustic Soda	11	55	605	lbs	0.18	ppb
Corrision Inhibitor	6	55	330	gal	0.38	gpb
Lime	55	55	3,025	lbs	0.91	ppb
Limestone	435	2,200	957,000	lbs	287.82	ppb
Salt (NaCl)	107	2,200	235,400	lbs	70.80	ppb
Starch LV	158	55	8,690	lbs	2.61	ppb
Soda ASH	18	55	990	lbs	0.30	ppb
Sodium Bicarbonate	14	55	770	lbs	0.23	ppb
H2S Scavenger	4	55	220	gal	0.25	gpb
O2 Scavenger	6	55	330	gal	0.38	gpb
Xanthan Gum	25	55	1,375	lbs	0.41	ppb

***CaCl2 was used for packer zone.**

Initial Volume	0.0 bbl
Total Volume Built	3,325.0 bbl
Volume received	1,419.0 bbl
Volume Returned	585.0 bbl
Total Volume Consumed	2,417 bbl
Volume Transfer to Next Section	1,742.0 bbl
Meter Drilled	0 m

Table 11-2: AZN-10 Drilling Fluid Type

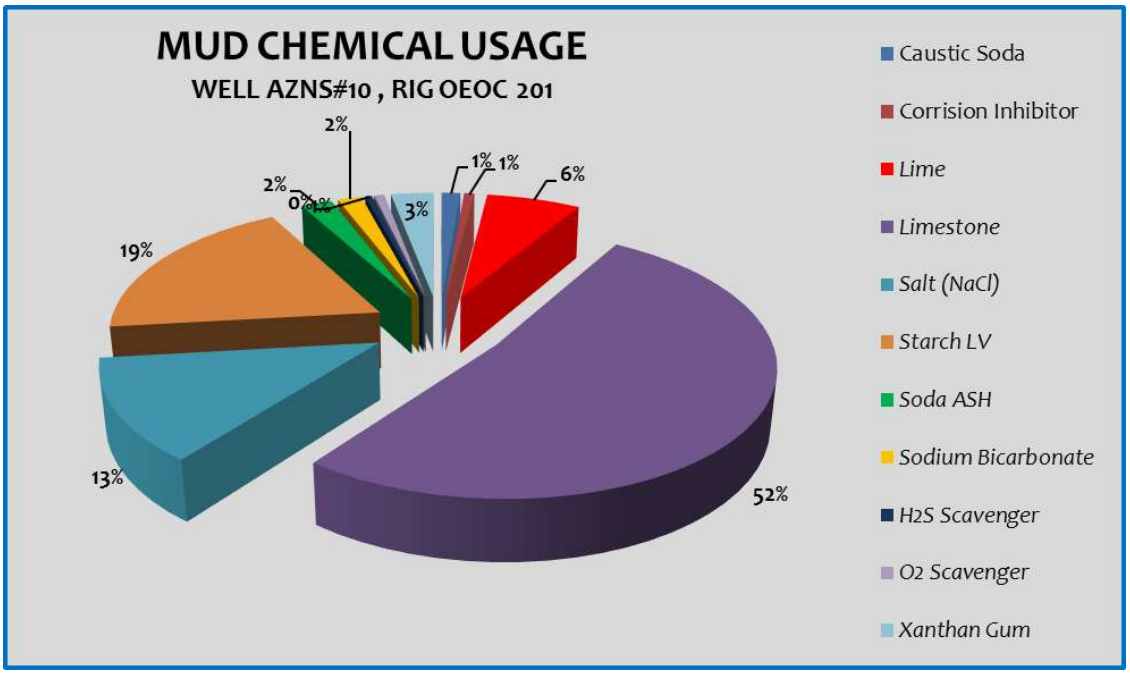


Figure 11-1: AZN-10 Mud Chemical Usage

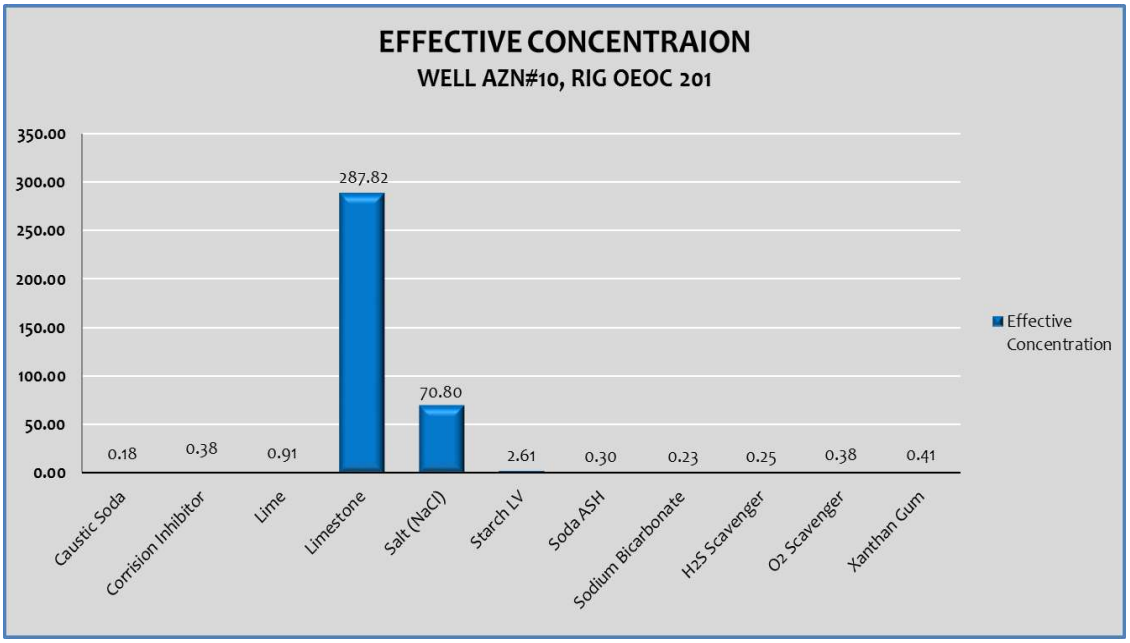


Figure 11-2: AZN-10 Effective Concentration

11.5 Volume Analysis

WELL AZN#10, RIG OEOC 201 VOLUME ANALYSIS		
Initial Volume	bbls	0.0
Received Volume	bbls	1,419.0
Volume Return	bbls	585.0
Volume Built	bbls	3,325.9
Surface Losses/Dumped	bbls	3,257.0
Formation Losses	bbls	0.0
Total Losses	bbls	3,257.0
Volume Transfer Out	bbls	903

Table 11-3: AZN-10 Volume Analysis

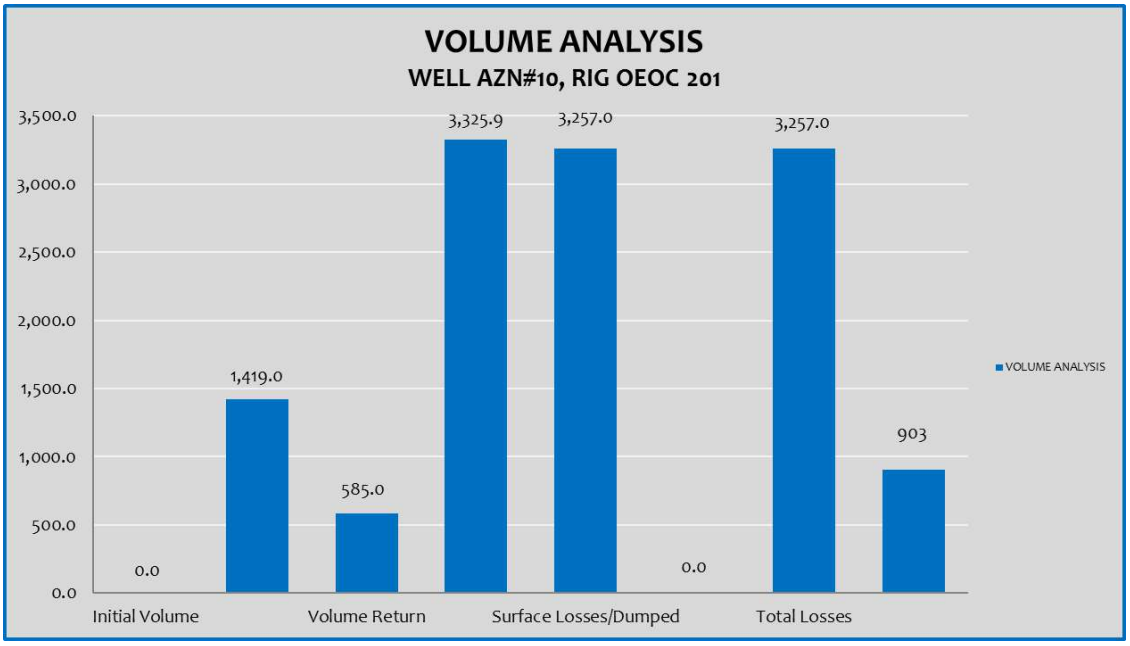


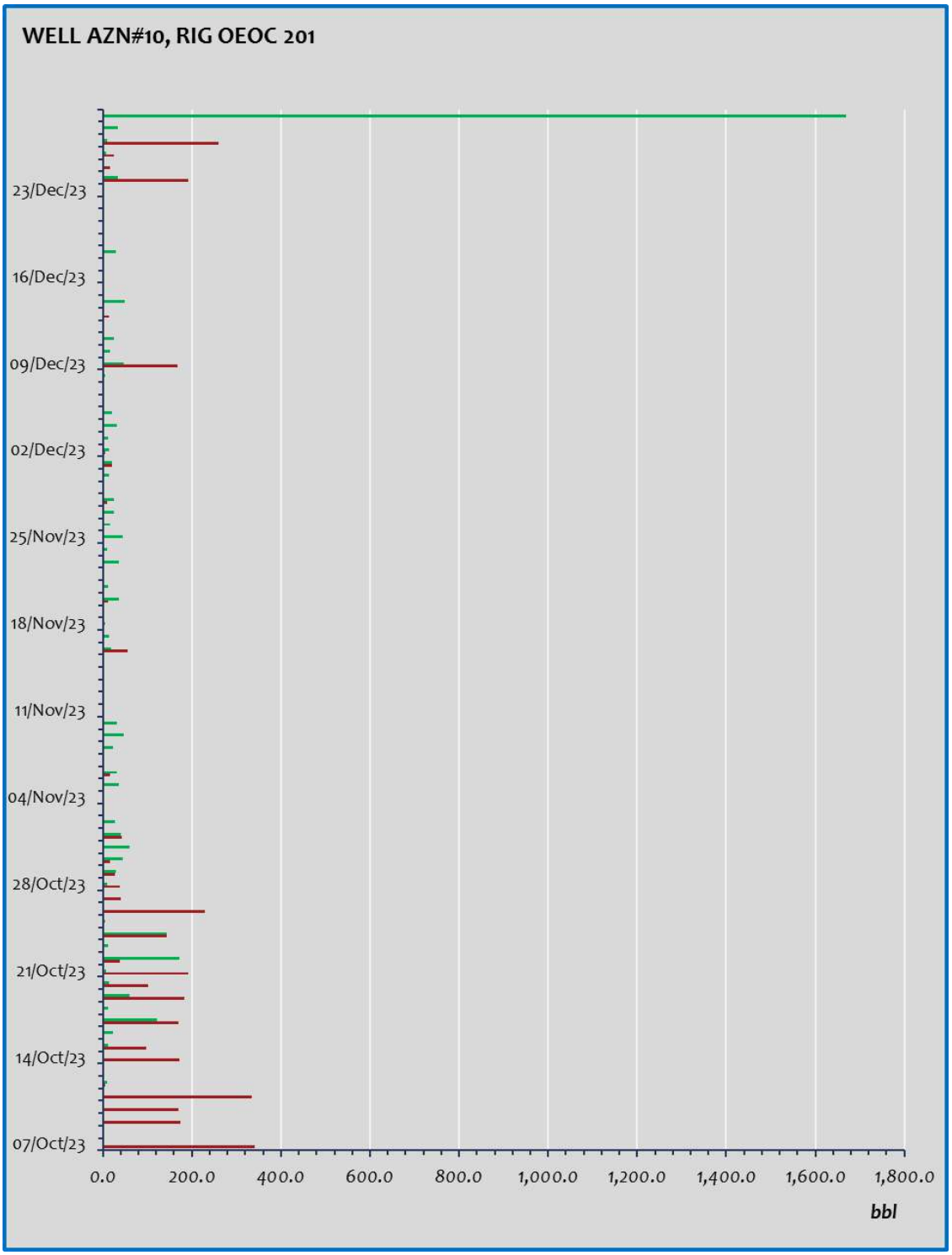
Figure 11-3: AZN-10 Volume Analysis

11.6 Daily volume Breakdown

Date/DMR.N		Volume of Built	Surface Losses	Formation Losses
7-Oct-23	01	341.6	0	0
8-Oct-23	02	0.0	0	0
9-Oct-23	03	173.2	0	0
10-Oct-23	04	170.9	0	0
11-Oct-23	05	335.6	0	0
12-Oct-23	06	4.7	9	0
13-Oct-23	07	0.0	2	0
14-Oct-23	08	172.8	2	0
15-Oct-23	09	97.8	12	0
16-Oct-23	10	0.0	23	0
17-Oct-23	11	170.5	121	0
18-Oct-23	12	0.0	12	0
19-Oct-23	13	182.2	60	0
20-Oct-23	14	102.1	13	0
21-Oct-23	15	191.7	7	0
22-Oct-23	16	38.6	171	0
23-Oct-23	17	0.1	12	0
24-Oct-23	18	142.7	143	0
25-Oct-23	19	0.0	5	0
26-Oct-23	20	229.1	0	0
27-Oct-23	21	40.9	0	0
28-Oct-23	22	37.2	10	0
29-Oct-23	23	27.9	30	0
30-Oct-23	24	16.8	45	0
31-Oct-23	25	0.0	60	0
1-Nov-23	26	42.8	40	0
2-Nov-23	27	0.6	28	0
3-Nov-23	28	0.0	0	0
4-Nov-23	29	0.5	0	0
5-Nov-23	30	0.0	35	0
6-Nov-23	31	17.0	32	0
7-Nov-23	32	0.0	0	0
8-Nov-23	33	0.0	23	0
9-Nov-23	34	1.2	47	0
10-Nov-23	35	0.6	31	0
11-Nov-23	36	0.0	0	0
12-Nov-23	37	0.0	0	0
13-Nov-23	38	0.0	0	0
14-Nov-23	39	0.0	0	0
15-Nov-23	40	0.0	0	0

16-Nov-23	41	56.6	18	0
17-Nov-23	42	0.0	15	0
18-Nov-23	43	0.3	5	0
19-Nov-23	44	0.0	0	0
20-Nov-23	45	11.6	35	0
21-Nov-23	46	0.0	12	0
22-Nov-23	47	0.4	0	0
23-Nov-23	48	0.1	37	0
24-Nov-23	49	0.0	10	0
25-Nov-23	50	0.2	45	0
26-Nov-23	51	0.0	16	0
27-Nov-23	52	0.1	26	0
28-Nov-23	53	9.5	26	0
29-Nov-23	54	0.0	0	0
30-Nov-23	55	0.0	15	0
1-Dec-23	56	21.1	20	0
2-Dec-23	57	4.7	15	0
3-Dec-23	58	0.1	12	0
4-Dec-23	59	0.0	32	0
5-Dec-23	60	0.3	20	0
6-Dec-23	61	0.0	0	0
7-Dec-23	62	0.3	0	0
8-Dec-23	63	0.0	6	0
9-Dec-23	64	167.2	46	0
10-Dec-23	65	0.6	16	0
11-Dec-23	66	0.0	24	0
12-Dec-23	67	0.0	0	0
13-Dec-23	68	14.6	0	0
14-Dec-23	69	3.9	50	0
15-Dec-23	70	0.0	0	0
16-Dec-23	71	0.0	0	0
17-Dec-23	72	0.0	0	0
18-Dec-23	73	0.0	30	0
19-Dec-23	74	0.0	0	0
20-Dec-23	75	0.0	0	0
21-Dec-23	76	0.0	0	0
22-Dec-23	77	0.0	0	0
23-Dec-23	78	0.0	0	0
24-Dec-23	79	191.5	33	0
25-Dec-23	80	16.8	0	0
26-Dec-23	81	26.0	8	0
27-Dec-23	82	260.9	10	0
28-Dec-23	83	0.0	34	0
29-Dec-23	84	0.0	1,668	0
Total (bbl)		Volume of Built	Surface Losses	Formation Losses
		3,325.9	3,257	0

Table 11-4: AZN-10 Daily Volume Breakdown


WELL AZN#10, RIG OEOC 201
23/Dec/23
16/Dec/23
09/Dec/23
02/Dec/23
25/Nov/23
18/Nov/23
11/Nov/23
04/Nov/23
28/Oct/23
21/Oct/23
14/Oct/23
07/Oct/23
0.0
200.0
400.0
600.0
800.0
1,000.0
1,200.0
1,400.0
1,600.0
1,800.0
bbl

11.7 Loss Distribution

Date/DMR.N		Volume of Built	Dump/Contamination	Shakers	Evaporation	Centrifuge	Trip	Mud cleaner	Pit Cleaning	Formation	Other
7-Oct-23	01	341.6	0	0	0	0	0	0	0	0	0
8-Oct-23	02	0.0	0	0	0	0	0	0	0	0	0
9-Oct-23	03	173.2	0	0	0	0	0	0	0	0	0
10-Oct-23	04	170.9	0	0	0	0	0	0	0	0	0
11-Oct-23	05	335.6	0	0	0	0	0	0	0	0	0
12-Oct-23	06	4.7	0	0	4	0	0	0	5	0	0
13-Oct-23	07	0.0	0	0	2	0	0	0	0	0	0
14-Oct-23	08	172.8	0	0	2	0	0	0	0	0	0
15-Oct-23	09	97.8	0	0	2	0	10	0	0	0	0
16-Oct-23	10	0.0	0	0	2	0	17	0	4	0	0
17-Oct-23	11	170.5	0	0	0	0	121	0	0	0	0
18-Oct-23	12	0.0	0	0	0	0	12	0	0	0	0
19-Oct-23	13	182.2	0	15	0	0	25	0	0	0	20
20-Oct-23	14	102.1	0	0	0	0	13	0	0	0	0
21-Oct-23	15	191.7	0	0	2	0	5	0	0	0	0
22-Oct-23	16	38.6	0	5	0	0	154	0	0	0	12
23-Oct-23	17	0.1	0	0	2	0	0	0	0	0	10
24-Oct-23	18	142.7	0	0	2	0	115	0	26	0	0
25-Oct-23	19	0.0	0	0	0	0	5	0	0	0	0
26-Oct-23	20	229.1	0	0	0	0	0	0	0	0	0
27-Oct-23	21	40.9	0	0	0	0	0	0	0	0	0
28-Oct-23	22	37.2	0	0	0	0	0	0	0	0	10
29-Oct-23	23	27.9	30	0	0	0	0	0	0	0	0
30-Oct-23	24	16.8	0	0	0	0	35	0	0	0	10
31-Oct-23	25	0.0	0	0	0	0	60	0	0	0	0
1-Nov-23	26	42.8	0	15	0	0	25	0	0	0	0
2-Nov-23	27	0.6	0	6	0	0	22	0	0	0	0
3-Nov-23	28	0.0	0	0	0	0	0	0	0	0	0
4-Nov-23	29	0.5	0	0	0	0	0	0	0	0	0
5-Nov-23	30	0.0	35	0	0	0	0	0	0	0	0
6-Nov-23	31	17.0	0	0	0	0	32	0	0	0	0
7-Nov-23	32	0.0	0	0	0	0	0	0	0	0	0
8-Nov-23	33	0.0	0	0	0	0	23	0	0	0	0
9-Nov-23	34	1.2	0	13	0	0	34	0	0	0	0
10-Nov-23	35	0.6	16	0	0	0	15	0	0	0	0
11-Nov-23	36	0.0	0	0	0	0	0	0	0	0	0
12-Nov-23	37	0.0	0	0	0	0	0	0	0	0	0
13-Nov-23	38	0.0	0	0	0	0	0	0	0	0	0
14-Nov-23	39	0.0	0	0	0	0	0	0	0	0	0
15-Nov-23	40	0.0	0	0	0	0	0	0	0	0	0

16-Nov-23	41	56.6	0	0	0	0	0	0	0	0	18
17-Nov-23	42	0.0	0	0	0	0	0	0	0	0	15
18-Nov-23	43	0.3	0	0	0	0	5	0	0	0	0
19-Nov-23	44	0.0	0	0	0	0	0	0	0	0	0
20-Nov-23	45	11.6	0	0	0	0	35	0	0	0	0
21-Nov-23	46	0.0	0	0	0	0	12	0	0	0	0
22-Nov-23	47	0.4	0	0	0	0	0	0	0	0	0
23-Nov-23	48	0.1	0	0	0	0	28	0	0	0	9
24-Nov-23	49	0.0	0	0	0	0	0	0	0	0	10
25-Nov-23	50	0.2	0	0	0	0	20	0	0	0	25
26-Nov-23	51	0.0	0	0	0	0	16	0	0	0	0
27-Nov-23	52	0.1	0	0	0	0	26	0	0	0	0
28-Nov-23	53	9.5	0	0	0	0	26	0	0	0	0
29-Nov-23	54	0.0	0	0	0	0	0	0	0	0	0
30-Nov-23	55	0.0	15	0	0	0	0	0	0	0	0
1-Dec-23	56	21.1	0	0	0	0	20	0	0	0	0
2-Dec-23	57	4.7	0	0	0	0	0	0	0	0	15
3-Dec-23	58	0.1	0	0	0	0	0	0	0	0	12
4-Dec-23	59	0.0	0	0	0	0	24	0	0	0	8
5-Dec-23	60	0.3	0	0	0	0	0	0	20	0	0
6-Dec-23	61	0.0	0	0	0	0	0	0	0	0	0
7-Dec-23	62	0.3	0	0	0	0	0	0	0	0	0
8-Dec-23	63	0.0	0	0	0	0	0	0	0	0	6
9-Dec-23	64	167.2	0	0	0	0	0	0	40	0	6
10-Dec-23	65	0.6	0	0	0	0	0	0	16	0	0
11-Dec-23	66	0.0	0	0	0	0	24	0	0	0	0
12-Dec-23	67	0.0	0	0	0	0	0	0	0	0	0
13-Dec-23	68	14.6	0	0	0	0	0	0	0	0	0
14-Dec-23	69	3.9	50	0	0	0	0	0	0	0	0
15-Dec-23	70	0.0	0	0	0	0	0	0	0	0	0
16-Dec-23	71	0.0	0	0	0	0	0	0	0	0	0
17-Dec-23	72	0.0	0	0	0	0	0	0	0	0	0
18-Dec-23	73	0.0	30	0	0	0	0	0	0	0	0
19-Dec-23	74	0.0	0	0	0	0	0	0	0	0	0
20-Dec-23	75	0.0	0	0	0	0	0	0	0	0	0
21-Dec-23	76	0.0	0	0	0	0	0	0	0	0	0
22-Dec-23	77	0.0	0	0	0	0	0	0	0	0	0
23-Dec-23	78	0.0	0	0	0	0	0	0	0	0	0
24-Dec-23	79	191.5	25	0	0	0	0	0	0	0	8
25-Dec-23	80	16.8	0	0	0	0	0	0	0	0	0
26-Dec-23	81	26.0	0	0	0	0	0	0	8	0	0
27-Dec-23	82	260.9	10	0	0	0	0	0	0	0	0
28-Dec-23	83	0.0	34	0	0	0	0	0	0	0	0
29-Dec-23	84	0.0	1,668	0	0	0	0	0	0	0	0
Total (bbl)		Volume of Built	Dump/Contaminate	Shakers	Evaporation	Centrifuge	Trip	Mud cleaner	Pit Cleaning	Formation	Other
		3,325.9	1,913	54	18	0	959	0	119	0	194.0

Table 11-5: AZN-10 Loss Distribution

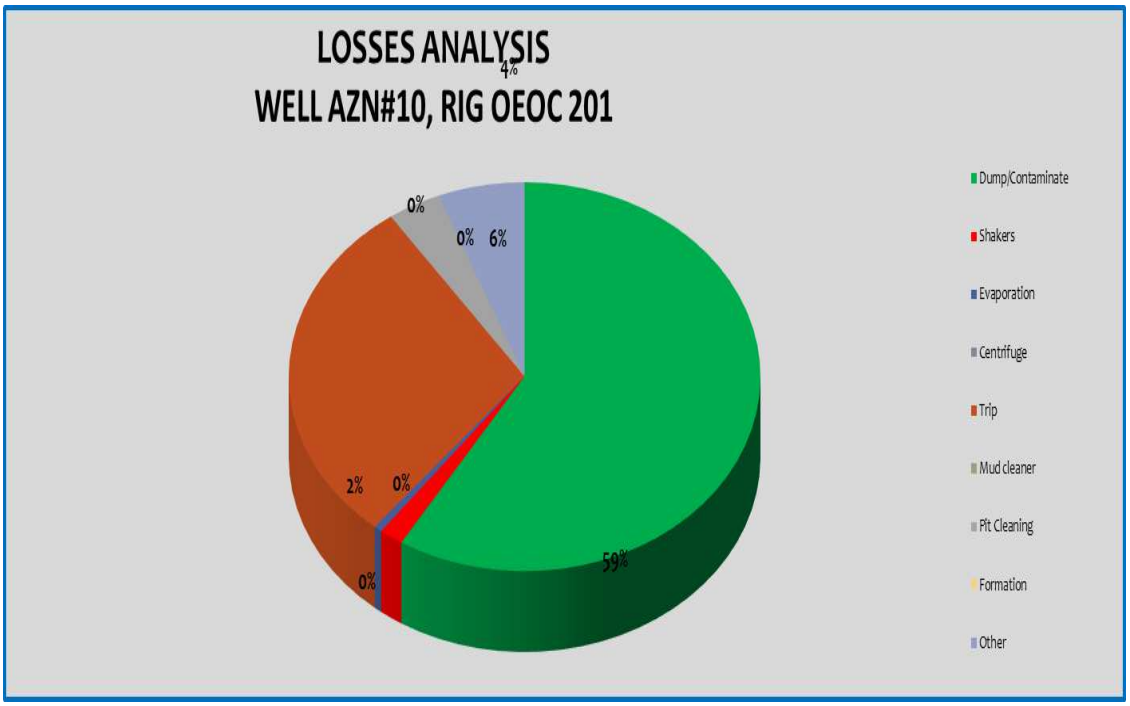


Figure 11-4: AZN-10 Loss Analysis

11.8 Solid Control Disscusion

Rig OEOC 201 has 3 shale shakers were dressed with:

Shaker#1: 120 Mesh Screen

Shaker#2: 120 Mesh Screen

Shaker#3: 120 Mesh Screen

WELL AZN#10, RIG OEOC 201	
Solid Control Equipment Working Hr.s	
Shaker # 01	62
Shaker # 02	330
Shaker # 03	28

Table 11-6: AZN-10 Solid Control

11.9 Completion Fluid Information

Item	Description		QTY (bbl)
1	Send to Well Site	$CaBr_2$ (106^{pcf})	720
2	Well Displacement	$CaBr_2$ (104^{pcf})	815
3	Used in Com. Fluid	$CaBr_2$ (104^{pcf})	695
4	Used in Com. Fluid	$CaCl_2$	120

Table 11-7: AZN-10 Completion Fluid Information

12. List of Contractors

Item	Service Item	Contractor
1	Drilling Rig, Mobilization and Demobilization the Rig, Rig Move Between Well Pads	OEOC
2	Drilling Fluids and Completion Fluids	VIRA
3	Waste Management	Pars Haffaran Arvand
4	Electrical Logging and Perforation	PANIZ
5	Mud Logging	MAPSA KISH
6	Directional Drilling (MWD)	VIRA
7	H2S Monitoring and Control	MAPSA KISH BONYAN IMEN
8	Casing and Tubing Running	OEOC
9	Cementing	SPAD ENERGY & Mehran
10	Dry Test, DST and RTTS	OEOC
11	Slick Line	OEOC
12	Stimulation (Coiled Tubing and Acidizing and Nitrogen Lifting)	OEOC
13	Surface Well Testing	OEOC
14	Liner Hanger	PARS POOLAD
15	Well Head	OEOC

Table 12-1: AZN-10 Contractors Involved

13. Appendix

13.1 DDR Summary

Item	Date	Remark
1	08 – Oct	Prepare & make tests on well control and mud Circ. Equ, Perform test on different BOP sections several time, BOP tests got success. P/U and M/U new BHA, Cont. RIH to 29^{m}.M/U and RIH new BHA, Strg got plug @138m, try ro remove plug, no success, POOH and remove the plug, RIH with same BHA (without jar due to failing) to 1120m, Start W&R/D to 1150m and D/O the settelment
2	09 – Oct	Cont. W&R/D to 1355mt the settelment , while reciprocate each STD . Tag the TOC @ 1381m . D/out hard CMT to 1405mt . due to low Rate (less than 4mph) in D/out CMT, decided pooh to surf. As chenge Bit .POOH to 910^{m}.
3	10 – Oct	POOH to surf. then M/Up & RIH with 8-1/2" used PDC bit to cont. D/out CMT plug #1 F/1405 to 1465^{m}.
4	11 – Oct	Cont. D/O Med to Hard CMT F/1465 to 1500 m . RIH w/ Wash Dn to 1542m w/ NG. Slip & Cut Drlg line while Cont. RIH w/ Wash Dn f/ 1542 to 2090^{m}.
5	12 – Oct	Cont. RIH with 8-1/2" D/O CMT BHA f/2090 to 3356m.
6	13 – Oct	Cont. RIH with 8-1/2" D/O CMT BHA to TOC @3629^{m}, Commence D/O CMT f/3629 to L.Lap @ 3706^{m}, Start POOH. Stop operation several times in D/O due to Elc. problem.
7	14 – Oct	Cont. POOH D/O BHA to surface, P/U & M/U 5-7/8" R. bit & RIH to 2612^{m}.
8	15 – Oct	Cont. RIH 5-7/8" BHA to 4663^{m} & took weight, Displace the well w/102^{pcf} mud, POOH to 3423^{m} due to failure of TDS torque wrench.
9	16 – Oct	POOH to surface, Change the Bit w/ C.C Bit, RIH to 3978^{m}.

10	17 – Oct	RIH w/5-7/8" C.C Bit to 4591m, prepare to perform CMT plug, pump 4bbl spacer ahead- 35/5bbl CMT slurry, 2bbl spacer behind, diplace w/ 196bbl WBM, POOH 12Stds, try to string, No success, POOH to 3676m, try to flush, no success, Start POOH.
11	18 – Oct	Cont. POOH C.C bit to surface while B/O & L/D 3-1/2" DPs, bit sub & C.C bit were plugged, unplugged 9 Jts & no success to unplug 64 Jts on rack, Unplugged C.C bit & bit sub.
12	19 – Oct	Cont. RIH w/ 5-7/8" C.C Bit f/ 3364 to 4158m, Took Wt @ 4158m, Wash down to 4167m, W&R to 4188m, D/O Med-Hard CMT to 4214m, D/O Hard CMT to 4250m, CBU, POOH w/ 5-7/8" C.C Bit f/ 4250 to 2050m till MN, NL-NG.
13	20 – Oct	Cont POOH to surface, RIH w/ 5-7/8" R.Bit & 7" x 9-5/8" Tandem scraper BHA to 3650^{m}.
14	21 – Oct	Cont POOH to surface, RIH w/ 5-7/8" R.Bit & 7" x 9-5/8" Tandem scraper BHA to 3650^{m}.
15	22 – Oct	Cont. POOH w/ 7" × 9-5/8" tandem scraper f/ 3673m to surface, while two times try to unplug string by apply pressure w/ No-Succ & Work on EZY TQ. Observe string plugged by a bolt (belong to TDS stabbing guide) in 4-3/4" B.Sub.
16	23 – Oct	P/U & M/U 7" x 9-5/8" Tandem scraper BHA to 600m, CBU, Cont. RIH to 2691m, CBU, Cont. RIH to 4150^{m}.
17	24 – Oct	Cont. RIH to 4250m, spot 70bbl CaCl2 & displace w/ 203bbl WBM, Start POOH to Surf.
18	25 – Oct	Cont. POOH to Surf. , L/D the BHA, Prepare to perform logging job, RIH Run#1&2
19	26 – Oct	RIH Run#3 and Log up to surface, the data not confirmed by witness, prepare for RIH Run#4 to 1200^{m}.
20	27 – Oct	Cont. logging job as per program, L/D logging Equ. & P/U and M/U Dry Test Assy., RIH to 159^{m}.

21	28 – Oct	Cont. RIH w/ 9-5/8" Dry Test BHA f/890 to 3058^{m}.
22	29 – Oct	Cont. RIH to perform Dry Test as per program, Balance the mud.
23	30 – Oct	Cont. Balance the mud, Unset the Packer and POOH f/ 3691 to 76^{m}.
24	31 – Oct	Cont. POOH & L/D 9-5/8" Dry Test BHA, Read the Memory Gauge Info, Obs. pressure increasing in Close situation in the Memory Gauge data, perform BOP test, wait on making decision meanwhile RIH Secure BHA to 500m, POOH to surface, P/U & M/U P.Mill BHA to dress 7" L.LNR
25	01 – Nov	CBU in TOL, Dress the 7" L.LNR, CBU, POOH f/ 3708 to 743^{m}.
26	02 – Nov	Cont. POOH Polish Mill Assy. to Surface, Check the LG mark on Polish Mill nose, Ok, P/U & M/U S.N PKR and RIH S.N PKR to 3547^{m}.
27	03 – Nov	Cont. RIH f/ 3547m to TOL and set that on 7" L.LNR, POOH f/ 3705 to Surf. & L/D setting tools then P/U and M/U Dry Test Assy. and RIH to 235^{m}.
28	04 – Nov	Cont. RIH to 724m, due to miss the Rabbit, POOH to 22m, retrieve the Rabbit, Cont. RIH f/22 to 2371^{m}.
29	05 – Nov	Cont. RIH f/ 2371 to 3687m and perform Dry Test as per program, Balance the mud, Unset the Packer
30	06 – Nov	POOH to surface and check the Memory Gauge Info and analyse the fluid sample, wait on client.
31	07 – Nov	Wait on instructions for the next operation, POOH securing BHA and RIH w/ 7-5/16" Polish Assy. to 2532^{m}.
32	08 – Nov	Cont. RIH w/ 7-5/16" Polish Assy. and perform CBU on top of 8-1/4" PBR, Polish inside PBR and make LG mark on the nose of P.Mill, POOH to Surf. and check the LG mark, Ok, P/U and M/U 9-5/8" Scrapping Assy. and RIH to 441^{m}.

33	09 – Nov	Cont. RIH w/ 9-5/8" Scrapping BHA to top of S.N PKR PBR and Scrap inside the 9-5/8" CSG w/ Wash Dn in the target interval, POOH 9-5/8" Scrapping BHA to 2578^{m}.
34	10 – Nov	Cont. POOH 9-5/8" Scrapping BHA f/ 2578m to surface, P/U & M/U new Securing BHA and RIH to 300m, wait on Rec. Tie-back facilities.
35	11 – Nov	CONT . Wait On Rec. Tie-back facilities & Slow Circulation @ 315 m W/ NL-NG.
36	12 – Nov	CONT . Wait On Rec. Tie-back facilities & Slow Circulation @ 315 m W/ NL-NG.
37	13 – Nov	CONT . Wait On Rec. Tie-back facilities & Slow Circulation @ 315 m W/ NL-NG.
38	14 – Nov	CONT . Wait On Rec. Tie-back facilities & Slow Circulation @ 315 m W/ NL-NG.
39	15 – Nov	Cont. Slow CIRC & RIH From 315M To 3012m While Fill Up STRG Every 15 STDs .
40	16 – Nov	CONT. RIH F/3012 M TO 3671.5 M WHILE FILL UP STRG EVERY 15 STDs. CONT. RIH F/ 3671.5 M TO 3699 M WITH W/REAM STD #128. TAKE WEIGHT @3699 @3691 @ 3683 M. C.B.U WHILE CIRCULATION DUE TO BALANCE MUD WEIGHT .
41	17 – Nov	Cont. Circulation Mud @ 3694 m W/ NL-NG.
42	18 – Nov	"Wait on REC Tie-back Facilities (Wait For CMT Service & Program). POOH F/3703m To 313 m While F/U The Well Across X-Well Head.
43	19 – Nov	Rec. Cmt Service, POOH f/ 313m to Surf. & P/U & M/U 7" Tie-back Assy. and RIH to 1069^{m}.
44	20 – Nov	Cont. RIH w/ 7" Tie-back Assy. f/ 1069 to Depth plan, preapre for set 7" Tie-back.

45	21 – Nov	perfom CMT job and set the S.N PKR, POOH Setting tools to surface, perform BOP test as per NIOC policy.
46	22 – Nov	P/U & M/U new 8-1-2" BHA and RIH to clean L.Lap [Soft CMT: 3442 - 3467m,Med. to Hard CMT: 3467 - 3527m], New 7" L.Lap: 3527^{m}.
47	23 – Nov	Perform CBU, Then POOH 8-1/2" BHA to 997m, Slip & Cut, Cont. POOH f/ 997m to surface, P/U & M/U new 6" BHA and RIH to 1633^{m}.
48	24 – Nov	Cont. RIH w/ New 6" BHA to L.Lap Depth,3527mt, Start D/O Hard CMT f/ 3527 to 3676m, While D/O Packoff Bushing, Wiper & Pump Down Plug and LDC.
49	25 – Nov	Cont. D/O Hard CMT W/ 6" BHA From 3676 to 3704 m .Pumped Hi-Vis pill & Circulation BOTTOMs UP . POOH 6" D/O BHA to Surf.
50	26 – Nov	P/U & M/U New 5-7/8" BHA and RIH to 3703m, took weight, Start D/O CMT f/ 3703 to 3711m, Cont. RIH f/3711 to 4242m on elevator without any drag, Start POOH F/ 4242 to 2784m on elevator.
51	27 – Nov	Cont. POOH f/ 2784m to surface, L/D BHA, P/U & M/U 9-5/8" * 7" Tandem Scraper BHA and RIH to 2347^{m}.
52	28 – Nov	RIH 9-5/8" x 7" Tandem Scraper BHA f/ 2347 to 3700m While Scrape inside 9-5/8" CSG & 7" LNR. CBU. POOH to 3523m. Spot Hi-Vis Pill above 7" Tie-Back L.Lap. POOH to Surface. M/U 9- 5/8" Dry Test Assy. w/ NL-NG.
53	29 – Nov	Cont. M/U 9-5/8" Dry Test Assy & RIH to 2226m. While F/U String w/ Cusion Water 80 Stds to apply 2000 psi drive force on 7" L.Lap.
54	30 – Nov	Cont. M/U 9-5/8" Dry Test Assy & RIH to 3500m. While F/U String w/ Cusion Water 80 Stds to apply 2000 psi drive force on 7" L.Lap. Perform Dry Test Operation as per program.
55	01 – Dec	Cont. POOH 9-5/8" Dty Test BHA to Surf.

56	02 – Dec	Cont. POOH 9-5/8" Dty Test BHA to Surf. The analyse the data, RIH W/R Tools as Dummy Run to 3527m, Try to RIH in 7" LNR, No success, POOH to 1880^{m}.
57	03 – Dec	Cont. POOH W/L Tools, M/U another Gun in the same W/L Tools and try to RIH and pass 7" L.Lap, No success, POOH to surface and P/U and M/U Magnet BHA and RIH to 3276m
58	04 – Dec	Cont. RIH w/ Magnet BHA f/ 3276 to 3700m, Break Circ. Then POOH f/ 3700 to 3527m and Work on Junk Sub in interval 3535-3519m by reciprocate the Strg, CBU, POOH to Surface, Recover 1lb metal debris, prepare for RIH W/L Tools..
59	05 – Dec	RIH Dummy Run to 4120m, POOH to surface, RIH GR-CCL and correlate the depth of perforation, POOH to surface, RIH w/ 1st perforation tools.
60	06 – Dec	POOH W/L Tools as #1 Perf. to surface. RIH W/L Tools as #2 Perf. to 4082m, Perform perforation (4082-4077m), POOH W/L Tools as #2 Perf. to surface, RIH W/L Tools as #3 Perf. to 4077m, Perform perforation (4077-4074m), POOH W/L Tools as #3 Perf. to 100m, wait on day light.
61	07 – Dec	POOH W/L Tools as #3 Perf. to surface. RIH W/L Tools as #4 Perf. to 4057m, Perform perforation (4057 - 4051m), POOH W/L Tools as #4 Perf. to surface, RIH W/L Tools as #5 Perf. to 4051m, Perform perforation (4051 - 4048m), POOH W/L Tools as #5 Perf. to 100m, wait on day light.
62	08 – Dec	POOH W/L Tools as #5 Perf. to surface. RIH W/L Tools as #6 Perf. to 4048m, Perform perforation (4048 - 4042m), POOH W/L Tools as #6 Perf. to surface, RIH W/L Tools as #7 Perf. to 4040.6m, Perform perforation (4040.6 - 4036.6m), POOH W/L Tools as #7 Perf. to 100m, wait on day lightPOOH W/L Tools as #3 Perf. to surface. RIH W/L Tools as #4 Perf. to 4057m, Perform perforation (4057 - 4051m), POOH W/L Tools as #4 Perf. to surface, RIH W/L Tools as #5 Perf. to 4051m, Perform perforation (4051 - 4048m), POOH W/L Tools as #5 Perf. to 100m, wait on day light

63	09 – Dec	POOH W/L Tools as #7 Perf. to surface. RIH W/L Tools as #8 Perf. to 4036.6m, Perform perforation (4036.6 - 4034m), POOH W/L Tools as #8 Perf. to surf., RIH W/L Tools as #9 Perf. to 4040.6m, Perform perforation (4040.6 - 4036.6m), POOH W/L Tools as #9 Perf. to surface, RIH w/ Mill
64	10 – Dec	Cont. RIH w/ Milling BHA to 4077m on elevator, Took Weight, Ream Dn f/ 4077 to 4101m, Resume RIH w/ Milling BHA f/ 4101 to 4177m on elevator, Took Weight, Ream Dn f/ 4177 to 4244m, Took Weight @ 4244m, POOH to 3700m, Perform CBU and Cont. POOH to 3100^{m}.
65	11 – Dec	Cont. POOH 5-7/8" Milling BHA f/3100 to surface, Perform Logging job (GR-CCL), R/D W/L Tools and change the BOP Top Rams, perform Press. Test on the same, RIH w/ 6-1/2" DCs + 5" HWDP + 5" DPs to 103^{m}.
66	12 – Dec	RIH to 469m and L/D Extra tubulars, P/U & M/U Scraper BHA and RIH to 1250, while L/D Extra Tubulars .
67	13 – Dec	'RIH w/ Scraper BHA to 4087m, Scrap PKR Depth several time, Pump Hi-Vis & CBU till clean shakers, balance 40bbl CaCl2 Brine to 3700m, Prepare CaBr2 Comp. Fluid [104pcf] and POOH f/ 4087 to 3921m and L/D extra tubulars jt by jt.
68	14 – Dec	Complete Disp. Well W/ CaBr2 Comp. Fluid [104pcf] and POOH to 2308 & L/D extra tubulars jt by jt .
69	15 – Dec	Cont. POOH Scraper BHA F/ 2308m To Surface on Elevator, While L/D extra tubulars jt by jt , While Fill Up The Well W/ 104 pcf Comp. fluid by T.Tank, W/ NL-NG.
70	16 – Dec	" R/U the Comp. Eqpt. & Running Comp. Strg to 262.4 as per tally
71	17 – Dec	Cont. Running Comp. Strg to 1276 m as per tally.
72	18 – Dec	Cont. Running Comp. Strg F/ 1276 m to 2528 m as per tally.
73	19 – Dec	Cont. R/U the Comp. Eqpt. & Running Comp. Strg to 3658 m, W/ NL-NG . Perfom Pressuer Test The Line and Hold For

		15min, Then Connect to Comp. Strg and Perform 7000psi Stepwise on Comp. Strg and Hold For 30min in Presence of Client Witness, Ok.
74	20 – Dec	Cont. Running Comp. Strg to 3670 m, W/ NL-NG . R/U Slick line and RIH With Slick line as Run#1 : Install 3.437" Protection Sleeve of SSSV , Run#2: Retrieve 2.75"WRB-2 & Check Valve From SSD , Run#3: Set 2.75" WRB-2 Check Valve in Landing Nipple.
75	21 – Dec	R/U Slick Line, Try To Set Prong In NO-GO Plug In Several Times "No SUCC. Work On Set Prong In NO-GO Plug . Final Set Protection Sleeve in TRSSSV, Retrieve Plug & Prong From NO GO Landing Nipple .Final COMP. String Press. Test "OK". Set TBG HGR "OK". L/D Slick Line Equip Till M.N.
76	22 – Dec	Cont. Slick Line Operation. (R/U Slick Line & Try Several Times To Pass SSD W/ Check Valve in SSD , NO Succ).
77	23 – Dec	Cont. R/U Slick Line & Try several times to pass SSD W/ Check Valve, No Succ., While Left The Check Valve and Ret. That, Redress The Check Valve and RIH to Try Again . Displace The Well in Progress.
78	24 – Dec	Displaced The Well With Mud 104 Pcf & Prepare for New Slick Line Job.
79	25 – Dec	SLICK LINE JOB.
80	26 – Dec	RIH w/2-3/4" plug & set into No-Go, POOH to surface & RIH w/equalizing prong & set into 2-3/4" plug, pressure test annulus w/700^{psi} without any succeed, RIH w/ 2-3/4" Check valve to set in SSD, POOH to surface, Press. Test Comp. Strg w/ 5000psi, RIH to Ret. Check valve, POOH to surface. Prepare Comp. fluid, while monitor the Well and Break Circ. And uniform the mud. 2 Hrs. waiting for Spad Energy due to failure in pump truck & 2 Hrs. NPT for OEOC Rig 201 due to lack of crane.
81	27 – Dec	Monitor the well. Displace well with COMP FLUID $CaBr_2$.

82	28 – Dec	Displace the Ann, RIH w/ Shifting tools & Close the SSD, Perform SSD negative, Perform Press test string, Ret. Protection sleeve, Land TBG HGR, Perform final Press test, Perform Press test Ann., Press test Tie-Down screw, Install 4"TWCV in TBG HGR & Test same.
83	29 – Dec	N/D 13-5/8" 10K BOP stack. N/U XMT & Related P/T; OK. Released the Rig 201 OEOC @ 1402-10-08, 23:59.

Table 13-1: AZN-10 Daily Drilling Report

14. Correction & Revision

Drilling Engineering							
1	**Well Information:** Target Coordinate با برنامه تعمیری تایید شده مغایرت دارد.	13	CM	• اصلاح گردید.			
2	**Bottom Hole Assembly Record:** • BHA No#1 با گزارش روزانه پیمان کار مغایرت دارد اصلاح شود گزارش پیمانکار طول کلی 164.17 گزارش شده در صورتی که در گزارش پایانی صفحه 46 ،طول کلی 155.29 اعلام شده است. • BHA No#2 با گزارش روزانه پیمان کار مغایرت دارد اصلاح شود ، گزارش پیمانکار طول کلی 164.21 گزارش شده در صورتی که در گزارش پایانی صفحه 46 ،طول کلی 155.29 اعلام شده است. (طول DC اشتباه میباشد) • تمامی اطلاعات وارد شده از صفحه 45 تا 63 مورد بررسی مجدد قرار بگیرد	45 تا 63	CM	• اصلاح گردید.			

Completion							
3	Completion Sketch عمق های اعلام شده ابزار رشته تکمیلی با تالی مغایرت دارد اصلاح گردد.	110	CM	• اصلاح گردید.			
4	Wellhead Schematic عمق های اعلام شده ابزار رشته تکمیلی با تالی مغایرت دارد اصلاح گردد. عمق لوله آستری "5 به شماتیک اضافه گردد.	130	CM	• اصلاح گردید. • اضافه گردید.			
5	Well montage به گزارش پایانی اضافه گردد.		CM	• با توجه به عدم حفاری حفره و به طبع آن عدم راندن لوله جداری و سیمان-کاری و همچنین عدم وجود مشکلات حفاری، تهیه Well Montage جهت چاه‌های تعمیراتی امکان‌پذیر نمی‌باشد. لذا شماتیک چاه قبل و بعد از تعمیرات قرار داده شده و مابقی اطلاعات مورد نیاز مانند اطلاعات سیال استفاده شده و نمودارگیری در			

				بخش‌های جداگانه لحاظ گردیده است.			
Fluid							
6	احجام مصرفی کلسیم برماید جدید و استفاده شده (منتقله به محل) در جدولی مجزا قید گردد. در این نسخه حجم مصرفی کلسیم برماید لحاظ نشده است.		CM	• اضافه گردید.			